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Faculty of Arts and Social Sciences

Redeeming the Pledge of Electoral Equity?

A Study of the Roads to and Effects of the Divisor-Based Biproportional Apportionment Method in Switzerland

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by
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Abstract

Electoral systems are the backbone of democratic governance. They structure political competition, distribute formal power, and influence voter behavior. The legal design of these systems is therefore central to the stability and functioning of democracy. However, electoral systems are not just mechanical frameworks; they are dynamic arenas where high democratic ideals often collide with the pragmatic interests of political actors, making them a captivating subject for research.

In Switzerland, the introduction of the biproportional apportionment method in cantonal parliamentary elections marks a significant innovation. This method, designed to enhance vote-seat proportionality while preserving existing constituency structures, has been implemented in nine Swiss cantons since 2005. Despite its widespread adoption, there has been a notable absence of research on its broader political implications. This study addresses this gap by investigating the impact of biproportional apportionment on the political landscapes of Swiss cantons. It examines whether the anticipated increase in proportionality has affected key political structures, including party competition, voter turnout, and the overall party landscape. Furthermore, the study delves into the political processes that shaped the reform, analyzing how the positions, alliances, and conflicts among political actors influenced the final design.

The findings reveal that the biproportional apportionment reform has indeed led to a significant improvement in proportionality of parliamentary elections. However, its impact on other political dimensions has been limited. While the reform succeeded in reducing electoral misrepresentation of parties, it did not produce a marked increase in voter turnout or significantly alter the competitive dynamics among parties as initially anticipated. Furthermore, the reform did not result in a proliferation of legislative parties as widely assumed. The analysis of the political processes behind the reform adoptions shows that the final reform designs were heavily influenced by the patterns of conflict during decision-making and often lead to compromises that mitigated the full reform potential.

Key words: electoral systems ▪ proportional representation ▪ electoral reform ▪ biproportional apportionment ▪ divisor method ▪ counterfactual estimation

To my mother Barbara and my stepfather Roland
and in memory of my friend Mikael

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This thesis is dedicated to my friend Mikael, who passed away on August 10, 2024. Rest in peace.

Zurich, August 2024
Sandro Lüscher

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Index of abbreviations [1]

AG	Aargau
AI	Appenzell Innerrhoden
a/o	and/or
AR	Appenzell Ausserrhoden
ATT	Average Treatment Effect on the Treated
BDP	Conservative Democratic Party of Switzerland
BE	Bern
BL	Basel Landschaft
BS	Basel Stadt
CSP	Christian Social People's Party
CVP	Christian Democratic People's Party of Switzerland
DN	Democratic Nidwalden
DSP	Democratic Social Party
EDU	Federal Democratic Union of Switzerland
EVP	Evangelical People's Party of Switzerland
FA	Federal Assembly
FDP	Free Democratic Party of Switzerland
FGA	Feminist-Green-Alternative Groups
FPS	Freedom Party of Switzerland
FR	Fribourg
FSC	Federal Supreme Court
FSO	Federal Statistical Office
GE	Geneva
GL	Glarus
GLP	Green Liberal Party
GPS	Green Party of Switzerland
GR	Graubünden
JU	Jura
LDU	Alliance of Independents
LPS	Liberal Party of Switzerland
LU	Lucerne
MR	Majoritarian Representation
NE	Neuchâtel
NW	Nidwalden
OW	Obwalden
PDA	Swiss Party of Labour
POCH	Progressive Organizations of Switzerland
PR	Proportional Representation
Rest	Residual Group
SD	Swiss Democrats
SG	St. Gallen
SH	Schaffhausen
SO	Solothurn
SVP	Swiss People's Party
SZ	Schwyz
TG	Thurgau
TI	Ticino
UR	Uri
VD	Vaud
VS	Valais
w/l	winner/loser
ZG	Zug
ZH	Zurich

[1] Party labels are explained in more detail in Table 39 in the appendix.

> "It is (…) necessary that the institutions be so constructed so as not to be biased in favor of any voter (or group of voters) or in favor of any alternative. We will find that this situation is the exception rather than the rule among countries that are commonly referred to as democracies." (McGann 2006: 5)

0. Introduction [2,3]

"Periodic, universal, free, equal, and secret" elections form the bedrock of liberal democracy (Kley 2016). Democratic elections are rule-based procedures that legitimize political authority, structure political competition and determine "who gets what and how" (Charvát 2023: 1). The "how" falls into the domain of electoral systems, which act as the manual for the election process. They are grounded in specific legal principles and define the mechanism by which formal political power is distributed among the competitors. They are "the cogs that keep the wheels of democracy properly functioning", as Farrell (2001: 2) allegorically put it. However, their impact extends beyond mere technicalities. They also have implications for the parties and the party system, for the formation of government and for voter behaviour (Cincea 2013: 176). This relevance explains why electoral systems are a focal point of intense political debate. As Sartori (1994: 27) wryly observed: "(...) if electoral systems were of little consequence why on earth would politicians fight so bitterly about them? And why would reformers fight so persistently to have them changed?"

0.1 Background

Modern parliamentary elections emphasize proportionality, grounded in the principles of electoral and political equity (Nohlen 2014: 43-45). This principle dictates that legislative seats should be distributed in proportion to the share of votes each party receives (Lijphart 1994, Cincea 2013, Nohlen 2014). Proportional representation is a multilayered concept with a hierarchical structure: votes can be counted, weighted, and translated into seats with varying degrees of equity. While all apportionment methods ensure equal vote counting as a fundamental prerequisite of representative systems ("one man, one vote"), only PR-based systems achieve higher levels of electoral equity, as illustrated in Table 1 (Kley 2016).

Equity in…	…count value	… voting power	… electoral success
MR/Plurality	Yes	No	No
Unidimensional PR	Yes	Yes	No
Bidimensional PR	Yes	Yes	Yes

Table 1: Representation formulae and the different layers of electoral equity.

Typically, electoral areas are subdivided into smaller districts to bring representatives closer to voters (Ramírez et al. 2008). However, this subdivision creates barriers, as parties have to obtain a certain vote share to be included in seat allocation. Votes for parties falling below these district-induced thresholds are effectively wasted, undermining proportional representation and conflicting with democracy's promise of equal political representation. This challenge highlights a fundamental conflict between two competing concepts of representation: the representation of parties versus the representation of territories (Pukelsheim 2017: 259).

The quest to resolve the dilemma of balancing district and party representation has engaged scholars and electoral engineers for decades, with the mixed-member electoral system emerging as the most prominent solution, often hailed as the "best of both worlds" (Shugart & Wattenberg 2001, Batto et al. 2016). In 2004,

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[3] Parts of this and the following chapter draw on two previously published papers (cf. Lüscher 2021, Lüscher 2023).

the German mathematician Friedrich Pukelsheim introduced a new approach in the context of subnational Swiss parliamentary elections. His biproportional (or double-proportional) divisor method with standard rounding, which debuted in Zurich and has since spread throughout Switzerland, ensures proportional representation of parties while preserving electoral districts (Pukelsheim & Schuhmacher 2004).

The new method ensures optimal vote-seat proportionality by closely mapping voter preferences onto parliamentary seats. It guarantees that every voter has nearly the same impact on the election outcome and the composition of parliament, irrespective of their constituency or party affiliation. According to its developers, this is achieved by cross-district vote tallying (Pukelsheim & Schuhmacher 2011). The new method refines traditional apportionment by shifting the initial allocation from the lower tier (district level) to the upper tier, thereby minimizing systemic biases and the threshold effects associated with small districts.[4] It employs the Sainte-Laguë divisor method with standard rounding of non-integer quotients, rather than rounding them down, to ensure equitable representation. Furthermore, the biproportional approach includes a sub-apportionment step that balances both party and territorial representation. In this respect, biproportionality is similar to mixed-member electoral systems, though it addresses the allocation problem differently. By closely approximating ideal vote-seat proportionality and minimizing the loss of votes in the allocation process, the biproportional method, along with mixed-member and at-large PR systems, stands out as one of the few apportionment methods that advances to the highest levels of political equity in democratic elections (as empirically possible). The combination of directives from the FSC and positive application experiences has led to the adoption of the new seat allocation method in nine cantons, starting in 2005 in the Canton of Zurich. In the Canton of Zurich, the method is also used for sub-cantonal elections to city parliaments.

0.2 Research gap, main arguments and research questions

Despite extensive research on the mathematical aspects[5] and simulative applications[6] of the biproportional divisor method, there is a lack of studies examining its broader implications beyond the distributional dimension. This is surprising given the high public expectations and hopes placed on the new apportionment formula. Specifically, small parties disadvantaged under prior electoral law were anticipated to benefit from the change, potentially stimulating party competition and spurring voter turnout. However, despite its use in over a third of all Swiss cantons, no studies have yet examined how the new procedure effectively impacts cantonal political systems.

This dissertation fills this gap by examining the effects of electoral reforms on political dynamics. It investigates whether the anticipated increase in overall proportionality has influenced political parameters, and if so, how and to what extent. The Swiss cantons provide a unique research opportunity for analyzing the short- and medium-term aggregate-level effects of biproportional reforms. Using Swiss cantons that have implemented the new method as a case group and the non-reformed as controls, I argue that the prospect of a more proportional translation from votes into seats boosts local democracy in a manner similar to MR-PR transitions. Drawing on a vast body of literature on electoral system effects rooting in the pioneering

[4] By "small districts" I refer exclusively to the number of allocated seats, not the geographical size.
[5] Cf. Bacharach 1970, Balinski & Demange 1989a, Balinski & Demange 1989b, Gassner 1991, Balinski & Young 2001, Pukelsheim 2004, Maier et al. 2010, Demange 2011, Pukelsheim 2017, Pukelsheim 2019.
[6] Cf. Balinski 2002, Bochsler 2005, Pennisi et al. 2006, Zachariassen & Zachariasen 2006, Ramírez et al. 2008, Oelbermann & Pukelsheim 2011, Hartmann 2023.

work of Maurice Duverger (1954), I hypothesize that party system fragmentation and voter turnout will increase, and more candidates and parties are encouraged to participate in elections.

To avoid endogeneity and acknowledge that electoral systems are not neutral but rather highly instrumental institutions shaped by political maneuvering, this dissertation also looks at the different contexts of reform formulation and implementation. In this part, electoral systems and their reforms are viewed as products of political decision-making and the efforts of actors seeking to influence these processes to their benefit. While existing research predominantly focuses on transitions from MR to PR, there is very limited research on "within-PR" transitions. Taking a process-oriented view, I argue that theories linking electoral institutions and their design to political decision-making processes – typically in the context of MR-PR transitions – are also suited to explain transitions from unidimensional PR to bidimensional PR. More specifically, I contend that the parties' stances on reform are largely determined by their anticipated post-reform status. Parties expecting an improvement in their electoral standing after reform are likely to advocate for radical reform, while those anticipating a decline in their post-reform electoral standing are expected to resist or dilute reform. Additionally, if the reform process was contentious – whether due to partisan opposition or direct democratic challenges – the final reform design is likely to be less distorted compared to a reform arising from a cross-party consensus, where the main parties may introduce distortive features as parts of political horse-trading.

Thus, this dissertation examines the innovation of biproportional apportionment from two different angles: as a consequence of political conflicts over electoral law and as a factor inducing changes in the political landscape. This approach combines two different branches in electoral systems research. First, the long-established tradition that views electoral systems as exogenously "given" and analytically focuses on how they affect political outcomes. However, given that electoral systems are shaped by the parties they are meant to regulate or the incumbents they help re-elect, there are doubts about the causal independence of these rules (Benoit 2007, Riera 2013). Since political actors are cognizant about the effects of electoral rules, they will seek to shape them to their advantage. Consequently, fully understanding the effects of electoral reforms requires considering and analyzing their antecedents. The second tradition of electoral systems research addresses this by treating electoral systems and their reforms as endogenous phenomena, focusing on the underlying factors that drive electoral system choice. Adopting this dual perspective, the thesis is guided by the following research questions:

1. How did political actors influence the reform process, what patterns emerged?
2. How did conflict patterns during decision-making affect the final reform design?
3. What is the political impact of the reforms in the cantons concerned?

0.3 Analytical scope of the study

The analysis focuses, although not exclusively, on Swiss cantons currently applying biproportional apportionment, which are (year of enactment in parentheses): Zurich (2005), Aargau (2008), Schaffhausen (2008), Nidwalden (2013), Zug (2013), Schwyz (2015), Valais (2017), Uri (2019), and Graubünden (2021).[7] The first empirical section of the dissertation examines the reform processes in these cantons, including those where

[7] See also Table 41 in the appendix.

the reform was proposed but not implemented (Obwalden, Bern, Thurgau). This part explores party positioning, conflict patterns, and the legal design of the reform policies, using a qualitative research design with document analysis and selective quantification of qualitative data.

The second part of the dissertation investigates the impact of electoral law reform on various aggregate-level variables, such as vote-seat disproportionality, the effective number of legislative parties, voter turnout, and political competition. This section makes causal claims about the reform effects and isolates non-reform-related changes through counterfactual estimation based on all cantons. Methodically, this approach roots in the difference-in-differences technique. This part follows a quantitative, large-N design and analyzes data from all cantonal parliamentary elections between 1976 and 2023. The second part is therefore a complete survey. Although the second empirical part includes constituency information, the data set does not provide constituency-level data. Consequently, certain potentially significant and insightful changes, such as strategic behavior by party actors in list submissions, local voter turnout, and other relevant details, are not captured. However, collecting this information for the entire study period and across all cantons would require an impractically high research effort, given the limited resources available.

0.4 Contributions

This study offers several contributions to the field:

First, it provides the first comprehensive analysis of the biproportional apportionment method, adopted by numerous Swiss cantons since the turn of the millennium to address demands for a fairer electoral system. The method is exclusive to Switzerland and is critically assessed in the dissertation, highlighting both its innovative aspects and the practical and conceptual challenges it presents. The study also includes a thorough documentation of cantonal reform processes, analyzing decision-making patterns and testing hypotheses to explain variations in reform designs.

Second, the study evaluates the potential political ramifications of electoral reforms within the PR family. While previous research has focused primarily on transitions from majoritarian to proportional systems, this study explores the less examined domain of reforms within different PR systems. By leveraging the biproportional method as a case study, it investigates how electoral reforms influence party competition and political participation, providing insights into how various PR formulas impact the political landscape.

Third, the research employs a novel counterfactual estimation method for causal inference in time-series cross-sectional data to assess the effects of electoral reforms on cantonal political systems. This advanced methodology is designed to address challenges such as incomplete data and varying intervention timings, offering a more precise analysis and advancing the field by providing deeper insights into the causal relationships between electoral reforms and political outcomes.

Fourth, the study introduces three new data sets with detailed information on cantonal elections from 1976 to 2023. The first data set (Lüscher 2024a) provides data on party representation in cantonal parliamentary elections. The second data set (Lüscher 2024b) offers aggregated data on all cantonal elections, covering aspects such as the electoral system and characteristics of the political landscape. The third data set (Lüscher 2024c) includes detailed information on constituency structures. While the second data set builds upon existing data with added variables, the third data set is entirely original, filling a crucial gap as, to the best of my knowledge, there are no similar publicly available data sets yet.

0.5 Main findings

This dissertation examines the adoption and implications of biproportional apportionment in Swiss cantons, providing an in-depth analysis of its achievements, challenges, and broader consequences for electoral fairness. Since its initial implementation in Zurich in 2005, biproportional apportionment has constituted a pivotal shift in Switzerland's electoral framework, aiming to achieve a more accurate reflection of voter preferences across constituencies while significantly reducing effective electoral thresholds. By enhancing proportionality, this reform has sought to advance Swiss elections toward a more representative and equitable system, particularly benefitting smaller parties that were previously marginalized under the former electoral rules.

However, the adoption of biproportionality has not been without significant challenges. The allocation process requires complex calculations. The high degree of technical complexity makes it difficult for the general public to understand the new method. Despite the development of tools such as the BAZI software, designed to facilitate the allocation process and uphold democratic scrutiny, the inherent complexity of the system raises concerns regarding its accessibility and the broader implications for public engagement and confidence in the electoral process. Additionally, the cross-district vote tallying integral to biproportional systems has led to instances of discordant seat allocations, where the distribution of seats within constituencies does not align with actual vote counts. This misalignment can be particularly problematic when it impacts influential candidates, potentially eroding voter trust and confidence in the electoral system.

The dissertation also provides a critical examination of the strategic behavior of political parties during the electoral reform process. The empirical findings largely support the hypothesis that political parties traditionally advantaged by existing electoral laws tend to resist or attenuate reform efforts, while those disadvantaged by the status quo advocate for more comprehensive changes. This strategic behavior underscores the complex dynamics of power within electoral politics, where parties maneuver to preserve or enhance their electoral advantages. Nonetheless, the study also identifies instances where parties deviate from expected patterns, influenced by factors such as legal imperatives, political ideals, and democratic norms. Notably, the research reveals that reforms imposed against the interests of advantaged parties often result in less distorted outcomes, indicating that political conflict, rather than consensus, can be a critical driver in achieving more equitable outcomes.

In evaluating the impact of these reforms on cantonal politics, the dissertation finds that while biproportionality has significantly improved vote-seat proportionality, the anticipated secondary effects – such as increased voter turnout, heightened political competition, and an expansion in the number of effective political parties – have not manifested as expected. Instead, the reforms have contributed to a reduction in the number of electoral lists submitted, reflecting strategic adaptations by political parties. This finding underscores the substantial strategic implications of seemingly technical adjustments in electoral systems, as political actors navigate these changes to optimize their resources. Furthermore, the study suggests that the expected psychological impact on voters, a critical component of the reform's objectives, has been limited, potentially due to a lack of awareness or other countervailing factors. In conclusion, while biproportional apportionment represents a significant advancement in the pursuit of electoral fairness in Switzerland, it also introduces new complexities and challenges.

0.6 Structure

The thesis is structured into three core chapters (Chapters 1-3), each crafted to be read independently, preceded by an overarching introduction (Chapter 0). The thesis' structure is shown in Table 2.

Chapter 1 lays the groundwork by providing an in-depth overview of the context that led to the development of the new seat allocation method. It thoroughly examines the technical and conceptual foundations of the method, highlights its advancements, addresses a key conceptual dilemma and critically assess the limitations associated with biproportional apportionment.

Chapter 2 investigates the underlying causes of the biproportional reforms. Drawing on detailed documentation of the cantonal reform processes, it explores the behavior of the involved actors and the characteristics of the decision-making processes, offering a cross-case analysis. This chapter delves into how partisan actors positioned themselves and influenced the reform process, as well as the conflict structures that emerged and their impact on the resulting reform designs.

Chapter 3 shifts focus to the political implications of the new method. It examines the practical importance of proportionality and evaluates how and to what extent specific parameters altered by the reforms are likely to affect the mapping accuracy of the electoral system. The chapter concludes with the use of counterfactual estimation models to assess the changes the reforms have induced within cantonal political systems.

Chapter 4 presents an overarching conclusion and provides a synthesis of the thesis' key findings. Chapter 5 compiles the references cited throughout the work, while Chapter 6, as the appendix, includes additional material and provides further context and supporting information.

Chapter	Subject	Remarks
0	Introduction	Establishes the overarching context and objectives of the thesis.
1	Foundational framework	Provides in-depth information on the new apportionment method, laying the groundwork for subsequent analyses.
2	Process-related analysis of cantonal reforms	Examines the reform processes in detail, including the formulation and testing of Hypotheses 1 and 2.
3	Outcome-related analysis of cantonal reforms	Focuses on the effects of the reforms, analyzing the outcomes in light of Hypotheses 3 to 7.
4	Conclusion	Synthesizes the thesis's key findings, offering a cohesive wrap-up that connects the insights across all chapters.
5	References	Compiles all sources cited throughout the thesis.
6	Appendix	Includes supplementary materials, providing additional context, data, and supporting information.

Table 2: Structure of the thesis.

Part I

⋆⋆⋆

1. Biproportional Apportionment: Context, Technicalities, Practice and Limitations

1.1 Introduction

Electoral systems are sets of rules and mechanisms that govern how social needs and interests are translated into formal political power. As societal values and priorities evolve, so does the design of electoral systems. Thus, electoral reforms reflect shifts in social concepts and aim to realign the system with contemporary ideas and values. In Switzerland, several cantons have made a notable shift from traditional PR- and MR-based apportionment methods to biproportionality. This change represents a significant and methodically unique advancement in enhancing the proportionality of vote-to-seat conversion. This chapter delves into the broader context of this transition and examines the implementation, practical application, and the challenges the new method presents in cantonal parliamentary elections.

The chapter unfolds in several parts: Initially, the political and legal backdrop that led to the development of the new seat allocation method are discussed. This section outlines the key phases of the FSC's evolving jurisprudence on electoral systems and its impact on cantonal reform efforts. Following this, the chapter offers a description of the new biproportional procedure, highlighting its conceptual and technical innovations compared to other/earlier methods. It then explores the practical implementation of the new method, detailing where it has been applied and the changes it has introduced to institutional structures. The chapter concludes with a critical assessment of the new procedure, addressing its technical complexity and the effects of cross-district vote tallying. It explores how these factors might affect public acceptance and discusses potential conflicts between practical demands and the method's intended goals. Finally, the chapter discusses the impact of additional design features, such as statutory quorums and majority clauses, within the context of biproportionality. It evaluates how these mechanisms, intended to mitigate political fragmentation and address representation imbalances, can conflict with the principles of pure, undistorted proportional representation.

By outlining these aspects, this chapter aims to deliver a comprehensive understanding of biproportional apportionment, acknowledging both its innovations and its limitations. This foundational discussion lays the groundwork for the subsequent sections of the dissertation (Chapters 2 and 3) by introducing the technical, practical, and theoretical dimensions of the new apportionment method.

1.2 Balancing autonomy and equity: The Federal Court's role in shaping cantonal electoral law

Traditionally, Swiss cantons wield considerable political autonomy including the design of their electoral law. However, since the turn of the millennium, evolving legal precedents have restricted this autonomy. These rulings now mandate that cantonal electoral systems adhere closely to the constitutionally implied principle of electoral equity. This shift in case law traces back to a ruling of December 18, 2002, concerning the Zurich City Council elections of March 3, 2002 (BGE 129 I 185). The decision concerns a voting rights complaint against the election procedure for the Zurich City Council. Based on population figures, two of the 125 seats were allocated to the smallest constituency (Constituency 1) and 19 to the largest (Constituency 11). The remaining seats were distributed among the other ten constituencies, with three having fewer than ten seats (four in Constituency 5, five in Constituency 7, and nine in Constituency 4).

In smaller constituencies, a list needs a large(r) share of the total vote to win a seat. The apportionment method used requires approximatively $1/(M+1)$ of the vote, where M is the district magnitude in terms of seats. For example, Constituency 1 with two seats requires 33% of the vote to secure a seat, whereas Constituency 11 with 19 seats only requires about 5%. This discrepancy contributed significantly to the Court's ruling. The Court criticized that voters do not contribute to the election results evenly due to significantly

varying district sizes. The Court's scrutiny was guided by constitutional principles of democratic equity and proportional representation, as articulated in Articles 8 and 34 of the Federal Constitution. According to the Court's interpretation, these articles guarantee that all citizens have the right to equal political participation and representation, ensuring that electoral systems do not favor or disadvantage any political group based on the size or location of their electoral base. The Court partially upheld the complaint and declared Zurich's constituency division unconstitutional (cf. Pukelsheim & Schuhmacher 2004). This marks the first of four turning points in the FSC's dynamic case law on cantonal PR-based electoral systems (cf. Grossenbacher 2018: 5-7):

First turning point – the ten percent rule: In its ruling, the FSC established a precedent by declaring that electoral thresholds exceeding 10% are incompatible with the principles of proportional representation enshrined in the Swiss Constitution.[8] Additionally, the size of each constituency must not vary by more than one-third from the average constituency size in the canton. The Court emphasized that high thresholds undermine the fundamental rights of voters and violate the principle of electoral equity, which forms the bedrock of democratic governance. As a result of this landmark decision, the Court mandated cantonal governments, including Zurich, to reform their electoral systems to rectify these disparities. The ruling set a precedent for electoral reforms across Switzerland and prompted legislative action to ensure that electoral districts are configured in a manner that respects federal law.

The Court instructed the city of Zurich to implement corrective measures but did not prescribe a specific method. In its statement, it suggested two options: a new constituency structure or the formation of constituency associations. Subsequently, the canton of Zurich mandated the mathematician and statistician Friedrich Pukelsheim to devise a comprehensive and minimally invasive solution for all parliamentary elections in the canton. He proposed a new method and termed it "biproportional divisor method with standard rounding".

Second turning point – admissible exceptions to the rule: Although the FSC stressed that any votes must have approximatively the same chance of success, i.e. of being effective, it also indicated in related rulings that this standard is not absolute. The Federal Constitution does not mandate a "pure" PR system. However, any deviation from this principle requires convincing objective justifications. Historically, from at least 1993 to 2014, the FSC deemed deviations from the equality of electoral success and proportional representation principles (i.e., surpassing the "ten percent rule") justified if they resulted from small constituencies based on traditional territorial organizations (BGE 131 I 85, BGE 136 I 352, BGE 136 I 376, BGE 140 I 107). Such constituencies could be considered unique due to historical, federalist, cultural, linguistic, ethnic, or religious reasons, thereby warranting minority protection and representation, even at the expense of optimal proportional representation. The stronger the constituency's identity, the more justified the deviation from axiomatic principles (BGE 129 I 185, BGE 131 I 74).

Third turning point – abrogation of the exemption clause: Over time, the FSC increasingly argued that alternative allocation methods like the biproportional apportionment method or constituency associations could realize proportional representation without altering traditional constituencies. This change of course became particularly clear in the decision of February 12, 2014, regarding constituency divisions in Valais (BGE 140 I 107). The FSC affirmed that through the biproportional apportionment method or constituency associations, a reasonably accurate reflection of party strength in parliament could be achieved, even in the

[8] These 10% equate to nine seats when applying the formula Threshold = $1/(M+1)$.

presence of small constituencies. The Court declared that if a canton opted not to use these corrective measures, small constituencies could no longer be justified under the principle of proportional representation, despite any historical or cultural significance. Essentially, the 2014 decision marked the Court's termination of exceptions allowing traditional constituencies to compromise proportional representation, favoring instead methods that maintain the integrity of constituencies while ensuring proportional outcomes (see also BGE 143 I 92).

Fourth turning point – expansion of equity requirements to non-PR systems: In its decision of September 26, 2014, regarding the parliamentary electoral system of Appenzell Ausserrhoden, the FSC clarified that Article 34 of the Federal Constitution mandates electoral equity across all systems – proportional, majoritarian, or mixed (BGE 140 I 394). While acknowledging that majoritarian systems may not fully achieve equality of electoral success, the Court recognized their validity under specific conditions. This expansion of equity requirements means that even in non-PR systems, the electoral processes must strive for fairness and equal representation, thus broadening the scope of constitutional mandates in cantonal electoral law.

It should be noted that the case law of the FSC, even though it accounted for cantonal particularities, at least until 2014, also caused discord. Some cantons felt affronted by the Court's decision, viewing it as an illegitimate and intrusive encroachment on their autonomy. This unease led the cantons of Zug and Uri to each submit a cantonal initiative in 2014, aiming to explicitly enshrine cantonal sovereignty over electoral law in the Federal Constitution. While both initiatives were ultimately rejected, they show that the judicial directives were not only met with approval on part of the cantonal authorities and legislatives.

As evident, the FSC played a pivotal role in driving electoral reform processes in the cantons. However, alongside this legal debate about cantonal electoral law, there is also a political dimension. The push for reform originates from a broader public dissatisfaction with traditional, unidimensional PR, coupled with a growing demand for electoral processes to be "fair", i.e., to generate politically unbiased results. As the Court highlighted, a constituency structure composed of small and variably sized districts hinders a proportional reflection of voters and their preferences in terms of seats and, resulting in a non-negligible portion of vote(r)s being ineffective and unrepresented. This is particularly (but not exclusively) evident in cantons with a highly fragmented electoral structure, characterized by a significant number of single-member or small multi-member districts. It is not implausible to assume that the Court's response is, at least to some extent, influenced by this public discontent.

1.3 How biproportional apportionment innovates conventional PR methods

To recapitulate: the core problem lies in a constituency structure characterized by numerous small districts, i.e. less than nine seats, and a high variance in district magnitudes, resulting in uneven voting power for voters depending on the constituency in which they are registered. Coupled with the Hagenbach-Bischoff apportionment formula, which systematically disadvantages smaller parties in favor of larger parties through the rounding off of fractions in determining (integer) seat entitlements, this setup means that small parties and their voters in small constituencies face lower chances of winning seats compared to a) larger parties in the same constituencies and b) small parties with identical voter shares in larger constituencies. This effect is amplified across the entire electoral area due to the district-based allocation of seats to parties.

Friedrich Pukelsheim proposed a solution that innovates traditional apportionment methods by optimizing the vote-seat proportionality across the entire electoral area while retaining the existing constituency structure. Two changes are at the core of this approach: firstly, splitting the seat allocation procedure into two

parts, and secondly, adopting the politically unbiased Sainte-Laguë allocation formula, which uses standard rounding for non-integer seat entitlements. The new procedure ensures that voters wield equal influence on the election result, irrespective of their constituency or the party they support, thus promoting fairness and equity in representation. Because the allocation of seats is based on both the territorial distribution of the electorate and the distribution of party votes (simultaneously rather than successively), the new method is termed bi-proportional.

The two-step allocation process represents the centerpiece of the new method. Initially, seats are allocated to parties proportionally based on their total votes across the entire electoral area, a step known as "upper-tier apportionment" or "super-apportionment".[9] This means if a party garners an overall vote share of 8%, it ideally secures about 8% of the available parliamentary seats. Given that voters in different districts wield varying numbers of votes corresponding to district size, standardizing these voting scales is crucial to ensure equal voting power (Pukelsheim 2017: 261). By shifting the initial level of seat allocation from individual districts to the entire electoral area and by applying the unbiased Sainte-Laguë formula, disparities between a party's vote share and its seat share are minimized, effectively neutralizing the distorting impact of small electoral districts.[10]

In the next step, the overall votes obtained by the parties get "re-distributed" or "sub-apportioned" to the parties at district level and must cumulatively satisfy the following conditions:

(i) The total seats allocated to a party across all districts match its total number of seats.
(ii) The total seats allocated to all parties in a district meet the district's size.
(iii) The total seats allocated across all districts match the overall number of party seats.
(iv) Proportionality is ensured among parties within each district and among districts within each party.

While the first three conditions pertain to the general setup of apportionment, the fourth condition adds proportionality as a criterion. The difficulty of allocating seats proportionally among different groups or constituencies based on multiple criteria simultaneously is known as the "biproportional apportionment problem" (Ramírez et al. 2008, Maier et al. 2010, Demange 2011, Serafini & Simeone 2012). Solving this problem manually is challenging, thus using a divisor-based iterative algorithm is typically recommended for efficiency.[11] The algorithm aims to identify divisors for lists and districts that best meet condition (iv) while satisfying conditions (i)-(iii).[12]

However, achieving perfect two-dimensional proportionality remains elusive. Divisor and multiplier methods cannot ensure district-level proportionality unless there are only two parties and two districts. This limitation sometimes leads to inconsistencies in specific districts, where the ranking of seat shares does not

[9] Strictly speaking, seats are allocated to electoral lists, not directly to parties. Not all lists constitute a political party. However, for simplicity and consistency with the literature, I will continue to refer to them as parties.
[10] Certainly, even under pure biproportionality, not all votes translate into seats. A minimum threshold of votes is required to secure a seat (approximately $\frac{1}{2} \times votes_{tot}/seats_{tot}$). However, this threshold is considerably lower compared to PR systems that allocate seats at the district level.
[11] In 2004, Friedrich Pukelsheim and his team developed a publicly accessible Java program (BAZI) to compute electoral input matrices. In 2016, a team of mathematicians at the Rosenheim University of Applied Sciences converted the program to a R-package (cf. Technische Hochschule Rosenheim 2024).
[12] A detailed step-by-step explanation of the method can be found in Pukelsheim 2017, Chapter 14.2, where he illustrates the approach using the 2016 Parliament Election in the Canton of Schaffhausen.

align with the ranking of vote shares, thereby violating criterion (iv). Such district-specific disproportionalities are commonly termed "discordant seat allocations" or "seat reversals", as described in detail by Maier et al. (2010: 375) and will also be addressed further in this first part.[13] Essentially, they are a consequence of a prioritization of achieving optimal overall proportionality over ensuring within-district proportionality.

But how significant is the impact of the apportionment formula on the election outcome? To illustrate the effect, I use the 2016 Cantonal Council elections in Schwyz as a case study, where the new biproportional procedure was applied for the first time. To highlight the effect, I compare the actual results with the hypothetical/counterfactual results using the previous Hagenbach-Bischoff formula. However, it is important to note that the counterfactual modeling of the election results according to Hagenbach-Bischoff has been simplified, abstracting from more complex circumstances. For instance, it does not assume any apparentements within the constituencies. Additionally, the hypothetical model does not account for how voters and parties might adapt their behavior to the incentives of the electoral system, which, very probably, affects the election outcome. Furthermore, the results are not presented at district level.

Party i	V_i in %	S_i in %			
		Biprop.	S_i - V_i in %pt.	Hag.-Bisch.	S_i - V_i in %pt.
SVP	33.21	33	-0.21	39	+5.79
CVP	27.24	27	-0.24	29	+1.76
FDP	21.70	22	+0.30	22	+0.30
SP-Gr-Ua[14]	15.29	15	-0.29	10	-5.29
GLP	2.56	3	+0.44	0	-2.56
Total	100	100	0	100	0

Table 3: Seat allocations for the 2016 Cantonal Council elections in Schwyz according to the de facto biproportional apportionment formula and the counterfactual Hagenbach-Bischoff formula. A 1% canton-wide quorum was introduced alongside the reform. V_i denotes the vote share, and S_i denotes the seat share of party i. The seat shares are integers since the Schwyz cantonal parliament consists of 100 members. Calculations are performed using the BAZI software.

As the table shows, the differences are quite substantial. While the biproportional formula accurately reflects the election outcomes in terms of seat shares (apart from inevitable rounding differences), the Hagenbach-Bischoff formula induces severe distortions. Overall, the change in formula, holding all else constant, accounts for eight seat shifts. It is also evident that electorally dominant parties, specifically the SVP and CVP, benefit the most under the previous formula, while smaller parties, such as the left-wing alliance and the GLP, are disadvantaged. The biproportional formula minimizes these vote-seat disparities inherently caused by the apportionment formula.

But why should a parliament ideally mirror the party preferences of the population? And, provocatively, why do we even need constituencies? The next chapter revisits the fundamental theoretical concepts underpinning modern notions of political representation and discusses how biproportionality relates to these concepts.

[13] These discordant seat allocations can manifest in two ways (vaguely mentioned by Maier et al. 2010: 380): Either within districts, where Party A gets more seats than Party B despite receiving fewer votes, or within parties, where a party receives more seats in District X than in District Y despite getting more votes in District Y than in District X.
[14] SP-Gr-Ua is a coalition of social democrats, greens and "independents".

1.4 Representing whom or what? [15]

Contemporary theoretical thinking on political representation is strongly influenced by the notion that parties act as the main link between the public and the political sphere (Pitkin 1967, Manin 1997, Rehfeld 2005, Dalton et al. 2011, Golosov 2017). The centrality of parties for the establishment and functioning of democratic institutions has been emphasized by various political theorists: Max Weber referred to parties as "the children of democracy" (1990: 35), John Aldrich described them as "endemic to democracy" (1995: 3) and for Richard Katz, party government is "a synonym for representative democracy" (1987: 2). However, the contemporary notion that parties are the main vehicle for formal political representation is historically quite young.

1.4.1 Territorial roots of modern representative government

Before the advent of parties in modern mass democracies, individual legislators located in small territorial constituencies served as primary linkages between the public and political decision-makers (Dalton 1985: 268-271, Pukelsheim 2017). The principle of territoriality, which finds expression in personalized elections within politically confined single-member constituencies, forms the conceptual origin of modern representative government (Urbinati and Warren 2008: 389). Democracy is organized and conceived as a local event, and voters are represented by an elect from their own "natural community of interest" (Golosov 2017: 119). Grounded in this territorial understanding of political representation, a long-lasting debate has raged over whether representatives should act as delegates or trustees (Pitkin 1967, Dalton 1985, Dovi 2009). This controversy was sparked by Edmund Burke's famous and disputed publication *Reflections on the Revolution in France* (1790).

In both models, constituency representation is realized by an act of authorization by the constituents at the outset of the election. This authorization enables the representative to act "[on] behalf of, in the interest of, as the agent of, someone else" (Pitkin 1967: 113). However, the models place conflicting demands on the legislative behavior of a representative, resulting in different degrees of political agency. The delegate model of constituency representation requires legislators to mirror the expressed policy preferences of their constituents. This leaves delegates no room for autonomy from their constituents. Their authority to act as representatives is limited to making the represented literally present again. Such notions of binding political allegiance found application in the Saddle period in parts of France, the United States, and beyond in the form of instructions, pledges, imperative mandates, and the practice of discretionary revocability of representatives (recall) (Manin 1997: 163-167).

The trustee model, in contrast, requires representatives to make decisions based on their judgment and in accordance with their own views of right action. Within the limits of authorization, the trustee enjoys full autonomy even if they act against the interests of the constituents. This does not mean that the representative is not held accountable for their actions. Voters still have the right not to reelect the representative. Apart from that, they should not be able to exert any influence on the trustee (Dovi 2009: 4-6). In Hanna Pitkin's account (1967), these two models of constituency representation should not be conceived as mutually exclusive. Instead, she suggests accepting the paradoxical nature of political representation and striving to safeguard both the autonomy of the representative and the autonomy of the represented.

[15] Parts of this subchapter have previously appeared in a similar form in a conference paper (Lüscher 2021).

Regardless of which model for describing citizen-elite relations seems empirically more accurate or appears favorable from a normative stance, they both point to the territorial patterns of early representative democracy. Electoral processes took place within territorially confined spaces and individual legislators established a corridor of formal representation between the constituents and the representative institutions of the polity.

1.4.2 Parties as the pillars of modern democracy

The national and industrial revolution at the end of the nineteenth century triggered a profound transformation of Western European politics. New political cleavages emerged, dividing society into different camps (Lipset & Rokkan 1967). At the same time, suffrage was extended to the working class and other segments of society. This dramatically changed the composition of electorates (Dalton 1985). The introduction of proportional representation resulting from mounting demands of these new voter segments for political recognition was another cornerstone that followed. All these episodes of modern democracy formation gave rise to the successful establishment and proliferation of mass parties.

While political representation was long guaranteed by and through local representatives, this attachment has lost importance over time and was largely substituted by political parties (Golosov 2017: 119). Parties offer voters manifestos with distinct policy goals. Voters, in turn, opt for those parties that best fit their policy preferences. This shift illustrates how the logic of electoral choice has moved from the direct nomination of a candidate to a comparative assessment of abstract party manifestos. Voters now think of formal political representation primarily in programmatic terms. Moreover, evaluations of the fairness of electoral systems are usually measured by the congruence of voters and parties in seats, ideology, policy preferences, or other indicators (Cincea 2013: 178-179). Thus, parties did not only monopolize political power; they also deeply affected how we think about politics, political representation, and political equity.

However, the territorial dimension of political representation is not a relic of the past. It still effectively shapes electoral processes today. Some countries, such as the United States or Great Britain, have never switched to proportional representation but still adhere to majoritarianism. In their candidate-centered voting systems, the principle of territoriality still prevails. Bicameral parliamentary systems devote a separate chamber of the national legislature to the territorial dimension of political representation. Additionally, countries such as Germany, New Zealand, Croatia, Albania, and Hungary have mixed their electoral systems by combining proportional with majoritarian or plurality voting elements to account for both dimensions of representation (Bochsler 2010a: 86-131).

1.4.3 The challenge of simultaneously meeting requirements of "both worlds"

The principle of territoriality persists subtly even within proportional electoral systems. Most democracies conduct elections at the district level, both nationally and sub-nationally, with few exceptions where legislative representatives are appointed through at-large elections. The purpose of electoral districts extends beyond organizational efficiency. They provide local populations with representation in the legislative body.[16] However, districts inherently conflict with the party dimension of political representation, often impeding fair, proportional representation, particularly for smaller parties. Research has consistently shown

[16] Even in countries like Israel and the Netherlands, where the entire nation serves as a single electoral district, the principle of territoriality remains important. Studies and polls indicate that geographic considerations continue to influence both voters and representatives, highlighting the ongoing importance of the territorial dimension of political representation even in the absence of specific electoral regulations and incentives (Nagtzaam & Louwerse 2023).

that the more parcelled an electoral area (i.e. the smaller the average electoral district), the greater the difficulty for small parties to pass electoral thresholds, resulting in disadvantages for them and their electorates (Duverger 1954, Rae 1967, Sartori 1976, Taagepera & Shugart 1989, Lijphart 1994). Following a more radical notion of electoral equity, voters not only deserve equality but also have the right to equal representation. This implies that their votes must be converted into seats with equal chances of success – a normative requirement incompatible with small districts in district-based elections.

In summary, electoral districts remain the primary geopolitical units for parliamentary seat allocation in modern democracies. From a theoretical perspective, districts embody the principle of territoriality, crucial to the genesis of modern democracy. However, they pose obstacles to achieving perfect party representation, sometimes leaving significant portions of the electorate unrepresented. There are three main solutions to this theoretical puzzle, differing in radicality:

- Abolish electoral districts and introduce a single constituency: This approach eliminates the territorial component and paves the way for a purely proportional representation system. However, it could neglect local interests and undermine the principle of territoriality.
- Mix the electoral system: Combining proportional representation with majoritarian or plurality elements can address both dimensions of political representation (Shugart & Wattenberg 2001). Examples include the electoral systems in Germany, New Zealand, and Hungary, which balance local representation with overall proportionality.
- Introduce a new apportionment method that reconciles the two antithetical concepts of representation: The biproportional apportionment method offers a novel solution. It ensures equitable representation across both political and geographical divisions. This method, implemented in several Swiss cantons, aims to reconcile the conflicting demands of party and territorial representation with minimal changes to the electoral system.

The main analytical focus of this dissertation is on the third solution, although the other solutions are also discussed peripherally. The next chapter examines the spatial and temporal adoption of biproportionality and assess whether the diverse responses of the cantons to the FSC's ruling align with the criticized aspects, particularly the small-scale and heterogeneous nature of constituency division.

1.5 Biproportional apportionment in practice

The cantons' responses to the new judicature by the FSC varied greatly. For instance, political stakeholders in the Canton of Zurich, the first affected, generally responded positively and pragmatically to the ruling. In contrast, a majority of political stakeholders in the Canton of Uri initially resisted the verdict and only reluctantly acknowledged the need for reform. However, it must be noted that cantonal political systems, despite many overlaps and similarities, exhibit significant differences and unique characteristics both politically and culturally. Smaller, catholic, and traditionally conservative cantons in the original part of Switzerland maintain a distanced relationship with the Confederation and its institutions. This is evident in historical voting patterns, such as the fact that the founding cantons of Switzerland (along with Appenzell Innerrhoden and Valais) rejected the Federal Constitution of 1848 and the two total revisions of 1874 and 1999 (Kley 2023). These cantons were correspondingly critical of the Court's rebukes.

Detailed analyses of individual cantonal reform processes (see Chapter 2) clearly show how closely cantons monitored both the dynamic case law and the responses of the reprimanded cantons. The media also played an important role, not from a legal perspective but in informing the public, shaping public opinion, and contributing to a discourse on electoral law. Over time, the issue became politicized as various parties advanced their agendas. This underscores the importance of viewing biproportional reforms as an outcome of a triangular interaction between legal jurisdiction, the media public sphere, and political decision-makers.

The debut of the refined method took place in February 2006 in the election of the council of the City of Zurich. One year later, the new method had its premiere in the Canton of Zurich (2007). Eight cantons followed suit: Schaffhausen (2008), Aargau (2009), Nidwalden and Zug (2014), Schwyz (2016), Valais (2017), Uri (2020) and most recently Graubünden (2022). As of 2023, a total of 25 elections have been conducted using the new method. Figure 1 illustrates the evolution of the application of different PR-based apportionment formulae in Switzerland from 1979 to 2023.

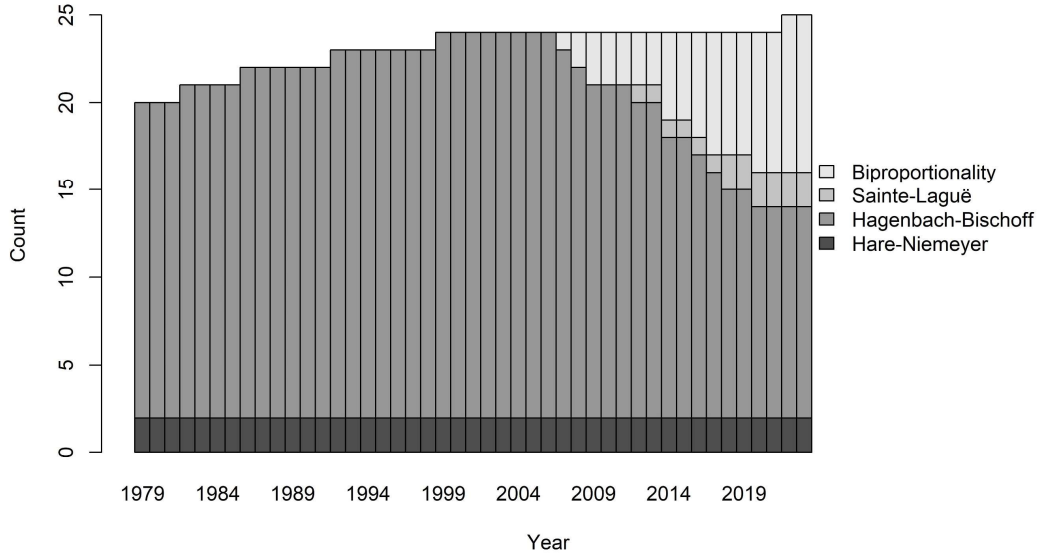


Figure 1: The cantonal application of various apportionment formulae between 1979 and 2023.

Since 1979, there has been a steady increase in the adoption of proportional representation for cantonal parliamentary elections, gradually replacing majority voting systems. From the turn of the millennium onwards, biproportionality primarily replaced the single-proportional Hagenbach-Bischoff method. The only exception to this pattern is the Canton of Graubünden, which transitioned directly from majoritarianism to biproportionality. This shift underscores the broader movement towards more equitable and representative electoral practices in Switzerland. Figure 2 illustrates the spatial dimension of biproportionality in the Swiss cantons as of 2023.



Figure 2: Spatial dissemination of biproportional apportionment in Switzerland.

Implementation of electoral reforms among reformed cantons varies significantly. While some have completely foregone introducing electoral hurdles that distort proportionality, the majority have implemented statutory quorums at either the constituency or cantonal level, or both (refer to Table 41 in the appendix). Additionally, the Cantons of Zug, Schaffhausen and Graubünden have included a "majority clause" in their electoral law, ensuring that the party with the most votes in a district secures at least one seat, regardless of district size. This clause resembles the revised German federal election model and applies only if the total cantonal seat count is not exceeded (no overhang and leveling seats). The goal is to guarantee that the leading party in the numerous single-member districts always secures a seat. However, this intervention also comes with implications, which are discussed in detail in the subsequent chapter on the limitations of the new procedure. Furthermore, in Uri, the new procedure applies exclusively to PR constituencies. Non-PR constituencies continue to vote under MR.

After nearly two decades of practical experience with the new allocation procedure prompted by FSC interventions, do we see evidence of reduced electoral thresholds resulting from changes in constituency divisions? Figure 3 illustrates that the cantons have responded to the FSC's mandates, directly or indirectly, resulting in a reduction of electoral hurdles that (implicitly) distort PR. This is evident in the significant decrease in the share of districts with fewer than nine seats (biproportional districts excluded). Since the 1980s, the proportion of small intra-cantonal constituencies in Switzerland has decreased significantly. Whereas this proportion used to be over a third of all cantonal constituencies, it has gradually declined to less than 10%. Conversely, this indicates that there are still cantons where the constituency structure for cantonal parliamentary elections does not comply with federal law. The figure on the right, concerning the effective electoral threshold, illustrates the reduced systemic hurdles that electoral systems impose on parties

aiming to gain parliamentary representation. While this value peaked in the mid-2000s, it has since halved to around 4%.

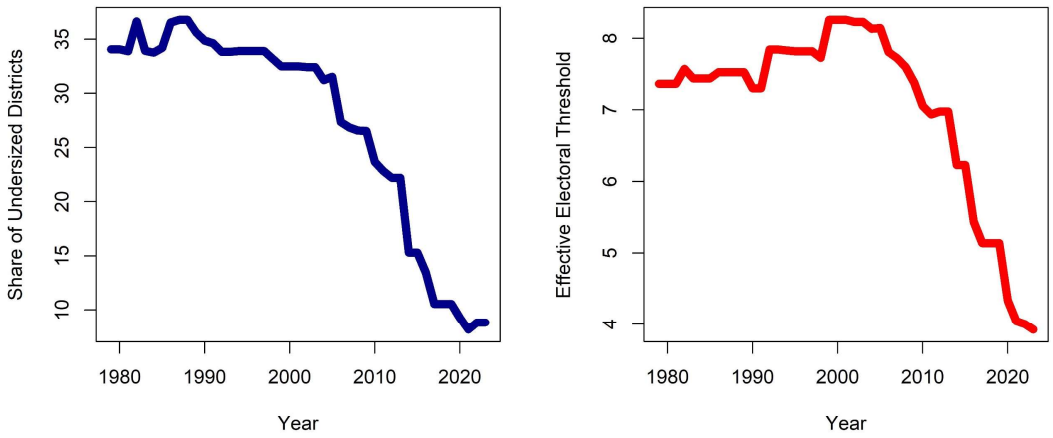


Figure 3: Left panel: Proportion of constituencies for PR-based cantonal parliamentary elections with fewer than nine seats (annual cantonal averages). Right panel: Effective electoral threshold (annual cantonal averages).

These consistent declining trends over the past two decades are attributable to a variety of factors, not solely biproportional reforms. They also reflect territorial reforms and other adjustments to the electoral systems. Nonetheless, they clearly demonstrate how the Swiss cantons have collectively responded to the guidelines set by the FSC.

1.6 All that glitters is not gold – biproportional apportionment and its limitations

The previous chapters have demonstrated that biproportionality mitigates some of the disadvantages inherent in unidimensional apportionment methods such as the Swiss Hagenbach-Bischoff method. Chief among these is the equalization of the effective voting power of voters of different parties in different constituencies. This leads to a more precise representation of voters' political preferences and eliminates the previous disincentive for those inclined to support fringe parties from abstaining. Additionally, implementing biproportionality does not require a change in the constituency structure. Therefore, can biproportionality be considered the pinnacle of electoral systems, reconciling both the territorial and political dimensions of electoral politics?

While biproportionality undoubtedly represents an attempt at optimization, the biproportional allocation method, like any other, comes with certain limitations. Electoral systems incorporate various principles, ideals and norms, including representation and minority protection. While territorial representation is most common, electoral systems also represent specific segments of society and the ideas, values, and ideals of parties. Other principles include governability, accountability, preventing excessive fragmentation and polarization, and ensuring simplicity, comprehensibility, and transparency in the conversion from votes into seats (Linhart et al. 2019: 23). Since not all principles can be maximized concurrently, trade-offs become inevitable (cf. Federal Chancellery 2013: 30). In the following sections, I will examine the most significant limitations, including some that have been previously discussed in the literature and others that have not yet been explored.

1.6.1 Technical complexity

One often-cited disadvantage of the new procedure, raised by both election experts and practitioners, is its technical complexity (see e.g. Demange 2011: 17-20). Optimally translating votes into seats both across the entire electoral area and within individual constituencies involves many calculation steps, especially for the distribution of seats within constituencies. The sub-allocation must always consider the applicable rounding rule and adhere to the prespecified marginals, i.e. the number of seats attributed to each constituency and the total number of seats for a party (Pukelsheim 2017: 264). Since adjusting the allocation in one constituency requires adjustments in others, an iterative algorithm is needed to efficiently navigate the complex calculations needed to resolve the allocation problem.

The BAZI software, specially programmed for the biproportional procedure, presents matrices with the quotients used for seat allocation within constituencies. These quotients are published by the responsible electoral authority along with the election results, enabling third parties to understand and verify the results. This transparency is crucial for democratic legitimacy and the acceptance of the election results.

However, critics argue that an election procedure should not be so complex that calculating the result requires specialized IT and expert knowledge. Despite this criticism, elections based on unidimensional PR are not necessarily less complex. For instance, apparentements, previously allowed in most cantons that transitioned to biproportionality, are among the most intricate elements of electoral systems. Before an election, it is impossible to predict the extent of their impact, or which allied party will benefit. Additionally, the distribution of seats within apparentements or sub-apparentements is unlikely to be familiar to the "average citizen". The fact that other apportionment methods or specific aspects of them can be similarly complex somewhat mitigates the criticism of excessive complexity of biproportional apportionment.

1.6.2 The pitfalls of cross-district vote tallying

The purpose of biproportional apportionment is clearly defined: to ensure that voters impact election results with nearly equal weight and similar chances of success, regardless of party preferences or constituency of residence. Biproportional reforms are thus rooted in a strong, radically democratic, voter-centered principle of equality. Biproportionality achieves this by tallying the votes across all constituencies to determine parties' seat entitlements. This imitation of a canton-wide constituency in the upper-tier allocation of seats to parties ensures that (1) all votes are accounted for in the vote tally, minimizing the proportion of "weightless votes", i.e., residual votes resulting from the rounding down of quotients and (2) that votes are mapped onto seats as proportionally as is empirically possible. However, the artificial expansion of the electoral area creates three side effects worthy of discussion.

1.6.2.1 Violations of within-district proportionality

One such issue is the phenomenon of "discordant seat allocations", which occurs when "the ranking of seat numbers within a district deviates from the ranking of vote numbers" (Maier et al. 2010: 375). In the sub-apportionment of seats to the parties within districts, it might occur that a party receives fewer seats than another party in the same district despite having more votes. Another scenario is that within a party, a list of that party in a district may receive more seats than the same list in another district despite having fewer votes proportionally. If the party with the most votes is affected, this is referred to as the "strongest seat discordant" (ibid. 2010: 386). These discordant seat assignments or "seat reversals" are an inevitable by-product of the biproportional apportionment problem. To achieve optimal overall proportionality, it may

be necessary to sacrifice optimal proportionality within individual districts. Maier et al. who evaluated these discordant assignments, laconically described them as "the price one has to pay to achieve proportionality over the whole electoral region" (ibid. 2010: 386).

In his monograph, Friedrich Pukelsheim (2017: 264) wrote that "discordant seat assignments are rare due to the excellent votes-seats image that is accomplished by double proportionality". However, various analyses, including his own from dealing with the Zurich Cantonal Council elections 2011 present a different perspective: "In the 2011 elections to the Zurich Cantonal Council, which were conducted on the basis of double proportionality, there are numerous discordant seat allocations" (Pukelsheim & Schuhmacher 2011: 1591, translated from German).[17] Similarly, the study by Maier et al., published a year prior, clearly illustrates that such seat reversals are not uncommon.[18] The frequency of districts affected by these reversals typically ranges between one and four per election, contingent on institutional factors, including the electoral system (Maier et al. 2010). Presenting an unpublished paper, Irina Lehner evaluated biproportional elections in the Swiss cantons and found that, on average, approximately every fifth district is affected by reversals.[19] But why should we be concerned about these seat reversals at all?

From a broader perspective, distortions in local proportionality may seem a minor trade-off for enhancing overall proportionality across the electoral region. However, from the voter's point of view, these distortions may raise concerns about the integrity of elections. In Switzerland, parliamentary elections are typically free list proportional elections. Thus, they entail a personal vote component, allowing voters to modify the predetermined party lists by crossing out, cumulating or adding candidates from all available lists.[20] This means that party votes also reflect voter preferences for specific candidates. Therefore, if in a constituency a popular candidate from the leading party loses to an outsider from another party due to a seat reversal, this outcome, despite its technical correctness, carries significant weight for both the candidate involved and the affected electorate.

This is why three cantons have co-implemented a "majority clause", ensuring the party with the most votes in a constituency gets at least one seat. While this addresses the most blatant cases of proportionality violations within districts, it does not resolve the underlying problem. The clause merely relocates reversals from smaller to larger constituencies. Inevitably, incompatibilities persist and require compensation in the form of reversals. This poses a political dilemma regarding where this compensation should occur.

1.6.2.2 Implications for territorial representation

The cross-district vote tallying introduces a lesser-known side effect that carries significant implications. In the process of super-apportionment, votes first become effective at the superordinate level. To aggregate these votes effectively, each party's vote count in constituencies is normalized by dividing it by the number

[17] Pukelsheim qualifies this observation by noting that similar deviations can occur in single-proportional elections under the Hagenbach-Bischoff system. However, applying the Hagenbach-Bischoff formula, these discordant seat assignments are not immanent to the apportionment method, but instead attributable to the feature of apparentements. Therefore, this comparison is not fully convincing.

[18] Cf. Figure 3 (Maier et al 2010: 386).

[19] Presented at the Conference "Die verflixte Arithmetik des Verhältniswahlverfahrens" on June 17, 2022, hosted by the Center for Democracy in Aarau.

[20] Electoral authorities also provide a blank ballot paper, where voters can write in the names of their preferred candidates. The number of pre-printed candidate lines matches the number of seats available in the corresponding constituency.

of seats attributed to that constituency (based on population figures). This normalization approximates the number of voters supporting each party, which is then aggregated at the cantonal level.

However, this approach, aimed at equalizing the influence of voters, renders the geographic origin of votes irrelevant in determining parties' seat entitlements. While this might appear fair and desirable at first, it becomes problematic when considering that constituencies with higher voter turnout and a greater proportion of eligible voters exert a disproportionately larger influence on the final electoral outcome than constituencies with lower participation rates. Although constituencies are upheld as a pillar of territorial representation, they play no role in the process of vote aggregation. This highlights a key aspect of the new seat allocation procedure: while constituencies retain a fixed seat entitlement based on census data, this entitlement is effectively undermined by cross-district vote tallying. Constituency electorates contribute to the election result not with the weight of the constituency's share of seats but its share of votes (cf. Lüscher 2023).

1.6.2.3 Persisting flaws in legislative representation

A further implication of the biproportional apportionment method relates to the equality of voting power and is closely linked to the effect previously described. This principle asserts that voters across the entire electoral area should ideally affect the composition of the parliament with the same weight. This requires a consistent ratio between the number of inhabitants and the number of seats in a constituency, known as the "representation figure", throughout the electoral area. This ensures that the parliament accurately represents the population.

Interestingly, the FSC has historically paid relatively little attention to this aspect in its case law on cantonal electoral systems, seemingly considering it secondary in the evaluation of an electoral system's equality/proportionality. This is despite the fact that the requirement was explicitly mentioned in the Court's landmark 2002 judgment concerning the City Council elections in the Canton of Zurich and reiterated in all subsequent judgments. It was not until the case involving the Canton of Graubünden (BGE 145 I 259) that the FSC examined this aspect in detail. This case involved the small municipality of Avers, with a Swiss resident population of 160 at the time. Due to statutory guarantees, forming its own constituency, the municipality was assured a seat in parliament, despite the cantonal representation figure being 1342. The discrepancy, with Avers' population being around one-eighth of the cantonal representation figure and one-fourth of the next smallest municipality, indicated substantial distortions in the equality of voting power. This distortion played a crucial role in the FSC's considerations and the eventual annulment of Graubünden's electoral system. But how does this relate to biproportional reforms?

The Canton of Graubünden switched from a majoritarian system to biproportionality in compliance with the FSC ruling. But what has actually changed in the Avers electoral district as a result of the reform? Because votes cast in Avers are primarily effective at the cantonal level and only subsequently at the constituency level, voters in Avers now have significantly less influence on the overall allocation of seats. Before the reform, they contributed around 0.83% to overall result.[21] In the first post-reform elections in 2022, this contribution dropped to approximately 0.16%.[22] This shift optimized the equality in voting power, aligning it more consistently across constituencies. However, the representation figure for Avers has remained nearly

[21] Corresponding to a constituency weight of 1/120.
[22] Corresponding to a voter number weight of 81/51714 (cf. Canton of Graubünden 2022).

the same, and due to a population decline by December 31, 2020, the parliamentary overrepresentation has even increased post-reform.

In traditional unidimensional PR elections, there is a parity between the electoral area and the representative body: elections occur within constituencies, and elected representatives mirror their constituency in its political composition. Voters' contributions to the overall result are proportionate to the constituency's weight. However, in biproportional elections, this clear parity is somewhat blurred. Constituencies remain, but votes are primarily allocated at the upper-tier level before being considered at the lower-tier level. This system aims to optimize equality in voting power and the efficacy of votes, but it leads to a conflation of political spaces with implications that extend beyond solving an allocation problem.

1.6.3 Statutory quorums – a step back?

In six of the nine cantons that transitioned to the new method, statutory quorums were established either concurrently with the reform or shortly thereafter. These quorums aim to prevent alleged excessive fragmentation of the party system, which, according to their proponents, could hinder the efficiency of council operations and complicate the process of reaching compromises.[23]

The introduction of quorums can be attributed to the first biproportional election in the canton of Aargau in 2009. This election, conducted without a threshold, resulted in four small parties – the BDP, EDU, GLP, and SD – gaining seats in the Aargau cantonal parliament for the first time. Much to the dismay of the established parties, who also feared that the new system would lead to further fragmentation of political forces (at their cost). Consequently, a quorum was introduced for the subsequent 2012 elections following a successful referendum, setting a precedent for other cantons. Although not all reforming cantons adopted a quorum, the issue was thoroughly debated in all legislative processes.

However, such quorums are fundamentally at odds with the system's intent. If the reform aims to optimize the accuracy of vote-seat conversion, as guaranteed by "pure", i.e., undistorted biproportionality, any political intervention that artificially diminishes this accuracy is detrimental to the goal. There are alternative methods to ensure efficiency in parliamentary operations – if this is the genuine concern: The key units of action in parliaments are parliamentary groups, which, in the Swiss context, generally require a minimum of five members. This requirement serves as a de facto quorum. Since parliamentary groups can include members from different parties, there is little justification for a priori excluding fringe parties from parliamentary participation and representation based on statutory quorum requirements.

Against this backdrop, increasing political fragmentation of legislatures can be viewed as not problematic in itself but rather a reflection of socio-political changes being mirrored in the party system due to the high permeability of the electoral system. A diverse party landscape is symptomatic of a diversification of political demands among voters, coupled with inclusively designed electoral rules. If a parliament should act as a mirror of society, a key principle of any PR-based electoral system, then deliberately obstructing an accurate translation of this diversity lacks theoretical justification.

[23] As will be demonstrated in Chapter 3, there is no evidence that the reform has led to party system fragmentation. Although there is a gradual inter-cantonal trend towards increased fragmentation, there is no indication that this trend is driven by biproportional reforms. However, it should be noted that statutory quorums in most reformed cantons might mitigate any potential positive effects.

As previously mentioned, majority clauses are another alien element designed to prevent discordant seat assignments in very small constituencies. These reversals can be politically sensitive and are thus transferred to larger constituencies where their impact is less significant (and less apparent). Although these clauses follow a majoritarian logic and interfere with unaltered biproportionality, they only affect sub-apportionment and do not compromise overall proportionality. Therefore, they are not considered a limitation here.

1.7 Conclusion

This chapter aimed to delineate the theoretical framework and practical backdrop of the biproportional apportionment method applied in nine Swiss cantons. Since its first introduction in Zurich in 2005, biproportionality has marked a significant shift towards a more representative electoral system in Switzerland. The reform has made strides in improving electoral fairness. By addressing some of the drawbacks of previous systems, biproportionality has enhanced the proportionality of election outcomes, ensuring that voter preferences are more accurately reflected across different constituencies. The notable reduction in effective electoral thresholds and the decline in constituencies with fewer than nine seats underscore the reform's success in fostering a more representative political environment.

Yet, biproportionality is not without its challenges. The complexity of the allocation process – requiring calculations and iterative adjustments – has introduced a level of technical difficulty that can undermine transparency and accessibility. Despite the advancements provided by tools like the BAZI software, which facilitates the allocation process and promotes democratic scrutiny, the inherent complexity of the system remains a concern, particularly in its impact on public understanding and engagement.

The cross-district vote tallying inherent in biproportional systems brings its own set of complications. Although designed to enhance overall proportionality, the method can lead to discrepancies known as discordant seat allocations. Such distortions occur when seat distributions within constituencies do not align with the actual vote counts, potentially undermining voter confidence, especially when influential candidates are affected. While majority clauses in some cantons attempt to address these issues, they merely transfer the distortions to larger constituencies. Although they address some of the more glaring issues of seat reversals, they often shift the problem rather than resolving it comprehensively.

Moreover, the impact on territorial representation presents another critical issue. By aggregating votes across constituencies, the geographic origin of votes becomes less relevant in determining seat allocations, which can diminish the role of individual constituencies in the electoral process. This approach, aimed at balancing voter influence, risks eroding the direct link between local representation and overall electoral outcomes.

The introduction of statutory quorums in several cantons further complicates the picture. These mechanisms, intended to prevent excessive fragmentation and ensure effective governance, distort the very proportionality that biproportionality seeks to achieve. While quorums, such as those introduced in Aargau following the 2009 elections, aim to stabilize the party system, they can compromise the accuracy of vote-to-seat translations.

In conclusion, while biproportional apportionment represents a significant leap forward in the pursuit of electoral fairness and proportionality, it also comes with its own set of challenges. The complexities of the system, the issues related to cross-district vote tallying, and the implications for territorial representation

highlight the ongoing difficulties in optimizing an electoral system that balances fairness, representation, and practical governance.

The potential for manipulation during reform processes must also be acknowledged. Electoral reforms are susceptible to political interference, as those who legislate the changes are also the ones primarily affected by them. This dual role provides incentives for parties to shape electoral laws to their advantage. Thus, to fully understand the design and effects of these reforms, it is essential to consider the interests and actions of the actors involved in the decision-making process, which will be addressed in the next part.

Part II

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2. Biproportional Reforms and Their Origins: A Process-Oriented View on Reform Designs

2.1 Introduction

There is a long and far advanced tradition in electoral systems research devoted to the study of the political effects of electoral rules (Duverger 1954, Taagepera & Shugart 1989, Cox 1990, Sartori 1997, Farrell & Gallagher 1998, Grofman & Lijphart 2003, Gallagher & Mitchell 2005, Klingemann 2009, Nohlen 2014 among others). In recent years, there has been a resurgence of interest in a field that can be denominated as the "politics of electoral reform". Instead of adhering to the traditional Duvergerian notion that essentially views electoral systems as determinative for the structures of a political system, this newer perspective suggests that understanding electoral institutions and reforms, along with their effects, requires considering the political configurations and strategic calculations of relevant party actors that have shaped them (Lundell 2005: 6-10, Charvát 2023: 15-18). Accordingly, the analytical focus lies in identifying and understanding the potential factors and mechanisms that trigger, facilitate, and motivate sudden changes in the "rules of the game" (cf. Cusack et al. 2007, Calvo 2009, Riera 2013, Leemann & Mares 2014, Walter & Emmenegger 2019). Drawing on this body of literature, the objective of this part of the thesis is to dismantle dynamics and patterns and shed a light on the different pathways that led to the adoption of biproportional apportionment in several Swiss cantons. Drawing on recent literature examining the adoption of PR in Swiss cantons during the late 19th and early 20th centuries, I develop a theoretical framework to explain the various paths to and different faces of biproportional reform in Switzerland.

Despite the successive establishment of the biproportional method as an alternative to (or supplement of) conventional, unidimensional apportionment formulae in Switzerland, there is notable variation in the specifics of individual reform designs across cantons. While some cantons have introduced artificial barriers to mitigate "undue" fragmentation of the party system, others have incorporated additional provisions to accommodate cantonal peculiarities. Evidently, these differences did not emerge randomly, but are the result of lengthy and often highly complex and non-linear political decision-making processes. However, the different outcomes cannot exclusively be ascribed to the involved decision-makers. They also mirror distinct cantonal trajectories of electoral system evolution and diverging ideas about which goals and ideals the cantonal electoral system should serve and which standards it should meet. Moreover, the differences must also be viewed in the broader legal and political context, whose dynamic nature adds to the complexity. Hence, numerous factors are assumed not only to influence the outcome of the reform but also the specifics of its resulting design. The following research questions are addressed in this part of the thesis: (1) What conflict patterns surfaced during the decision-making processes, and (2) how did these patterns impact the resulting reform design?

I argue that conflict patterns were predominantly influenced by parties' self-interest in maximizing their seat shares. Parties anticipating a potential decline in their electoral status due to the reform are expected to resist or aim to mitigate its impact, while those poised to benefit are expected to advocate for more substantial changes. Additionally, I argue that reform processes characterized by conflict yielded less distorted reform designs compared to those following a consensual pathway.

The chapter is structured as follows: After this introduction, the subsequent chapter presents a theoretical framework on the basis of which the hypotheses are derived. This is followed by the specification of variables and discussion on data and methodology. The following chapter provides a comprehensive, process-oriented documentation, covering successful reform processes as well as cases where reform attempts were

either rejected after a lengthier process or at an early stage of the decision-making process.[24] Subsequently, I consolidate the findings and contrast them with the assumptions to disentangle the links between conflict patterns during decision-making, partisan strategizing, and the variations in biproportional reform designs.

2.2 Taking a step back – the prosaic rationale behind electoral reforms

When Arend Lijphart (1995a: 412) succinctly stated that "if one wants to change the nature of a particular democracy, the electoral system is likely to be the most suitable and effective instrument for doing so", he highlighted two key aspects of an electoral system. First, he emphasized its significance in shaping the nature of democracy, and secondly, he pointed to its susceptibility to manipulation. Given the awareness among party strategists and electoral engineers regarding the political implications of electoral law, they are inherently incentivized to customize electoral regulations in harmony with their interests (Kedar et al. 2016). Consequently, understanding of the effects of such reforms requires a thorough examination of their determinants (Riera 2013).

One of the most momentous political renovations in the early 20th century was the adoption and successful dissemination of electoral systems belonging to the PR family (Grofman & Lijphart 2003, Calvo 2009). There is no doubt that contemporary party systems belonging to this family would look radically different today had this change not taken place. To carry the thought a bit further, it can also be argued that a wide range of political outcomes (social spending, social welfare, inequality, etc.) can at least to some extent be linked to these reforms and their repercussions, directly or indirectly. Recent comparative research has thus intensely focused on unravelling the motives behind such electoral reforms (Leemann & Mares 2014).

Historically, the adoption of PR intertwined closely with the expansion of suffrage, particularly to previously disenfranchised groups like workers and women. These expansions mirrored societal changes and, in turn, led to the emergence of new political cleavages and parties. In 1970, Stein Rokkan presented two complementary hypotheses aimed to dissect the origins of PR adoption, proposing the "minority protection hypothesis" and the "socialist threat hypothesis". The former posits that in culturally, ethnically, and denominationally diverse European nations (he mentions Denmark, Switzerland, Belgium, Moravia, and Finland), majority elections risked disrupting the political system. To fortify territorial unity within these diverse and politically fragile settings, the incorporation of measures to protect minority interests served as a strategic mechanism preserving the system's continuity (Rokkan 2009: 157). Conversely, the latter contends that the ascent of the working class before and during World War I posed an existential threat to established parties, prompting them to confederate with the working class and other proponents of PR to safeguard their political dominance. Though these hypotheses reflect different social dynamics, both underscore mounting pressures that weakened the resistance of established parties, primarily driven by the prosaic motive of preserving their political power.

Rokkan's hypotheses did not lose popularity since his (re)formulation. Quite on the contrary, numerous recent studies inquiring the origins of proportional representation build upon his theoretical legacy and refine it with new empirical data (Calvo 2009, Ahmed 2013, Leemann & Mares 2014, Walter & Emmenegger 2019 to mention only a few). While most of these studies investigate why reforms towards PR occurred, Walter and Emmenegger (2019) approached the subject from a new angle, considering how these reforms

[24] When the term "successful" is used to describe a reform process, it carries no inherent judgment. It does not imply desirability of the outcome but rather indicates that the reform has been effectively implemented. This interpretation remains consistent throughout the entire work.

were conducted – with or against the support of advantaged parties – and the degree of systemic distortions they yielded. They postulated that when PR adoption had the support of advantaged parties, resulting reforms exhibited higher vote-seat disparities. On the contrary, reforms forced upon these advantaged parties led to more proportional electoral systems (ibid. 2019: 66). Applying these hypotheses to Swiss cantons, their findings turned Rokkan's minority protection hypothesis upside down, suggesting that PR adoption with cross-party consensus wasn't primarily about minority inclusion but rather about safeguarding the political interests of parties benefiting from the prior electoral law.[25] These findings align with research on the nexus between party systems and electoral systems. In his study analyzing 219 elections across 87 countries since the 19th century, Colomer (2005) concludes that party systems are not simply determined by electoral rules, as prominently suggested by Maurice Duverger. Rather, political parties often chose electoral rules that solidified or strengthened existing party configurations. His findings underscore how political parties strategically utilize electoral rules as instruments to achieve benign political outcomes.

While the study of Walter and Emmenegger (2019) centered on transitions from majoritarian to proportional representation in Swiss cantons since 1890, the aim here is to adapt their theoretical framework to transitions towards biproportionality since its initial adoption in Switzerland in 2005. However, the contexts in which unidimensional and bidimensional proportionality were adopted are vastly different. The social dynamics underpinning the party system in contemporary Switzerland deviate considerably from those of the late 19th and early 20th century. Consequently, Walter and Emmenegger's socialist threat hypothesis isn't pursued in this analysis. Their modified minority protection hypothesis, however, remains a viable explanatory framework applicable to more recent electoral reforms, I argue.

The biproportional reforms in Switzerland are set within a specific legal context. Notably, the FSC's landmark decision in 2002 prompted PR cantons with small districts to adapt their electoral laws.[26] Moreover, the presence of direct democracy as a means of bottom-up mobilization and radical transformation emerges as another pivotal contextual factor. Walter and Emmenegger (2019) highlighted the crucial role of direct democracy in PR adoption, particularly in cantons where staunch resistance from beneficiaries of the status quo thwarted reform efforts. In circumstances where conventional approaches proved ineffective, direct democracy emerged as the final recourse for instigating change. According to the authors, this conflictual path towards proportional representation, against resistance from parties benefiting from the former majoritarian rules, led to the establishment of electoral systems with lower levels of distortion (ibid. 2019: 67). In light of this context, I argue that direct democracy emerges as a historical continuum in Swiss history, playing a pivotal role in both "waves of proportionalization". Splitting Walter and Emmenegger's "majority protection" argument into two parts, the first part (Hypothesis 1) emphasizes conflict patterns in the decision-making process, while the second part (Hypothesis 2) addresses the design of reform policies.

H1: Parties previously favored by electoral law tended to oppose or dilute reform during the decision-making process, while potential reform beneficiaries aimed for the most radical reform.

[25] The widespread reception of Rokkan's minority protection hypothesis, interpreting the inclusion as an act of altruism of traditional elites to protect local minorities, is one way of reading him. Strictly speaking, he does not elaborate the motives of incumbents to improve minority representation. In my interpretation, traditional parties in multicultural contexts had vital, self-directed interests to move away from MR when political upheavals were a likely scenario.
[26] However, the Court did not say how the requirement of nine seats per districts should be met. Extending the sizes of districts or legislatures or even introducing a single constituency (as, for example, the Canton of Neuchâtel did) are alternative routes to meet the requirement.

H2: Reforms enforced against the will of advantaged parties (either through partisan opposition or popular initiatives) resulted in less distorted policies compared to reforms arising from cross-party consensus.

2.3 Variables, data and methods

This section delineates the variables, data sources, and methodological approaches employed to test the hypotheses 1 and 2.

Since the biproportional method is only implemented in Switzerland, the analysis concentrates on the nine cantons where this method is currently in use (ZH, SH, AG, NW, ZG, SZ, VS, UR, GR). In the initial part of the analysis, which examines the extent of parties' strategic maneuvering (Hypothesis 1), three additional cases where attempts to adopt biproportionality were unsuccessful are included (OW, BE, TG). This expansion not only increases the case count but also allows for hypothesis testing regardless of the reform outcome (successful reform versus reform rejection). However, Graubünden as a former majoritarian canton is excluded from testing Hypothesis 1, and Valais, being a highly unique case that defies comparison for several reasons, is also omitted.[27] The subsequent part, investigating the link between conflict structures during reform processes (Hypothesis 2) and the reform design, is limited to the nine cantons where reform adoption was successful.

The relatively small sample size restricts the viability of a quantitative, large-N research design. Instead, a qualitative examination of cantonal reform processes is conducted. This offers a nuanced understanding of the factors and complexities driving the individual reforms. An in-depth description of the biproportional reform processes follows in the next chapter. These case descriptions then also provide the data source for testing the hypotheses. This approach aligns with the methodology adopted in analogous studies and adheres to the recommended research practices for studying electoral reforms (Charvát, 2023: 21):

> "The politics of electoral reform (...) aims to test the relationship between the party and electoral systems in the broader analytical and contextual framework, also including the process of selecting and changing electoral institutions by existing political actors, in particular political parties. As a result, the politics of electoral reform is studied as a complex process, which is determined and shaped by a range of factors, e.g., historical, cultural, institutional or contextual. It cannot be understood without a previous analysis of the mechanisms by which these factors interact, and which then lead to specific outcomes (...). Studies on electoral reforms are therefore not largely affected by statistical data, focusing rather on a detailed process tracing, and primarily consist of an in-depth examination of a small number of cases. Hence, qualitative analyses appear more beneficial, while quantitative studies remain rather an exception to this approach."

Hypotheses 1 and 2 explore common conflict patterns during decision-making processes and their influence on the design of reforms. Hypothesis 1 suggests that support for reform (y) primarily depends on parties' self-directed utility expectations (x). Parties previously favored by electoral laws are likely to use their political leverage to either block reform altogether or shape it to their advantage (e.g., by advocating for legal thresholds), while those disadvantaged are expected to push for the most radical reforms possible. In contrast, Hypothesis 2 focuses on the nature of the reform design itself, positing that reforms enacted without the support of previously advantaged parties (x) result in less distorted outcomes (y) compared to those achieved through cross-party consensus. Data for this section of the thesis, with the exception of Table 30

[27] Detailed explanations for the omissions can be found in the case descriptions and in the evaluation section.

regarding the (dis)proportionality of cantonal parliamentary elections where pre-existing data sets were used, was independently collected and quantified or visualized as appropriate.

The testing of the first hypothesis follows a multi-step analytical strategy:

1. Estimate the parties' positions on the reform based on their legislative (mis)representation prior to the reform (i.e., their presumed positions).
2. Collect data on the parties' actual positions regarding the reform as a whole or specific components of it (i.e., their true positions).
3. Compare the presumed and true positions for each party to determine if there is alignment and assess whether parties actively attempted to influence the electoral law to their advantage.

Before examining whether historically privileged parties show greater resistance to biproportional reform compared to those previously disadvantaged by electoral regulations, it is crucial to establish clear criteria for identifying a party as a "profiteer" or a "victim" of the existing electoral law. A party is considered a beneficiary when its seat share consistently exceeds its vote share. Conversely, when a party's seat share is significantly and consistently lower than its vote share, it is disadvantaged by the electoral system. Given the volatility of elections, it is assumed that the direction, magnitude, and consistency of legislative misrepresentation influence a party's self-assessment as a "winner" or "loser". Therefore, the classification of a party is based not on the most recent election before the reform process, but on the last four to five elections. When trends are weak (i.e., minor vote-seat differences) or inconsistent over time, making it difficult to estimate a party's stance, the label "unclear" is used.[28]

In the next phase, the parties' true stances on electoral reform are determined. To uncover these positions, content analyses of plenary minutes are conducted, specifically using valence analysis (Budge et al. 2001, Krippendorff 2004, Mayring 2010).[29] Valence analysis is a content-based method that scales relevant text components according to two or three levels (Mayring 2010: 15f.). Each cantonal reform process is examined by systematically gathering the positional data of the involved parties (pro/contra/neutral). When available, voting tables are utilized to simplify position attribution. In other cases, plenary minutes are analyzed, and when debates were only available as audio, transcriptions are produced.

The third step involves testing the first hypothesis. If the assumed and actual party positions align, it supports the notion that party stances on reform are primarily driven by self-directed utility expectations, aiming to maximize their seat share. However, the diverse affiliations of parliamentary groups add complexity, as these often vary between cantons due to the heterogeneity of cantonal party systems. For instance, some cantons have multi-partisan groups, such as the Greens and the SP forming a joint group in Uri due to their relatively low seat count. To maintain comparability and coherence while addressing cantonal peculiarities, the mapped positions are shown for individual parties wherever feasible. The results are presented in a table.

The second hypothesis explores the relationship between conflict patterns in the decision-making process and the final reform design, specifically focusing on the level of distortion within the reform. Building on Walter and Emmenegger (2019: 65f.), I hypothesize that reforms enacted despite opposition from previously favored parties or through direct democratic means will contain fewer distortive elements benefiting

[28] Consistent legislative underrepresentation = assumed reform support; consistent legislative overrepresentation = assumed reform rejection; weak or inconsistent trend in legislative misrepresentation = unclear assignment.
[29] A compilation of all primary sources consulted during this step can be found in Table 40 in the appendix.

these privileged parties. Conversely, if a reform results from cross-party consensus, this consensus is likely achieved through concessions as part of political horse-trading, leading to a final reform design with elements that distort optimal proportionality.

The analysis centers on the extent to which the final reform design is distorted. Existing literature on the selection of electoral systems during the initial "wave of proportionalization" views electoral distortions as deviations from ideal vote-seat proportionality. These distortions are influenced by factors such as the total number of mandates, constituency characteristics (size and shape), artificial electoral thresholds, and the number and strength of competing parties/lists (Taagepera & Shugart 1989, Boix 1999, Colomer 2007, Carey & Hix 2011). Various metrics have been developed to quantify these disproportionalities, including empirical measures like the Least Squares Index (Gallagher 1991) and the Effective Threshold proposed by Taagepera (1998), which assesses the "mechanical permeability" of electoral systems. While these metrics are useful for cross-sectional and longitudinal comparisons across different electoral environments, they offer limited insights into the specific mechanisms generating distortions. Additionally, metrics such as the Least Squares Index are sensitive to contextual factors not directly related to electoral design, such as the number of competing parties or the size of the parliament. Therefore, they do not isolate the effects of individual reform policies effectively. Given these limitations, I opt for a qualitative assessment of distortion for each case, rather than relying on a statistical formula that may include significant extraneous information.

In biproportional reform designs, distortions in vote-seat proportionality can arise from various elements, including local and global quorums, either separately, in conjunction, or in disjunction, as well as other features adversely affecting proportionality. The assessment of distortion in reform designs uses a scoring system (see Table 4). Points are allocated as follows:

- 1 point for a statutory quorum of 5% or less.
- 2 points for a statutory quorum exceeding 5%.
- 2 points for designs incorporating a subdivision of the electoral area for super-apportionment.
- 2 points for designs that include or maintain majoritarian elements, such as plurality elections in specific electoral districts.

The total score is used to measure the overall distortion of the reform design, categorized into a four-level hierarchical gradation.

Score(s)	Distortedness
0	absent
1	low
2-3	moderate
4-6	high

Table 4: Scoring system for the assessment of the distortedness of a reform design.

This classification precedes an evaluation of the reform process, distinguishing between conflictual processes (involving opposition from advantaged parties or direct democracy) and consensual processes (achieved through cross-party agreement). A subsequent comparison assesses whether the final reform reflects the conflict structures observed during the reform process. The second hypothesis is confirmed if there is a consistent link between the conflict structure and the degree of distortion in the final reform design. Results from these analyses are presented in tabular form.

2.4 Successfully reformed cantons

2.4.1 Canton of Zurich

The Canton of Zurich, located in northern Switzerland, is a critical hub of political activity and innovation, characterized by its high degree of urbanization and significant economic and cultural influence. Its diverse political landscape, encompassing a wide range of political parties, non-partisan groups, and social movements, reflects the canton's complex and dynamic social structure. Zurich's prominence in the national context often positions it as a trendsetter in political developments and a birthplace for new political parties.

Political profile of the Canton of Zurich

Population (in 1000)	1,580 (as of 12-31-2022)
Eligible voters (in 1000)	967 (as of 12-31-2020)
Electoral system (parliament)	proportional representation
Apportionment method	biproportional apportionment
Year of reform adoption	2005
Year of first application	2007
Elections under the new law	5
Level of reform initiation	executive
Statutory basis of reform	law on political rights
Popular vote on reform bill	no
Prior reform attempts	no
Size of the legislature	180
Legislative period	4 years
Number of electoral districts	18 (min. size 4, max. 18, median 11)
Candidacies per seat	9.4 (2023 elections)
Effective number of parties	6.1 (2023 elections)
Disproportionality (LSq-index)	1.9 (2023 elections)

Table 5: Descriptive statistics for the Canton of Zurich.

As the most populous canton in Switzerland, Zurich hosts the country's largest cantonal legislature, with 180 seats. The SVP holds the relative majority of these seats, followed by the SP and the FDP (see Figure 4). Elections are conducted across 18 electoral districts using a biproportional representation system, a method Zurich was the first canton in Switzerland to implement. This system was introduced following a 2002 voting rights complaint concerning the election process for Zurich's City Council, which led the FSC to partially uphold the complaint. The Court criticized the existing electoral districting as unconstitutional due to the small size of certain districts, a decision that has had lasting implications (BGE 129 I 185).

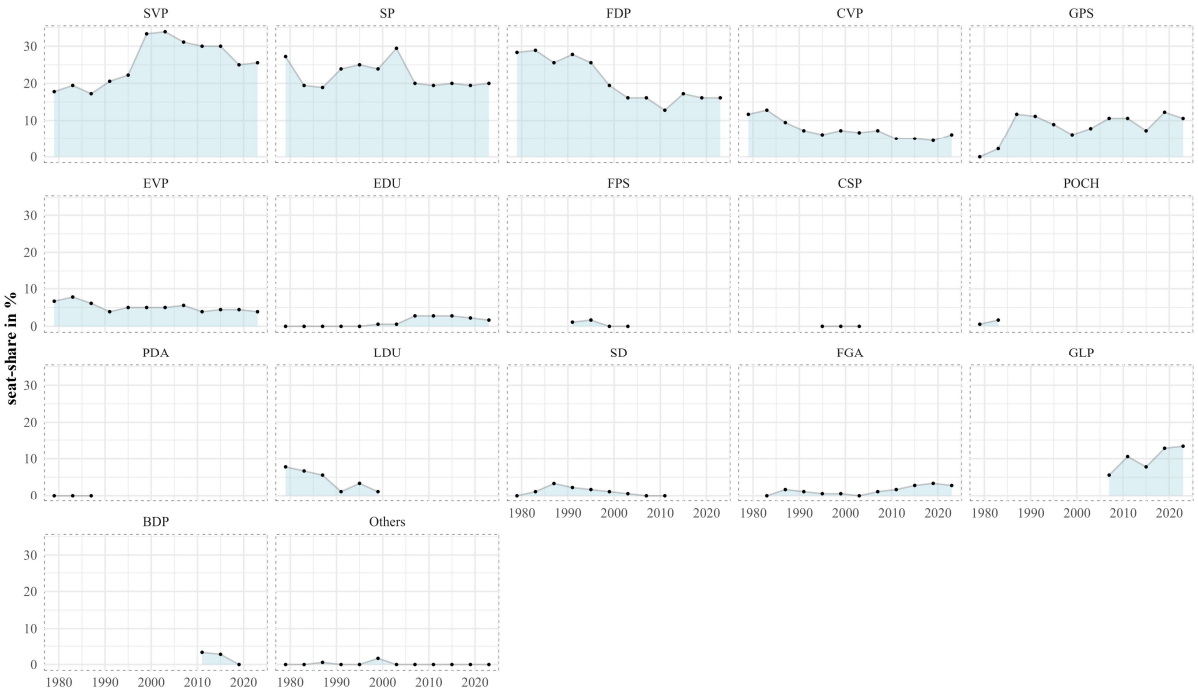


Figure 4: Parties' seat-shares in parliamentary elections in the Canton of Zurich, 1979-2023.

The issue identified in the Court's ruling was that, in smaller districts with high electoral thresholds, the principle of vote equality was compromised. Votes for smaller parties that did not secure a mandate in their district were effectively "weightless" and excluded from the final election results. The Court established a

maximum threshold of 10% for securing a seat within a district, corresponding to a minimum of nine seats per district. However, the Court did not prescribe specific measures to achieve this, leaving the political authorities with the discretion to devise appropriate compensatory measures. In response, the city of Zurich expanded the size of its electoral districts for municipal elections. This step, however, might have been unnecessary given that the biproportional apportionment procedure subsequently developed was specifically designed to address disparities in electoral success probabilities due to varying district sizes.

Cornerstones during the reform process in the Canton of Zurich

August 28, 2002	The Government Council of the Canton of Zurich initiated a comprehensive amendment to the law on political rights.
December 18, 2002	The FSC declared the division of electoral districts for elections to the Zurich City Council unconstitutional.
April 7, 2003	The Cantonal Council of Zurich addressed the new law in its first reading. In response to the FSC decision, the Council decided to remove the issue of electoral districting from the bill and deal with it separately.
September 1, 2003	Second reading on the new law on political rights and first reading on the introduction of a new apportionment system (two related issues but treated separately).
November 17, 2003	Second reading on the introduction of the apportionment method later referred to as "Doppelter Pukelsheim" or in more scientific terms: the biproportional method.
February 27, 2005	Revisions to the cantonal constitution concerning the election of the Cantonal Council passed the popular vote.
March 22, 2021	As a minor adjustment, the Cantonal Council decided to correct a formula error in the upper-tier allocation.
July 1, 2021	Enactment of the new provisions regarding the electoral threshold (a global 3% quorum was added to the 5% local quorum).

Table 6: Key reform stages in the Canton of Zurich.

Simultaneously, the Cantonal Council initiated a comprehensive revision of the cantonal law on political rights (KR-No. 4001a), which also governs the electoral provisions for all city parliaments within the canton, including the Zurich City Council.[30] In response to the Federal Court's ruling in late 2002, the Cantonal Council decided in April 2003 to separate the politically sensitive issue of electoral districting from the original legislative proposal. In a subsequent session, the Council opted to maintain the existing division of electoral districts while revising the seat allocation procedures under the new political rights law (KR-No. 4001c). The new procedure, based on biproportionality, was developed to ensure equal voter efficacy and was ultimately adopted.

The reform was debated in two key sessions, on September 1, 2003, and November 17, 2003. Remarkably, the commission's draft was widely accepted within the Cantonal Council, with broad cross-party support not only for the necessity of electoral reform but also for the introduction of the biproportional method (see Figure 5). The prevailing view was that the existing electoral system leaves a substantial portion of the electorate underrepresented. The new method, while a relatively modest adjustment to the existing electoral framework, significantly enhances overall proportionality and eliminates the need for apparentements, which had been a source of criticism. These alliances, formed for strategic rather than substantive reasons, were seen as obscure tools of electoral competition. As Green MP Daniel Vischer observed, "Until now, apparentements were a necessary evil, a political nonsense, which will be rendered obsolete by the new method."[31]

[30] According to §111 GPR ZH, a slightly simplified apportionment procedure applies for elections to single-district city parliaments. For multi-district city parliaments, however, cantonal provisions apply analogously (with the exception that they are free to deviate from the quorum regulation, see §111(4) GPR ZH).

[31] Source: Minutes of the 12th meeting of Council, September 1, 2003, p. 913. Translated from German.

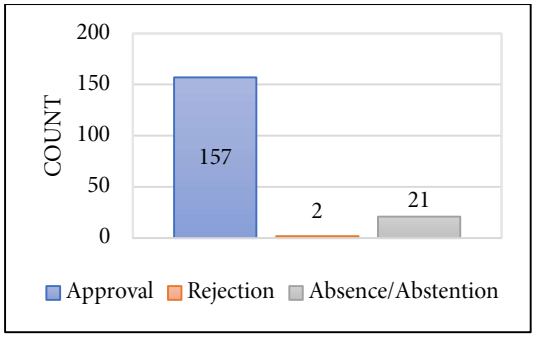


Figure 5: Positions on the final reform bill to introduce biproportional apportionment for parliamentary elections in the Canton of Zurich.[32]

The introduction of an electoral threshold, however, sparked a contentious debate. Five different models were proposed and rigorously discussed:

1. No quorum.
2. A local 3% quorum.
3. A local 5% quorum, with an alternative global 2% quorum.
4. A local 5% quorum (as recommended by the commission).
5. A local 5% quorum combined with a global 3% quorum.

This debate was marked by a thorough examination of the potential electoral implications of each model. Smaller parties and the Greens advocated for a system with minimal barriers to entry, supporting models 1, 2 and 3. They argued that the primary advantage of the new biproportional method was its ability to translate voter preferences into seats with minimal distortion. Introducing a quorum, they contended, would undermine the proportionality and representational accuracy of the new system. Concerns were also raised that any form of quorum would particularly disadvantage smaller groups, including youth parties, especially if apparentements were abolished. Despite these arguments, models 1 and 2 were eventually rejected.

Larger parties, by contrast, argued for the necessity of electoral thresholds to prevent further fragmentation of the party system and to restrict the entry of small, single-issue groups.

Quorum Model	**Model 1**	**Model 2**	**Model 3**	**Model 4**	**Model 5**
	No quorum	3% local	5% local or 2% global	5% local	5% local and 3% global
Party/Faction	EDU, EVP, FDP, GPS, SD	EVP, FDP, GPS, SP	GPS, CVP, EVP, SD, EDU	FDP, SP, SVP	SVP

Table 7: Party positions on the different quorum models in the Canton of Zurich.

The FDP's divided stance on the issue is particularly noteworthy. While the majority of the party supported the commission's model (model 4), a minority favored eliminating the quorum entirely, arguing instead for a 3% local threshold. The CVP initially backed model 4 but was persuaded by Martin Bäumle (GPS) to support model 3, which he argued was necessary to avoid discrimination against small parties with dispersed electorates. Bäumle's model, which required exceeding either a 5% local or 2% global threshold, did not gain majority support. The SVP then introduced a stricter model 5, which required parties to meet both the local

[32] Note: According to the procedural rules in force at the time, a roll-call vote was only held upon request. For this reason, a classification by party/faction is not possible. In addition, the minutes do not allow us to determine who among the non-voters actively abstained from voting and who did not vote due to physical absence.

and global thresholds, making it the most restrictive option. Ultimately, model 4, endorsed by the dominant parties (FDP, SP, SVP), was adopted.

The issue of the electoral threshold resurfaced in 2016 when MPs Claudio Schmid (SVP) and Martin Romer (FDP) proposed replacing the local 5% quorum with a canton-wide 3% threshold (KR-No. 110/2016). The proponents argued that this change would prevent the entry of small parties with limited cantonal relevance, thereby enhancing the efficiency of the council. The Government Council, however, recommended maintaining the existing system. The responsible commission proposed a compromise solution, combining both thresholds in an "either-or" clause, identical to the ones applying in the cantons of Aargau and Zug. This proposal aimed to reduce electoral barriers for parties with dispersed electorates, counter to the original proponents' intent. The Cantonal Council adopted this compromise on November 30, 2020 (KR-No. 110a/2016), and the new regulation came into effect on July 1, 2021.

Although the reform was subject to a facultative referendum, no popular vote was held. Nonetheless, as part of the 2005 revision of the cantonal constitution, voters approved two amendments related to the Cantonal Council's election (Art. 51(2-3) KV/ZH), thus indirectly legitimizing the new electoral system.

In 2021, the Cantonal Council addressed an error in the upper-tier seat allocation process. It was noted that rounding voter numbers during super-apportionment could result in slight distortions in seat distribution (KR-No. 118/2018). The Council unanimously agreed that super-apportionment should be based on unrounded voter numbers, with rounding applied only to the final quotient.

2.4.2 Canton of Aargau

Political profile of the Canton of Aargau	
Population (in 1000)	711 (as of 12-31-2022)
Eligible voters (in 1000)	443 (as of 12-31-2020)
Electoral system (parliament)	proportional representation
Apportionment method	biproportional apportionment
Year of reform adoption	2008
Year of first application	2009
Elections under the new law	4
Level of reform initiation	legislative
Statutory basis of reform	electoral law and constitution
Popular vote on reform bill	yes (76.2% resp. 78.0% approval)
Prior reform attempts	no
Size of the legislature	140
Legislative period	4 years
Number of electoral districts	11 (min. size 7, max. 30, median 10.0)
Candidacies per seat	7.3 (2020 elections)
Effective number of parties	5.0 (2020 elections)
Disproportionality (LSq-index)	0.6 (2020 elections)

Table 8: Descriptive statistics for the Canton of Aargau.

The party system in the Canton of Aargau closely mirrors the national political landscape of Switzerland, with nearly identical parties represented in both the cantonal and national parliaments, and their relative strengths similarly aligned. Electoral trends in Aargau also strongly reflect those observed at the national level. As illustrated in Figure 6, which details the parties' seat shares over time, Aargau's party system is notably fragmented. Two dominant parties occupy the extremes of the political spectrum: the SVP and the SP, the latter in coalition with the Green Party. The canton's 140-member parliament is elected based on biproportionality, a reform that places Aargau alongside the Canton of Schaffhausen in Switzerland's second wave of adopting biproportional representation.

Aargau stands out for its vigorous pursuit of electoral reforms over a relatively short period. The implementation of biproportional apportionment must be understood within this broader context of institutional reformism, which encompasses both intra-cantonal vertical and inter-cantonal horizontal dimensions. Table 9 summarizes the various reform steps undertaken during this period.

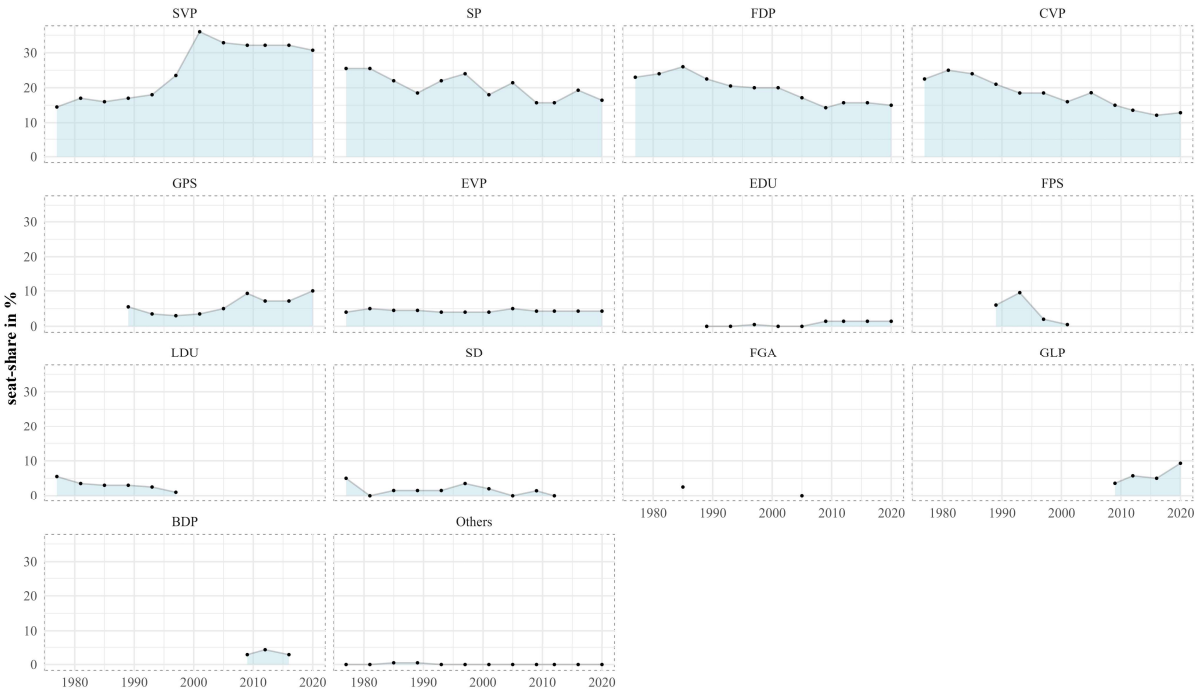


Figure 6: Parties' seat-shares in parliamentary elections in the Canton of Aargau, 1977-2020.

The period of heightened reform activity began in May 2003, when the size of the cantonal parliament was reduced from 200 to 140 members. This reduction was the result of a successful popular initiative by the FDP titled "Slimming down the Grand Council!" The initiative, motivated by the desire for a "lean, efficient, modern, and cost-effective parliament" (as quoted in the official voting documents), passed after previous parliamentary efforts to achieve the same had failed. However, reducing the size of the parliament inevitably led to higher electoral thresholds in smaller districts, a consequence that was well anticipated by policymakers. To mitigate this, a second constitutional amendment was proposed, allowing for the merger of small districts. The proposals for constitutional revision were approved by a majority of voters, with 63.8% in favor. Despite this, the representatives chose not to exercise the option to merge smaller electoral districts, effectively accepting a violation of the principle of equality in electoral success, which is only tolerated in "objectively justifiable cases" according to legal jurisprudence. As a result, the reform led to electoral thresholds exceeding 10% in the districts of Laufenburg, Muri, and Zurzach, marking a significant setback in terms of electoral equity.

Cornerstones during the reform process in the Canton of Aargau

Mai 18, 2003	A popular initiative to reduce the size of parliament from 200 to 140 seats and to allow for district mergers passed the popular vote.
June 8, 2004	In preparation of the Implementation Act of a new electoral law, the Grand Council decided to dispense with the legal possibility of creating district mergers and changing the apportionment formula as proposed by the Government Council.
July 21, 2004	The Greens and other co-petitioners have filed an appeal with the FSC against the above-mentioned decisions of the Grand Council.
September 26, 2004	A legislative bill to make minor adjustments to the electoral law for cantonal elections passed the popular vote.
October 27, 2004	Verdict of the FSC that declared parts of the cantonal law for parliamentary elections as unconstitutional.
March 20, 2007	First parliamentary reading on an electoral reform.
September 18, 2007	Second reading & final vote on the issue.
February 24, 2008	Legislative and constitutional bill to introduce the biproportional apportionment method for cantonal elections passed the popular vote.
November 27, 2011	Legislative bill to introduce electoral quorums for parliamentary elections passed the popular vote.

Table 9: Key reform stages in the Canton of Aargau.

To address the challenges posed by the reduction in parliament size while still complying with the Federal Constitution, the cantonal government proposed two models to the Grand Council. The first model envisioned merging districts, while the second proposed changing the apportionment method, as suggested by Friedrich Pukelsheim. Ultimately, the Grand Council decided to retain the existing regulations and refrain from changing the apportionment method. Nonetheless, the Grand Council pursued further electoral reforms, comprising three smaller revisions – none of which addressed the core issue of high district-induced electoral thresholds.[33] This legislative package passed a popular vote in September 2004, with 74.7% approval.

Just a month later, in October 2004, the FSC issued a verdict on a constitutional complaint filed by the Green Party and other co-petitioners from Aargau (BGE 131 I 74). The complaint challenged the inequalities in voting effectiveness caused by the small size and significant variance in district sizes, a status quo that allegedly favored larger parties at the expense of smaller ones. The Court upheld the plaintiffs' position and mandated the creation of a "constitutional electoral order" for future parliamentary elections, laying the groundwork for subsequent reforms.

[33] The proposed changes included a switch from a list-based to a candidate-based voting system (1); a new method for allocating remainder seats that enables parties without a full seat entitlement to participate in the process (2) and a ban of candidate cumulations on pre-printed lists (3) (source: official voting documents).

In September 2007, after two intense readings, the Grand Council decided to introduce biproportional apportionment. Figure 7 shows the party positions on the final reform bill, revealing that a razor-thin majority (66:63, with 1 abstention and 10 absences) voted in favor of the bill, which included both an amendment to the law and the constitution. The figure also shows an unexpected coalition formed by the left (SP, GPS) and center parties (CVP, EVP) against the right-wing parties (FDP, SVP).

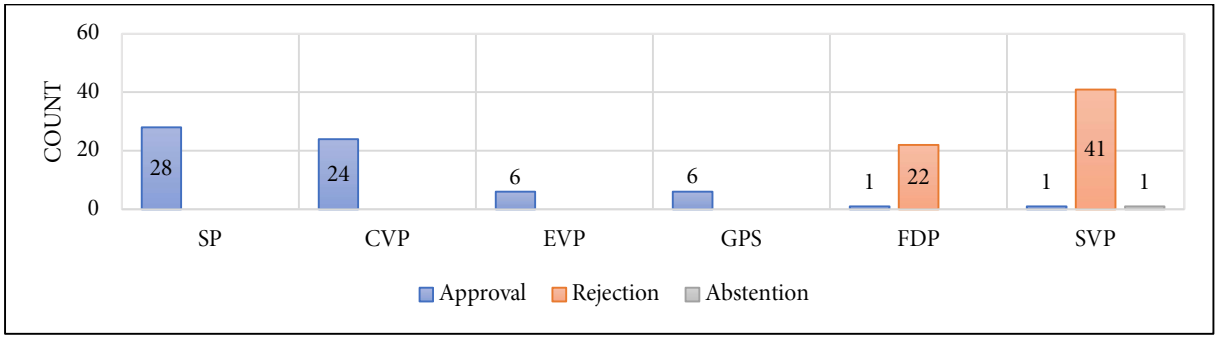


Figure 7: Party positions on the final reform bill to introduce biproportional apportionment for parliamentary elections in the Canton of Aargau. The were no differences in the voting pattern between the legislative vote and the constitutional vote.

Contrary to what the final vote results might suggest, the reform was only marginally contentious during the plenary sessions, making it difficult to discern the exact motives behind the parties' (often unexpected) positions. The most heated debates centered on the implementation of electoral thresholds and the potential for apparentements within the same party family. One faction argued that quorums are fundamentally incompatible with the principle of proportionality. A second faction was willing to accept quorums as part of a political compromise, while a third viewed them as a necessary institutional safeguard. This third camp argued that quorums prevented the emergence of extremist or single-issue groups, "lone wolves", and also mitigates party fragmentation, thereby enhancing legislative efficiency, cost-effectiveness, and governability.

During the debate on the introduction of electoral quorums, a related discussion emerged concerning whether political parties should be allowed to engage in internal apparentements (e.g., Young Socialists aligning with the SP). Some parliamentarians argued that, if quorums were to be established, allowing apparentements would be prudent as it would facilitate the entry of youth parties into the parliament through electoral cooperation with their parent party. Additionally, this mechanism could provide a way for groups and parties that do not meet the quorum to circumvent it effectively. Ultimately, the Grand Council refrained both from establishing quorums and from establishing apparentements.

However, the debate on quorums resurfaced following the first election held under the new electoral law on March 9, 2009. On June 1, 2010, the FDP submitted a motion to introduce a local 5% quorum, citing concerns that a proliferation of political parties would reduce the Council's efficiency. The Government Council countered with a proposal for a combined quorum: either 5% within a single district or 3% across all districts. In the final vote on June 7, 2011, the Grand Council endorsed the government's combined quorum proposal by 68 to 53 votes (with 2 abstentions and 17 absences). Except for the Social Democrats, the largest parties voted in favor of the amendment, while smaller parties opposed it (see Figure 8). Notably, the CVP, which had previously opposed quorums, shifted its position and supported the introduction, tipping the balance. The amendment to the law was approved by 58% of voters on November 7, 2011.

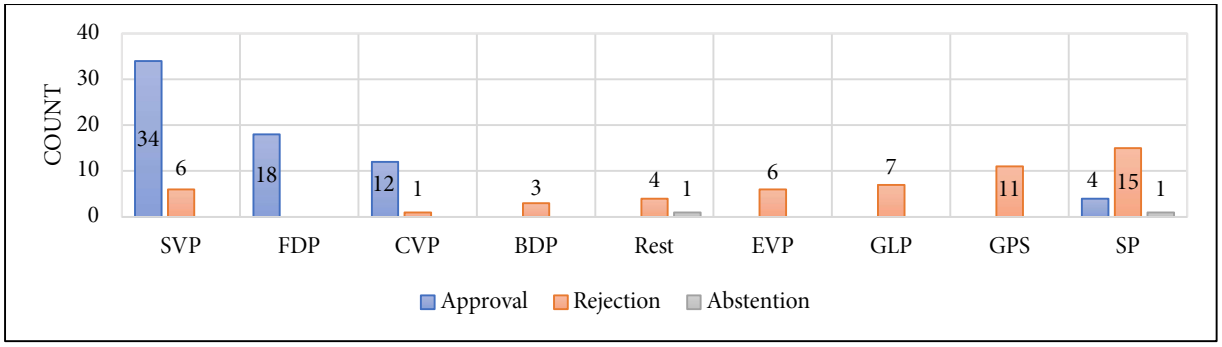


Figure 8: Party positions on the legislative bill to introduce a combined quorum (either at least 5% in one district or at least 3% across all districts) for cantonal elections. The "Rest" group consists of SD (2), EDU (1), SLB (1) and an Independent who abstained from the vote.

2.4.3 Canton of Schaffhausen

Political profile of the Canton of Schaffhausen	
Population (in 1000)	85 (as of 12-31-2022)
Eligible voters (in 1000)	54 (as of 12-31-2020)
Electoral system (parliament)	proportional representation
Apportionment method	biproportional apportionment
Year of reform adoption	2008
Year of first application	2008
Elections under the new law	4
Level of reform initiation	executive
Statutory basis of reform	electoral law and constitution
Popular vote on reform bill	yes (61.1% resp. 62.8% approval)
Prior reform attempts	no
Size of the legislature	60
Legislative period	4 years
Number of electoral districts	6 (min. size 1, max. 27, median 7.5)
Candidacies per seat	9.2 (2020 elections)
Effective number of parties	5.2 (2020 elections)
Disproportionality (LSq-index)	1.0 (2020 elections)

Table 10: Descriptive statistics for the Canton of Schaffhausen.

Despite the relatively small size of Schaffhausen's cantonal parliament, with only 60 members, it exhibits considerable party diversity and a highly competitive electoral environment, reflecting a vibrant democratic culture. The two dominant parties in this context are the SVP and SP, with other significant parties including the FDP, GLP, GPS, and the "Alternative Liste" (AL), a progressive left-wing party active across northwestern Switzerland and included in the FGA group. Legislative elections in Schaffhausen are held within districts under a proportional representation system, with seats allocated through the biproportional apportionment method. Alongside the Canton of Aargau, Schaffhausen was part of the second wave of national reforms introducing biproportional representation, following Zurich. Figure 9 illustrates the seat shares of all parties contesting for the cantonal legislature between 1976 and 2020.



Figure 9: Parties' seat-shares in parliamentary elections in the Canton of Schaffhausen, 1976-2020.

The initial conditions in Schaffhausen closely mirrored those in Aargau. The FSC's landmark decision concerning the constitutionality of Zurich's City Council elections in 2002 was pivotal, emphasizing that electoral districts of varying sizes, particularly very small ones, must adhere to legal equality only if these districts represent units with a shared sense of belonging, whether for historical, cultural, linguistic, or other significant reasons. Concurrently, a popular initiative titled "60 members of the Cantonal Council are enough!", launched in November 2003 and approved by voters in August 2004, reduced the Cantonal Council's size from 80 to 60 members, with 70.2% voter support. These two factors created significant pressure for electoral reform.

Cornerstones during the reform process in the Canton of Schaffhausen

August 29, 2004	A popular initiative to reduce the size of parliament from 80 to 60 seats passed the popular vote.
August 30, 2004	Just one day after the vote, a legislative motion was submitted (07/2004) that aimed to "improving the electoral system to better protect small parties".
November 17, 2004	Submission of a second legislative motion (02/2004) that requested a reform of the electoral law.
April 24, 2007	Report and proposal of the Executive Council for a reform of the electoral system.
July 6, 2007	Release of the report of the Special Commission appointed by the Government Council.
September 3, 2007	First parliamentary reading. Sequel September 17, 2007.
October 29, 2007	Second reading & final vote on the issue.
February 24, 2008	Legislative and constitutional bill to introduce the biproportional apportionment method for cantonal elections passed the popular vote.
October 26, 2015	Cantonal Council adds a "majority clause" to the electoral law.

Table 11: Key reform stages in the Canton of Schaffhausen.

The arguments for reducing the size of the cantonal legislature in Schaffhausen paralleled those in Aargau, with proponents suggesting that a smaller assembly would enhance efficiency and reduce costs, as demonstrated in similar reforms in Vaud and Lucerne. However, it was acknowledged that such a constitutional revision would not only require changes to parliamentary procedures but also necessitate a comprehensive overhaul of the electoral system. Two reform paths were outlined: one involving district mergers (for allocation purposes) and the other focusing on biproportional apportionment.

Following the plebiscite on reducing the cantonal parliament's size and the submission of a legislative initiative, the cantonal government proposed a reform draft. Consultation with stakeholders revealed broad support for biproportional apportionment, leading the Government Council to dismiss alternative reform models and advocate solely for the biproportional method, a decision endorsed by the "special commission" appointed by the government.

The Cantonal Council discussed various reform concepts in two separate sessions. Plenary minutes indicated a broad, cross-party consensus on the need for electoral reform, though disagreements persisted on the design of the reform. A majority favored the biproportional model, considering it the most suitable solution for ensuring accurate voter representation while preserving the existing district structure. A minority, primarily from the right-wing SVP, proposed a less extensive regionalized model that merged small districts rather than simulating a single constituency, as in biproportional representation. The SVP opposed the bi-

proportional model, arguing it could produce anomalous results, reduce parliamentary efficiency, undermine regional representation, and centralize election campaigns, thereby threatening regional party sections. They preferred a model addressing the issue of small districts through specific mergers.[34]

A secondary discussion focused on whether an electoral quorum should be introduced. Despite intense deliberations in the special commission, the majority of the parliament found no compelling reason to introduce any form of electoral hurdle. Florian Keller, an MP from the Alternative Liste, expressed the prevailing sentiment: "There is no reason to deny entry to the Cantonal Council to anyone who can muster enough electoral votes. A possible fragmentation of the parties reflects the voters' will, and any attempt to suppress it with quorums is undemocratic."[35] The SVP was the only party advocating for a canton-wide 5% quorum, arguing it would prevent party fragmentation and discourage the entry of individuals with extremist views, while also enhancing the Council's productivity and efficiency. However, this proposal was ultimately rejected by the majority of the Council.

Following the extended debate on quorums, the final vote on adopting biproportional apportionment took place. Figure 10 and Figure 11 depict the final voting results on the bill and the constitutional amendment to introduce biproportional apportionment, with eight absentees in each vote.

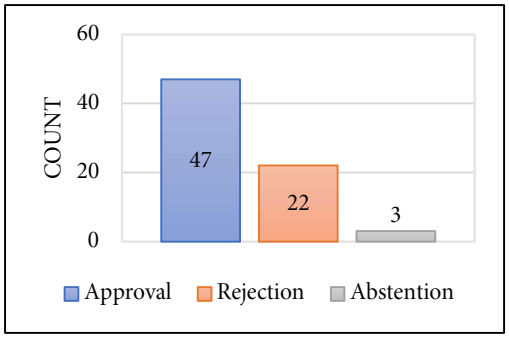


Figure 10: Positions on the constitutional amendment for the adoption of biproportional apportionment for parliamentary elections in the Canton of Schaffhausen.

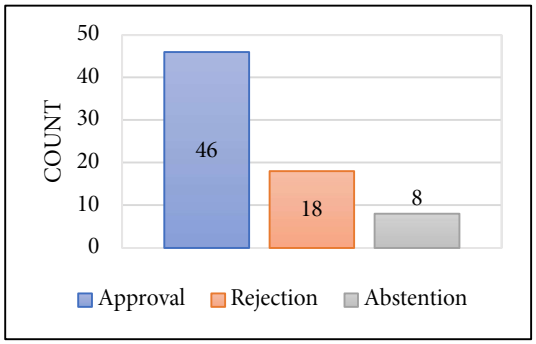


Figure 11: Positions on the reform of the electoral law for the adoption of biproportional apportionment for parliamentary elections in the Canton of Schaffhausen.

On October 26, 2015, the Cantonal Council made minor amendments to the Electoral Act, including a "majority clause" designed to ensure that the strongest party always secures at least one mandate in a constituency, provided it is entitled to at least one seat under the overall allocation. This broadly worded addition aimed to prevent discordant seat allocations in the single-member constituency of Buchberg-Rüdlingen, and it garnered widespread support across all parties, with 41 supporting the changes and only one dissenting vote.[36]

[34] The SVP was not entirely clear on whether these should be actual mergers of districts or just hypothetical mergers for the purpose of seat apportionment. Some of the statements made in this regard were contradictory.
[35] Source: Minutes of the 14th meeting of Council, 17 September 2007, p. 697. Paraphrased quote.
[36] Cf. Beriger 2018: 120 and Glaser 2018: 295.

2.4.4 Canton of Nidwalden

The cantonal legislature of Nidwalden, consisting of 60 seats, transitioned from a majoritarian to a proportional representation system in 1981. Despite this change, the formerly dominant CVP and FDP parties retained significant influence. By the 2000s, the SVP emerged as a third major force, and later the GPS and GLP entered the political scene. The SP, with its marginal vote share, formed a faction with the GPS. Since 2014, seat allocation has been determined by the biproportional apportionment method. Figure 12 illustrates the development of seat shares from the first proportional elections in 1982 to the latest in 2022.

Political profile of the Canton of Nidwalden

Population (in 1000)	44 (as of 12-31-2022)
Eligible voters (in 1000)	32 (as of 12-31-2020)
Electoral system (parliament)	proportional representation
Apportionment method	biproportional apportionment
Year of reform adoption	2013
Year of first application	2014
Elections under the new law	3
Level of reform initiation	executive
Statutory basis of reform	electoral law
Popular vote on reform bill	yes (60.4% approval)
Prior reform attempts	no
Size of the legislature	60
Legislative period	4 years
Number of electoral districts	11 (min. size 2, max. 11, median 5.0)
Candidacies per seat	3.7 (2022 elections)
Effective number of parties	4.6 (2022 elections)
Disproportionality (LSq-index)	1.3 (2022 elections)

Table 12: Descriptive statistics for the Canton of Nidwalden.

The reform of the seat allocation process was initiated by a complaint from a group associated with the Green Party of Nidwalden regarding the 2010 elections. The Government Council's decision on seat distribution across districts was challenged as unconstitutional due to the varying sizes of districts, some of which were relatively small. Although the Cantonal Constitutional Court initially rejected the complaint, stating that the division of electoral districts was consistent with the constitution given historical circumstances, the FSC overturned this decision. The Court ruled that the political communes did not justify significant deviations from proportional representation, compelling the canton to revise its electoral law.

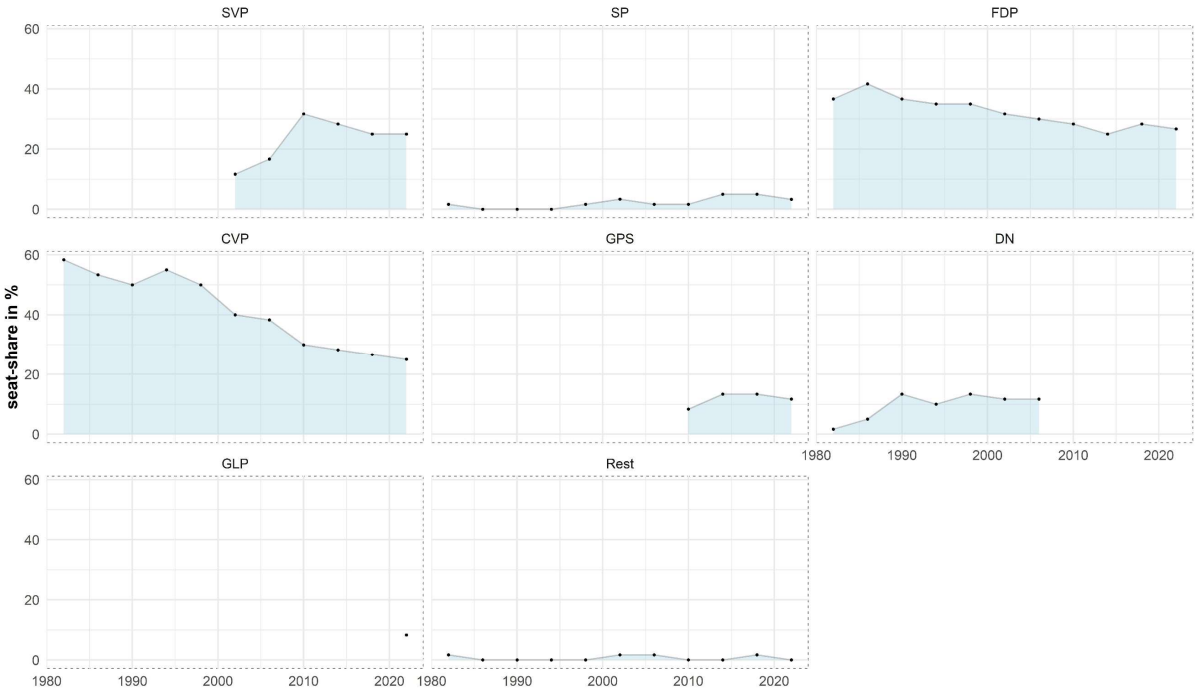


Figure 12: Parties' vote-shares in parliamentary elections in the Canton of Nidwalden, 1982-2022.

In response, the Government Council proposed several options for constitutional compliance, including a return to majoritarianism, district mergers, and the adoption of the biproportional divisor method. A majority during the consultation process supported the biproportional method. Consequently, the Government Council drafted a bill adopting this method and submitted it for further consultation, which saw broad support except for the planned introduction of a 5% quorum.

Cornerstones during the reform process in the Canton of Nidwalden	
July 7, 2010	Ruling of the FSC that the electoral law of the Canton of Nidwalden is not in conformity with the Federal Constitution.
March 28, 2012	First parliamentary reading on the reform of electoral law.
April 25, 2012	Second parliamentary reading and final vote on the reform bill.
September 22, 2013	Popular vote on the introduction of biproportional apportionment.

Table 13: Key reform stages in the Canton of Nidwalden.

The Cantonal Council debated the bill in two readings, where arguments mirrored those in other cantons: the biproportional method was praised for better reflecting voter preferences without altering constituency structures, thus preserving municipal autonomy. The SVP's alternative model of a regionalized Pukelsheim system, combined with the Hagenbach-Bischoff method, did not gain traction. The SVP and parts of the CVP criticized the biproportional system for its complexity and perceived distance from citizens. Despite this opposition, the biproportional method was ultimately adopted (see Figure 13).

The initial bill included a 5% quorum, but this was removed in the revised version after failing to gain support during consultations. The two readings in the Cantonal Council did not see significant controversy over the quorum issue, as proponents agreed that no artificial electoral hurdle was necessary.

Two "constructive referenda" were launched against the partial revision of the electoral law: one by the SVP, advocating district mergers, and another proposing a return to majoritarianism.[37] However, both counterproposals were defeated by the biproportional variant, which received 60.4% approval.

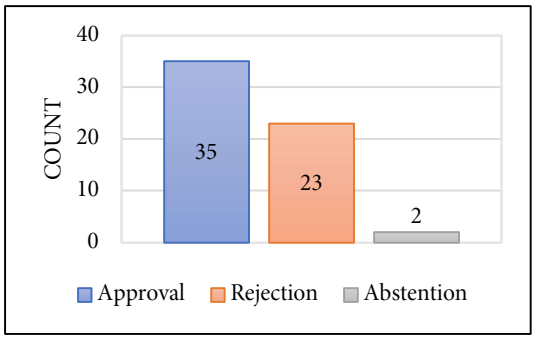


Figure 13: Positions on the final reform bill to introduce biproportional apportionment for parliamentary elections in the Canton of Nidwalden.

[37] 250 eligible voters can challenge a passed legislation by formulating a counterproposal. The draft bill and the counterproposal are then put to a popular vote (see Art. 25 of the Political Rights Act of the Canton Nidwalden). It is also possible that a legislation is challenged by multiple referenda.

2.4.5 Canton of Zug

The Zug cantonal parliament consists of 80 members and cantonal elections take place on the basis of proportional representation. Since 2014, the distribution of the seats to the parties follows the biproportional method.

Political profile of the Canton of Zug

Population (in 1000)	131 (as of 12-31-2022)
Eligible voters (in 1000)	79 (as of 12-31-2020)
Electoral system (parliament)	proportional representation
Apportionment method	biproportional apportionment
Year of reform adoption	2013
Year of first application	2014
Elections under the new law	3
Level of reform initiation	executive
Statutory basis of reform	electoral law and constitution
Popular vote on reform bill	yes (80.6% approval)
Prior reform attempts	yes (2006)
Size of the legislature	80
Legislative period	4 years
Number of electoral districts	11 (min. size 2, max.19, median 6)
Candidacies per seat	4.0 (2022 elections)
Effective number of parties	5.4 (2022 elections)
Disproportionality (LSq-index)	1.0 (2022 elections)

Table 14: Descriptive statistics for the Canton of Zug.

Until the 2006 elections, two parties formed the left camp: on the one hand, the Social Democratic Party and, on the other, the Alternative Left Party. In 2009, the Alternative Left Party together with the Green Party formed an alliance party "Alternative – die Grünen Zug" (henceforth classified as GPS). Apart from this change and the entry of the Green Liberal Party in 2010, the party landscape has remained relatively stable over the last decades. Figure 14 illustrates the seat shares of the cantonal parties between 1978 and 2022.

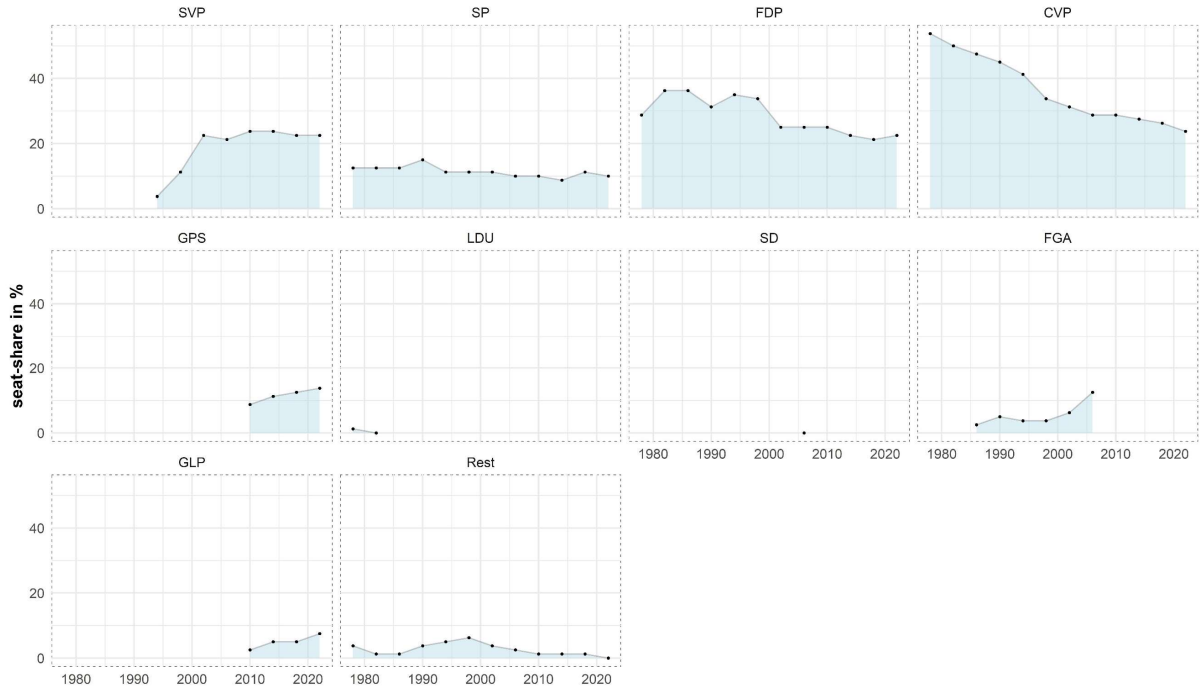


Figure 14: Parties' vote-shares in parliamentary elections in the Canton of Zug, 1978-2022.

Cornerstones during the reform process in the Canton of Zug	
July 6, 2006	Minority request by the Advisory Committee to consider the biproportional apportionment method for parliamentary elections was rejected by the Cantonal Council.
December 20, 2010	Ruling of the FSC that the electoral law of the Canton of Zug is not in conformity with the Federal Constitution.
January 31, 2013	First parliamentary reading on the reform bill.
Mai 2, 2013	Second parliamentary reading and final vote on the reform bill.
July 10, 2013	Ruling of the FSC, which overturned the decision of the Cantonal Council to oppose the biproportional apportionment method with a second proposal (status quo).
September 22, 2013	Popular vote on the introduction of biproportional apportionment.

Table 15: Key reform stages in the Canton of Zug.

A first substantive point of contact with the biproportional seat allocation procedure already occurred in 2006. On 6 July 2006, the Cantonal Council discussed the total revision of the cantonal electoral law. A minority of the Advisory Committee proposed that the matter be rejected in order to examine in detail the method of biproportional apportionment. However, this proposal was rejected by the Cantonal Council by 54 votes to 15 (with 3 abstentions).

Then, about four years later on 7 February 2010, the cantonal parties of the Alternative Green Party, the Social Democratic Party, the Christian Social Party and the Independent Green Forum Hünenberg, as well as six private individuals, lodged an appeal with the FSC against the decision of the cantonal council of 28 January 2010 concerning the number of Cantonal Council mandates for each municipality (which constitute the electoral districts) as well as the decision of the Cantonal Council of 27 August 2009 concerning the amendment of electoral law of 28 September 2006.

On 20 December 2010, the FSC upheld the appeal on the main point, i.e., with regard to the Cantonal Council's decision on the number of Council mandates attributed to the municipalities/districts (BGE 136 I 376). In line with its previous jurisprudence, the Court reasoned that the realization of proportional representation strongly depends on the size of the electoral district. Districts with a low number of seats correspond with a high natural quorum and any vote cast for a party or group below that quorum would be ineffective. Under such conditions, the chance of success of a vote (in the sense that it counts towards the result) varies between the districts and, consequentially, between voters. However, according to the Court, this infringes the constitutional requirement of article 34 paragraph 2 of the Federal Constitution, according to which "the free formation of will and uncorrupted voting are protected by the warranty of political rights".

In response to the Court's decision, the Government Council drafted an amendment to the legal basis of the election procedure and sent it out for consultation. Subsequently, the government presented three different models to the advisory commission and the cantonal parliament:

- Adjustment of the district structure (model 1): The first model envisaged a redistricting. To meet the FSC's requirement of a natural quorum of less than 10%, existing districts would have had to be merged into districts of similar size.
- Creation of constituency associations (model 2): The second proposed the formation of constituency associations and a change in the distribution procedure. Instead of distributing the legislative seats on district level, they should first be distributed at the level of constituency associations and then, in a second step, at the level of districts. Through this artificial partition of the seat allocation procedure, high district quorums can be effectively circumvented.
- Change of the apportionment formula (model 3): The third model envisaged the adoption of the biproportional apportionment method. This model is essentially very similar to the second one, as

it proposes an adjustment of the apportionment procedure. But instead of determining seat entitlements at the level of constituency associations, the results at the level of the entire electoral area serve as the basis for seat allocation. In addition, a slightly different formula applies (divisor method with standard rounding). Since this model requires only a minimal intervention in the existing electoral system, it was clearly favored by both the government and the consultation participants.

The commission in charge was also in favor of the third model but suggested two changes to the government's proposal: First, a statutory quorum should be introduced. The commission opted for an "either-or" clause, as was already in force in then Canton of Aargau: To participate in the seat allocation, an electoral list must either garner 5% of the electoral votes in at least one district or 3% of all the votes cast throughout the entire electoral territory. A second amendment concerns the single-member districts. A "majority clause" is to ensure that in each district the list with the largest number of votes receives at least one seat. Due to discordant seat allocations, there is the possibility that the list with the largest number of votes in a single constituency would not win the seat. As this would be hard to comprehend and accept from the voters' point of view, it was decided to include this additional clause in the commission's draft.

The reform of Zug's electoral system was influenced by another decision-making process that took place in parallel, which shall be briefly described. At that time the guarantee of the constitution of the Canton of Schwyz by the National Council was pending. This was relevant because the constitution of the Canton of Schwyz contained provisions on the election of the Cantonal Council that conflicted with jurisdiction of the FSC. Specifically, the revision provided that the municipalities form the electoral districts, that each electoral district is entitled to at least one parliamentary seat and that the cantonal councilors are elected within the electoral districts. As these provisions were not compatible with the latest jurisdiction of the FSC, the Federal Council recommended to the FA not to guarantee the conflicting constitutional provisions. However, in its decision of 27 November 2012, the Council of States guaranteed the constitution integrally. The integral guarantee of the constitution of the Canton of Schwyz by both councils would have been interpreted as a strong political signal in favor of cantonal electoral autonomy and as a relativization of the previous jurisdiction of the FSC. Against this background, the Cantonal Council has formulated two antithetical requests, depending on the outcome:

- If the Schwyz Constitution was to be guaranteed, the introduction of biproportional apportionment should be constitutionally excluded.
- If the Schwyz Constitution was not to be guaranteed, the Commission proposes the introduction of its own version of the biproportional apportionment.

On 31 January 2013, the Cantonal Council commenced deliberations on revision of the electoral system. In the first session, the majority of the legislative assembly essentially followed the proposal of the Government Council. The CVP and SVP factions were fundamentally opposed to the biproportional apportionment method. In view of the SVP faction, the FSC's decision interfered too much with cantonal autonomy. For this reason, they opposed both the revision of the law and the revision of the constitution. The CVP faction criticized the new procedure also on a material level: the new procedure presumably contained serious deficiencies and led to even greater injustices than the election procedure that had applied until then. In particular, they were dissatisfied by the occurrence of discordant seat allocations. The FDP faction also had reservations about the biproportional method, but unlike the SVP and CVP saw a need for revision. How-

ever, if the Schwyz cantonal constitution should be guaranteed by the FA, the FDP would reject the biproportional method. Otherwise, it would support the draft submitted by the Advisory Committee, albeit less out of conviction than because of the futility of the situation.

SP, GPS and GLP were on the side in favor of the new apportionment system. The arguments among the proponents were largely identical: Each vote should be considered and given equal weight in the outcome. And the only method that can optimally fulfill this condition without changing the district structure is biproportional apportionment. However, the SP also expressed criticism of the biproportional method. The lack of transparency in the allocation of seats poses a major disadvantage.

Regarding the introduction of a statutory quorum, the Council dissented from the governmental position. In contrast to the government, a majority of the Cantonal Council saw the need to implement an electoral threshold. Thus, the Cantonal Council followed the commission's proposal to introduce a quorum following the example of the Canton of Aargau. I.e., a party needs at least either 3% of all party votes in the canton or 5% of the party votes in a district in order to be included in the seat allocation process. Remarkably, there was little debate on the proposed statutory quorum, with only the SVP and SP commenting on it. The SVP's stance was notably surprising; their criticism of the quorum was typically associated with left-leaning parties. They argued that the quorum system, as outlined in § 52c para. 4, would disadvantage smaller parties in favor of larger ones.[38] This perspective, articulated by SVP MP Beni Riedi, reflects a view usually found on the left, which traditionally opposes mechanisms perceived as favoring major parties at the expense of minorities.

Since, apart from the SVP and the SP, who opposed the implementation of a statutory quorum for identical reasons, the minutes do not provide any information on the other parties' stances on that matter, no further conclusions can be drawn from the voting result shown in Figure 15.

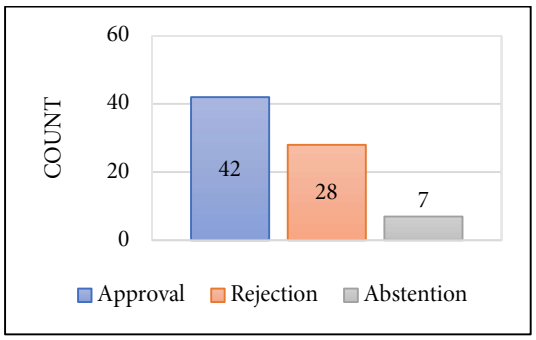


Figure 15: Positions on the proposal of the advisory commission to introduce a paragraph providing for a statutory quorum (3% globally or 5% locally) for parliamentary elections in the Canton of Zug.

A few months later, on 18 March 2013, it was the turn of the National Council to decide on the guarantee of the Schwyz Cantonal Constitution. The Council rejected the guarantee of the article in question for the second time, making the non-guarantee final. The decision was close however, with 100 votes against 91. For the abovementioned reasons, this decision also had an impact on the second reading of the Cantonal Council, which took place on 2 Mai 2013.

Since the general discussion on the introduction of the biproportional apportionment method had already taken place in the first reading, the second reading was primarily concerned with clarifying the further

[38] Source: Minutes of the 41st meeting of Council, 31 January 2013, p. 1361. Translated from German.

course of action. It was clear, however, that a constitutional revision that contradicted federal law would have very little chance of being guaranteed by the FA. In the second reading, two new proposals were added to the draft from the first reading. It was then decided to vote on the three proposals at hand and to submit the two winning proposals to a popular vote. If no proposal receives an absolute majority of votes in the first round of voting, a vote shall be taken to determine which of the two proposals receiving the fewest votes shall be eliminated. The two remaining proposals are then the variants that are presented to the voters.[39] The following models were available for selection:

- Model 1: Draft according to the first parliamentary reading (biproportional apportionment with quorums and a majority clause).
- Model 2: Status quo plus an explicit constitutional prohibition of biproportional apportionment.
- Model 3: Mixed electoral system: MR for municipalities/districts with fewer than 10 seats. PR/biproportional apportionment for municipalities/districts with 10 seats or more.

Figure 16 and Figure 17 show the results of the runoff vote.

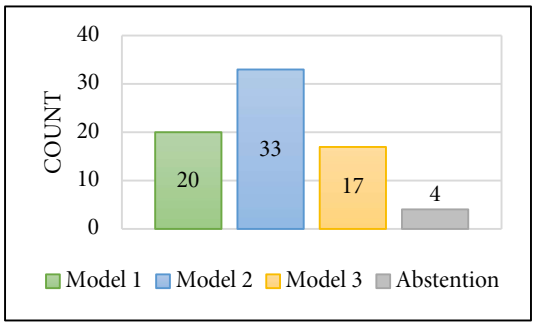


Figure 16: First round of voting on the three variants under discussion to amend the constitution of the Canton of Zug.

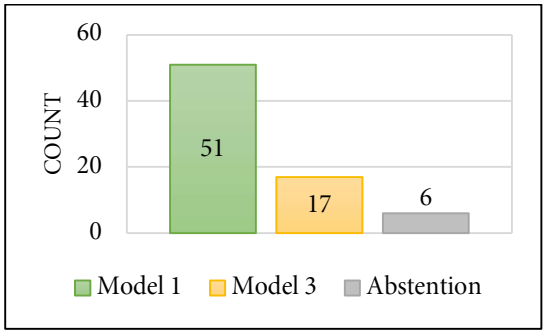


Figure 17: Second round of voting between the two defeated variants, Canton of Zug.

Model 3 was thus eliminated, leaving Models 1 and 2 to be submitted to a popular vote. However, things took a different turn. A collective around the Green Alternative Party of Zug filed an appeal with the FSC against the voting proposal passed by the Zug Cantonal Council. According to their complaint, the controversial voting proposal (Model 2) is inadmissible because it is designed to introduce PR-based electoral system that conflicts with the Federal Constitution. The Court upheld the appeal and annulled Model 2, which was scheduled for vote (BGE 139 I 195). As a result, the Government Council decided to withdraw Model 2 as a voting option and to submit only Model 1. In the voting documents, the Government Council explained its decision as follows:

> "The Cantonal Council wanted to submit two variants to the voters for a vote. Because the FSC has already declared one of these two variants to be unconstitutional following a new appeal, the voters will now be presented with the constitutional proposal 'Yes to the new allocation of seats for the Cantonal Council' (sic!)."[40]

[39] It is noticeable that the presiding cantonal councilor did not reckon with the eventuality of a proposal actually achieving an absolute majority in the first voting round. In that case, voters would not have been able to vote on two variants (as intended), but according to the rules of procedure only on the one winning variant (binary yes/no vote) (cf. § 76(3) GO KR).

[40] Source: Official documents for the votes of 22 September 2013, p. 4. Translated from German. The proposal's highly suggestive title given by the government is quite remarkable.

On 22 September 2013, the voters approved the proposal by a large majority. The vote only included the constitutional amendments, as the statutory amendments were subject to a facultative referendum and no such referendum was called.

2.4.6 Canton of Schwyz

The Cantonal Council of Schwyz has 100 members who are elected according to proportional representation. In 2016, the first election took place following the biproportional procedure. The Canton of Schwyz is the only canton in which electoral reform was triggered not by the legislature or the executive, but by popular initiatives. This, of course, also affected the reform process, which was eminently multi-layered, with the various levels intertwining in complex ways.

Political profile of the Canton of Schwyz

Population (in 1000)	165 (as of 12-31-2022)
Eligible voters (in 1000)	108 (as of 12-31-2020)
Electoral system (parliament)	proportional representation
Apportionment method	biproportional apportionment
Year of reform adoption	2015
Year of first application	2016
Elections under the new law	2
Level of reform initiation	plebiscitary
Statutory basis of reform	electoral law and constitution
Popular vote on reform bill	yes (53.5% approval)
Prior reform attempts	no
Size of the legislature	100
Legislative period	4 years
Number of electoral districts	30 (min. size 1, max. 10, median 2)
Candidacies per seat	3.9 (2020 elections)
Effective number of parties	4.2 (2020 elections)
Disproportionality (LSq-index)	0.6 (2020 elections)

Table 16: Descriptive statistics for the Canton of Schwyz.

The party landscape in the Canton of Schwyz is dominated by the SVP, FDP, CVP and the SP (cf. Figure 18). In 2016, the GLP ran for the first time and entered parliament with 3 seats. In the last elections, they were able to double their seat share. The electoral landscape, with its 30 districts and a median district size of 2, is very markedly parcelled.

As mentioned in the previous chapter on the reform process in the Canton of Zug, the reform of the Schwyz cantonal electoral system was preceded by a protracted and interruptive process.

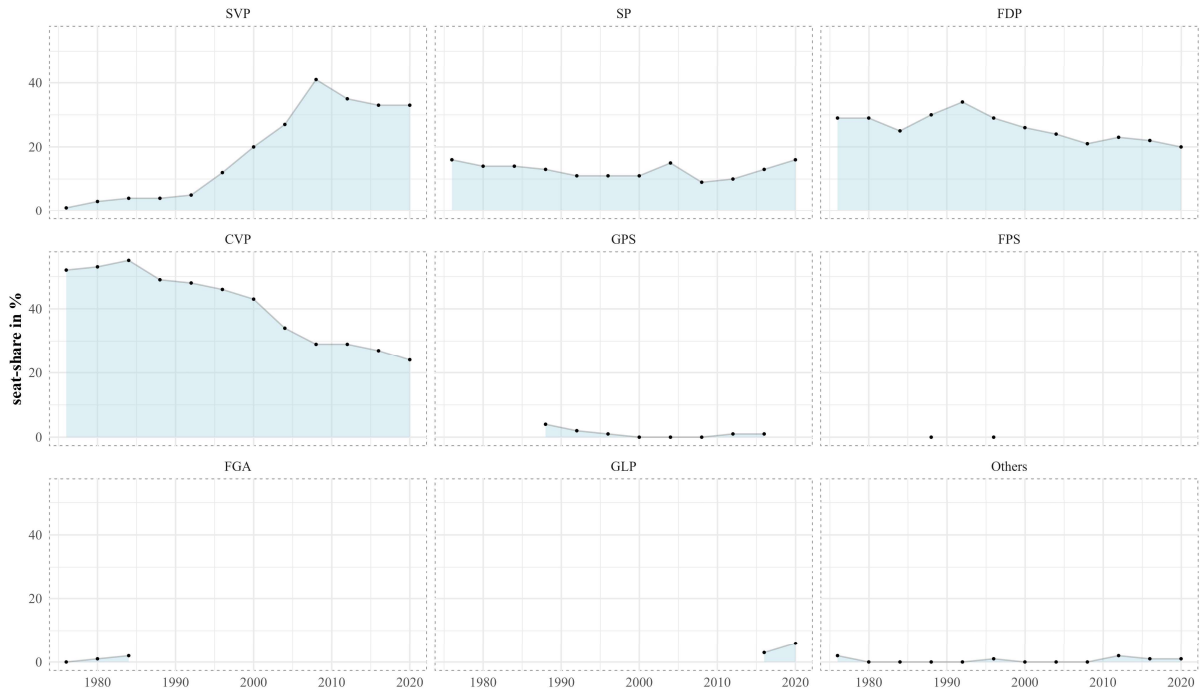


Figure 18: Parties' seat-shares in parliamentary elections in the Canton of Schwyz, 1976-2020.

Cornerstones during the reform process in the Canton of Schwyz	
Mai 15, 2011	Approval of the new cantonal constitution by the voters. Among other things, the new constitution envisaged the retention of the electoral system in force at the time.
September 16, 2011 & October 10, 2011	Filing of three independent complaints against the decree of the Government Council and the decision of the Government Council concerning the cantonal elections.
March 19, 2012	Ruling of the FSC that the electoral law of the Canton of Schwyz is not in conformity with the Federal Constitution.
March 18, 2013	Decision of the FA not to guarantee provisions of the constitution of the Canton of Schwyz conflicting with constitutional law.
August 30, 2013	Members of SVP submitted a popular initiative calling for the introduction of single-round plurality voting for cantonal parliamentary elections.
October 17, 2013	A multipartisan committee submitted a popular initiative for the introduction of a "fair" PR-based electoral system for cantonal parliamentary elections.
November 19, 2014	First parliamentary reading on the popular initiatives.
December 17, 2014	Second parliamentary reading on the popular initiatives.
March 8, 2015	Popular vote on the reform of the electoral system for cantonal parliamentary elections.

Table 17: Key reform stages in the Canton of Schwyz.

The reform process in the Canton of Schwyz was initiated in response to a federal dispute, highlighting the interconnectedness and reciprocity of the involved processes. To fully understand the context of this reform, it is essential to examine the broader legislative and judicial dynamics that influenced its initiation.

In 2011, the Canton of Schwyz adopted a new constitution, which was approved by 59.8% of voters in a referendum held on May 15. This constitutional revision retained the existing electoral system for the Cantonal Council. As part of the preparations for the 2012 cantonal parliamentary elections, the Government Council issued the necessary electoral decrees. However, these decisions faced three separate legal challenges before the FSC. On March 19, 2012, the Court ruled that the cantonal electoral law was not in compliance with the Federal Constitution.

Subsequently, the FA reviewed the constitution of the Canton of Schwyz and opted to reject certain provisions related to cantonal elections. Specifically, the FA declined to guarantee § 48(3) of the cantonal constitution, which mandated proportional representation for cantonal parliamentary elections. This decision was influenced by § 48(2) of the constitution, which stipulates that each municipality forms its own electoral district and that seats are allocated proportionally based on population, with each municipality guaranteed at least one seat. As a result, 13 of the 30 municipalities, which each receive a single seat, effectively experienced first-past-the-post elections. This practical outcome diverged from the proportional representation requirement outlined in § 48(3), leading to the FA's decision not to guarantee this provision.[41]

Political responses to the FA's decision were swift. On August 30, 2013, the cantonal branch of the SVP introduced a popular initiative titled "For a Simple and Comprehensible Electoral System", advocating for the adoption of single-round plurality voting in cantonal parliamentary elections. This initiative sought to simplify the electoral process by eliminating proportional representation.

In contrast, on October 17, 2013, a multi-party committee representing the EVP, GLP, SP, GPS, and the BDP presented a competing initiative titled "For Fair Proportional Representation – Yes to Diversity of Opinion in the Schwyz Cantonal Council". This initiative aimed to establish a fair, proportional representation system for cantonal elections. It proposed removing the municipalities as defined electoral districts

[41] The decision caused great stir inside and outside of Schwyz. For it was only the fifth time in the entire history of the Swiss Confederation that the FA did not guarantee a cantonal constitution (or parts of it) (cf. Biaggini 2007: 567, Auer 2016: 240f.).

from the cantonal constitution and repealing the constitutional guarantee of seats for municipalities. However, the initiative did not specify the precise design of the proposed electoral procedure.

The Schwyz Government Council reviewed the proposed initiatives and recommended that the Cantonal Council reject them, advising against their approval in a potential popular vote. The government supported maintaining proportional representation but found the PR initiative's proposed constitutional changes too radical. It viewed the minimum seat guarantee for each municipality as a fundamental constitutional principle to be preserved.

Given the lack of legal grounds to present the two initiatives to voters simultaneously, the Government Council decided to counter the single-round plurality initiative with a direct counterproposal. Anticipating a possible withdrawal of the PR initiative, the counterproposal incorporated aspects favored by its proponents. Specifically, it proposed adopting the biproportional method for party seat allocation while preserving the existing district structure and the guaranteed minimum seats for municipalities. The majority of the advisory commission endorsed the government's counterproposal and recommended the rejection of both popular initiatives.

The deliberation of the popular initiatives and the government's counterproposal occurred in two separate readings. The first reading involved an initial discussion and detailed consultation of both the initiatives and the counterproposal. The second reading focused on decisions regarding the validity of the initiatives, recommendations for the popular vote, and legislative amendments related to the single-round plurality initiative and the government's counterproposal.

The vote on the PR initiative was scheduled for a subsequent ballot, and it remained uncertain whether its proponents would maintain their stance, leading to no implementation law being drafted for it. The primary debate occurred during the first reading on November 19, 2014. This debate was broad in scope and included aspects beyond the focus of this study. Consequently, the following summary of the parliamentary deliberations concentrates specifically on the discussion related to the apportionment method.

Following the FA's partial rejection of provisions from the Schwyz Constitution and related decisions by the FSC on cantonal electoral reforms, there was broad consensus among political parties that a change was necessary. However, significant disagreements emerged regarding the design of the new electoral system.

All parties except the SVP supported proportional representation and largely endorsed the government's proposed changes to the electoral law, which included the implementation of biproportional representation. The debate centered on the Government Council's counterproposal, which allowed for significant legal adjustments and political influence.

Proponents of proportional representation generally agreed that the parliament should reflect current political trends in the electorate. Within this group, there were differing opinions on the biproportional apportionment method. The FDP preferred a system incorporating constituency associations with guaranteed seats for municipalities. The SP, while not enthusiastic about the biproportional method, accepted it as a compromise when combined with a seat guarantee for constituencies.

In contrast, the SVP, through its popular initiative, advocated for a majoritarian voting system rather than a PR-based system. The SVP viewed the FA's partial rejection as an affront to cantonal autonomy and argued that a majoritarian system would provide greater transparency and simplicity, aligning better with the needs of the Canton of Schwyz.

In addition to the core debate over the appropriate electoral system for the cantonal parliament, a further controversy emerged regarding the introduction of a statutory electoral quorum. The advisory commission proposed two options: the majority request called for a global 3% quorum, while the minority request suggested a combined 1% global quorum with a 5% local quorum.

The Government Council, guided by various legal opinions, opposed any form of quorum and advocated for maintaining an open electoral system. This stance was echoed by the FDP parliamentary group, which opposed an electoral quorum on principle. Christoph Pfister, a member of the FDP, argued that such thresholds are mechanisms used by powerful entities to sustain their dominance: "Let's not delude ourselves, an electoral quorum is a tool used by the powerful to maintain power. That's wrong."[42]

At the same time, some FDP members supported a statutory quorum of 3-5%. Xaver Schuler from the SVP cited evidence from the Canton of Aargau, arguing that a quorum enhances council efficiency.[43] The SP, Greens, and Independents, while critical of quorums as they "arbitrarily set lower limits for entry into parliament" and are thus "undemocratic" (Luka Markić, MP SP), were not entirely opposed to the concept.[44] They indicated a willingness to consider a constitutional provision for a quorum in the future if deemed necessary. The CVP and GLP supported a minimum threshold of 1% for a party or list to gain entry to parliament. Ultimately, four models were compared and subjected to a runoff vote:

1. No quorum (as proposed by the government)
2. 1% global quorum (proposed by MP Markus Ming)
3. 1% global quorum combined with a 5% local quorum (minority proposal by the advisory commission)
4. 3% global quorum (majority proposal by the advisory commission)

As shown in Figure 19, MP Ming's (GLP) motion prevailed in a three-round runoff vote. Hence, the majority of the council opted for the canton-wide quorum of 1%.

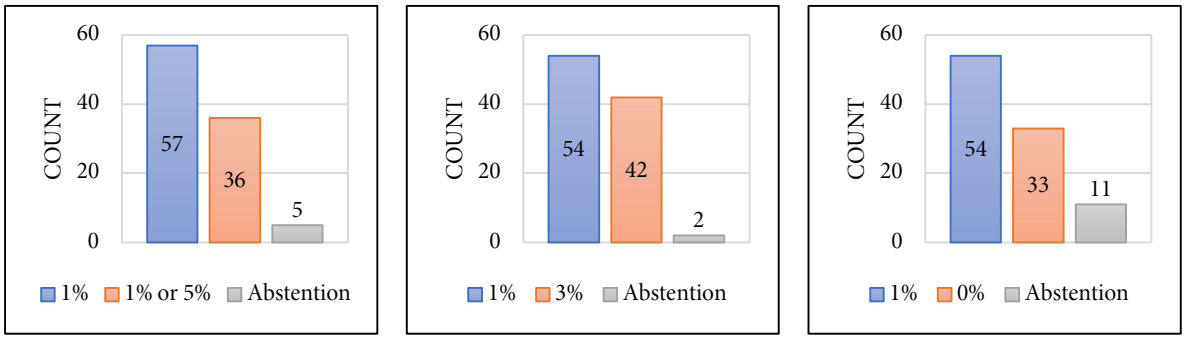


Figure 19: Runoff vote on the different quorum models brought up during consultation, Canton of Schwyz.

[42] Source: Extraordinary meeting of Council, 19 November 2014, p. 999. Translated from German.
[43] Source: Ibid., pp. 998-999.
[44] Source: Ibid., p. 1006. Translated from German.

In the second reading on December 17, 2014, the Cantonal Council conducted a detailed examination and final voting on all relevant issues, including the MR initiative, the Government Council's counterproposal, and the PR initiative. The MR initiative was first debated, confirmed as valid, and its implementation law was adopted. It was then decided to recommend the rejection of the MR initiative to the voters. Following this, the Council voted on whether to oppose the single-round plurality initiative with a counterproposal; this proposition was approved, and the implementation law for the counterproposal was adopted. The PR initiative was then discussed, confirmed as valid, and it was decided to recommend its rejection in a potential popular vote, contingent on whether the initiative would be withdrawn by its authors (see Figure 20 and Figure 21 for voting results).

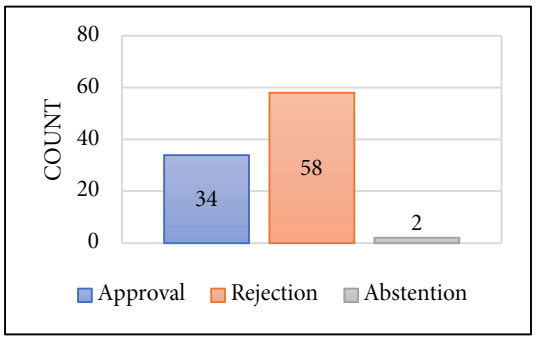


Figure 20: Positions on the voting recommendation on the single-round plurality initiative, Canton of Schwyz.

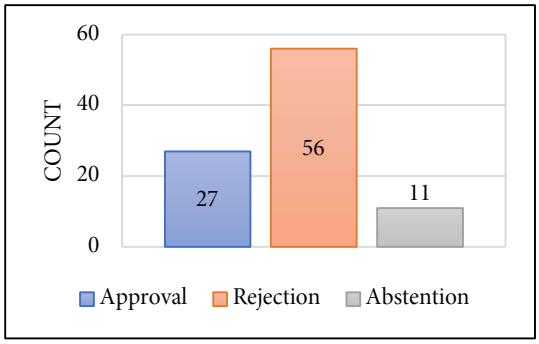


Figure 21: Positions on the voting recommendation on the "fair PR" initiative, Canton of Schwyz.

The referendum held on March 8, 2015, presented voters with two choices: the single-round plurality initiative and the counterproposal from the Cantonal and Government Councils. The single-round plurality initiative was rejected by 55.7% of voters, while the counterproposal was accepted with 53.6% of the vote. Although the ballot included a runoff question for a scenario where both proposals received a majority, this did not occur. Consequently, the initiative committee for the PR initiative opted to withdraw their proposal.

2.4.7 Canton of Valais

Valais, located in the southwestern part of Switzerland, is a bilingual canton with French and German as its official languages. This linguistic divide significantly influences cantonal politics. The canton has a strong historical connection to catholicism, which underscores the prominence of the CVP and its local branch CSP.

Political profile of the Canton of Valais

Population (in 1000)	357 (as of 12-31-2022)
Eligible voters (in 1000)	233 (as of 12-31-2020)
Electoral system (parliament)	proportional representation
Apportionment method	biproportional apportionment
Year of reform adoption	2017
Year of first application	2017
Elections under the new law	2
Level of reform initiation	executive
Statutory basis of reform	decree and law on political rights
Popular vote on reform bill	yes (but failed to find a majority)
Prior reform attempts	yes (2015)
Size of the legislature	130
Legislative period	4 years
Number of electoral districts	14 (min. size 1, max. 18, median 8.5)
Candidacies per seat	2.3 (2021 elections)
Effective number of parties	4.3 (2021 elections)
Disproportionality (LSq-index)	1.4 (2021 elections)

Table 18: Descriptive statistics for the Canton of Valais.

The Grand Council of Valais, comprising 130 members, is elected through PR using the biproportional apportionment method. The electoral reform of 2017, subsequently codified in the cantonal electoral law, introduced several changes, including a redesign of the electoral district structure. Previously, electoral districts aligned with administrative units between communes and the canton. Under the new system, these districts were restructured into 14 sub-districts, which were then organized into six broader districts.

A distinctive feature of the Valais electoral system is that upper-tier apportionment is conducted not at the cantonal level but within the six newly formed districts. This parallel and separate apportionment process at the district level represents a unique approach among the cantons applying biproportionality.

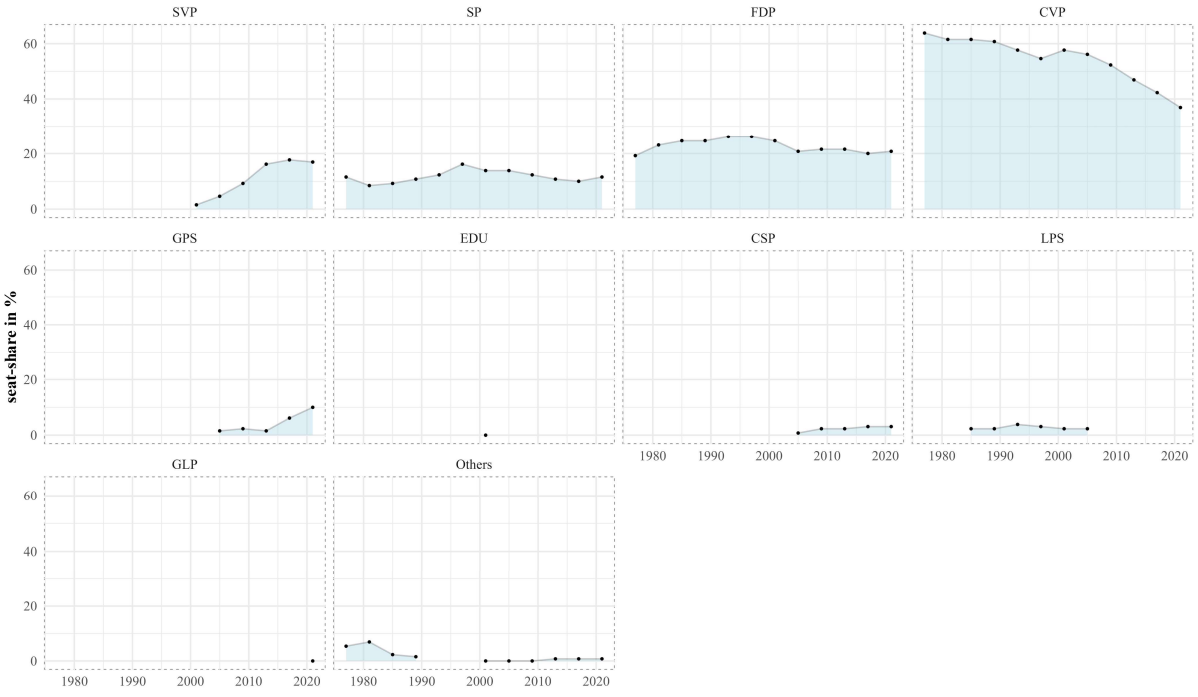


Figure 22: Parties' seat-shares in parliamentary elections in the Canton of Valais, 1977-2021.

Cornerstones during the reform process in the Canton of Valais	
October 27, 2004	FSC ruling that elections to the Valais Grand Council comply with federal constitutional law.
December 6, 2010	Submission of the popular initiative "Every Vote Counts" by an alliance of different political groups. Among other things, the initiative envisaged the introduction of biproportional apportionment.
October 3, 2012	Final report of the extra-parliamentary "commission R21" on the reform of cantonal structures and institutions.
February 12, 2014	Decision of the FSC that the electoral system for the Grand Council does not comply with requirements of the Federal Constitution.
September 8, 2014	First reading on the amendment of the constitutional provisions concerning the territorial organization and the cantonal institutions (introductory debate).
March 12, 2015	Second reading on the amendment of the constitutional provisions concerning the territorial organization and the cantonal institutions.
June 14, 2015	Popular vote on the constitutional reform "R-21". The reform fell just short of the required majority.
November 29, 2015	Vote on the popular initiative "Every Vote Counts". The initiative was rejected by a vast majority of the voters.
March 9, 2016	Adoption of a decree providing for an adjustment of the electoral system according to the biproportional model.
December 14, 2017	The Grand Council adopts an amendment of the legal basis for elections to the Grand Council (in the spirit of the abovementioned decree).

Table 19: Key reform stages in the Canton of Valais.

For a long time, the CVP (and CSP) maintained an absolute majority in the cantonal legislature of Valais. However, with the rise of the GPS and SVP since the early 2000s, alongside the nationwide decline of Christian democracy, the CVP lost this absolute majority in the 2013 elections.

The Canton of Valais has a complex reform history. The debate over a new electoral system began in 2004. On May 24, 2004, Susanne Hugo-Lötscher, then a member of the Grand Council for the Social Democrats, lodged a constitutional complaint with the FSC (BGE 131 I 85). Hugo-Lötscher and her co-petitioners contested a decision by the Council of State from April 7, 2004, which had fixed the number of parliamentarians to be elected from each district for the upcoming 2005-2009 legislature.

The complaint argued that some electoral districts were excessively small and that the variance in district sizes was too pronounced. This led to significant quorum thresholds in certain districts, effectively excluding not only small splinter groups but also substantial minorities from representation in the Grand Council. The petitioners contended that such practices were incompatible with the principle of equal representation, asserting that proportional representation was impractical in small districts with high quorums.

On October 27, 2004, the FSC ruled that the electoral system for the Valais Grand Council met federal requirements for cantonal parliamentary elections. The Court found that the constituency division in Valais was historically justified and that deviations from "perfect" vote-seat proportionality were acceptable given the historical context. Furthermore, key provisions of the election procedure were enshrined in the cantonal constitution and thus beyond the Court's review. This ruling effectively marked the provisional conclusion of the debate on electoral law in Valais.

In response to the legal ruling and ongoing dissatisfaction with the electoral system, a multi-party alliance from the German-speaking part of the canton (Upper Valais) – including representatives from the SP, SVP, GPS, and FDP – initiated a political reform effort. On December 6, 2010, they proposed the popular initiative "Every Vote Counts", aiming to amend the constitution.[45] The initiative called for the creation of three

[45] During the reform period, the Grand Council was composed of the following factions: Alliance de Gauche / Linksallianz (AdG/LA), CVP Oberwallis, (CVPO), PDC Valais Central (PDCC), PDC Bas-Valais (PDCB), CSP Oberwallis (CSPO), PLR, SVP Oberwallis (SVPO), UDC, Les Verts.

electoral districts and the implementation of the biproportional seat allocation procedure. Under this proposal, seats would first be apportioned at the level of the three new districts and then sub-apportioned within the existing districts.

The initiative sparked a broad reform debate and prompted the government to consider a fundamental reorganization of cantonal institutions and territorial structure. Despite the initiative's significance, it was not put to a vote until November 29, 2015. Both the parliament and the government opposed the initiative, and no legislative counterproposal was presented to avoid duplication with a parallel, more comprehensive reform effort. The initiative failed to pass, receiving 32% of the vote in favor, 64% against, and 4% blank or invalid. This outcome demonstrated clear opposition from the electorate to the proposed reform.

On June 6, 2011, the Valais government created the extra-parliamentary commission "Territory and Institutions of the 21st Century". The government saw a need for a comprehensive reform of the canton's territorial and institutional organization. The newly found commission was tasked with preparing and presenting a report with specific propositions how to modernize the communal and cantonal institutions as well as the territorial organization. Due to its mandate to develop reform proposals, the commission's name became established as "R21" ("Reforms in the 21st Century").

On October 3, 2012, the commission presented the final report with a variety of concrete reform recommendations. A subset of these recommendations concerned the elections of the Grand Council. In the report, the commission pointed to the previous case law of the FSC regarding the design of PR-based cantonal electoral systems for parliamentary elections. According to the FSC, a PR-based system is characterized by the prospect of representation for the various parties in proportion to their share of the electorate. Since the implementation of this requirement depends primarily on the design of the electoral districts, the structure of the electoral area was a key issue in the commission's report. Although the FSC had already ruled that the electoral system for the Grand Council in Valais was in conformity with the federal provisions (see above) and that there was therefore no need for an electoral reform (at least from a legal point of view), the commission was nevertheless dissatisfied with the current district structure. The commission argued that the will of the electorate is not adequately reflected by the current system due to the relative smallness of many districts:

> "The R21 Commission is guided by [the] considerations and the case law of the Federal Supreme Court. From a political point of view, it is not correct to simply nip in the bud the will of the voters expressed in a small constituency by means of a high natural quorum, without the possibility of asserting this will to vote in a larger constituency."[46]

In the section on electoral districts and the electoral system, the commission presented three different models to remedy the undesired allocation effects arising from small districts:

1. Formation of six electoral districts with a minimum size of nine seats each.
2. Retention of the current electoral districts and simultaneous introduction of (three) superordinate electoral regions (biproportional model).
3. Formation of six electoral districts, each comprising two or more sub-districts (alternative biproportional model).

[46] Source: Report of the "R21" commission of October 3, 2012, p. 75. Translated from German.

The first model was excluded from the outset because, in the opinion of the majority of the commission, it pays too little attention to the peripheral areas. Two-thirds of the commission were in favor of the third model, as this option takes better account of regional units. The remaining third supported the second model. The commission suggested to uphold the district-based statutory quorum of 8%. The Government Council welcomed the changes proposed by the majority of the commission and requested the Grand Council to approve the proposed reform so the changes could be subjected to a mandatory constitutional referendum. However, before the Grand Council could launch the debate on the planned constitutional reform, the FSC issued a new ruling in electoral law matters.

In its decision of February 12, 2014, the FSC ruled that the current electoral system for the Grand Council does not comply with the Federal Constitution, as it does not meet its requirements (BGE 140 I 107). Some of the electoral districts are too small and as a consequence, proportional representation cannot be realized in the districts concerned. The FSC has thus ordered the cantonal authorities to introduce a different electoral system for the Grand Council which complies with the Federal Constitution. This is quite astonishing, since the Court had ruled that the election procedure was permissible ten years earlier. In its ruling, the FSC also pointed out this contradiction with its previous case law and justified the deviation with the circumstances that have changed since then. On the one hand, it referred to the new possibilities of allowing the inadequacies of small districts to be compensated for with measures that do not require a change in the constituency structure.[47] On the other hand, the Court referred to the cantonal efforts for territorial reorganization, which relativize the importance of the political districts (as historical relics) as a basis for the electoral districts.[48] Although this Court decision did not fundamentally change the state of affairs, it did give urgency to the cantonal government's aspiration to modernize the cantonal institutions.

The Government Council supported the changes as proposed by the R21 commission and applied to the Grand Council to partially change the cantonal constitution in accordance with the propositions. The Grand Council accepted the partial amendment to the constitution in two separate readings on September 8, 2014, and March 12, 2015, with some substantive changes.

During the parliamentary deliberations, a variety of reform variants were proposed and discussed. In addition, other elements were added to the reform, which were neither foreseen in the Commission's draft nor proposed by the government: For example, the attempt to reduce the council to 100 members or the introduction of a guarantee of 35 Grand Council seats for the Upper Valais as a means of minority protection for the German-speaking part of the canton, as proposed by the CSPO and CVPO. Although the latter element of linguistic minority protection had no substantive connection with the reform of the allocation method itself, it still found a parliamentary majority and was thus included in the reform bill. This addition sparked a controversial discussion in the Grand Council and was also to play a decisive role in the ensuing popular vote.

The centerpiece of the electoral reform was Article 42 of the cantonal constitution, which is to determine the method of seat-allocation as well as the future district structure. There was a broad consensus among all political factions that an electoral reform was inevitable. In light of the upcoming elections in 2017, it was

[47] Here the Court mentioned the two possibilities of either creating constituency associations or introducing the bi-proportional apportionment method.
[48] Here the Court referred to the reform project R21.

therefore considered a high priority issue on the political agenda to lay the groundwork for legitimate parliamentary elections and to prevent the reform from failing at all costs. Accordingly, all factions signaled a high willingness to compromise. The CVPO and CSPO tabled two identical amendments which, in addition to guaranteeing seats for Upper Valais, also provided for the division of the electoral area into districts and sub-districts. The existing districts would continue in the form of sub-districts and be grouped into new districts on the upper-tier level. Additionally, the seats should be allocated based on "inversed biproportionality".[49] However, the idea of an inversed biproportional method was discarded during the second reading. Besides the CVPO and CSPO, also the UDC, PDCB and PDCC supported the idea of dividing the electoral area into districts and sub-districts. They emphasized the importance of maintaining the existing and historically deeply anchored district structure. In their view, only the biproportional method could sustain this continuity since one-dimensional PR would require a change in the number and size of the electoral districts. The other factions, SVPO, PLR, AdG/LA, Les Verts, voted for a model that proposed the creation of six new districts around the urban centers in the canton and the retention of the one-dimensional PR apportionment formula (i.e., district-based elections).[50] They asserted that this model is easier for the electorate to understand and that it is more in line with the concept of proportional representation (since one-dimensional PR does not allow for the occurrence of locally disproportionate results, they argued).

By a narrow majority, the Grand Council followed the CVPO/CSPO proposal to establish a guarantee of 35 seats for the Upper Valais with 66 votes in favor, 58 against and 4 abstentions. With regard to the district structure, the Grand Council followed the proposal of the Commission, which explicitly defined the six upper-tier districts and the 14 sub-districts. In addition, the biproportional procedure should be explicitly mentioned in the constitution (instead of implementing it "solely" as law).

Figure 23 shows the positions of the parliamentary groups on the final reform bill: 71 supported the partial constitutional revision, 53 opposed it and 4 abstained from the vote.

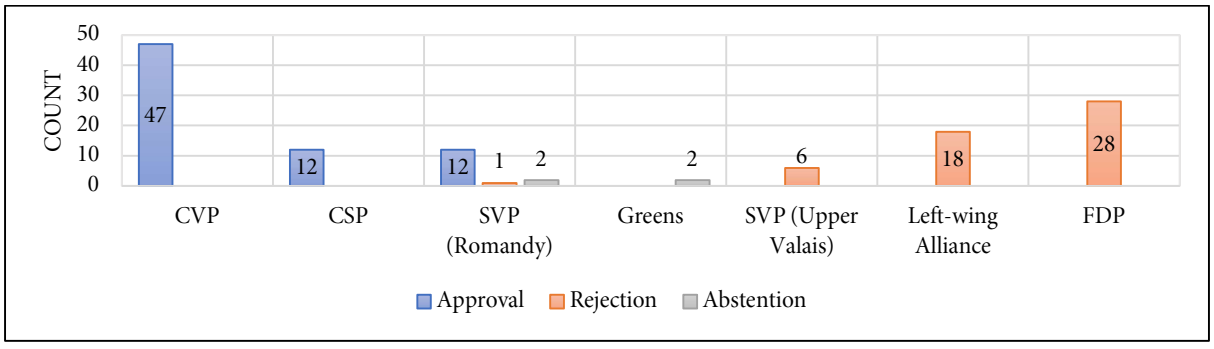


Figure 23: Positions on the final reform bill, which, among other things, provided for the introduction of the biproportional apportionment method for parliamentary elections in the Canton of Valais.

However, the figure illustrates the final vote on the overall reform conducted on March 12, 2015, rather than specifically on the allocation procedure. The reform itself was highly contentious within the Grand Council, as it encompassed a range of elements, each of which was controversial in its own right. Additionally, these elements highlighted shifting fault lines among the factions. The complexity and multidimensionality of the

[49] According to the inverted biproportional method, the first allocation should take place on the level of sub-districts, the allocation of remainder seats should then be performed on level of the six districts. The details of the apportionment method should be defined in law.

[50] These six electoral districts should be centered around the cities of Brig, Visp, Sierre, Sion, Martigny and Monthey.

reform made it challenging to discern the factions' stances on each component individually and to fully grasp the considerations and arguments that influenced their final positions in the vote.

Nevertheless, the controversial bill found a parliamentary majority and, as a constitutional bill, was subject to a popular referendum. To avoid any extraneous influence with regard to the sensitive and controversial issue, the parliamentarians decided that the composition and electoral system of the Grand Council should form the subject of a separate issue, independent from other reform elements concerning the organization of the Valais authorities.

The popular vote was held on June 14, 2015. 45% of the voters voted in favor of the amendment, 46% voted against, and 9% cast either a blank or invalid ballot. The constitutional amendment thus failed to win the required absolute majority (and also fell short of a relative majority).[51] In a later message to the Grand Council, the Government Council commented on the voting result with the following words: "It is not unreasonable to assume that the guarantee clause of 35 seats in favor of the two Upper Valais constituencies [Brig and Visp, S.L.] played a role in the result that occurred."[52] Given the scope and urgency of the reform, the failure of the reform bill constituted a significant political setback. Additionally, with the popular initiative "Every Vote Counts" failing to secure a majority in the vote held on November 29, 2015, the canton found itself without a legally valid framework for conducting cantonal parliamentary elections.

In response to the deadlock, the Government Council enacted a temporary decree to establish a legal framework for the 2017 Grand Council elections in compliance with federal law. If the Grand Council had rejected the decree or if a subsequent referendum had annulled it, the only recourse would have been to adopt a government resolution as a last resort. However, the Grand Council unanimously approved the decree on March 9, 2016. The decree, which mirrored the R21 proposal but excluded the seat guarantee for Upper Valais, was intended for the 2017 elections and had a provisional validity of four years.

To ensure the electoral system's compliance with federal law beyond the 2017 elections, the provisions of the decree were incorporated into a revision of the Cantonal Law on Political Rights. The Grand Council approved this revision on December 14, 2017, with a vote of 83 in favor, 42 against, and one abstention.

[51] According to the Cantonal Law on Political Rights Art. 4(2) and the Cantonal Constitution Art. 106, the absolute majority of the citizens participating in the vote decides on constitutional revisions. This peculiar regulation means that both blank and invalid (!) ballots are added to the no-votes, resulting in a strong braking effect.

[52] Source: Message of the Government Council to the Grand Council on the issuance of a decree regulating the cantonal parliamentary elections, December 17, 2015, p. 4. Translated from German.

2.4.8 Canton of Uri

Uri, a rural-alpine canton in central Switzerland, features a fragmented electoral landscape due to its mountainous geography. The canton's 64-member cantonal council elections are conducted across 20 municipalities, which also serve as electoral districts. The electoral system is mixed: proportional representation (PR) applies to districts with five or more seats, while plurality voting is used in districts with fewer than five seats. As of the 2020 elections, 16 districts utilized plurality voting and four districts employed PR. Following the 2019 electoral reform, PR districts now use the biproportional apportionment method.

Political profile of the Canton of Uri	
Population (in 1000)	37 (as of 12-31-2022)
Eligible voters (in 1000)	27 (as of 12-31-2020)
Electoral system (parliament)	mixed electoral system
Apportionment method	biproportional apportionment (for the "PR-districts")
Year of reform adoption	2019
Year of first application	2020
Elections under the new law	1
Level of reform initiation	executive
Statutory basis of reform	electoral law and constitution
Popular vote on reform bill	yes (68.4% resp. 57.5% approval)
Prior reform attempts	yes (2016)
Size of the legislature	64
Legislative period	4 years
Number of electoral districts	20 (min. size 1, max. 14, median 1.5)
Candidacies per seat	2.3 (2020 elections)
Effective number of parties	3.5 (2020 elections)
Disproportionality (LSq-index)	2.5 (2020 elections)

Table 20: Descriptive statistics for the Canton of Uri.

Politically, Uri is conservative, with approximately three-quarters of electoral votes going to the SVP, FDP, and CVP. However, party performance in majority/plurality voting districts – comprising 27 of the 64 parliamentary seats – can be misleading, as party alignments are less significant in these districts. Party vote shares have been officially recorded by the FSO since the 2008 cantonal elections, although PR has been in place in certain districts since 1992.

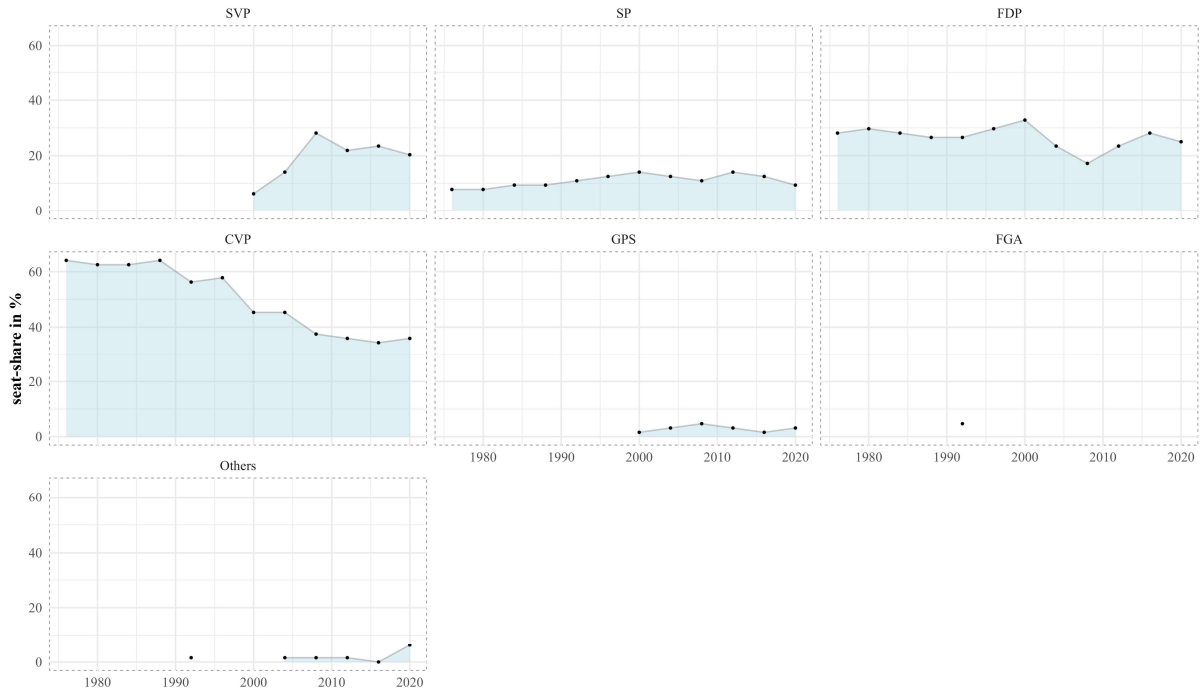


Figure 24: Parties' seat-shares in parliamentary elections in the Canton of Uri, 1976-2020.

Cornerstones during the reform process in the Canton of Uri	
September 23, 2012	Vote on a popular initiative calling for the (re)introduction of majority/plurality voting for cantonal parliamentary elections. The initiative found no popular majority.
October 12, 2016	FSC ruling that Uri's Cantonal Council elections do not meet the requirements of the Federal Constitution.
November 16, 2016	Submission of a parliamentary motion calling for a profound redesign of the district structure. The motion was subsequently rejected by parliament.
February 13, 2019	Parliamentary reading on an electoral reform.
May 19, 2019	Popular vote on the reform bill. Both the constitutional amendment and the reform of the electoral law found a clear popular majority.

Table 21: Key reform stages in the Canton of Uri.

In 1989, Uri's electorate approved a partial introduction of PR for cantonal parliamentary elections. The new system was implemented in the 1992 elections, applying PR to districts with three or more seats while maintaining MR in districts with one or two seats. Uri remains one of the few cantons with a mixed electoral system, alongside Basel Stadt and Appenzell Ausserrhoden.

In 2011, the CVP youth section proposed a cantonal popular initiative titled "Election of Heads Instead of Parties", aiming to reduce party influence by advocating for a return to majoritarianism. The initiative was decisively rejected in a popular vote held on September 23, 2012, with 57.5% of voters opposing the change. This outcome reaffirmed the electorate's preference for retaining the PR system in larger municipalities.

In preparation for the 2016 cantonal parliamentary elections, the Government Council issued a resolution on September 29, 2015, specifying the number of seats allocated to each municipality. This decision was subsequently challenged by a coalition of left-wing politicians from the GPS and SP, who filed an appeal with the FSC (BGE 143 I 92). The appellants criticized the electoral district division, arguing that the small size of some districts created unacceptable distortions in electoral equity, contrary to FSC case law. They contended that the seat guarantee for municipalities lacking an arithmetical entitlement further exacerbated these distortions. Additionally, they argued that party affiliation plays a significant role even in majoritarian districts, countering the Government Council's claim that party influence was negligible in personality-based elections.

On October 12, 2016, the FSC upheld the appeal. While acknowledging that a mixed electoral system is permissible, the Court found that the district-induced quorum in PR districts with fewer than nine seats was excessively high. The Court noted that deviations from proportional representation are only opportune in "objectively justifiable cases" and did not find any such justification in this instance, thus rejecting the significant intrusions into proportional representation (BGE 143 I 92, p. 100).

A few weeks after the Court ruling, the SP and GPS proposed a parliamentary motion for a comprehensive electoral district reform. They advocated for either creating a single constituency or establishing new districts that would align with future population trends and federal constitutional principles. This motion, however, failed to secure a parliamentary majority and was therefore rejected. Consequently, the responsibility to draft a revised legal framework for cantonal parliamentary elections fell to the Government Council.

The Government Council submitted a draft reform to the Cantonal Council to implement the FSC's decision, which proposed introducing the biproportional seat allocation method in PR electoral districts. However, discontent with the FSC's practice led the Cantonal Council of Uri, mirroring earlier actions by the Canton of Zug, to submit an initiative to the Federal Council on July 7, 2014. This initiative aimed to safeguard cantonal sovereignty over electoral law by amending the Federal Constitution to grant cantons the authority to determine their electoral systems and district boundaries. As the issue remained unresolved,

the Cantonal Council of Uri instructed the Government Council to suspend the revision bill pending clarity at the federal level.[53] On December 14, 2018, the National Council rejected the constitutional amendment bill by 90 votes to 103 (with 3 abstentions). Consequently, with the upcoming 2020 cantonal elections, new legal foundations needed to be established promptly.

In response to the FSC ruling, the Government Council conducted a consultation process on the electoral law, engaging political parties and municipalities. All respondents supported the introduction of biproportional apportionment in municipalities using PR. However, the CVP, FDP, parts of the SVP, and the Municipal Councils of Attinghausen, Flüelen, Schattdorf, and Seedorf advocated extending the majoritarian/plurality voting system to municipalities with three and four seats (Attinghausen, Flüelen, Seedorf, and Silenen). They argued that, due to the small population size and the minimal influence of party affiliation, the party dimension was less significant, and local parties had a low presence. They also highlighted that the small number of seats posed a high risk of distorting results due to seat reversals necessary for maintaining overall proportionality. Thus, they proposed that plurality/majority voting should be used for municipalities with three or four seats, while biproportional apportionment would apply to those with five or more seats. Despite these recommendations, the Government Council, in its January 8, 2019, statement, rejected extending majoritarianism to municipalities with three and four seats, citing both substantive reasons and the increased risk of litigation.

In contrast to the Government Council, the Judicial Commission supported extending majority/plurality voting to municipalities with three and four seats, incorporating this into its draft bill. Consequently, there were two reform proposals: one from the Government Council, advocating for biproportional apportionment across all PR municipalities, and another from the Judicial Commission, which also endorsed biproportional apportionment but limited it to municipalities with five or more seats. The primary difference between the proposals was the scope of the new seat-allocation method.

On February 13, 2019, parliament debated the future apportionment method in a single reading.[54] The FDP and CVP supported the Judicial Commission's draft, arguing that it better addressed the interests of Uri's municipalities and had a lower litigation risk, as the FSC had previously upheld the mixed system. However, not all members of these parties were fully supportive. An FDP deputy expressed concern that the biproportional method could lead to a situation where the party with the most votes in a PR district might not secure a seat, thereby lacking parliamentary representation. Another FDP member advocated for the Government Council's proposal, which was seen as legally safer. Similarly, within the CVP, there was criticism of the biproportional method, with a representative acknowledging its imperfections but deeming it the best available option.

On February 13, 2019, the debate on the apportionment method highlighted stark contrasts in parliamentary positions. The SP and Greens supported the Government Council's draft, criticizing the Judicial Commission's proposal to extend majoritarianism to additional municipalities. They argued that this extension was an unjustified attempt by the bourgeois parties to consolidate their power, undermining electoral equity

[53] Source: Minutes of the Cantonal Council's meeting of March 21, 2018, p. 6. Translated from German.
[54] A point of order was approved that called for waiving a second reading.

and political efficacy. Christoph Schillig, MP from the SP, emphasized the importance of maintaining proportional representation to ensure that every vote is meaningful, and that the cantonal parliament accurately reflects political forces:

> "This agenda item is by far the most important since I am part of the Council. The right to vote is an enormously important topic. (…) With the proposal of the Judicial Commission, I would in fact be deprived of the right to vote in elections to the County Council. Flüelen would vote by majority and my vote would in all likelihood never count again in the elections to the Cantonal Council. I would also no longer be able to vote, as there would no longer be any suitable candidates. My feeling of being able to shape the process would be greatly diminished. The cantonal parliament would no longer reflect the political forces as accurately as possible."[55]
>
> Christoph Schillig, MP SP

In the debate on February 13, 2019, concerning electoral reform in the Canton of Uri, the SP and Greens expressed strong support for the Government Council's proposal, which advocated for the implementation of the biproportional seat allocation method in municipalities with five or more seats. This method was viewed as essential for ensuring fair representation and maintaining electoral equity. The SP and Greens also shared the Government Council's concerns about the potential for legal challenges. They feared that adopting the Judicial Commission's draft, which extended majoritarian voting to municipalities with three and four seats, might not withstand scrutiny from the FA. They argued that such a reform could leave the canton without a constitutionally compliant electoral system and require a new reform process.

The SVP, on the other hand, was divided on which draft should be prioritized. A narrow majority within the SVP favored the Government Council's draft. They referenced the previous popular vote against a return to canton-wide majoritarianism, stressing that this decision should be respected. Concerns about the possibility of legal challenges and the risk of the reform being overturned were also significant for the SVP. One SVP representative warned that extending majoritarianism could jeopardize the diversity of the party landscape, potentially benefiting established parties at the expense of smaller or new ones. While the arguments of the minority within the SVP supporting the Commission's draft were not detailed, it is likely that their reasoning aligned with that of the CVP and FDP, which had advocated for extending majoritarianism to smaller municipalities.

Despite these internal differences, all parties agreed on the need for electoral reform and supported the introduction of the biproportional method for proportional representation municipalities. The consensus was that the Government Council's draft, despite its limitations, offered a more stable and legally secure path forward compared to the Judicial Commission's proposal, which risked further legal complications and potential invalidation of the electoral system.

In the detailed consultation process, a decisive vote determined the priority between the two reform drafts. The Commission's draft, which proposed extending majoritarian voting to municipalities with three and four seats while applying biproportional apportionment to larger municipalities, emerged victorious. It was favored by 45 votes to 16 for the electoral law and 47 votes to 14 for the constitutional amendment. Subsequent to this decision, both drafts were subject to final votes, which saw broad approval from the deputies, as illustrated in Figure 25 and in Figure 26.

[55] Source: Meeting of Council, February 13, 2019, audio file. Translated from German.

Following the parliamentary approval, the amendments to the cantonal electoral law and the constitutional amendment were put to a referendum on May 19, 2019. Both bills were approved, with the law amendment receiving 68.4% support and the constitutional amendment 57.5%. The FA later confirmed the constitutional amendment after contentious debates and a narrow vote. Despite some concerns about potential future litigation, the 2020 elections proceeded under the new biproportional system without any legal challenges.

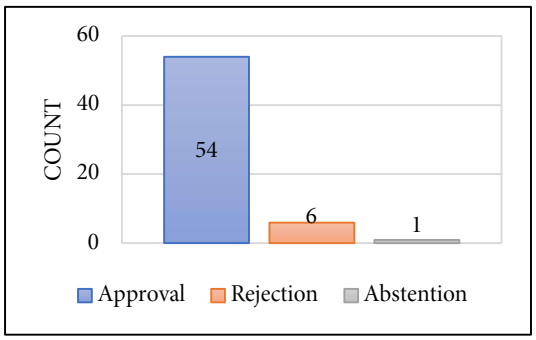


Figure 25: Positions on the reform bill as proposed by the commission, Canton of Uri.

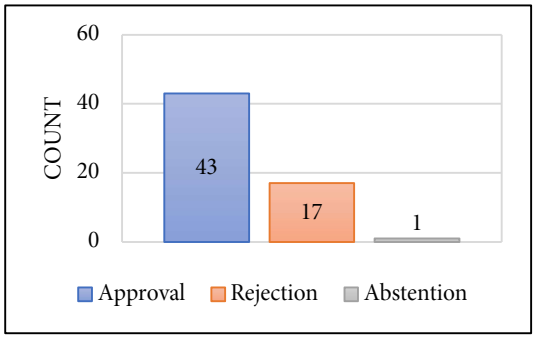


Figure 26: Positions on the constitutional amendment as proposed by the commission, Canton of Uri.

2.4.9 Canton of Graubünden

The cantonal legislature of Graubünden, which consists of 120 members, transitioned to PR in 2022. Prior to this, Graubünden, along with Appenzell Innerrhoden, was among the last cantons in Switzerland to use majority voting for parliamentary elections. The electoral system in Graubünden is highly fragmented, with 39 constituencies, making it the most divided canton for cantonal parliamentary elections. Seats are now allocated using the biproportional method, which includes a majority clause applicable to the 20 single-member constituencies. Graubünden's shift marks not only its adoption of a new PR-based electoral system but also its distinction as the sole canton among those that reformed to transition from a purely majority/plurality voting system.

Political profile of the Canton of Graubünden

Population (in 1000)	203 (as of 12-31-2022)
Eligible voters (in 1000)	142 (as of 12-31-2020)
Electoral system (parliament)	proportional representation
Apportionment method	biproportional apportionment (with a majority clause)
Year of reform adoption	2021
Year of first application	2022
Elections under the new law	1
Level of reform initiation	executive
Statutory basis of reform	electoral law and constitution
Popular vote on reform bill	yes (78.9% approval)
Prior reform attempts	no (however, several PR reform attempts)
Size of the legislature	120
Legislative period	4 years
Number of electoral districts	39 (min. size 1, max. 21, median 1)
Candidacies per seat	4.1 (2022 elections)
Effective number of parties	4.4 (2022 elections)
Disproportionality (LSq-index)	0.5 (2022 elections)

Table 22: Descriptive statistics for the Canton of Graubünden.

Despite Graubünden's historical adherence to majoritarianism, its party system has exhibited significant dynamism, as illustrated in Figure 27. The introduction of PR in 2022 did not result in the election of any new parties to the cantonal parliament, which is not surprising given the brief interval of only one year between the adoption of PR and its first implementation. This period is generally too short to observe substantial changes in party system dynamics.

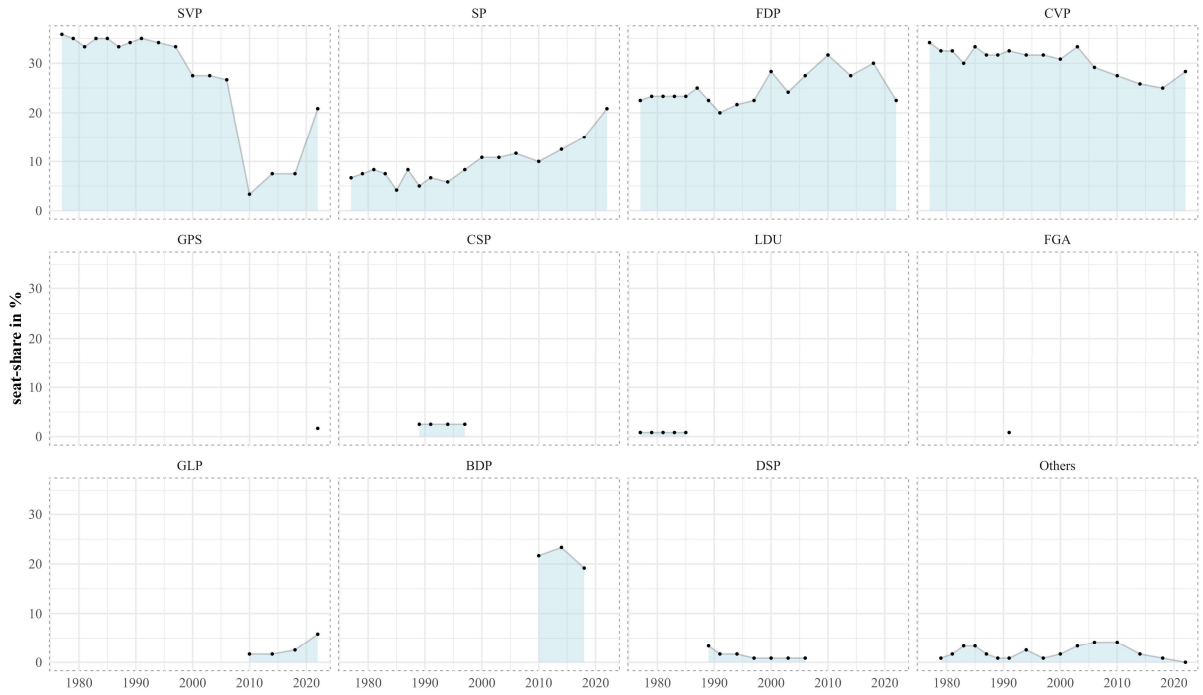


Figure 27: Parties' seat-shares in parliamentary elections in the Canton of Graubünden, 1977-2022.

Cornerstones during the reform process in the Canton of Graubünden	
July 29, 2019	Ruling of the FSC that the cantonal parliamentary electoral system is incompatible with the Federal Constitution.
February 15, 2021	First parliamentary reading on the electoral reform.
February 16, 2021	Second parliamentary reading on the electoral reform.
June 13, 2021	Popular vote on the reform bill. A vast majority of the voters supported the constitutional amendment.

Table 23: Key reform stages in the Canton of Graubünden.

Nevertheless, the electoral reform has had an immediate effect on the distribution of seats, reflecting broader electoral trends beyond the reform itself. Notably, the decline in the SVP's seat-share between the 2006 and 2010 elections can be attributed to its rebranding and subsequent split in federal politics in 2008. The SVP's offshoot, the BDP, eventually merged with the CVP to form the centrist "Die Mitte" in 2021.

Before delving into the reform process, it is essential to understand the context in which the electoral reform in Graubünden took place. Unlike other Swiss cantons that have undergone electoral reforms, Graubünden's reform represented the most significant departure from the existing electoral system. Located in the Alps, Graubünden is Switzerland's largest canton by area. Its rugged alpine terrain has led to a dispersed settlement pattern, contributing to the development of distinct language cultures in its remote valleys. As a result, political identities in Graubünden are deeply rooted in territorial affiliations, making the territorial dimension a critical aspect of the canton's politics.

This emphasis on territorial representation is pervasive in Graubünden's political sphere. There is a strong belief among both the political elite and the general population that ensuring territorial representation is crucial for maintaining the delicate political balance between the various regions of the canton. This belief is reflected in the structure of Graubünden's electoral system, which includes 20 single-member constituencies. Many of these constituencies, particularly those in more peripheral regions, benefit significantly from the legal guarantee of a parliamentary seat, even though they would not be entitled to one based purely on population numbers.

This seat guarantee is a political concession to the more remote regions of the canton, ensuring their political inclusion despite their smaller population sizes. In this context, the guarantee of parliamentary seats in even the smallest constituencies is considered more important than strict adherence to principles of electoral equity.

The reluctance to depart from majoritarian rule in Graubünden, coupled with concerns about altering the constituency structure, stemmed from fears that such changes could weaken the entrenched protection of local minorities. As a result, all eight previous attempts to introduce a PR-based system for cantonal parliamentary elections failed (Rathgeb & Lanz 2023: 60-62). However, these numerous reform attempts also indicate that there had been a prolonged and advanced public debate on electoral law and districting issues in the canton.

Despite the continuing importance of constituencies as guarantors of local representation, the majoritarian system began to be seen as outdated. Over time, there was a growing public desire for a "fairer" electoral system. This shift was driven by increasing disparities between party strengths in PR-based federal elections to the National Council and the majoritarian system used for cantonal elections to the Grand Council (ibid. 2023: 64-66). These discrepancies were interpreted as signs of the distorting effects of the majoritarian system, leading to doubts about whether the composition of the Grand Council accurately reflected the political preferences of the electorate (Ehrat & Eigenmann 2018: 185).

Following the last unsuccessful attempt to reform the cantonal electoral system in 2008, the SVP launched a popular initiative in 2009 titled "For fairer elections to the Grand Council". The initiative aimed to introduce a unidimensional PR system for all electoral districts with more than one seat. The fact that the SVP, a party generally restrained about electoral reforms, spearheaded this initiative is indicative of the growing and widespread dissatisfaction with the majority voting system in Graubünden.[56] This movement for reform reflected a broader sentiment among the electorate, who increasingly viewed the existing system as failing to ensure fair representation of their political preferences.

In response to the government's determination of the number of deputies to be elected in each constituency for the 2018-2022 legislative period on September 12, 2017, 54 individuals and five political parties active in Graubünden filed a joint appeal against the decision with the FSC on September 18, 2017. The complainants argued that the majority voting system used for the election of the Grand Council violated Articles 8 (equal rights) and 34 (political rights) of the Federal Constitution. They requested that the competent cantonal authorities be directed to establish a legal framework for an electoral system that would conform to the Federal Constitution, to be implemented in time for the 2022 parliamentary elections.

In its decision of July 29, 2019 (BGE 145 I 259), the FSC partially upheld the appeal against the existing electoral system in Graubünden. However, the Court stopped short of declaring the majoritarian system fundamentally unconstitutional. Instead, it provided a nuanced assessment, focusing on specific districts and the disproportionality caused by the current system.

Firstly, the Court identified the district of Avers as problematic due to its extremely small population of 160 inhabitants. Avers fell far short of the required representation figure by more than eight times, a discrepancy that the Court found unjustifiable under the principle of electoral equality. The Court suggested that Avers could be merged with neighboring constituencies to remedy this distortion.

Secondly, the Court examined the applicability of the majoritarian system in the canton's most populous districts – Chur (20 seats), Fünf Dörfer (12), Oberengadin (8), Rhäzüns (7), Davos (6), and Ilanz (6). It concluded that there were no valid reasons to maintain majoritarian rule in these districts due to their size and the significant impact on electoral fairness. Conversely, for the remaining, less populous constituencies, the Court deemed plurality or majority voting as justified and consistent with the constitutional requirements.

While the Court did not provide a detailed prescription for how the Graubünden authorities should reform the electoral system for the Grand Council, it did highlight potential adjustments, such as the merger of very small districts like Avers with adjacent ones and a reconsideration of the voting system in larger constituencies.[57] This left the cantonal authorities with the responsibility to create a revised electoral system that would comply with the Federal Constitution and address the specific issues identified in the Court's ruling.

[56] The SVP initiative was subsequently invalidated by the Grand Council in its session of April 18, 2011, by 109 votes to 4, as it conflicted with provisions of the Federal Constitution. Just six months after the unsuccessful reform attempt, the SVP, together with the GPS, GLP, SP, EDU, and EVP, submitted a new initiative "For fair elections", demanding, among other things, the adoption of PR. However, this attempt failed as well – this time at the ballot box in the vote of March 3, 2013. 56% of the voters opposed the initiative.

[57] The Court was extremely careful not to give the impression that it was imposing rules on the cantonal authorities regarding the organization of cantonal elections. It merely added that if the Canton of Graubünden wishes to retain the pure majoritarian method for elections to the Grand Council, this would require a revision of the constituency

The FSC ruling, which declared the allocation of half of the parliamentary seats unconstitutional, initiated a new electoral reform process in Graubünden. In response to this ruling, the Grand Council tasked the Government Council during the December 2019 session with developing a proposal to revise the cantonal electoral system. To this end, the government launched a consultation process involving all political parties and municipalities. The consultation presented nine different models, ranging from purely majoritarian systems to mixed models combining proportional and majoritarian elements, as well as a biproportional model.

Although the government expressed a preference for the biproportional model, the majority of the consulted municipalities and the three strongest political parties, which together held over 50% of the seats in the legislature, favored a mixed electoral model. In light of these responses, the Government Council decided to also explore the mixed model further in the reform process, expressing readiness to support it if it garnered sufficient parliamentary backing. The draft law based on the mixed model proposed dividing the canton into 43 electoral districts, with 41 districts electing representatives through majority voting and the two most populous districts, Chur and Fünf Dörfer, through unidimensional PR.[58]

Simultaneously, the SP had a pending initiative that proposed reducing the size of the parliament from 120 to 90 seats and restructuring the constituency boundaries. The SP strategically offered to withdraw this initiative on the condition that the biproportional model was adopted in the parliamentary deliberations.[59] Given that a similar SP initiative had only narrowly missed approval a few years earlier, securing 49.1% of the vote, this maneuver allowed the SP to significantly influence the reform process.

The parliamentary deliberations on the revision of the cantonal constitution began on February 15, 2021, and were organized into four parts: an opening debate, a detailed discussion of the mixed model, a detailed discussion of the biproportional model, and the final vote. The first two parts were completed on the first day of the session, while the discussions and final vote on the biproportional model took place on the following day.

During the opening debate on the electoral reform, it became evident that apart from the CVP, all political factions supported the biproportional model, though this support was more of a political compromise than a result of strong conviction. The FDP, for example, highlighted the importance of retaining the existing constituency structure as a significant argument in favor of the biproportional model. They appreciated that the majority clause, which guarantees the strongest list in a district at least one seat, would maintain a majoritarian element within the system. This viewpoint was shared by the BDP, which saw the continued parliamentary representation of the numerous single-member constituencies as a key criterion for approving the new electoral law. Similarly, the SVP backed the compromise proposal, seeing it as a way to end the prolonged struggle over electoral reform. The SP, despite securing many of its primary goals, was still compelled to make political concessions to ensure broader support for the reform.

structure. As conceivable solutions, the Court mentioned the division of the largest constituencies into smaller constituencies. Alternatively, a mixed system in which PR applies in the larger constituencies could be introduced (cf. BGE 145 I 259, p. 280).

[58] Of these 43 constituencies, 34 correspond to the previous districts. The constituencies of Ilanz, Rhäzüns, Oberengadin and Davos must be divided into nine new constituencies, and the district of Avers must be integrated into the Rheinwald constituency (cf. report of the Government Council of August 24, 2020).

[59] Source: Media release of the SP cantonal section of January 21, 2021.

The CVP, however, remained the sole faction advocating for the mixed model. They argued that previous votes on the introduction of PR had been rejected by voters, and they believed that the people of Graubünden preferred to maintain a majoritarian system. The CVP contended that the mixed model would best accommodate this preference, representing the smallest possible change to the status quo that still complied with the FSC's guidelines.

The second reading, held the following day, focused on a detailed deliberation of the biproportional model and the final vote on both models. A significant point of contention was the proposal to introduce a quorum. While most MPs supported a 3% threshold to prevent excessive fragmentation of political forces and to ensure parliamentary effectiveness, there were dissenting voices. Critics argued that a parliament should reflect societal political dynamics as closely as possible, and some were concerned that a quorum would disadvantage independent candidates, who are becoming more numerous and lack party affiliation. This concern was particularly relevant in single-member constituencies, where strong candidates might be excluded despite the majority clause being designed to ensure their representation.[60] Some MPs, including those from the CVP, expressed concerns about this potential exclusion, while others, particularly from the FDP, argued that the 3% threshold was too low and proposed raising it to 5% to ensure that elected parties could form factions in the parliament.

Ultimately, the proposal for a canton-wide 3% quorum, as recommended by the responsible commission and the government, was supported by a significant majority, with 84 votes in favor, 28 against, and no abstentions. Another contentious issue was whether to offer voters a choice between the biproportional and mixed models in a variant vote. However, the majority decided against this and opted to propose only the biproportional model in the referendum, as illustrated in Figure 28.

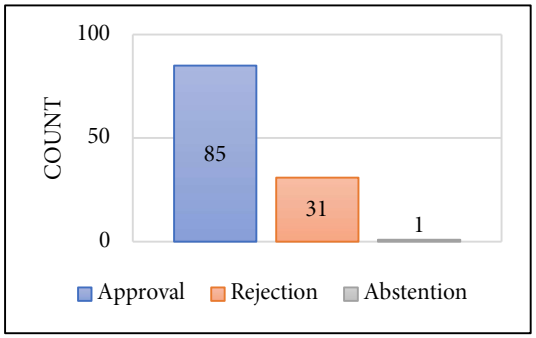


Figure 28: Positions on the ballot to propose only the biproportional model for popular referendum, Canton of Graubünden.

The constitutional and legislative amendments were overwhelmingly adopted in the final vote, with 85 and 84 votes in favor, respectively, and no votes against, though there were 32 abstentions. The constitutional amendments were later put to a popular vote on June 13, 2021, and were approved by a large margin, with 79% in favor. The legal amendments were not subjected to a popular vote.

[60] Following the same logic, a claim to ten seats determined in the super-apportionment would not be superseded even if eleven or twelve candidates were theoretically entitled to a seat by majority clause – the number of seats would remain at ten (similar to the electoral law for German Bundestag elections as amended in 2023).

2.5 Cantons with unsuccessful reform attempts

2.5.1 Canton of Obwalden

Political profile of the Canton of Obwalden	
Population (in 1000)	39 (as of 12-31-2022)
Eligible voters (in 1000)	28 (as of 12-31-2020)
Electoral system (parliament)	proportional representation
Apportionment method	Hagenbach-Bischoff
Level of reform initiation	parliamentary
year(s) of attempted reform(s)	2020
Popular vote on reform bill	no
Size of the legislature	55
Legislative period	4 years
Number of electoral districts	7 (min. size 4, max. 15, median 7)
Candidacies per seat	3.4 (2022 elections)
Effective number of parties	4.3 (2022 elections)
Disproportionality (LSq-index)	5.9 (2022 elections)

Table 24: Descriptive statistics for the Canton of Obwalden.

The canton of Obwalden, a rural area centrally located in Switzerland, ranks among the smallest cantons in terms of population. Since 1990, the number of seats in its cantonal parliament has been fixed at 55, in contrast to earlier practices where the size of the legislature varied, with one seat allocated per 500 inhabitants. PR has been used for parliamentary elections in Obwalden since 1986, with elections conducted within the canton's seven municipalities. According to Article 66(2) of the cantonal constitution, each municipality is guaranteed a minimum of four seats, though this provision has practically affected only Lungern, the smallest municipality, which received an additional seat due to this entitlement.

Obwalden was among the last Swiss cantons to transition from a majoritarian to a proportional representation system. Prior to the end of the 20th century, the CVP enjoyed an absolute majority in the cantonal parliament. However, since the turn of the millennium, the CVP's seat share has significantly decreased, with the SVP gaining a substantial number of those seats. The current parliamentary representation includes not only the SVP but also the SP, FDP, CSP, and more recently, the GLP, as depicted in Figure 29.

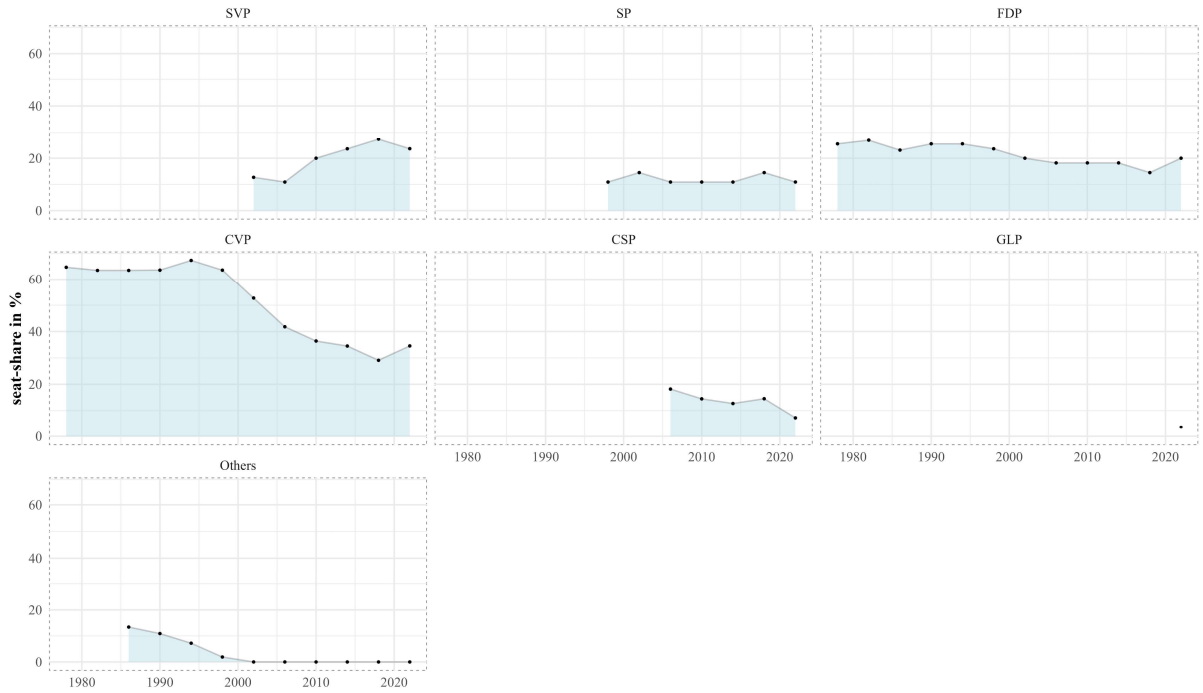


Figure 29: Parties' seat-shares in parliamentary elections in the Canton of Obwalden, 1978-2022.

Obwalden presents a unique case for electoral system analysis due to its partial non-compliance with FSC rulings. In the 2022 elections, four of its seven constituencies each had fewer than nine seats, exceeding the permissible 10% quota for effective representation as defined by the Court. This discrepancy highlights a potential contradiction between Obwalden's electoral system and federal case law.

As early as 2005, influenced by the FSC's landmark decision of 2002 regarding the Zurich City Council, the Obwalden government assessed its electoral system and identified that some constituencies were too small. The government of Obwalden defended its small and varied constituencies by citing their historical origins and strong local affiliations, emphasizing that the cantonal constitution, ratified by the FA in 1990, should take precedence over FSC rulings. The government believed that its electoral law complied with constitutional requirements and could withstand legal scrutiny.

However, since the government's 2005 assessment, the FSC has increasingly reprimanded similar deviations from PR. The Court has acknowledged that deviations may be justified by historical, federal, cultural, religious, linguistic, or ethnic factors intended to protect local minorities. Nonetheless, it has stressed that such deviations must be extremely well justified, with the burden of proof becoming heavier as the deviation from PR increases.[61] The only exception granted was to the Canton of Valais in 2004 (BGE 131 I 85), which underscored the Court's restrictive approach.

In response to this stringent interpretation, the Obwalden government re-evaluated its electoral system. In 2012, it conducted a consultation with all cantonal political parties to gauge the political feasibility of various measures designed to align the electoral system with federal constitutional standards.

The majority of political parties in Obwalden opposed both altering the constituency structure (whether by creating a single constituency, redesigning existing ones, or forming constituency associations) and changing the apportionment method (whether to a majoritarian system or a biproportional approach). This resistance led the Government Council to conclude that the current electoral system had broad support among the political elite. The Council maintained its belief that the existing legislation would stand up to scrutiny by the FSC.[62]

Moreover, at the time, several neighboring cantons – Zug, Schwyz, Uri, and Nidwalden – were undergoing their own electoral reforms, and it was uncertain whether their proposed changes would meet federal judicial standards. Given these factors, the Government Council decided that pursuing electoral reform in Obwalden was not advisable at that moment.

In subsequent rulings on cantonal electoral systems, the FSC tightened its stance on permissible deviations from the 10% benchmark for implicit district quorums. The Court ruled in the Canton of Valais (BGE 140 I 107) that deviations from this benchmark were no longer acceptable, even with substantial historical, federal, cultural, linguistic, or religious justifications. It emphasized that maintaining small constituencies to

[61] Cf. BGE 136 I 376 p. 384, BGE 129 I 185 p. 190, BGE 131 I 74 p. 79, BGE 131 I 85 p. 87.

[62] Source: Media release of November 7, 2012, on the decision of the Government Council of October 23, 2012. Contrary to the Government Council's assessment, however, it is more likely that the FSC would rule that the current electoral system is unconstitutional upon appeal (cf. Fuhrer 2018: 72-73).

protect local minorities could still align with accurate representation of party strengths through measures such as combining small districts for seat allocation or adopting the biproportional apportionment method, which had been effectively implemented in various cantons.[63]

Episodes in the cantonal tug-of-war over future electoral law

October 23, 2012	Report of the Government Council on the assessment of the need for electoral reform.
Mai 28, 2015	Query by Max Rötheli and others to the government inquiring about the need for reform.
October 22, 2020	Parliamentary motion by Max Rötheli and others calling for the introduction of biproportional apportionment for cantonal parliamentary elections.
January 28, 2021	Parliamentary motion by the CVP calling for a mixed electoral system.
Mai 26, 2023	Parliamentary motion by the SP, SVP and CSP calling for the introduction of biproportional apportionment.

Table 25: Key reform stages in the Canton of Obwalden.

This evolving legal context prompted Max Rötheli, a member of the SP, along with five co-signatories, to submit a query to the Government Council on May 28, 2015, seeking an evaluation of the need for electoral reform. In its response on August 18, 2015, the Government Council cited pending cantonal initiatives from Uri and Zug, which sought greater cantonal autonomy in electoral law and amendments to the federal constitution. The government referred to the FSC's decision regarding the Canton of Appenzell Ausserrhoden (BGE 140 I 394) to highlight that the legal clarity on majoritarian elections was still evolving. The government argued that the current electoral system was politically uncontroversial and not comparable to the more extreme cases in Schwyz and Nidwalden. It concluded that the "residual risk" of potential legal challenges was acceptable.[64]

However, on October 22, 2020, Rötheli and seven co-signatories introduced a parliamentary motion advocating for the adoption of the biproportional apportionment method. This followed the rejection of cantonal initiatives from Uri and Zug by the National Council in 2018, which had sought increased cantonal autonomy in electoral law. The motioners argued that the FSC's rulings and the National Council's decisions reinforced the view that Obwalden's electoral system violated federal law. They emphasized that the biproportional method, which preserves the existing district structure while ensuring proportional representation, had been successfully implemented in other cantons.

Acknowledging the updated legal framework and the legislative confirmation of the FSC's case law, the Government Council, albeit reluctantly, recognized the necessity for reform. It recommended to the Cantonal Council the adoption of the biproportional method as proposed.[65]

The Cantonal Council considered the motion for biproportional representation on January 28, 2021. Only the SP faction supported the motion, arguing that the biproportional method aligns with the FSC's rulings and represents only a minor adjustment to the electoral system. The SP also deemed a mixed electoral system unsuitable due to potential legal uncertainties. Opposition came from the SVP, which found the biproportional procedure overly complex and viewed the Court's rulings as an infringement on cantonal autonomy. The CVP argued that proportional representation distorts voter preferences in Obwalden, where personal candidate attributes are significant. Instead of biproportionality, the CVP proposed a mixed system

[63] Cf. BGE 140 I 107, BGE 143 I 92.

[64] Source: Response letter of the Government Council to the query of Max Rötheli and co-signatories dated August 18, 2015, p. 2. Translated from German.

[65] Literally, the Government Council wrote in its statement on the Rötheli motion (pp. 7-8): "The Government Council regretted this constriction of the sovereignty of the cantons in regulating the exercise of political rights in cantonal affairs." (Translated from German.)

combining majoritarian representation in small districts with unidimensional PR in larger districts. The FDP and CSP factions did not express explicit opinions during the debate, but the voting results suggest they opposed the motion (see Figure 30). At that time there were no nominal voting records, but the minutes suggest that only the SP faction, which held eight seats at the time of the vote, supported the motion.

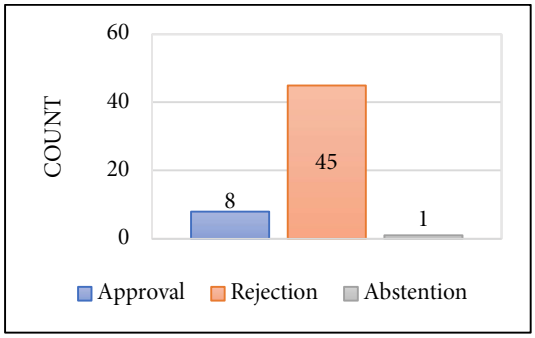


Figure 30: Positions on the parliamentary motion instructing the government to draft a bill for the introduction of biproportional apportionment in cantonal parliamentary elections, Canton of Obwalden.

With the motion's rejection, biproportional representation was temporarily removed from the agenda. However, on January 28, 2021, the CVP also introduced a motion for a mixed electoral system. Despite a recommendation from the Government Council to reject this motion, the Cantonal Council approved it on May 27-28, 2021. The motion was, however, subsequently dismissed later that year.

The SP and SVP, joined by the CSP, renewed their push for biproportional representation with a new motion on May 26, 2023. The support from SVP and CSP suggests decreasing opposition to biproportionality, potentially increasing its chances of passing in the Cantonal Council. Nonetheless, with the FDP and CVP holding a combined majority, they could block this latest reform effort. The ongoing risk of a FSC appeal, due to the current system's non-compliance with PR requirements, could make further electoral reform inevitable.

2.5.2 Canton of Bern

In 2002, the Grand Council of Bern was reduced from 200 to 160 members through a popular vote. This reduction was accompanied by a revision of the electoral district structure, consolidating the previous 27 constituencies into eight larger ones. A second reform in 2010 increased the number of constituencies to nine. Elections are conducted using unidimensional proportional representation (PR) within these nine districts. The 2022 elections in Bern were highly competitive, with nearly 14 candidates per seat.

Political profile of the Canton of Bern

Population (in 1000)	1'051 (as of 12-31-2022)
Eligible voters (in 1000)	747 (as of 12-31-2020)
Electoral system (parliament)	proportional representation
Apportionment method	Hagenbach-Bischoff
Level of reform initiation	legislative
Popular vote on reform bill	no
Size of the legislature	160
Legislative period	4 years
Number of electoral districts	9 (min. size 12, max. 25, median 17)
Candidacies per seat	13.8 (2022 elections)
Effective number of parties	6.1 (2022 elections)
Disproportionality (LSq-index)	2.4 (2022 elections)

Table 26: Descriptive statistics for the Canton of Bern.

Bern is a bilingual canton with German and French as official languages. The cantonal constitution ensures protection for the French-speaking minority. The Bernese Jura constituency is allocated twelve seats, and proportional representation is guaranteed for the French-speaking population of the Biel-Seeland constituency (Art. 73(3) of the cantonal constitution). Additionally, the Bernese Jura is assured one seat in the seven-member government (Art. 84(2)).[66]



Figure 31: Parties' seat-shares in parliamentary elections in the Canton of Bern, 1978-2022. The seat-share of "Die Mitte" (merger of CVP and BDP in 2021) is subsumed under "CVP" for the year 2022.

[66] The municipality of Moutier, formerly incorporated in the Bernese Jura constituency, decided in a popular vote in 2021 to switch to the Canton of Jura. As a result, the political weight of the Bernese Jura, which is (now even more) overrepresented in parliament and government, has been called into question. It is likely that the special constitutional status of the Bernese Jura will be renegotiated.

The Bernese Jura, a region with significant linguistic and cultural diversity, has its own regional parliament, the Bern Jurassic Council, established in 2006. This body, although possessing limited decision-making authority, focuses on promoting local culture and aligning political interests within the region. It helps the Bernese Jura to manage its unique political dynamics more effectively, despite the region's diverse political landscape and low electoral volatility, where no single party or political camp dominates the Grand Council.

Episodes in the cantonal tug-of-war over future electoral law	
January 22, 2007	Parliamentary postulate demanding the examination of the biproportional method for cantonal parliamentary elections.
June 12, 2007	1st parliamentary motion demanding the implementation of the biproportional method for cantonal parliamentary elections.
June 4, 2014	2nd parliamentary motion demanding the implementation of the biproportional method for cantonal parliamentary elections.
December 10, 2019	3rd parliamentary motion demanding the implementation of the biproportional method for cantonal parliamentary elections.

Table 27: Key reform stages in the Canton of Bern.

Unlike Obwalden, the electoral regulations in Bern have consistently adhered to federal law, and there have been no judicial challenges. Reforms in 2002 and 2010 aligned the electoral districts with the principles of PR as mandated by the FSC, ensuring that all districts meet the nine-seat requirement.

Despite the legal conformity of the Bernese electoral system, there have been internal pressures for reform. Since 2007, four parliamentary attempts to implement the biproportional apportionment method have been made. The most recent attempt, initiated on December 10, 2019, was supported by a coalition of seven MPs and 45 co-signatories. The proposal emphasized that the biproportional method would make the electoral process fairer and more representative. However, on May 13, 2020, the Government Council recommended rejecting the motion. It cited that both the biproportional and unidimensional methods have their own sets of advantages and disadvantages. The Government Council argued that the current unidimensional PR system, with its large electoral districts, aligns with FSC jurisprudence and is widely accepted both nationally and within the canton. Therefore, the Council maintained its position against adopting the biproportional method.

The Grand Council debated the motion on August 31, 2020. Supporters included the EVP, GLP, EDU, GPS, and BDP, who argued that the biproportional method would more accurately reflect the voters' will and promote a more inclusive political environment. They suggested that the reform could stimulate the participation of smaller parties and enhance the overall representativeness of the legislative process.

Opposition came from larger parties, including the FDP, SP, SVP, and smaller groups like the Alternative Left (AL) and the Autonomous Socialist Party of Southern Jura (PSA), which are part of the SP parliamentary group. These opponents contended that the current system was already compliant with FSC case law and that the proposed changes were neither legally necessary nor urgent.[67] A representative of the FDP argued that the reform was not compelling and did not address any immediate or material issues.

The SP's opposition is particularly notable. Despite generally supporting such reforms in other cantons, SP representative Urs Graf criticized the biproportional method as potentially exacerbating existing electoral distortions. He argued that allocating seats to fringe parties could lead to inefficiencies and that consolidating political interests at the electoral level might be preferable.

[67] Source: Minutes of the Grand Council meeting of August 31, 2020, p. 4. Translated from German.

The smaller parties, AL and PSA, also voiced concerns that the biproportional method might disadvantage them since their voter base is confined to the city of Bern, as opposed to parties with broader geographic appeal. They feared that the reform, combined with a canton-wide statutory quorum and the elimination of apparentements, could threaten their viability.

Ultimately, the motion was defeated, with 47 councilors in favor, 98 against, and four abstentions. The failure of the motion reflects a clear divide between the smaller parties advocating for reform and the larger, established parties that resisted it.[68] The detailed breakdown of the voting results is depicted in Figure 32.

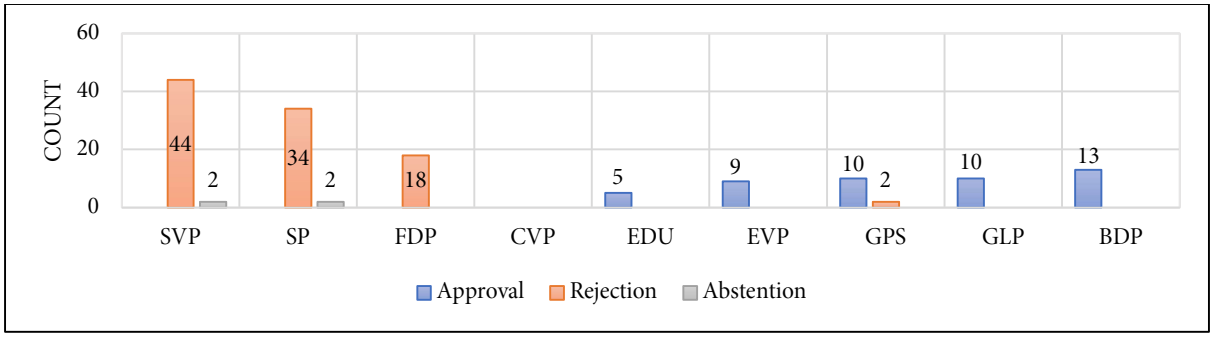


Figure 32: Party positions on the motion calling for the introduction of biproportional apportionment for parliamentary elections in the Canton of Bern. No values are shown for the CVP because the only CVP representative at the time, MP Mohamed Hamdaoui, missed the vote.

[68] At the time of the decision, the GPS and GLP still had a seat share of less than 10%.

2.5.3 Canton of Thurgau

Political profile of the Canton of Thurgau

Population (in 1000)	290 (as of 12-31-2022)
Eligible voters (in 1000)	179 (as of 12-31-2020)
Electoral system (parliament)	proportional representation
Apportionment method	Hagenbach-Bischoff
Level of reform initiation	parliamentary and plebiscitary
Popular vote on reform bill	yes
Size of the legislature	130
Legislative period	4 years
Number of electoral districts	5 (min. size 22, max. 32, median 26)
Candidacies per seat	7.6 (2020 elections)
Effective number of parties	5.3 (2020 elections)
Disproportionality (LSq-index)	3.4 (2020 elections)

The Canton of Thurgau, located in the Swiss midlands and bordering Germany to the north, has a Grand Council consisting of 130 members elected from five constituencies under unidimensional PR. This structure emerged from a 2009 reform that consolidated the previously eight electoral districts into five. This restructuring aimed to modernize the constituency framework and align it with federal law to avoid potential legal challenges. The political landscape in Thurgau is diverse, with the SVP being the dominant party, holding over 30% of the seats since the 2000 elections (see Figure 33).

In parallel with the district reform, there were two notable attempts to introduce biproportional apportionment in Thurgau. The first effort occurred on March 12, 2008, when councilors Anita Dähler (CVP) and Matthias Müller (EVP) proposed a motion to adopt a "fairer seat allocation procedure" in PR-based elections. This motion was driven by a desire to improve the fairness of seat distribution within the Grand Council.

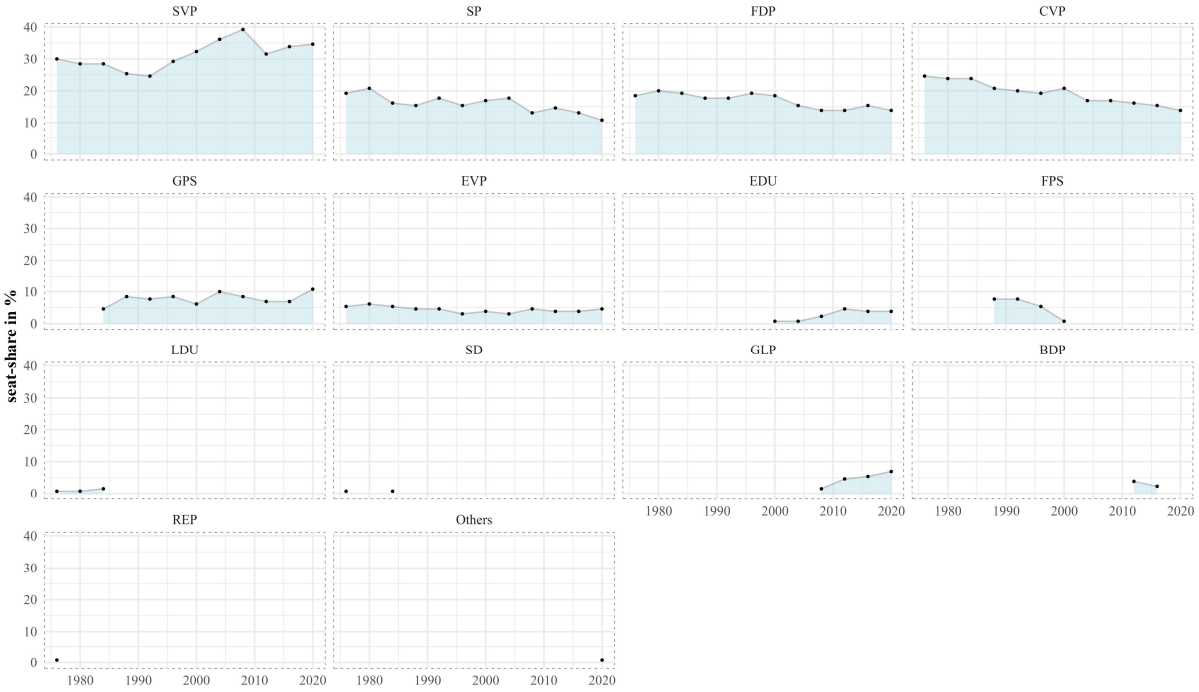


Figure 33: Parties' seat-shares in parliamentary elections in the Canton of Thurgau, 1976-2020.

A subsequent reform effort to amend the apportionment method emerged through a general popular initiative dated March 1, 2010.[69] This initiative, titled "Fair Electoral System for the Elections to the Grand Council," was endorsed by the BDP, EDU, EVP, and GPS, as evidenced by their party logos on the signature sheet. To maintain clarity, a detailed chronological account of the reform processes is omitted. Instead, the discussion begins with the reform process initiated by the parliamentary motion, followed by an examination of the reform process instigated by the popular initiative.

At the time the motion was submitted, the issue of reforming political and electoral districts was already on the agenda but had not yet been resolved. Prior to the reform, the electoral area was divided into eight unevenly sized districts (see Table 28). The proponents of the motion cited the district of Diessenhofen, which, with only four seats, fell significantly short of the 10% threshold. They argued that the existing system resulted in unequal voter influence and numerous "lost" votes in smaller constituencies due to the relatively high implicit quorum, which adversely affected the electoral success of smaller parties. To address these issues and ensure the constitutionality of the electoral system, they proposed either adjusting the number and size of constituencies or adopting the biproportional apportionment formula.

Before (as of 2008)		**After (as of 2012)**	
District	Seats	District	Seats
Arbon	21	Arbon	27
Bischofszell	17	Frauenfeld	32
Diessenhofen	4	Kreuzlingen	23
Frauenfeld	25	Münchwilen	22
Kreuzlingen	18	Weinfelden	26
Münchwilen	21		
Steckborn	10		
Weinfelden	14		

Table 28: Seats per electoral district in the Canton of Thurgau before and after the constituency reform.

In its statement regarding the motion submitted to the Grand Council on February 24, 2009, the Government Council recommended rejecting the proposal. While acknowledging the biproportional method's benefits, such as improved proportionality and enhanced electoral opportunities for smaller parties, the government also highlighted significant drawbacks. These included limited guarantees for regional representation and the potential for increased political fragmentation. Additionally, the government pointed to the concurrent redistricting process, which was already well advanced, as a means to achieve a fairer seat allocation. From the Government Council's perspective, the proposed reform of the seat allocation procedure was therefore deemed unnecessary.

[69] March 1, 2010, is the date of submission, the initiative was launched six months earlier, on August 28, 2009. A "general popular initiative" is the formally less binding counterpart to an elaborated draft and serves as a general suggestion. If a majority of deputies support the initiative, the necessary amendments are made. If these amendments entail a partial revision of the constitution, the bill is put to a vote. A vote is also held if a majority of the deputies do not support the initiative (cf. § 26(3) of the cantonal constitution and § 81-83 of the electoral law).

Episodes in the cantonal tug-of-war over future electoral law	
March 12, 2008	Submission of a legislative motion calling for the introduction of biproportional apportionment for cantonal parliamentary elections.
November 29, 2009	Popular vote on a redistricting bill that aimed at reducing the number of electoral districts from eight to five. The constitutional amendment was approved by 61.8% of voters.
February 17, 2010	Parliamentary reading on the parliamentary motion.
March 1, 2010	Submission of a general popular initiative calling for the introduction of biproportional apportionment for cantonal parliamentary elections.
November 24, 2010	Parliamentary reading on the general popular initiative.
May 15, 2011	Popular vote on the initiative calling for the introduction of biproportional apportionment. The initiative was rejected by 62% of voters.

Table 29: Key reform stages in the Canton of Thurgau.

The parliamentary consideration of the motion was deferred to await the outcome of the popular vote on the redistricting bill. The constitutional amendment, which aimed to alter the political and electoral districts, was approved in the popular vote held on November 29, 2009. Since this redistricting met the federal requirements for cantonal electoral laws, the legal impetus for further reforms was obviated. Despite this, the proponents of the motion persisted with their proposal, and parliamentary deliberation occurred on February 17, 2010.

The motion received support from the EVP/EDU and GPS factions, which argued that the biproportional method would offer the highest level of mapping accuracy, ensuring that each party would receive seats proportionate to its share of the electorate across districts. This approach was said to eliminate the need for adjustments to the seat allocation procedure based on policy reasons. Additionally, proponents noted that the new method would remove unpredictable apparentements and expressed a willingness to compromise by introducing a quorum, such as 2%, to prevent excessive party fragmentation.

In contrast, the SVP, FDP, SP, and CVP/GLP factions opposed the motion. These groups argued that the completed district reform had already addressed the primary concern of achieving a fairer seat distribution. They also criticized the biproportional method for potentially undermining the legitimacy of constituencies as recognized territorial units. Figure 34 illustrates that a clear majority opposed the reform. Although detailed voting records are unavailable to attribute votes to specific groups, the fact that the EVP/EDU and GPS factions, which together held 20 seats, struggled to garner additional support indicates their difficulty in convincing other parliamentary members.

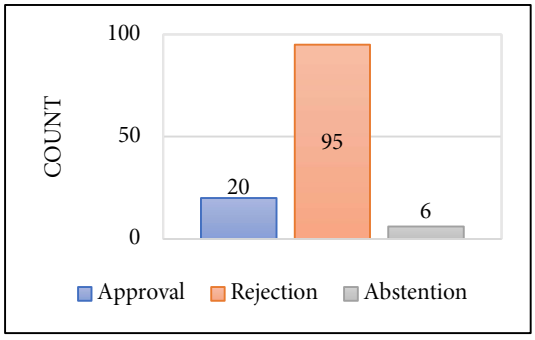


Figure 34: Positions on the parliamentary motion instructing the government to draft a bill for the introduction of biproportional apportionment in cantonal parliamentary elections, Canton of Thurgau.

The decision-making process on the motion to introduce biproportional apportionment in the Canton of Thurgau coincided with a cantonal popular initiative aimed at the same goal. Launched in August 2009 and successfully submitted to the cantonal authorities in March 2010, the initiative was backed by a cross-party

committee comprising the BDP, EDU, EVP, and GPS. Notably, BDP was not represented in parliament at the time, but all parties supporting biproportionality during the motion deliberations were involved.

The parliamentary debate on the initiative occurred on November 24, 2010. Both the responsible commission and the Government Council recommended rejecting the initiative without a counterproposal. They argued that the recently implemented redistricting already addressed concerns of fairness and that introducing biproportionality simultaneously would reduce the proximity between voters and elected representatives, weaken regional seat distribution, and complicate the seat allocation process.

The positions of the parties were consistent with those expressed during the earlier debate on the parliamentary motion. The EVP/EDU and GPS factions continued to advocate for biproportionality, arguing it would rectify the electoral disadvantage faced by smaller parties and eliminate the need for apparentements, which obscure the impact of individual votes. They also noted that unidimensional PR systems could still result in seat reversals due to apparentements.

Conversely, the SP, FDP, and CVP/GLP factions, aligned with the Government Council's recommendation, opposed the initiative. They argued that the district reform had already addressed the issues of vote weight and that further changes were unnecessary. According to them, the support for the initiative was an act of political obduracy. As shown in Figure 35, the Grand Council's rejection of the initiative was even more pronounced than the earlier vote on the motion. The initiative was subsequently subjected to a popular referendum on May 15, 2011, where it was defeated by 62% of the vote with a low turnout of 27.2%. Thus, Thurgau remains the only canton to have rejected the introduction of biproportional apportionment by a popular verdict.

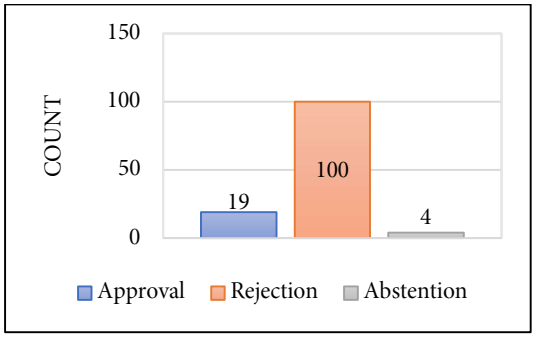


Figure 35: Positions on the cantonal popular initiative "Fair electoral system for elections to the Grand Council" in the Canton of Thurgau, which called for the introduction of biproportional apportionment.

2.6 Reform attempts in other cantons

This subsection provides an overview of biproportional reform attempts in various Swiss cantons that either did not come to fruition or are still in progress. Unlike the previously discussed unsuccessful reforms, the cases examined here were identified as unfeasible proposals early in the process, each for distinct reasons. These include the cases of Lucerne, Fribourg, Basel Stadt, Solothurn, and Vaud. In the case of Basel Landschaft, the decision-making process is still ongoing. The following sections offer a concise presentation of these individual cases, focusing particularly on the elements related to the apportionment method. The section begins with the similar cases of Lucerne and Fribourg, followed by the cases of Basel Stadt, Basel Landschaft, Solothurn, and, finally, Vaud.

2.6.1 Canton of Lucerne

In Lucerne, a reduction in the size of the cantonal parliament sparked a significant debate over the future structure of electoral law. Specifically, in 1998, the legislature was reduced from 170 to 120 seats without making corresponding adjustments to the constituencies. This reduction, combined with the 2007 total revision of the cantonal constitution and the FSC's ruling on the city and Canton of Zurich, heightened the urgency of restructuring the electoral territory (Fuhrer 2018: 65).

During the preliminary discussions on constitutional revision, various models were considered. These included proposals for a single cantonal constituency, constituencies with five, eight, or ten members, the use of biproportional apportionment, and territorial mergers. Given the political sensitivity surrounding electoral districting, the decision was made to avoid explicitly defining constituencies within the constitution. Instead, a minimum of five constituencies and adequate regional representation were specified, leaving the detailed division of electoral districts to be addressed later at the legislative level. Lucerne voters approved the draft constitution on June 17, 2007.

Following this, Michael Töngi (GPS), a member of the cantonal parliament, submitted a motion on January 21, 2008, advocating for the adoption of biproportional apportionment. In response, the Government Council suggested converting the motion into a postulate, signaling its intent to consider biproportionality during the upcoming legislative process for amending the electoral law. However, the Cantonal Council rejected the proposal in both forms.

On January 25, 2010, the Cantonal Council approved a draft revision related to the territorial organization of cantonal parliamentary elections. This revision involved minor adjustments to the electoral law, most notably the formation of a constituency association between the neighboring districts of Entlebuch (7 seats) and Willisau (16 seats), effectively simulating a joint single constituency. The methodology for determining election results mirrored the biproportional process, with super-apportionment at the simulated single constituency level and sub-apportionment at the district level.

In response to the Council's decision, the SVP launched a referendum, arguing that the small district of Entlebuch, with its long-standing tradition as an autonomous valley, should remain an independent constituency. The SVP also criticized the new seat allocation procedure, claiming it unfairly favored smaller parties by lowering electoral thresholds. However, in the referendum held on September 26, 2010, over 70% of voters supported the proposed changes to the electoral law. The first elections under the new law took place in 2011, and the revised electoral system has remained legally unchallenged since then.

2.6.2 Canton of Fribourg

In an amendment to the cantonal constitution dated May 16, 2004, several significant changes to the electoral law in Fribourg were introduced, including a reduction in the number of seats in the Grand Council from 130 to 110. This downsizing occurred without a corresponding reform of the electoral districts, leading to an implicit quorum exceeding 10% in the Glâne constituency (8 seats) and the Vevey constituency (6 seats). On September 4, 2012, the Cantonal Administrative Court ruled that the legal framework for Grand Council elections was inconsistent with the Federal Constitution (Fuhrer 2018: 73f.).

In response, three different models were thoroughly examined:

1. biproportional apportionment method
2. municipal merger of the Glâne and Vevey constituencies
3. creation of a constituency association between Glâne and Vevey (forming sub-constituencies with seat allocation based on the biproportional method)

The biproportional model faced criticism for its complexity. Major political parties, including the FDP, SP, SVP, and CVP, either rejected the biproportional model outright or demanded the introduction of statutory quorums up to 7.5% at the cantonal level as a condition for their support. In contrast, smaller parties like the CSP, BDP, GLP, and GPS supported biproportionality without statutory quorums.

The second proposal, which involved politically merging the Glâne and Vevey municipalities, was almost unanimously rejected by both the affected municipalities and the broader political landscape.

After two rounds of consultation, a comprehensive legal analysis, and two extensive parliamentary deliberations, the third model emerged as the most viable alternative. This model was ultimately approved by the Grand Council on October 7, 2014, with 85 council members voting in favor, none opposing, one abstaining, and 24 absent. The decision mirrored a similar approach previously adopted in Lucerne to address the issue of undersized constituencies. These changes were first implemented in the 2016 cantonal parliamentary elections.

2.6.3 Canton of Basel Stadt

In contrast to the prevalent electoral practices in several other Swiss cantons that use a one-dimensional PR system, specifically the d'Hondt/Hagenbach-Bischoff method, which rounds down non-integer seat allocations, the Canton of Basel-Stadt employs the Sainte-Laguë method, which uses standard rounding. Notably, the smallest electoral district, Bettingen, benefits from a constitutional seat guarantee and uses a plurality voting system, making Basel-Stadt one of the few Swiss cantons with a mixed electoral system.

In 2005, the reduction in the size of the Grand Council from 130 to 100 seats prompted discussions about electoral reform in Basel-Stadt. This reduction did not alter the constituency structure, leading to an increased implicit electoral threshold within constituencies. However, except for the single-member district of Bettingen, the resizing did not result in any of the other four constituencies exceeding the 10% threshold (Beriger 2018: 148-152).

In 2009, Remo Gallacchi, a CVP member of the Grand Council, and co-signatories proposed a parliamentary motion to introduce the biproportional apportionment method for cantonal parliamentary elections. A Special Commission was appointed to examine this proposal. In 2011, the Commission presented its report, which, while recognizing the advantages of biproportionality in accurately converting votes into seats,

ultimately rejected its adoption due to the complexity and potential paradoxes in seat distribution. Instead, the Commission recommended the following reforms:

- Maintaining the one-dimensional PR system with standard rounding of seat quotients.
- Retaining the existing constituency structure.
- Reducing the district-based quorum from 5% to 4%, applicable individually to each constituency.
- Abolishing both horizontal (between party lists) and vertical (within party lists) apparentements.

On June 8, 2011, the Grand Council approved all these proposed changes, with the statutory quorum being a particularly contentious issue. The new electoral procedure was first implemented in the 2012 Grand Council elections. In 2016, the Grand Council decided to abolish the statutory quorum, a move that was approved by voters the following year (Beriger 2018: 150)

2.6.4 Canton of Basel Landschaft

The electoral system for cantonal parliamentary elections in Basel Landschaft is organized into twelve constituencies, each allocated between six and ten seats, with a minimum guarantee of six seats per constituency. These twelve constituencies are grouped into four electoral regions, each with 17 to 30 seats, ensuring that the implicit quorum stays below the 10% threshold across all regions, thus complying with federal legal requirements.

In 2015, Klaus Kirchmayr, a GPS councilor, initiated an interpellation requesting the cantonal government to examine the comparative effects of the current electoral system versus a biproportional model. The following year, the Government Council responded without identifying any need for reform.

Kirchmayr later introduced a procedural postulate advocating for the formation of a special commission to thoroughly review the electoral law. This commission would develop various reform proposals aimed at improving the proportional representation of party strengths within the Cantonal Council, while preserving the constituencies as local representation guarantors. Although the Cantonal Council rejected this proposal on October 17, 2019, the concern it raised gained traction. Parliament instructed the parliamentary executive board to collaborate with external experts in devising alternatives to the current electoral system, focusing on achieving greater proportionality while maintaining the local connection of elected representatives.

On July 1, 2022, the parliamentary executive board submitted a report recommending a comprehensive electoral system reform, favoring a canton-wide biproportional apportionment system while retaining constituencies. The proposal included replacing the six-seat guarantee with a one-seat guarantee. On September 15, 2022, the parliament approved this proposal with a vote of 45 to 32, instructing the government to draft a revision of the electoral law in line with the proposed biproportional method. Apart from the SVP and FDP, all other factions supported the electoral reform. The matter is currently pending with the cantonal government.

2.6.5 Canton of Solothurn

In the Canton of Solothurn, the reduction of the cantonal parliament from 144 to 100 seats led to significant deliberation on the design of electoral legislation. A popular initiative, aimed at reducing the size of the parliament and proposing a maximum of seven constituencies of comparable magnitude, was approved by voters on March 3, 2002. However, the Cantonal Council's counterproposal, which retained the five existing

administrative districts as constituencies despite their varying sizes (ranging from 13 to 29 seats), was ultimately endorsed by the voters.

In 2011, Markus Knellwolf from the GLP initiated an investigation into the potential adoption of the biproportional apportionment method for cantonal parliamentary elections. The following year, a coalition of the CVP, EVP, and GLP requested the Government Council to assess the biproportional method for both municipal and cantonal elections through a parliamentary motion. This proposal was partly driven by a concurrent motion from the FDP, SP, and SVP seeking to prohibit cross-party apparentements, which could significantly impact smaller parties. The proponents viewed biproportionality as a compensatory measure.

The Government Council, after extensive deliberation, acknowledged the advantages of the biproportional method but also pointed out its complexities and potential risks, such as discordant seat allocations and party fragmentation. It found the existing constituency structure to be compliant with federal law and recommended rejecting the motion. After a contentious debate, the Grand Council rejected the motion, with support only from the CVP, EVP, GLP, and GPS. Since this rejection, there have been no further attempts to reform the electoral system in Solothurn.

2.6.6 Canton of Vaud

In the Canton of Vaud, an attempt was made to introduce biproportional apportionment in response to concerns about high implicit electoral quorums in smaller constituencies. The Canton of Vaud, along with Ticino, is unique in Switzerland for using the Hare-Niemeyer method for seat allocation. As part of a constitutional revision, the number of seats in the Grand Council was reduced from 180 to 150, with the electoral districts reorganized into ten constituencies, three of which include two sub-constituencies. Despite some constituencies having only eight seats, which falls below the FSC's nine-seat minimum guideline, the electoral system is considered likely to remain compliant due to district mergers and the accurate representation achieved by the Hare-Niemeyer method (Beriger 2018: 160-161).

On June 23, 2009, Bernard Borel of the PST-POP party ("Parti Ouvrier et Populaire") proposed a parliamentary initiative to introduce biproportional apportionment to address the disproportionately high electoral thresholds in smaller constituencies. However, Borel later withdrew the initiative in favor of a motion submitted on November 1, 2009, which also supported maintaining a 5% electoral threshold, which needed to be surpassed in at least one constituency.

This motion was eventually turned into a postulate and debated in the Grand Council on February 8, 2011. Despite support from some members, the postulate was rejected. Opponents cited concerns about the complexity and transparency of the proposed system, potential disruptions to the constituency structure, and the risk of inconsistent seat allocations across constituencies. Since then, there have been no further attempts to implement biproportional apportionment in Vaud.

2.7 Parties between strategical maneuvering and idealism?

In the context of electoral reform, political parties are uniquely positioned as both architects and primary stakeholders of electoral laws. This dual role creates strong incentives for strategic behavior, as parties aim to maximize their electoral advantages. But are they exclusively guided by strategic considerations or are there other factors that moderate their reform positioning?

This chapter starts with addressing the question whether parties that previously benefited from electoral design tend to resist biproportional reform. Conversely, it explores whether parties previously "penalized" by electoral regulations are more inclined to strive for comprehensive biproportional reform adoption. If parties seek to maximize their share of seats and use their political leverage to twist the "rules of the game" in their favor, their actions in decision-making during electoral reform processes will largely hinge on the expected improvement or worsening of their position relative to the status quo. According to the first hypothesis, parties advantaged by electoral regulations in force will resist any electoral reforms or elements thereof that endanger their advantageous position. Conversely, if the current electoral law adversely affects their legislative representation, it is reasonable to expect these parties to push for potentially radical reforms to the electoral system.

Various factors influence the extent of a party's misrepresentation relative to its vote share, thereby affecting the overall proportionality of an electoral system. Key factors include the specific formula used to translate votes into seats, the total number of legislative seats, and constituency characteristics such as number, size, and shape. Consequently, changes to the seat allocation method and other mechanical features of the electoral system are likely to produce varying distributional effects on different parties, both in direction and magnitude (Kedar et al. 2016).

Understanding a party's position and behavior during the electoral reform process requires an analysis of its electoral performance prior to the reform. Parties that have historically experienced significant misrepresentation, with a substantial portion of their votes failing to convert into seats, are likely to support reforms that promise to improve their parliamentary representation. In contrast, parties benefiting from the existing electoral system are expected to oppose changes that could undermine their current advantages.

In cantons that underwent biproportional reform, most previously used unidimensional PR (ZH, AG, SH, NW, ZG, VS). However, Uri and Schwyz adopted a mixed electoral system (MR in smaller districts and PR in larger multi-member districts), while Graubünden exclusively used MR in its former cantonal parliamentary elections. For Graubünden, party representation estimates are not indicated due to the electoral system, and for the cantons of Schwyz and Uri, estimates should be viewed with caution. Cantons rejecting biproportional reform (OW, BE, TG) all persist in electing their parliaments using unidimensional PR. Table 30 forms the basis for estimating the parties' stances on biproportional reform. It shows the degree and direction of the competing parties' legislative misrepresentation for all elections between 1976 and 2023, expressed in vote-seat disproportionality. To ensure comparability across cantons, these values were standardized using the total number of parliamentary seats per canton. Further specifics are available in the table's notes.

The table illustrates the anticipated effect where the big, established parties (cf. columns 1-4) tend to experience higher levels of overrepresentation, while smaller fringe parties consistently face underrepresentation. It's notable that these disproportionalities are considerable, particularly during specific (pre-reform) periods and in certain cantons. In the cases of Uri and Schwyz, the electoral systems prior to the reform

featured pronounced majoritarian elements, as MR was (and in the case of Uri still is) used in small constituencies/municipalities, resulting in greater disproportionalities.

One source of systemic distortions besides the district structures and the electoral system/formula lies in the legal provision that allows electoral alliances between different lists, which is provided in most cantons that elect their parliaments based on unidimensional PR. Although smaller parties can also benefit from these alliances, empirical evidence indicates that they primarily favor larger parties or the dominant partner within the alliance (Bochsler 2010b), thereby further perpetuating systemically induced disproportionalities.

Certain anomalies can be observed, such as the 2006 elections in Zug, where the Alternative Green Party (classified as FGA in the table) as part of an apparentement doubled its seats from five to ten. However, this was partly at the expense of its coalition partner, the SP. This demonstrates that parties in a marginalized position can not only minimize systemic disadvantages through strategic electoral alliances but also leverage them to their advantage. Nevertheless, a broader analysis of the data suggests that such anomalies are relatively rare occurrences.

The successfully reformed cantons generally exhibit a clear "proportional effect", where the disparity between votes and seats – after the implementation of the reforms – typically remains within the threshold of +/- one parliamentary seat, with few exceptions. The case of Uri, however, warrants discussion, as significant distortions persist even after the reform. This can be attributed to the mixed electoral system in place, where, as part of the reform, the majoritarian system was extended to four additional municipalities that previously used PR. As the results are aggregated across the entire area, including municipalities voting according to MR, the overall disproportionality has actually slightly increased following the reform.

Table 30: The (dis)proportionality of party representation in parliaments of (un)successfully reformed cantons 1976-2023. The cells show vote-seat-differences expressed in parliamentary seats. Reading example: In the 1991 Cantonal Council Elections in Zurich, the SP was overrepresented by 8.26 parliamentary seats.

Canton	Year of El.	FDP	SP	CVP	SVP	GPS	GLP	EVP	BDP	FGA	SD	EDU	CSP	Others
ZH	1979	6.35	3.82	1.21	0.50	-0.54		-0.95			-4.50	-0.54		-1.44
	1983	7.54	-0.83	2.66	4.39	-1.76		0.28		-4.68	-5.04	-0.18		-3.96
	1987	4.61	3.04	1.15	2.56	2.29		-0.09		-2.93	-4.86	-1.08		-2.74
	1991	7.88	8.26	-0.86	2.09	1.46		-1.51		-0.88	-3.92	-1.44		-1.98
	1995	5.51	6.30	-0.52	2.02	0.16		-0.50		-1.69	-2.21	-2.16	-0.54	-0.90
	1999	-0.29	3.22	0.94	7.97	0.74		0.00		-1.15	-2.34	-2.23	-0.54	-1.66
	2003	-1.06	5.29	0.49	6.28	-0.22		-0.10		0.00	-1.51	-2.77	-0.18	-5.94
	2007	0.20	0.95	-0.05	1.17	0.29	-0.43	0.33		-0.29	-2.39	-0.07		
	2011	-0.22	0.25	0.18	0.72	-0.07	0.45	0.09	-0.31	0.13	-0.72	0.32		-1.03
	2015	-0.14	0.54	0.18	0.00	0.04	0.32	0.14	0.32	-0.40		0.14		-1.21
	2019	0.74	0.25	0.25	0.90	0.58	-0.22	0.24	-2.70	0.41		-0.14		-0.18
	2023	0.38	1.26	0.20	1.19	0.27	1.14	-0.04		0.31		-0.42		-4.32
SH	1976	1.80	1.40	-0.84	-0.24			0.68			-0.80			-1.28
	1980	2.40	1.28	-1.32	-0.56			-0.28						-1.40
	1984	1.52	0.52	-0.76	-0.24			-0.96						-0.32
	1988	-0.04	2.40	-0.16	0.60			0.32		-0.52		-0.56		-0.44
	1992	1.16	-0.16	0.52	1.24			-0.52		-0.68		-0.64		-0.48
	1996	0.28	0.84	-0.40	1.24	0.00		0.48				-0.80		
	2000	-0.70	-0.30	0.67	0.10	-0.03		0.42						-0.18
	2004	0.32	2.24	-0.68	1.52	-0.56		-0.12		-0.60		-0.80		-1.30
	2008	0.68	0.19	-0.07	-0.56	0.04		-0.41		0.53		-0.40		
	2012	0.02	0.44	0.06	-0.16	-0.44		-0.32		0.50		-0.10		
	2016	0.70	0.32	-0.22	-0.06	-0.34		-0.44		-0.26		-0.34		
	2020	-0.46	0.18	0.08	-0.04	-0.34		0.44		0.34		-0.04		
AG	1977	4.00	1.80	-1.40	0.00			-2.20			-2.20			-0.20
	1981	3.20	2.80	0.80	0.20			-1.00			-4.60			-0.80
	1985	4.60	3.00	1.40	0.20			0.20		-5.00	-3.20			0.00
	1989	4.60	1.60	0.60	2.80	-2.00		-0.40			-3.20	-0.60		-0.20
	1993	1.80	4.20	1.40	1.80	-1.80		-1.60			-2.20	-0.80		-2.00
	1997	0.80	4.60	2.44	3.22	-1.00		-0.68			0.66	-1.64		-2.58
	2001	1.96	-1.12	2.04	5.06	-1.04		-1.70			0.48	-2.08		-0.62
	2005	0.31	2.53	1.53	3.64	-2.44		-0.97		-0.62	-1.51	-0.94		-1.54
	2009	-0.01	0.01	0.00	0.34	0.55	0.10	-0.29	-0.34		0.32	-0.52		-0.14
	2012	0.43	0.71	0.38	0.20	-0.36	0.29	0.55	-0.15		-0.98	-0.38		-0.70
	2016	-0.41	0.55	0.06	0.34	0.06	-0.42	0.41	0.22			-0.52		-0.28
	2020	0.28	0.18	0.08	0.85	0.28	0.13	0.13				-0.66		-1.26
NW	1982	-0.56	-2.30	3.20										-0.08
	1986	0.58	-2.22	3.14										
	1990	-0.38	0.00	3.12										-0.72
	1994	-0.78	0.00	4.26										-0.42
	1998	-1.20	-3.68	1.50										
	2002	0.52	-3.28	1.62	-0.44									-0.14
	2006	0.83	-0.87	2.39	-1.70									-1.53
	2010	0.32	-0.26	-0.66	3.04	-2.14								-0.30
	2014	0.48	0.12	-0.46	-0.46	0.32								
	2018	0.20	0.30	-0.08	-0.54	-0.04								0.16
	2022	0.24	0.18	-0.54	0.42	0.30	0.18							-0.78
ZG	1978	-0.60	-1.60	5.00										-1.64
	1982	2.04	-2.48	4.24										-3.32
	1986	1.64	-1.36	2.64						-0.56				-2.36
	1990	0.12	-1.75	4.24						-0.56				-2.05
	1994	2.48	-1.74	5.08	-3.88					-1.00				-0.94
	1998	0.84	1.44	0.12	-1.08					-0.89				-0.43
	2002	-0.88	-0.14	1.80	0.96					-0.02				-1.71
	2006	-0.54	-2.46	-0.04	1.85					3.03	-0.39			-1.45
	2010	1.44	-0.24	1.80	0.84	-1.32	-1.12							-1.40
	2014	0.32	-0.44	0.56	0.12	-0.28	0.00							-0.28
	2018	0.28	-0.44	0.52	0.08	-0.16	-0.24							-0.02
	2022	0.48	-0.16	-0.20	0.72	-0.12	0.00							-0.72

Canton	Year of El.	FDP	SP	CVP	SVP	GPS	GLP	EVP	BDP	FGA	SD	EDU	CSP	Others
SZ	1976	-1.20	-2.00	5.60	-1.30					0.00				-1.10
	1980	0.00	-1.50	3.30	-1.30					-0.50				
	1984	-2.20	-0.50	7.00	-0.10					-4.20				
	1988	1.70	0.00	4.80	-1.70	-2.70								
	1992	1.20	-1.10	5.70	-2.60	-1.60								-1.60
	1996	-0.20	-4.21	8.10	-3.20	-0.21								0.62
	2000	-0.70	-1.30	4.30	-0.90	0.00								-1.40
	2004	1.00	-0.20	2.00	-2.10	0.00								-0.70
	2008	1.40	-2.00	-0.80	3.50	-0.80								-1.30
	2012	0.20	-2.40	-0.10	1.00	0.80								0.50
	2016	0.40	0.10	-0.20	-0.10	0.20	0.50							-0.90
	2020	-0.25	-0.70	0.00	-0.20		0.20							0.95
VS	1977	-0.09	-2.42	4.75										-2.24
	1981	-0.94	-4.99	5.64										0.29
	1985	-0.49	-1.39	4.86										-2.07
	1989	1.07	-2.64	3.21										-0.47
	1993	2.80	-2.46	2.46										
	1997	1.78	-1.17	1.26										
	2001	2.80	-0.47	3.83	-3.98							-0.29		-0.57
	2005	0.27	3.35	3.83	-2.31	-2.77							-0.06	-0.04
	2009	-0.99	-1.03	4.95	-2.95	0.14							0.92	-1.03
	2013	-0.08	-1.47	4.06	-1.37	-2.29							1.05	0.21
	2017	-0.26	-0.65	0.79	1.68	-0.98							1.27	-1.90
	2021	1.39	1.87	-1.66	0.03	0.65	-0.78						1.66	-3.16
UR	2008	0.12	-1.25	1.79	-1.45	1.08								-0.28
	2012	2.01	1.25	-0.42	-1.74	-0.49								-0.60
	2016	0.79	-0.32	1.97	-0.42	-0.54								-1.47
	2020	2.48	-2.81	3.21	-2.75	0.12								-0.26
OW	1986	-0.53		0.36										0.16
	1990	0.20		0.26										-0.46
	1994	0.17		1.05										-1.22
	1998	-0.68	0.36	0.01										0.31
	2002	-0.51	-0.43	1.29	-0.40									0.05
	2006	-1.49	-1.21	3.65	-1.59								0.96	-0.32
	2010	0.15	-0.01	1.42	-0.62								-0.53	-0.42
	2014	0.67	-1.07	1.33	-0.48								-0.49	0.05
	2018	-1.47	-0.32	-0.39	1.52								0.86	-0.20
	2022	1.11	-0.89	3.55	-1.05		-1.27						-1.45	0.00
BE	1978	4.19	-0.80	1.10	2.75			0.09		0.45	-1.41	0.26		-5.15
	1982	3.20	0.20	1.20	3.40			-1.20		-1.60	0.20	-0.40		-1.60
	1986	2.40	3.20	0.60	6.00	-1.60		-2.60		-1.80	-1.80	-2.60		-1.00
	1990	1.00	7.60	-0.20	6.40	-4.80		-2.80		0.40	-0.80	-2.60		-3.80
	1994	0.40	5.80	-0.40	6.00	-3.20		-1.00		1.40	-2.00	-3.60		-2.60
	1998	1.92	6.16	0.20	2.60	-0.60		-0.90		1.00	-1.60	-5.20		-2.28
	2002	1.00	5.80	0.00	3.40	-1.00		-1.00			-0.20	-5.40		-2.60
	2006	-0.21	3.55	-1.81	3.14	-1.63		1.25			-1.92	-1.65		-0.08
	2010	0.53	4.77	-1.07	1.44	-0.16	-2.56	0.56	-0.59		-0.64	-2.03		-0.24
	2014	-0.11	2.45	-1.28	2.61	-0.16	0.29	1.76	-3.92	-0.64	-0.48	-1.55		1.36
	2018	1.28	2.32	-1.12	3.12	-2.16	-0.03	0.08	-1.39	0.21	-0.32	-0.91		-0.59
	2022	-0.02	1.78	0.24	2.67	-1.24	0.30	0.09		-0.01	-0.22	-0.38		-1.93
TG	1976	0.86	-0.09	0.55	3.90			-0.81			-0.30			
	1980	1.30	0.48	-0.33	2.66			-1.23			-0.78			
	1984	0.69	-0.07	1.24	3.26			-1.46			-0.69			-0.78
	1988	1.55	1.92	0.61	2.48	-0.70		-1.27			-0.78			-1.82
	1992	1.03	3.11	0.78	2.92	-0.53		-1.14			-2.99			-1.30
	1996	0.64	0.68	1.13	3.15	1.24		-3.15						-1.74
	2000	-1.43	1.61	1.95	2.58	-1.34		-1.89			-0.36	-0.12		-0.16
	2004	-0.45	2.76	-0.70	3.36	0.34		-2.54				-2.56		-1.22
	2008	-0.85	0.36	1.33	2.83	-0.70	-0.60	-0.49				-2.07		-0.65
	2012	-0.46	1.59	2.54	1.04	-1.01	-1.66	-1.11	-1.24			0.03		
	2016	-0.28	-0.03	2.57	1.25	-0.62	0.23	-1.37	-1.03			-0.97		-0.13
	2020	0.19	-1.08	0.32	2.32	-0.56	-0.88	-0.24	-0.59			-0.72		0.55

Notes:

- Cell values: Each cell represents the difference between a party's share of seats and its share of votes expressed in parliamentary seats. This metric ensures inter-cantonal comparability by considering varying parliament sizes. For instance, a 5.0 percentage point difference between seat and vote shares translates to 3 seats in the Nidwalden cantonal parliament (with 60 seats), whereas the same variance corresponds to 9 seats in the Zurich cantonal parliament (with 180 seats), demonstrating a more substantial effect. Hence, a standardized metric is used that considers varying parliament sizes. Changes in parliament size within certain cantons over time are also considered.
- Votes cast for a joint list comprising multiple parties (apparentement) are proportionally distributed among the respective parties.
- The formula used for calculation: $\Delta_i = (\text{seat share}_i - \text{vote share}_i) \times \frac{\text{parliament size}}{100}$ for party i for a given election.
- Year of election: The initial election applying the new apportionment method is highlighted in yellow.
- Color scheme: Differences between 0 and 1 seat are not highlighted, as they fall within an expected margin of error. Variances exceeding ±1 seat are highlighted with graduated colors (thresholds: ±1, ±2, ±3).
- Parties: Only parties active during the reform periods are considered. Therefore, LDU, POCH, PDA, FPS, DN, and LPS are excluded (and are not part of the "others" group). As a result, the row totals do not equal 0 in all cases.
- In the case of the Canton of Uri, the FSO only reports election results from 2008 onwards.
- Data sources: Calculations derived from the "Patterns of Democracy" data set (Vatter et al. 2020), supplemented, and selectively revised with data from the FSO and primary sources from cantonal statistical offices.
- OW/BE/TG: Cantons of Obwalden, Bern, and Thurgau are examined as cases where legislative efforts were made to implement biproportionality. However, in all instances, these attempts were unsuccessful.

Table 31 provides an overview over the assumed party positions in all cantons concerned.[70] In line with their parliamentary over-representation, it can be expected that, in many cases, the SP, CVP and SVP and to a lesser extent the FDP as established and dominant parties will reject a reform or reform elements aimed at a more proportionate representation of the parties. However, there are exceptions observed in certain cantons, particularly in central Switzerland, where the trend is not distinctly clear to estimate their stance. Particularly, the case of the FDP is noteworthy. Despite being a prominent beneficiary of the electoral law for an extended period and, in some cantons, being the strongest party at times, it has been experiencing a consistent decline in vote share. This downward trajectory is observable on a national scale as well. In line with this electoral decline and the consequential loss of its status as a dominant party, the vote-seat-disparities in various cantons exhibit significant volatility, rendering it difficult to estimate the parties' reform stances. The smaller parties including the GPS and GLP, on the other hand, which face a legislative underrepresentation in many cantons, are presumably mostly in favor of reform.

[70] Due to ambiguities in the Valais minutes regarding the mapping of parties' reform stances and for further reasons rendering it a unique case, this canton is excluded from hypothesis testing (cf. remarks in the case study of Valais).

	FDP	SP	CVP	SVP	GPS	GLP	EVP	BDP	FGA	SD	EDU	CSP	Others
ZH	?	Opp.	?	Opp.	?		?		Supp.	Supp.	Supp.		
SH	?	Opp.	?	Opp.	?		?		?		?		
AG	Opp.	Opp.	Opp.	Opp.	Supp.		Supp.			Supp.			
NW	?	Supp.	Opp.	Opp.	Supp.								
ZG	?	Supp.	Opp.	?	Supp.	Supp.							
SZ	?	Supp.	?	?	?								
UR	?	?	Opp.	Supp.	?								Supp.
OW	?	?	Opp.	?								?	
BE	?	Opp.	Supp.	Opp.	Supp.	?	?	Supp.	?	Supp.	Supp.		
TG	?	Opp.	Opp.	Opp.	?	?	Supp.	Supp.			Supp.		

Table 31: Assumed reform positioning of parties based on their pre-reform vote-seat (in)congruence. Supp. = Support; Opp. = Opposition; ? = unclear assignment.

In the next step, the anticipated positioning is juxtaposed with the observed positioning. The assessment of the actual positioning draws from the case studies delineated in the previous subchapters. Instances where a deviation occurs between the expected and observed positioning are highlighted in the table (cf. table notes). As shown in Table 32, there are deviations in seven cases out of a total of 71, with estimations not possible due to unclear trends in legislative misrepresentation in 31 cases. Astonishingly, in all seven deviant cases, the parties expressed support contrary to expectations – there is no case where a party unexpectedly opposed reform. Broadly, the parties' observed and their expected positions coincide in most cases. However, drawing the conclusion that expected improvement or deterioration in the parties' post-reform status governs the parties' positioning would be unwarranted. Looking at the parties' positioning reveals discernible cross-cantonal patterns that suggest an additional dimension alongside the "winner-loser" antagonism. As anticipated, the GPS consistently supported the reform across all cantons under scrutiny, while the SP showed higher support than expected. In three cases, the majority favored the implementation of biproportional reform despite potential seat losses. The SP took a negative stance only in two cases, where the reform attempt failed (BE, TG).

	FDP	SP	CVP	SVP	GPS	GLP	EVP	BDP	FGA	SD	EDU	CSP	Others	Total
ZH	Supp.	**Supp.**	Supp.	**Supp.**	Supp.		Supp.		Supp.	Supp.	Supp.			9 vs. 0
SH	Supp.	**Supp.**	Supp.	Opp.	Supp.		Supp.		Supp.		Supp.			7 vs. 1
AG	Opp.	**Supp.**	**Supp.**	Opp.	Supp.		Supp.			Supp.				5 vs. 2
NW	Supp.	Supp.	**Supp.**	Opp.	Supp.									4 vs. 1
ZG	Supp.	Supp.	Opp.	Opp.	Supp.	Supp.								4 vs. 2
SZ	Supp.	Supp.	Supp.	Opp.	Supp.									4 vs. 1
UR	Supp.	Supp.	**Supp.**	Supp.	Supp.								Supp.	6 vs. 0
OW	Opp.	Supp.	Opp.	Opp.								Opp.		1 vs. 4
BE	Opp.	Opp.	Supp.	Opp.	Supp.	Supp.	Supp.	Supp.	Supp.	Supp.	Supp.			8 vs. 3
TG	Opp.	Opp.	Opp.	Opp.	Supp.	Opp.	Supp.	Supp.			Supp.			4 vs. 5
Total	6 vs. 4	8 vs. 2	7 vs. 3	2 vs. 8	9 vs. 0	2 vs. 1	5 vs. 0	2 vs. 0	3 vs. 0	3 vs. 0	4 vs. 0	0 vs. 1	1 vs. 0	

Table 32: Assumed vs. true reform positioning of parties. Cases in which the true positioning contradicts the expected positioning are highlighted in bold and yellow. All other cases do not contradict the assumed positioning (highlighted in blue). The totals show the row and column totals for support and opposition.

In these cantons, it is important to recognize that despite the potential political feasibility of electoral reform, there was or is no legal imperative for such changes, as the existing electoral laws already comply with federal regulations. Notably, support for reform was not limited to the usual proponents like the GPS and SP; other left-leaning parties, including the FGA, also advocated for electoral reform. This highlights that beyond the typical "winner-loser" antagonism, reform positions have an ideological dimension, with left-wing parties generally pushing for cantonal reforms toward biproportionality.

The minutes from legislative discussions support this assumption. Representatives from the SP frequently underscored their commitment to establishing a more equitable electoral system, even if it might result in potential seat losses for their party. Furthermore, smaller parties such as the EVP, BDP, SD, EDU, and the "others" group in Uri also demonstrated support for the reform, aligning with the anticipated coalition of proponents.

The SVP generally resisted biproportional reform across most cantons, with notable exceptions in Zurich and Uri, where it supported the reform. In Zurich, this support, though unexpected, was partially driven by legal pressures that necessitated adjustments to the electoral system to comply with constitutional requirements. In Uri, the FSC similarly compelled the canton to reform its electoral law. However, in Uri, the SVP's support for biproportional apportionment was accompanied by successful advocacy for the extension of the plurality voting system to municipalities with three and four parliamentary seats.

Given that the SVP had only recently established itself within Uri's party landscape and thus held a "fringe party" profile during the reform process, their support for the reform is less surprising. This context underscores how the party's relatively weaker position at that time may have influenced its strategic alignment with the broader reform efforts.

The positions of the FDP and CVP exhibit less consistency in their stance on biproportional reform compared to other parties. The expected alignment is largely confirmed for the FDP across all cantons, with the notable exception of Aargau, where the party opposed reform despite its eventual adoption. In other cantons that successfully implemented biproportional reform, the FDP generally supported it.

Similarly, the CVP mostly favored reform across the successfully reformed cantons, with Zug being the only exception. However, in Aargau, Nidwalden, and Uri, the CVP's support deviates from expectations. In Aargau, the surprise lies not only in the CVP's unexpected support for reform but also in the enthusiasm with which the party championed biproportional reform. In Nidwalden, the CVP was internally divided: while the majority supported the reform, a vocal minority criticized it for its perceived impracticality and complexity, aligning with the SVP in opposing the reform.

In Uri, the CVP's support was part of a broader cross-party agreement. After the FSC invalidated key elements of the cantonal electoral law, reform became a pressing political issue. The biproportional method was identified as the preferred solution, though debates arose over its implementation across different municipalities. A compromise was reached, proposing a mixed model where smaller municipalities would use plurality voting, and larger ones would adopt the biproportional method. This compromise, supported by both the FDP and CVP, gained broad parliamentary approval.

The FDP and, to a greater extent, the CVP are considered key actors in the "consensus-oriented political center", a role that enables them to significantly influence policy and decision-making processes (Schwarz 2009: 7). Their support or opposition is crucial, particularly in cantons where these parties traditionally hold

a substantial share of parliamentary seats. When they withhold consent, as seen in failed reform attempts, securing majority support for electoral reforms becomes challenging, if not impossible. This underscores their pivotal role in shaping the outcomes of reform initiatives, particularly in regions where they have traditionally wielded significant power.

While left-leaning parties and groups generally exhibit a strong inclination towards electoral reform, the SVP, in contrast, typically opposes biproportional reform. However, the motivations behind the SVP's opposition may not always be as genuine as they appear. Their arguments might serve as strategic justifications aimed at preserving the status quo, which often works to their advantage. Similarly, fringe parties that advocate for radical reforms may also be driven primarily by utility considerations, with their arguments serving as a mere facade to justify their stance.

Given this, it is challenging to definitively ascertain the sincerity of the arguments presented by any party, regardless of affiliation. It is plausible that both ideological beliefs and strategic, utility-driven considerations influence party positions, with the emphasis on each varying depending on the specific circumstances. This observation suggests a broader hypothesis: across the political spectrum, parties weigh ideological and utility-centered considerations differently, depending on the situation and potential conflicts between these factors. This dynamic is illustrated in Figure 36, which demonstrates how these mechanisms interplay in shaping party positions on electoral reform.

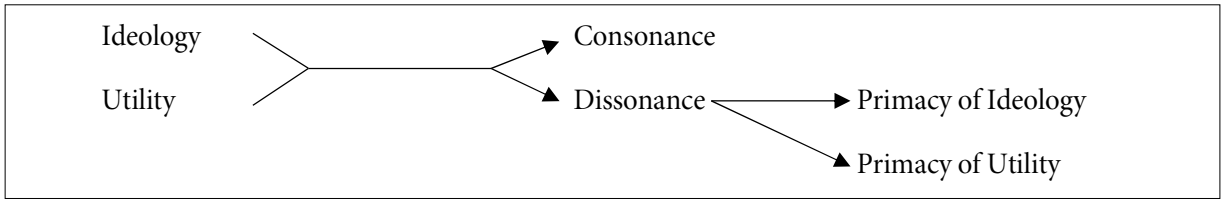


Figure 36: Theoretical interaction between ideological and utilitarian considerations of reform positioning.

Supporting or opposing the final reform policy is the most direct and visible way for parties to shape the reform process, yet it is not the only avenue. Biproportional reforms often consist of various components, allowing parties to exert their political influence on each element individually. Statutory quorums, in particular, became a focal point of intense political contestation. These quorums were extensively debated in the parliaments of nearly every canton. Consequently, the following examination of the more nuanced attempts by parties to influence the reform process will concentrate on statutory quorums, as they provide a clearer insight into how parties sought to subtly shape the outcomes of electoral reforms.

In most cases, decisions regarding statutory quorums aligned with the broader reform process, as evidenced in Table 33. However, Aargau stands out as an exception, where no statutory quorums were initially included in the original electoral design but were introduced only years later through a subsequent decision. In contrast, Schaffhausen, Nidwalden, and Uri have never seen sufficient support for establishing statutory quorums.

The SVP consistently advocated for the introduction of quorums across nearly all cantons. Similarly, the FDP and CVP, as established forces in Swiss politics, largely supported the implementation of quorums. Conversely, left-wing and smaller parties consistently opposed quorums across the board. Even the SP, which might have benefited from statutory quorums in certain contexts, rejected them on ideological

grounds, arguing that such measures conflicted with the core principle of biproportional apportionment – namely, to enhance vote-seat proportionality and reduce ineffective votes. An exception to this trend occurred in Zurich, where the SP agreed to a quorum as part of a broader cross-party agreement.

The CVP's stance toward statutory quorums, however, was more ambiguous, as it supported them in some instances while rejecting them in others. This variability highlights the CVP's role as a key majority provider: whenever a quorum was successfully introduced, it was with the CVP's support. Conversely, in cases where CVP backing was absent, no majority could be secured for the implementation of quorums.

Canton	Pro Quorum	Contra Quorum	Outcome	Remarks
ZH	SP, GPS, CVP, FDP, SVP	-	5% local	3% global quorum added in 2021 (either/or clause)
SH	SVP	FDP, SP, CVP, GPS, EVP, FGA, EDU	no statutory quorum	majority rejected quorums
AG	SVP, FDP	SP, CVP, GPS, EVP, SD	no statutory quorum	3% global and 5% local quorums were added in 2011 (either/or clause)
NW	-	FDP, SP, CVP, SVP, GPS	no statutory quorum	unanimous rejection of quorums
ZG	SVP, FDP, CVP	SP, GPS, GLP	3% global and 5% local (either/or clause)	
SZ	SVP, FDP, CVP	SP, GPS	1% global	
VS	unclear	unclear	8% local	no point of contention; quorum already existed prior to reform
UR	-	-	no statutory quorum	was never subject to discussion
GR	FDP, CVP, (BDP, SVP, GLP)	(SP)	3% global	vast majority voted for quorum; stances of parties in brackets not conclusively clear

Table 33: Party positions on statutory quorums during the cantonal reform processes and the resulting outcome.

From an inter-cantonal perspective, the emerging patterns align with the assumption that larger, established parties tended to support the concurrent implementation of statutory quorums. In some cases, the introduction of a quorum was a prerequisite for their endorsement of the overall reform, effectively serving as a political bargaining chip. Conversely, left-wing and fringe parties only reluctantly agreed to quorums as a necessary concession to secure broader support for the overall reform.

Consistent with the findings on overall reforms, left-wing parties exhibited fundamental reservations about quorums. Generally, it was these left-wing parties, alongside smaller fringe groups, that championed the least distorted reforms possible. This stance aligns with a "natural inclination" for self-preservation among smaller parties, as the decision to implement quorums or not can have critical implications for their survival. However, left-wing parties sometimes faced the prospect of a potential decline in their electoral standing due to the introduction of electoral hurdles. Despite this risk, they consistently opposed quorums, invoking principles such as fairness, equality, and integrity, which they prioritized over maintaining electoral advantages. While other parties were not entirely uninfluenced by such egalitarian concepts, their impact was generally more moderate and less frequent. In contrast, right-wing parties, particularly the SVP, and to some

extent the FDP and CVP, prioritized stable governance and operational efficiency over concerns about electoral fairness and party diversity, arguing from a more pragmatic standpoint.

These findings support the emphasis by earlier authors on the significant role of ideas, ideologies, and values in shaping electoral institutions (Blais & Massicotte 1997, Katz 2005, Bowler et al. 2006, Blais & Shugart 2008, Bowler & Donovan 2013). In their comparative study on the dissemination and underlying patterns of electoral formulas across the world, Blais and Massicotte (1997: 117) concluded that "ideas count as much as interests" and that "politicians sometimes make choices primarily on the basis of their views about what is good, just, or efficient", rather than solely based on strategic considerations. The adoption of biproportional apportionment in several Swiss cantons corroborates this perspective. While parties generally advocate for rules that they perceive as advantageous, there is substantial evidence that political idealism also significantly influenced the positions taken during the reform processes, albeit with varying levels of emphasis among the involved parties.

2.8 Policy-level implications of process-related conflict structures

Political parties employ a range of tools and strategies to shape electoral reform designs in line with their interests and principles. Consequently, the final design of such reforms often reflects the conflict dynamics inherent in the reform process. As discussed in the previous sections, there are situations where parties may opt to align with electoral reforms despite holding fundamental criticisms. Rather than outright opposing the reform, they may choose to influence it more subtly at the policy level. This approach is particularly relevant when parties are either judicially compelled to undertake electoral reform or when they perceive that the proponents of the reform are gaining momentum, making the passage of the reform likely despite their opposition. In such scenarios, parties may strategically shift from their original position, abandon outright opposition, and engage in political bargaining. This proactive approach allows them to address potential electoral threats strategically rather than passively accept an unfavorable outcome (cf. Walter & Emmenegger 2019: 66).

With a nuanced understanding of their electorate and the subtle but powerful effects that electoral regulations can produce, political parties are well-positioned to strategically shape institutional rules during electoral reforms to their advantage, potentially at the expense of rival parties. The more powerful the party and the stronger its parliamentary alliances, the greater its capacity to influence policy-making. Considering the multiple ways in which parties can influence the reform process, it is essential to closely examine the process itself to understand the resulting reform design.

In this context, this section examines reform elements that emerged from political negotiations, highlighting instances where the reform was diluted, leading to distorted designs. Broad cross-party support for a reform can often indicate such distortions, while reforms achieved without the support of advantaged parties are presumably less prone to distortion. Within the framework of biproportional reforms, institutionally induced distortions of PR can empirically occur through various mechanisms, including:

- statutory quorums
- subdivision of the electoral area instead of a single constituency for the super-apportionment
- retention of elements of the electoral system that distort PR (i.e. mixed electoral system)

These institutional features distort PR by interfering with the optimal conversion of votes into seats.[71] Since the resulting disproportionalities typically favor larger, more established parties at the expense of smaller ones, it is plausible that such parties are motivated to incorporate elements into the reform design that inherently distort proportionality (second hypothesis).

Six of the nine reformed cantons have implemented statutory quorums (ZH, AG, ZG, SZ, VS, GR), with most introducing these quorums alongside the biproportional reform. In the remaining three cantons, no quorums have been introduced to date (SH, NW, UR). These quorums can be categorized into three groups:

1. Exclusively global quorums (SZ, GR),
2. Exclusively local quorums (VS),
3. Hybrid approach combining global and local quorums (ZH, AG, ZG).

The electoral impact of quorums depends not only on their size but also on post-election factors, such as the geographic distribution of party electorates. Global quorums disadvantage parties that are locally strong but weak overall, while local quorums have the opposite effect. The hybrid approach, which employs an either/or formulation, helps mitigate the biases associated with the interplay between global and local quorums and the geographic distribution of votes. However, determining the extent of distortion caused by statutory quorums is challenging. The resulting vote-seat disproportionalities are influenced by several factors, including the quorum magnitude, vote shares, geographic distribution of party votes (concentration vs. dispersion), and the size of the legislature. For simplicity, the following analysis distinguishes between the absence of statutory quorums, quorums of 5% or less, and quorums exceeding 5%.

An additional legal deviation from "pure" biproportional apportionment is the subdivision of the electoral area into electoral regions for the super-apportionment. Instead of treating the entire canton as a single constituency, the electoral area is divided into regions where the super-apportionment is conducted independently. Since the implicit quorum is inversely proportional to the total number of seats, any artificial division of the electoral area represents a deviation from optimal proportionality. Empirically, Valais is the only reformed canton that divides the electoral area into six independent electoral regions. In all other cantons, the entire canton serves as the relevant level for seat allocation to parties.[72]

The retention of majority or plurality voting elements, representing only a partial implementation of biproportional apportionment, is another method through which reform designs can be distorted. Uri is the sole canton with a mixed electoral system that combines biproportional and majority/plurality voting elements. As part of its reform, plurality/majority voting was even extended to four additional municipalities that had previously used PR.

The "winner-take-one modification", also known as the majority clause, guarantees the strongest party in a district at least one seat (Pukelsheim 2017: 265). This can be seen as a majoritarian tool that distorts optimal vote-seat proportionality. However, in all instances where it is applied (SH, ZG, GR), the clause is legally designed to prioritize optimal vote-seat proportionality over any seat claims arising from the majority clause, thereby overriding such claims in case of conflict. Consequently, empirical evidence indicates that

[71] The "optimal" conversion of votes into seats implies that the overall proportion of seats corresponds perfectly to the overall proportion of votes, apart from mathematically unavoidable inaccuracies.
[72] One exception is the Canton of Uri, where all constituencies applying PR form the upper-tier level of seat allocation.

optimal vote-seat proportionality remains unaffected by majority clauses, and they are therefore not considered a significant source of distortion in PR.

Table 34 provides a detailed assessment of the distortive elements present in the electoral systems of all successfully reformed cantons, evaluating the overall degree of distortion in their reform designs. Two cantons (SH, NW) exhibit no distortion in their reform designs. Notably, these two have the smallest cantonal legislatures among the reformed cantons, each comprising 60 seats. The limited size of their legislatures rendered additional statutory quorums unnecessary.

Most reformed cantons (ZH, AG, ZG, SZ, GR) incorporate a statutory quorum of 5% or lower, which results in a slight degree of distortion. Uri's reform design is moderately distorted due to the retention and expansion of majority/plurality elements within its electoral system. Valais, in contrast, displays the highest level of distortion among the reformed cantons. This is due to its implementation of a local statutory quorum of 8% and the subdivision of its electoral area into six separate regions, with each region independently allocating seats regardless of the results in other regions.

Canton	Policy Element				Distortedness	
	Statutory Quorum		Subdivision of Electoral Area	MR elements	Total Score	Level
	≤ 5%	> 5%				
(Score/s)	(1)	(2)	(2)	(2)		
ZH	x				1	low
SH					0	absent
AG	x				1	low
NW					0	absent
ZG	x				1	low
SZ	x				1	low
VS		x	x		4	high
UR				x	2	moderate
GR	x				1	low

Table 34: Assessment of the distortedness of biproportional reform designs. Levels of distortedness: Total score of 0 = absent; 1 = low; 2-3 = moderate; 4-6 = high.

The next step is to assess whether the extent of distortion in biproportional reforms can be linked to the conflict structures within the decision-making process, as proposed in Hypothesis 2. Specifically, the hypothesis suggests that more conflictual reform processes are likely to produce less distorted reform policies, while consensual processes may lead to greater distortions.

The data shows that most reform designs exhibit minimal distortion, achieving near-optimal proportionality. This outcome is attributable to the FSC's strong disciplinary role, which includes intervening in some cases already during policy formulation (e.g., in the Canton of Zug) to uphold proportionality standards. While the Court has allowed moderate quorums, substantial deviations from proportionality are permitted only in exceptional circumstances, and its stance has become stricter over time. In Uri and Valais, where distortions are more pronounced, the Court justified these deviations by the unique territorial characteristics of these cantons. Therefore, the legal framework largely accounts for the minor degrees of distortion observed.

However, legal constraints are not the only explanatory factor. It is hypothesized that the dynamics of conflict during the reform process also influence the degree of distortion. Conflict-ridden processes, where reforms face significant opposition from dominant political forces, are expected to result in less distorted reforms. Conversely, consensual processes are more likely to produce reforms with distorting elements.

Referring to Table 35, six out of nine reform processes were marked by significant conflict and were adopted despite political resistance. The Canton of Schwyz implemented reform through a popular initiative, showcasing a direct democratic approach. In Zurich and Uri, reforms were achieved through consensus, with all parties supporting the reform bills. Valais presents a complex case. Despite political resistance, the opposition was mainly directed at comprehensive territorial changes rather than the biproportional reform itself. This resistance often crossed party lines and was not spearheaded by dominant political forces. Instead, the reform was supported by the main political forces, including the CVP/CSP and the majority of the SVP. Given that the need for reform was widely recognized and the dominant forces were proponents, Valais is categorized as having a "consensual" reform path.

Canton	Conflict-ridden road to reform?	Distortedness	Confirmation of H2?
ZH	no	low	no
SH	yes	absent	yes
AG	yes	low	yes
NW	yes	absent	yes
ZG	yes	low	yes
SZ	yes	low	yes
VS	no	high	yes
UR	no	moderate	yes
GR	yes	low	yes

Table 35: The link between conflict structures during reform processes and the distortedness of reform designs.

The relationship between conflict structures during the reform process and the distortedness of the reform design reveals that more contentious reform paths tend to result in less distorted designs. Table 35 shows that in cantons where the reform faced significant resistance from dominant political forces, the resulting designs are generally less distorted compared to those in cantons where the reform was supported by key political actors.

An exception is the Canton of Zurich, where cross-party consensus led to only marginally distorted reforms. This is partly because Zurich was the first canton to implement biproportional apportionment, and critiques of an electoral reform subsided following a leading FSC decision. Additionally, besides statutory quorums, no other policies restricting biproportional representation were ever considered in Zurich.

Empirically, reforms backed by dominant parties often resulted in more distorted designs, featuring high statutory quorums, artificial subdivisions of the electoral area, and/or the retention of majoritarian elements. Conversely, reforms imposed against the will of dominant parties generally exhibited lower levels of distortion, challenging the established advantages of larger parties. This pattern suggests that, in addition to the FSC's influence, the conflict structures in political decision-making processes significantly impacted the extent of distortion in reform designs.

Larger parties, even when consenting to reform, often used their political influence to introduce elements that reduced their potential losses, effectively "watering down" the reforms. In contrast, smaller parties, which were negatively affected by systemic distortions, were more motivated to advocate for reforms that would enhance proportionality.

2.9 Conclusion

Drawing on a comprehensive documentation of biproportional reform trajectories in the Swiss cantons, this chapter, structured around two central hypotheses, shows how parties operate within a highly complex and sensitive policy field, where they simultaneously act as both lawmakers and principal recipients of the outcomes of these laws.

The analysis confirms that parties that have traditionally benefited from existing electoral laws exhibit a tendency to oppose or dilute reform drafts, seeking to preserve their advantageous position. This behavior is evident in various cantons where established parties have assisted in implementing statutory quorums, or other immanently distortive elements, such as a subdivision of the electoral area, or the retention of majoritarian elements. These maneuvers align with H1, showing that parties have a vested interest in maintaining their electoral advantages and will resist reforms that threaten their status. On the other hand, parties disadvantaged by these laws generally push for more comprehensive reforms to create a more equitable system. This strategic calculus underscores the balance of power within electoral politics, where parties strategically navigate reform processes to either maximize their relative electoral advantage or to minimize potentially adverse effects (Kedar et al. 2016). However, deviations from this pattern also emerge. In some cantons, certain parties, notably the Social Democrats, unexpectedly supported reforms despite potential bonus losses. This finding challenges a notion that views electoral systems primarily as tools of partisan politics, where "the election of a particular voting system is dictated more by partisan interests and political opportunism than by the principles governing electoral equity" (Cincea 2013: 182). Instead, the findings suggest that beyond maximizing strategies, legal requirements, political idealism, and democratic principles also impact party positions (cf. Blais & Massicotte 1997).

H2 posits that reforms imposed against the interests of advantaged parties tend to result in less distorted reform designs compared to those achieved through cross-party consensus. The results support this hypothesis by demonstrating that contentious reform processes, marked by significant parliamentary opposition or direct democratic initiatives, often lead to less distorted reform outcomes. In contrast, reform processes marked by cross-party consensus are more prone to including distortive elements such as higher statutory quorums, an artificial subdivision of the electoral area or the retention of majoritarian elements. The main explanation for this phenomenon is that cross-party consensus often comes at a political cost (cf. Walter & Emmenegger 2019: 71). In the context of electoral reforms, this means that larger, traditionally advantaged parties are more likely to support a reform if it includes elements that secure and stabilize their power position. In contrast, dissent signals a contentious reform process, where smaller fringe parties may successfully unite to advance a reform that aligns with their interests through parliament. In other words, the support of major parties carries a political price, and depending on the political configuration, it may be necessary to "purchase" their agreement to the overall reform at a significant political cost.

In conclusion, the study affirms that advantaged parties typically resist reforms to protect their positions, while reforms imposed against these parties tend to result in fairer electoral systems. These findings highlight the intricate nature of electoral reforms, where strategic interests intersect with institutional constraints

and legal mandates. Further research is needed to disentangle the complex dynamics at play and refine the understanding of electoral reform processes and their underlying drivers. Comparative studies across diverse political contexts can help increase the generalizability of the findings. Moreover, investigating public opinion formation in the realm of democratic norms, the interplay between these norms and jurisdiction, and their implications in electoral reform processes also provide potential avenues for future research.

Part III

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3. Reforms and Their Effects:
A Comparative View on Different Reform Outcomes

3.1 Introduction

While the second part of the thesis focused on the dynamics of cantonal reform processes towards biproportionality and their effects on reform designs, this part is dedicated to the consequences of the reforms on the structures of cantonal political systems. Here, the electoral system is treated as an explanatory variable, while various macro-level variables, such as characteristics of the party system, are regarded as dependent variables. The research tradition devoted to this "effect-side" of electoral institutions traces its origins to Duverger's seminal work "Les Partis Politiques" (1951) and has since become the dominant approach in the study of electoral systems (Charvát 2023: 15-18). Against this background, the primary objective of this third part is to assess the extent and magnitude of effects that can be causally attributed to biproportional reforms.

Despite extensive research on the mathematical properties of the biproportional divisor method (cf. Bacharach 1970, Balinski & Demange 1989a, Balinski & Demange 1989b, Gassner 1991, Balinski & Young 2001, Pukelsheim 2004, Maier et al. 2010, Demange 2011, Pukelsheim 2017, Pukelsheim 2019), as well as a series of simulations (cf. Balinski 2002, Bochsler 2005, Pennisi et al. 2006, Zachariassen & Zachariasen 2006, Ramírez et al. 2008, Oelbermann & Pukelsheim 2011, Hartmann 2023), there is yet no research that looks at the effects of biproportional apportionment beyond the distributional dimension. Consequently, the potential polity-level ramifications of the new method in the nine Swiss cantons where it currently applies are still largely obscure. This represents a surprising oversight given the practical relevance and the public interest in this new seat allocation procedure. Therefore, and in view of growing advocacy for the adoption of biproportionality for Swiss National Council elections, this section aims to offer initial insights into the impact of this new apportionment method on various characteristics of the cantonal democracies concerned.[73] The corresponding research question is: To what extent have biproportional reforms affected changes in cantonal political landscapes?

Building on existing research on party system evolution, I argue that the introduction of biproportional apportionment has increased party system fragmentation, manifesting in a larger number of effective legislative parties. Furthermore, I expect voter turnout and electoral competition to intensify, which is reflected in higher and an increased number of electoral lists and candidacies. However, concerning the expected impact on the number of lists, an alternative mechanism is also conceivable. Strategic considerations may drive parties to shift away from district-specific voter targeting strategies and opt to compete in all districts instead. This shift could potentially lead to a reduction in the overall number of submitted lists and ultimately contribute to a "de-regionalization" or "cantonalization" of the party system.

The chapter is structured as follows: It begins with an introduction to the core concept of proportionality in electoral systems, setting the stage for understanding how biproportional reforms impact vote-seat disproportionality. The chapter then moves to examine the varying levels of proportionality resulting from these reforms and their subsequent political effects, including party system fragmentation, electoral turnout, and electoral competition. Following a discussion of data sources, variables, and analytical strategies, the chapter

[73] Beyond academic discourse, calls for electoral reform of elections to the National Council are a recurring phenomenon in the aftermath of national elections. These calls are typically voiced by parties that stand to benefit from an electoral reform. Local and national media outlets frequently cover the topic, publishing articles on the alleged necessity of reform and exploring the potential distributional effects of various reform proposals (e.g., Tagesanzeiger 2023 or Neue Zürcher Zeitung 2024). The successful implementation of biproportionality at the cantonal level often serves as justification for advocating its introduction at the national level.

presents results on how biproportional reforms influence these outcomes. It concludes by summarizing key findings and reflecting on the broader implications of biproportional reforms for political systems.

3.2 Proportionality as the fundamental concept in electoral system analysis

Vote-seat proportionality and the number of parties are among the most frequently employed metrics for evaluating electoral systems and democratic competition (Shugart & Taagepera 2017). Despite being widely established in the field of electoral research, vote-seat proportionality is not as straightforward a concept as it might initially appear. As a metric based on the actual voting results, vote-seat proportionality is not only tied to the institutional rules in place but also depends on a set of other factors such as the actual distribution of votes between contesting parties (Cincea 2013: 179). It thus combines both endogenous (pre-election) and exogenous (post-election) information about an electoral system. The proportional properties of an electoral system have been identified as relevant variables in comparative studies across political science disciplines, including political economy, civil conflict literature, and comparative politics. For example, they have been found to affect the stability of government coalitions, the responsiveness of politics to shifts in public opinion, and political behavior (Bochsler 2024: 1). Proportionality thus captures more than certain effects of an electoral system. It serves as an important link in a longer causal chain, directly or indirectly affecting a wide array of electoral outcomes, thus assuming the role of an intermediate variable.

The goal of the subsequent subchapter is to show how proportionality intersects with electoral systems. I start with a working definition of "proportionality" and then delve into its connection to electoral rules. I then introduce the key indices and measurements devised to capture the distortive nature of an electoral system. These indices fall into two categories: those that gauge the proportionality of an electoral system irrespective of election results and strategic (re)actions of voters and parties, and empirical metrics that measure disproportionalities based on the actual election results.

3.2.1 Proportional representation and its determinants

The concept of proportionality is rooted in a normative notion of political representation according to which voters should be represented in rough proportion to their share of votes, as succinctly articulated by John Stuart Mill (1861: 85): "In a really equal democracy, every or any section would be represented, not disproportionately, but proportionately." The normative connotation of the concept also surfaces in the work of Arend Lijphart (1984: 140), who viewed proportionality as "virtually synonymous with electoral justice". Hence, the more precisely an electoral system maps the party preferences of the electorate in terms of seat shares, the more proportional and, consequently, the fairer it is. However, due to the "indivisible nature of seats" (Charvát 2023: 44), achieving perfect proportional representation, where seat shares precisely match vote shares, remains an empirically unattainable ideal. Therefore, proportionality can only be achieved in an optimal sense by approximating the "perfect" or "ideal" seat allocation.

Yet, the inevitability of leaving certain votes unrepresented due to mathematical constraints is only one aspect of how electoral systems inherently conflict with "perfect" proportional representation. Other, more decisive parameters of electoral systems found to affect proportionality include the district magnitude, defined as "the number of representatives elected in a district" (Lijphart 1994: 10), the electoral system (majority, plurality, or proportional representation), the mathematical formula used for vote-seat conversion, the assembly size, and legal thresholds (Charvát 2023: 46-50). In the following review, I will focus on these explanatory factors identified as central in the relevant literature. The review extends to "apparentements", a feature exclusively relevant in PR systems, that enables parties to "formally (…) link or connect their lists"

(Lijphart, 1994: 15).[74] This often "neglected feature of electoral systems" (Bochsler 2010b: 617) will be included in the discussion, given its relevance in the context of this thesis. While there are indeed numerous other factors at play, the goal here is not to provide an exhaustive examination of all possible determinants (an impossible undertaking anyway). Rather, the aim is to point out the inherent link between seat-vote proportionality and various characteristics of electoral systems. As previous research has demonstrated, the "mapping accuracy" of an electoral system holds significant implications for various political outcomes.

Among all parameters, the district magnitude has been identified as "the decisive factor" in affecting electoral proportionality (Taagepera & Shugart 1989: 112, Grofman & Lijphart 2003: 53). James Hogan (1945: 13), one of the pioneering scholars in the field and among the first to recognize the significant role of district magnitude in shaping electoral system effects, noted that "the decisive point in P.R. is the size of the constituencies: the larger the constituency, that is, the greater the number of members which it elects, the more closely will the result approximate to proportionality". Subsequent empirical studies have supported his finding, demonstrating that the proportionality of seat allocation correlates significantly and positively with larger district magnitudes. However, research suggests that this relationship is not linear but rather curvilinear or degressively proportional, meaning that the proportionality of seat allocation increases at a decreasing rate with larger district magnitudes (Rae 1971, Gallagher 1991, Lijphart 1994, Powell 2000, Gallagher & Mitchell 2005, Flick-Witzig & Vatter 2017). Within typical multiparty systems, district magnitudes of five or six have been identified as a lower boundary "to provide relatively proportional outcomes" (Taagepera & Shugart 1989: 114). Notably, various studies have pointed to the inconsistency of the effect across different electoral system types. In fact, the effect may reverse under plurality/majority rules, where proportionality tends to decrease with increasing district magnitudes (Blais & Carty 1987, Taagepera & Shugart 1989, Lijphart 1999, Benoit 2001, Farrell 2001). This indicates a more complex and less straightforward relationship between district magnitudes and vote-seat proportionality when electoral rules are factored into the equation.

Introducing an upper tier of seat allocation within a broader region or across the entire electoral territory can further enhance the overall vote-seat proportionality as corresponding district magnitudes and, consequentially, the mapping accuracy increases. With higher district magnitudes, the implicit threshold as an "entry barrier" for parties is lower and votes can be converted into seats more accurately since the vote quota required to win a seat decreases (Farrell 2001, Blais & Massicotte 2002, Charvát 2023). Additionally, higher district magnitudes can also reduce the number of "wasted votes" for parties whose share of votes falls below the implicit threshold needed to gain parliamentary representation, thereby fostering proportionality (Duverger 1964: 216-228, Gallego et al. 2012: 165). The same logic extends to the proportionality effect of assembly sizes: larger legislative bodies with more available seats tend to exhibit higher levels of proportionality (Lijphart 1999, Farrell 2001, Gallagher & Mitchell: 611). However, this effect is tempered by the number of electoral districts, as a greater number of electoral districts implies smaller districts and thus negatively affects the relationship between assembly size and proportionality (in PR systems).

Previous studies have consistently found that vote-seat proportionality is significantly higher under PR electoral rules compared to non-PR rules (Mackie & Rose 1991, Lijphart 1994, Gallagher et al. 1995, Norris 1997). In his seminal work, Rae (1971: 124) concluded that alongside district magnitude, electoral formulae

[74] According to Nohlen (2014: 113), "apparentement" is a term borrowed from French as there is no equivalent English term. However, the alternative spelling "apparentment" can also be found in the relevant literature (e.g. Bochsler 2010b or Flick-Witzig & Vatter 2017). In acknowledgment of its French origin, I will adhere to the former spelling.

are "among the most important aspects of electoral laws" influencing the proportionality of seat allocation.[75] Even if this nexus appears rather trivial since PR systems are quite literally devised to represent voters proportionally, there can be considerable differences when comparing the different apportionment formulae used in PR elections. Within the PR family, three different formulae are typically distinguished: the highest averages or divisor methods (like d'Hondt or Sainte-Laguë), the largest remainders or quota methods (like Hare, Droop, or the two Imperiali quotas), and the single transferable vote (STV) (Lijphart 1994: 23, Riera 2013: 54-55). In his empirical analysis, Lijphart (1994: 159) finds that "LR-Hare and pure Sainte-Laguë (...) are the most proportional formulas; the d'Hondt and the two LR-Imperiali formulas are the least proportional; and LR-Droop, STV (...), and modified Sainte-Laguë are in an intermediate category."[76]

Other important devices of PR-based electoral systems that directly affect the vote-seat proportionality are legal thresholds. Legal thresholds can broadly be defined as a minimum level of electoral support a party needs to gain parliamentary representation, or inversely, as a "cut-off point below which parties contesting the elections are not awarded any seats" (Charvát 2023: 49).[77] Such an additional criterion can either apply at the national, regional, or at the district level. Legal thresholds can be specified in terms of absolute votes, as a percentage of votes or as some other criterion (Lijphart 1994: 11-12). It is important to distinguish between these formal thresholds and electoral thresholds implied by the territorial structure, i.e. the district magnitude, as they former are explicitly stipulated in the electoral law (unlike implicit thresholds) and exhibit slightly different effect mechanisms.[78] Given the presence of implicit thresholds, explicit formal thresholds are only effective if they significantly exceed the former. Although Lijphart (1994: 12) initially found that most electoral systems lack such formal thresholds, subsequent research reached a different conclusion, suggesting that they are relatively common (Charvát 2023: 49). Gallagher and Mitchell (2005: 13) asserted that "[v]irtually every PR system employs some kind of threshold that prevents the smallest parties getting the 'fair' share of the seats." A conclusion supported by Riera (2013: 58), who noted that "[a]lmost every PR system employs some kind of threshold."

The possibility to conduct inter-party a/o intra-party apparentements is another aspect pertinent to the overall proportionality and closely intertwined with the issue of electoral thresholds. They are a peculiarity of PR systems and enable parties to "pool their votes for seat allocation purposes" (Gallagher & Mitchell 2005: 631). Apparentements can involve two or sometimes more electoral lists and serve as a potent tool, especially for smaller parties whose voter share often hovers around or falls below the "threshold of representation". Empirical studies indicate that apparentements are used by parties of varying sizes and types (see e.g. Pukelsheim & Leutgäb 2013). While their specific effects are among the most cryptic aspects of electoral

[75] By "electoral formula", Rae means the main rationale of vote-seat conversion and distinguishes between majority, plurality, and proportional representation types of electoral formulas. However, other authors refer to "electoral formulae" as "the mathematical mechanisms governing the transformation of votes into seats" (Benoit 2000: 381). Given this definitional ambiguity, I adhere to the distinction proposed by Charvát (2023: 48), who defines "electoral formula" as the "way (default logic) in which votes are to be counted in order to allocate seats, and the mathematical formula as the method by which votes are translated into seats within electoral districts under PR.

[76] For a comprehensive discussion on the ranking of the different formulas in terms of their estimated effect on vote-seat proportionality, see Gallagher 1991: 33-51, Lijphart 2003: 170-179, or Gallagher & Mitchell 2005: 584-589.

[77] Relevant literature refers to legal thresholds also as "formal" or "statutory" thresholds (cf. Charvát 2023: 49 or Gallagher & Mitchell 2005: 70).

[78] The effects of formal thresholds and implicit thresholds differ because the impact of formal thresholds is transparent and predictable, whereas the impact of implicit thresholds imposed by the district magnitude depends to a certain extent on the mathematical formula used for vote-seat conversion (cf. Gallagher & Mitchell 2005: 607-608).

systems, the mechanism behind them is rather straightforward: parties involved in the apparentement are considered as a single list in the super-apportionment of the seats to parties. Then, in the subsequent step, these seats are sub-apportioned to the parties involved in the apparentement based on their respective share of votes. While there remains much ambiguity, partly due to limited research on the topic and the dependency of apparentements' effects on specific electoral contexts, the few existing studies offer some valuable insights (i.e., Lijphart 1994, Bochsler 2010b, Pukelsheim & Leutgäb 2013).

Firstly, parties involved in an apparentement cannot fare worse than if they competed independently with a single list.[79] This is because participating in an electoral alliance allows for the aggregation of "remainder votes", which would otherwise be ineffective, potentially leading to additional seat gains. Or, as Pukelsheim and Leutgäb (2013: 128) put it: "Lists who prefer to maintain their independence and do not join an apparentement must, on average, endure a seat deficit so that their competitors may be served with a seat surplus." However, it would be erroneous to conclude that the participation in an apparentement guarantees seat gains. The potential benefits depend on various factors, including the apportionment formula, the actual vote shares of the competing lists, the respective position within the apparentement in terms of votes, the presence of other apparentements between different parties, and numerous other aspects.

Secondly, apparentements prove to be particularly effective when used within highly fractionalized political blocs comprising numerous small parties (Bochsler 2010b: 625). However, as Bochsler (2010b) highlights, this does not necessarily mean that fractionalized parties reap the primary benefits. In his empirical analysis of Swiss parliamentary elections spanning from 1995 to 2007, the findings suggest quite the opposite. He argues that "[e]ven if alliances of small parties are among the main beneficiaries of apparentments, the gains of these apparentments are more likely to be accorded to large parties within these apparentments" (Bochsler 2010b: 625). However, it is important to note that in Swiss National Council elections, the D'Hondt method is employed. This method is known to have mathematical properties that disadvantage small parties (e.g. Benoit 2000). Since the D'Hondt method is also used for the sub-apportionment of seats within the apparentements to individual list partners, the adverse effect on the "junior partner" is exacerbated. When apparentement partners are unequally sized in terms of vote shares, Pukelsheim and Leutgäb (2013: 128) denote the resulting advantage for the larger partner within the apparentement thus as a "double bonus".

Lastly and most importantly, while the initial belief was that apparentements had a consistently positive effect on overall vote-seat proportionality (Lijphart 1994, Nohlen 2014), more recent studies paint a more nuanced picture. Although Pukelsheim and Leutgäb (2013) find that in many cases, apparentements indeed lead to a reduction in the overall bias in most of the cases, they also found the bias to increase "in more than one third of all cases" (2013: 130) and conclude that "[i]t is a widespread belief that apparentements *always* dampen the total bias. This belief is erroneous (...)" (ibid: 2013: 130). Similarly, Bochsler (2010b: 625) finds that "apparentments, especially when they unite parties of different sizes, might rather increase than decrease the seat bias under D'Hondt". In summary, apparentements have the potential to enhance proportionality by providing smaller parties with a chance to have their votes represented in parliament. However, under certain conditions, they can also have the opposite effect, benefiting larger parties and contributing to a decrease in the overall proportionality.

[79] At least not under the assumption that a list does not incur any "electoral damage" from forming an alliance with a partner considered unsuitable by its own electoral base.

To summarize, the (in)capacity of an electoral system to allocate the seats proportionately to the parties' vote shares is conditioned by various institutional features. These technical aspects of electoral systems collectively set the limits of how (in)accurately votes are translated to seats. The less restrictive specific regulations are, i.e. no legal hurdles, low implicit thresholds, and a politically unbiased apportionment formula, the higher the inherent propensity for a proportional translation from votes into seats. Apparentements can also contribute to an improvement of vote-seat proportionality. However, this effect is contingent upon other aspects of the electoral system as well as the specific electoral circumstances.

The resulting effects on proportionality can be straightforward and direct, but they can also be a product of more complex, cumulatively reinforcing, weakening, or neutralizing interactions. In small electoral districts, e.g., the choice of mathematical formula can significantly impact the proportionality of seat allocation, whereas in larger electoral districts, the influence of different formulas may be relatively small (Charvát 2023: 48). As Gallagher (1991: 50) put it: "When district magnitude is large, either because constituencies are large or because of the use of higher tier seat allocation, the proportionality of election outcomes is high, regardless of the formula or the index of measurement employed." Or, to provide another example: In a multi-district electoral context without a superordinate allocation tier, the factors that hinder proportional vote-seat transmission at the district level can exacerbate across the entire electoral area, culminating in a substantial impairment of the overall vote-seat proportionality.

Deviations from perfect vote-seat proportionality, known as "disproportionalities" or "distortions", indicate systematic biases in an electoral system that favor certain parties (Grofman 1983, Gallagher 1991, Lijphart 1995b, Calvo 2009, Nohlen 2014, Cox 2006). Lijphart (1994: 58) described proportionality and disproportionality as "two sides of exactly the same coin". Evaluating electoral systems often involves examining their inherent tendencies to obstruct optimal proportionality. Maurice Duverger (1954) was among the first to formalize the disproportional effects of electoral systems, conceptualizing these effects as "mechanical effects" that distort vote-to-seat conversion due to electoral rules in combination with election outcomes (Benoit 2006, Riera 2013, Flick-Witzig & Vatter 2017). The next section discusses the key indices and concepts developed to quantify these distortions and the extent to which electoral systems skew proportionality.

3.2.2 Measurements of mechanically induced disproportionalities

A variety of measures have been developed to assess the extent to which electoral rules interfere with the proportional conversion of votes into seats. The quest to measure the degree of distortion caused by electoral rules has diverged into two distinct methodical approaches. One set of measures, categorizable as "a priori" indices, is theory-based and attempts to predict the extent to which electoral rules inherently affect proportionality, regardless of actual election outcomes. In contrast, "a posteriori" measurements are empirically grounded and assess the actual disparities between votes and seats, thus also conveying information about the actual election results (Bochsler 2024: 2).

3.2.2.1 "A priori" measurements

The a priori measurements that I will briefly outline have the common goal of predicting how electoral rules inherently influence an electoral system's (dis)proportionality. They do so by approximating a theoretical value referred to as the "effective threshold", which delineates the percentage of votes a party needs to secure

a seat.[80] The corresponding formula is based on a single predictor: the district magnitude. This should not come as a surprise in light of the vast body of research showing that district magnitude is the primary institutional feature influencing an electoral system's (dis)proportionality (cf. Taagepera and Shugart 1989, Lijphart 1994, Boix 1999, Lublin 2015, Flick-Witzig and Vatter 2017). As Lijphart (1994: 12) highlighted, "low magnitudes have the same effect as high thresholds: both limit proportionality and the opportunities of small parties to win seats; as magnitudes increase and thresholds decrease, proportionality and the chances for small parties improve." The concept of the effective threshold thus helps us to understand and quantify the "degree of squeeze [electoral systems, S. L.] put on representation of small parties" (Taagepera 1998: 393).

However, the effective threshold implied by district magnitude does not reflect a fixed, objectively assessable percentage. Rather, the estimate encapsulates a spectrum of potential values between the "threshold of representation" and the "threshold of exclusion", which set the lower and upper theoretical boundaries (Lijphart 1994: 25, see also Gallagher & Mitchell: 607-611). While the threshold of representation (or inclusion) indicates the vote share enabling a party to win a seat under the most favorable conditions, the threshold of exclusion reflects the scenario where a party garners a large vote share and yet fails to win a seat due to extremely unfavorable circumstances (Rokkan 1968, Rae et al. 1971, Laakso 1979, Gallagher 1992, Lijphart 1994, Taagepera 1998). An exception arises when a formal national threshold applies. In such cases, the effective threshold equals the formal threshold, resulting in a precise value rather than a rough estimate (Lijphart 1994: 29).

Figuratively speaking, if a party's vote share falls below the "lower" boundary, it cannot win a seat. If it exceeds this threshold, winning a seat becomes possible (if other parameters align). And if it surpasses the "upper" boundary, it will certainly secure a seat. The effective threshold lies between these boundaries and therefore provides a more practical univariate estimation of the exclusionary or reductive nature of electoral systems (Taagepera 1998: 395). The relationship is depicted in Figure 37 on the next page.

Lijphart (1994) went a step further and looked at the correlation between the effective threshold and vote-seat disproportionality. He regressed the percentage of disproportionality (using the least-squares index) on the effective threshold for two different groups: a pool of PR electoral systems with formal thresholds (1) and a group without formal thresholds where the effective thresholds were inferred directly or partly from the district magnitudes (2). He repeated the regression for the second group, adding the mathematical formula and the (logged) assembly size to the equation (3). He found slopes of 0.42 for equation (1), 0.40 for equation (2) and 0.42 for equation (3). The results indicate that for every percentage increase in the effective electoral threshold, disproportionality increases on average by 0.42 (Lijphart 1994: 27-28).[81]

Building on the seminal contributions of Rokkan (1968), Rae et al. (1971), Taagepera & Shugart (1989), and Lijphart (1994), recent research has made significant strides in accommodating the complexities of contemporary electoral systems. For the a priori estimation of the institutional "permissiveness" or "openness" of electoral systems for (small) parties, more recent studies include factors such as a multi-tier structures of the

[80] An alternative and essentially equivalent concept is the "effective magnitude". According to Taagepera and Shugart (1998: 398) the effective threshold and the effective magnitude "are the reciprocals of one another". However, in my brief outline, I focus solely on Lijphart's perspective, which puts emphasis on the effective threshold and, in my view, offers a more intuitive approach.

[81] Unfortunately, he did not report any numbers on the robustness of his findings.

electoral system, formal thresholds, and bonus seats (Neto & Cox 1997, Clark & Golder 2006, Lublin 2015, Renwick & Pilet 2016, Shugart & Taagepera 2017, Bochsler 2024).

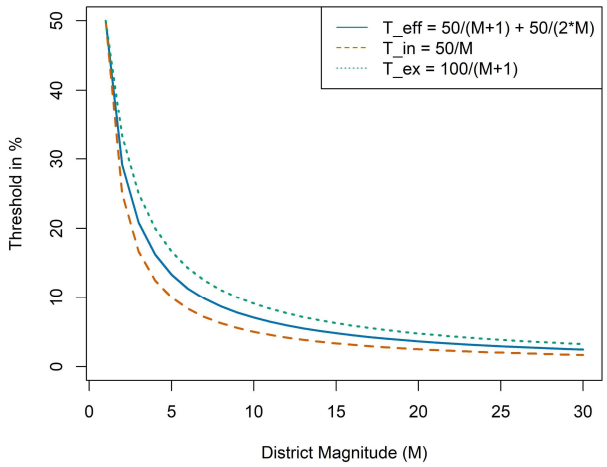


Figure 37: Relationship between district magnitudes and the effective threshold according to Lijphart (1994: 25-30). T_eff = effective threshold; T_in = threshold of inclusion; T_ex = threshold of exclusion.

3.2.2.2 "A posteriori" measurements

While a priori measurements exclusively take institutional parameters into account and are only indirectly linked to disproportionality, a posteriori measurements also take actual electoral results into consideration and are therefore designed to measure the actual disproportionality of an electoral system. The higher the summed deviations between votes and seats for all parties in a particular election, the greater the value of disproportionality, indicating a more substantial departure from "ideal proportionality". While disproportionality as a measurement of partisan misrepresentation in terms of votes and seats appears to be a rather simple and straightforward concept, the methodology for its measurement is more complex (Lijphart 1994: 57-59). The different approaches led to the development of various indices. The one devised by Gallagher (1991) emerged as the most widely applied measure in relevant literature. The index squares the vote-seat share differences for each party, totals these differences, divides the total by two and takes the square root of the resulting value (Lijphart 1994: 60-61). In his evaluation of four measures of disproportionality, Lijphart (1994: 62) describes the Gallagher index as "the most sensitive and faithful reflection of the disproportionality of election results".

In their evaluation of nineteen different indices established in the field, Taagepera and Grofman (2003: 659) arrived at the same conclusion as Lijphart (1994) and stated that the Gallagher index "offers the most desirable combination of features". Gallagher's least-squares index offers the advantage of not just measuring vote-seat disparity, but also taking into account how the competing parties "contribute" to this disparity. It is designed to be sensitive to (a small number of) large seat-vote disparities while being relatively insensitive to (a large number of) small disparities. Taagepera and Grofman (2003: 668-669) argue that the problem with (over)sensitivity to smaller deviations is that this sensitivity also applies to the "lumping of residuals",

which arise randomly rather than systematically, resulting in an overestimation of the overall disproportionality (see also Lijphart 1994: 60-61 and Gallagher & Mitchell 2005: 602-606).[82]

3.3 Moving beyond (dis)proportionality: Varying levels of proportionality and their political effects

Understanding the political ramifications of electoral systems requires more than the examination of their institutional features. Duverger (1954) introduced the concept of "psychological effects" to complement the mechanical effects. These psychological effects include the responses of parties and voters to the mechanical constraints imposed by the electoral system. Voters interpret the electoral rules and adjust their behavior in anticipation of their expected effects. In a formal sense, strategically minded voters seeking to influence electoral outcomes are likely to deviate from their preference order if voting for their preferred options carries a high risk of ineffectiveness due to slim chances of success. Instead, they opt for alternative choices that are close to their ideal point but offer better electoral prospects. This strategic behavior is well-captured by the rational choice model (Downs 1957, Grofman and Lijphart 2003, Aldrich et al. 2018, Harfst et al. 2018). Similarly, parties may choose to participate in elections only if they expect intact prospects of electoral success. Otherwise, if they deem their chances futile and when the projected costs outweigh potential benefits, they may refrain from candidacy (Bochsler 2005). Duverger (1954) suggested that the interaction between the static mechanical effects and the dynamic psychological effects of electoral rules significantly influence the character of a party system. In essence, he argued that plurality/majority systems tend to favor bipartism (or "party dualism" in Duverger's own words, ibid. 1954: 247) due to both mechanical and psychological factors. In contrast, PR-based systems tend to foster the emergence of multipartism. The regularities described by Duverger have shown high empirical validity, although the precise mechanisms behind these effects remain the subject of ongoing debate.[83] Subsequent studies have built upon Duverger's rich theoretical legacy and aimed to disentangle and formalize the relationship between electoral rules and the structures of political systems.

Following this line of research, this section of the thesis delves into the potential macro-level effects induced by cantonal biproportional reforms. The central theoretical premise is that while biproportional reforms primarily alter the mechanics of the electoral system, they also engender dynamic responses from both voters and partisan elites (Sartori 1997: 32). These actors interpret the institutional changes, anticipate their potential ramifications, and adapt their behavior accordingly. The behavioral adaptations of voters and parties are expected to materialize in observable and quantifiable changes in the political landscape.

The illustrated mechanism (Figure 38) suggests that modifications to the electoral system's mechanics may alter the perceived electoral permeability. Voters are anticipated to respond to these changes by adjusting their voting behavior, while political parties may reassess their electoral strategies in light of the new system. It is expected that the combined and mutually influencing reactions of voters and partisan actors contribute

[82] For these reasons, I will also apply Gallagher's least-squares index for the measurement of vote-seat disproportionality in the further course of the thesis.

[83] The question about nature of the relationship between electoral systems and characteristics of party systems in Duverger's postulates has sparked controversy and has been the subject of numerous inquiries (e.g. Sartori 2003: 43-68). I prefer to cautiously describe it as a strong relationship, given its high empirical plausibility, without delving into the specifics of whether it adheres to a strict law or lawlike pattern (or neither).

to a change in the structure of a political system. However, the evolution of the political landscape is influenced by various factors beyond electoral rules and reactions to them, most notably by socio-structural factors.

Moreover, changes in political structures may provoke calls for further electoral reforms, suggesting a recursive relationship between electoral rules and political structures. This model highlights the dynamic nature of this relationship. It illustrates how alterations in electoral rules can trigger a cascade of responses from voters, political actors and institutions and ultimately shape the trajectory of the political system over time.

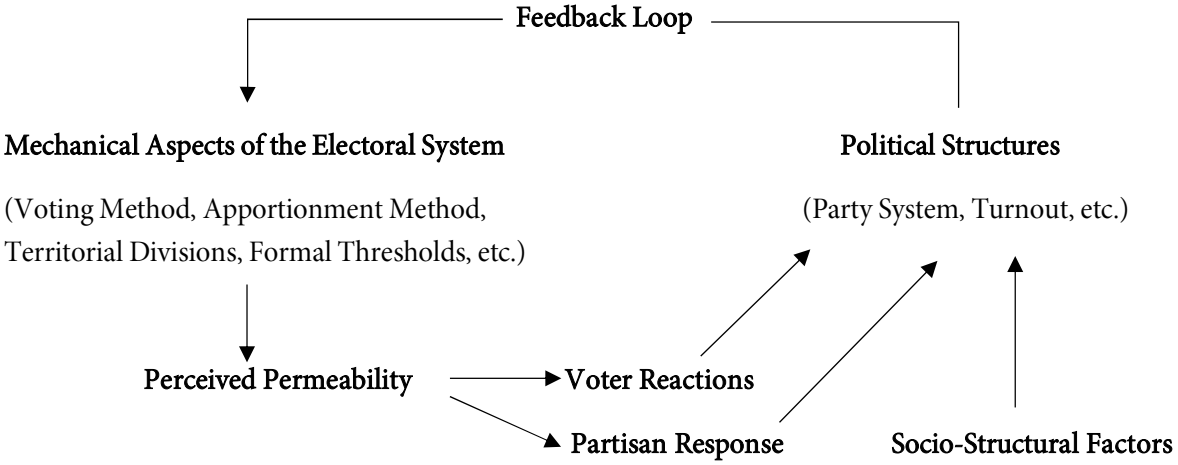


Figure 38: Theoretical framework of the interplay between electoral mechanics, psychological effects, socio-structural factors, and political structures.

In the following subchapters, I will discuss various political parameters that are presumably influenced by the transition to biproportional apportionment and derive corresponding hypotheses. These parameters include the number of parties, voter turnout rates, and the number of lists/candidacies. However, before delving into the assessment of these outcomes, as a preliminary step, it is essential to determine the extent to which the accuracy of the electoral system in translating votes to seats has been affected. This is important because electoral rules do not directly affect political outcomes. Rather, their impact hinges on the degree to which they facilitate or constrain proportional representation. Therefore, if the mapping accuracy would remain untouched by the electoral reforms, hypotheses about changes in the political landscape would be unfounded and implausible.

3.3.1 Reform effects on disproportionality

Vote-seat disproportionality is both an outcome influenced by multiple factors and a driver of various consequences. Exclusive electoral rules inevitably lead to high vote-seat disproportionality. Conversely, inclusive rules are a precondition for low disproportionality. However, low disproportionality does not necessarily imply a mechanically inclusive system. Even in highly constraining systems that impose significant electoral hurdles, disproportionality can be low if voters are aware of the systemic effects induced by the electoral rules. By strategically assessing their options, voters may preemptively discard choices that are unlikely to become effective. This behavior reflects an adaptive response to the mechanical constraints imposed by the electoral system. Similarly, parties or less institutionalized electoral groups may be dissuaded from

electoral participation by high electoral barriers and abstain from candidacy. The proactive alignment with mechanical constraints ultimately leads to a reduction in vote-seat disproportionality, despite the exclusivity of the electoral rules. Bipartism can be viewed as another epitome of the same effect.

Although biproportional reforms in Switzerland vary in their implementation pathways and designs, the applied formula is consistent over all cantons. In a first step, seats are allocated between the districts proportionally to population figures and between parties proportionally to their cross-district, canton-wide vote count. In the following step, seats are distributed simultaneously by district and party (Pukelsheim 2017: 259). The first step is pivotal, particularly in its impact on disproportionality. By super-apportioning seats across districts in the first step, parties secure a seat share reflective of their canton-wide vote share. This approach effectively counters the constraining effect of electoral districts, creating an artificial, canton-wide unitary district. In cases where the formula applies to the entire canton, the magnitude of this artificial district matches the size of the cantonal assembly. In instances where the formula is limited to a subset of electoral districts, the size of this new fictional district corresponds to the sum of seats allocated to these districts (as is the case in the Canton of Uri).

Expanding the electoral area mitigates the mechanical effect arising from the division of the electoral area into districts. If implemented purely, i.e. without additional provisions that adversely affect vote-seat conversion, disproportionality should take a value close to ideal proportionality. In addition, the possibility to create apparentements was not provided in any of the reform designs. Neither "horizontal" apparentements between parties or list groups, nor "vertical", within-list apparentements between different subgroups of parties ("sub-apparentements"). As such alliances offer smaller parties, particularly those with a vote share below or around the threshold of representation the opportunity to circumvent the adversity of mechanical effects by collaborating with other parties, there was long the erroneous belief that the possibility to form such alliances immanently reduces overall disproportionality. The main argument is that the participants of such alliances, or at least the junior partners, would not manage to enter parliament on their own, i.e., as a single list, without this electoral tool, contributing to an increase in disproportionality. However, empirical research casts doubt on this view. In his study of apparentements in Swiss National Council elections, Bochsler (2010b) showed that seat gains through apparentements tend to work to the benefit of large parties already favored by the d'Hondt formula. This amplifies disproportionality rather than curbing it. In fact, according to Bochsler, apparentements would only reduce disproportionality under specific conditions. He concluded (ibid. 2010b: 618): "The model established in this paper suggests that the frequent believe that apparentments correct the disproportionalities of the D'Hondt rule cannot be maintained." Since biproportional apportionment empirically does not include the provision of apparentements, this potential source of disproportionality is eliminated.

Altogether, the implementation of biproportional apportionment is thus expected to result in a significant reduction of vote-seat disproportionality, irrespective of the specific reform design. Despite the resulting effect being purely mechanical, a reassessment of the systemic incentives might prompt changes in the voter and partisan behavior that potentially mitigate this mechanical effect. The prospect of a more accurate vote-seat translation may encourage voters to vote more "sincerely", i.e. in accordance with their "true preferences". This could lead to increased voter support for fringe parties that ultimately still fall below the threshold of representation. While votes for such parties may not suffice for a parliamentary seat, they contribute to an increase in disproportionality, albeit marginally, as the LSq-index is relatively insensitive to small vote-seat differences. Nonetheless, significant reductions in vote-seat disproportionality can be expected in the

wake of biproportional reforms, given that this is an essential goal of reform. The first hypothesis thus suggests:[84]

H3: The adoption of biproportional apportionment corresponds with a significant decrease in vote-seat disproportionality.

Setting aside any case-specific peculiarities, a common feature among the cantons transitioning to biproportional apportionment is the presence of electoral districts failing to meet the FSC's stipulation of at least nine seats for PR-based elections. As per the Court's interpretation, proportional representation is not effectively realized in districts with magnitudes lower than nine seats. These districts imply a hidden but substantial effective threshold, which poses obstacles to fair representation for smaller parties and their electorates. The Court concluded that the greater the deviation from the nine-seat requirement, the more pronounced the departure from the principle of proportional representation. Table 36 illustrates the count of districts within the group of reformed cantons that did not conform to the judicature before the transition (t-1). The numbers indicate large variances, both in absolute and relative terms. These differences can be largely explained by the different territorial structures of the cantonal electoral systems.

Canton	Number (and share) of non-conforming districts at t-1
SH	2 (33.3%)
AG	3 (27.3%)
UR	6 (75.0%)
ZG	7 (63.6%)
VS	7 (50.0%)
ZH	7 (38.9%)
NW	10 (90.9%)
SZ	14 (46.7%)

Table 36: Number and share of non-conforming districts prior to the reform. GR that switched from plurality voting is excluded. UR numbers relate to districts applying PR only. Source: Own data collection (Lüscher 2024b & 2024c).

Building upon H3, which suggests a substantial decrease in disproportionality following the reform and considering that district magnitude is recognized as an important institutional driver of disproportionality, it can be assumed that the resulting declines in disproportionality will be more pronounced in cantons with a large proportion of small electoral districts before the reform. This leads to the following hypothesis:

H4: The decrease in disproportionality following a reform is more significant in cantons with lower district magnitudes prior to the reform.

3.3.2 Party system fragmentation

Assessing party system fragmentation or concentration through the number of parties has become the gold standard of electoral system analysis. It is generally accepted that party system fragmentation is shaped by the interplay of institutional and socio-structural factors (e.g. Ordeshook & Shvetsova 1994, Neto & Cox

[84] The numbering of the hypotheses is not specific to individual sections but applies to the entire thesis. Therefore, the first hypothesis is continued at number three.

1997, Flick-Witzig & Vatter 2017, Taagepera & Shugart 1989). In my short outline, I will first focus on the institutional factors and then discuss the socio-structural determinants.

Duverger's distinction between static mechanical and dynamic psychological effects (1954), akin to Rae's concept of "proximal" and "distal" effects (1967), has emerged as the dominant institutional framework for understanding how electoral rules affect the number of parties.[85] Mechanical or proximal effects highlight the inherent tendency of an electoral system to grant bonus seats to larger parties, thereby overrepresenting them in parliament at the expense of smaller parties (Taagepera & Shugart 1989: 61-66). Because these effects contribute to a reduction in the number of parties, they are often labelled as "reductive" or "constraining" in relevant literature. Their magnitude varies depending on the specific design of the electoral system and the degree to which it facilitates (or restricts) a proportional translation from votes to seats. In contrast, the psychological or distal effects refer to strategical considerations of voters, parties, and other stakeholders. Anticipating this mechanically induced discrimination against smaller parties and seeking to avoid the risk of wasting their votes (on them), they may strategically opt for larger, more promising parties instead, even if this implies a deviation from the first preference (Lijphart 1994: 71-72). This realism-driven "pre-filtering" amplifies preexisting biases in favor of larger, established parties and further reduces the number of legislative parties. Duverger famously posited that these effects accentuate under plurality rule and tend to either establish or solidify a two-party system, while proportional representation allows for multipartism (see also Riker 1982).

The socio-structural approach to understanding party fragmentation originates from Lipset and Rokkan's cleavage theory, which identifies socio-structural divisions as the primary determinants of party system characteristics. Lipset and Rokkan (1967) identified four dominant cleavages, two arising from historical national revolutions and two from the industrial revolution. Essentially, they argue that these four cleavages structure political conflict in a political entity. Political parties are seen as mechanisms to regulate these conflicts. The more salient these cleavages are within a country, the higher the number of legislative parties, resulting in a higher fragmentation of the party system. Initially, the cleavage concept was operationalized using the raw number of issue dimensions within a society (Taagepera & Grofman 1985, Lijphart 1994). However, this approach was later supplanted by indicators of social heterogeneity, which were identified as the true causal factors driving social divides (Ordeshook & Shvetsova 1994, Flick-Witzig & Vatter 2017).

In their study, Neto and Cox (1997) explored the relationship between electoral institutions and cleavage structures as two distinct approaches to understanding the determinants of the number of effective parties. Their findings suggest an interactive rather than a summative effect of socio-structural and institutional factors. In their own words, "the effective number of parties appears to depend on the product of social heterogeneity and electoral permissiveness, rather than being an additive function of these two" (ibid. 1997: 166). Put differently, bipartism, or low party fragmentation, can arise due to either a "strong electoral system or because it has few cleavages" (ibid. 1997: 167). This indicates that neither an open electoral system nor the presence of numerous potentially exploitable cleavages alone is enough to guarantee multipartism. Instead, multipartism emerges (only) when these two factors align. Table 37 shows the theoretical relationship

[85] There are various ways in which parties can be counted (see e.g. Lijphart 1994: 77-91). The most common distinction lies between "elective" or "electoral" parties and "effective", "parliamentary", or "legislative" parties. In my discussion, I will exclusively focus on the latter category, as it aligns with the relevant dimension of my empirical measurement. I will denote this construct as the "effective number of legislative parties".

between the mechanical inclusivity of electoral rules, socio-structural heterogeneity, and the expected degree of party system fragmentation. By integrating both institutional and sociological perspectives, we can expect an increase in the number of parties in settings characterized by high socio-structural heterogeneity, provided that the institutional conditions allow for the mechanical translation of this diversity.

Mechanical Inclusivity	Socio-Structural Heterogeneity	
	Low	High
Low	Very Low Fragmentation	Low Fragmentation
High	Low Fragmentation	High Fragmentation

Table 37: Theoretical relationship between mechanical inclusivity, socio-structural heterogeneity, and party system fragmentation.

Switzerland, when viewed through a macroscopic lens, reveals a landscape rich in socio-structural contrasts: four national languages and a history marked by enduring political tensions between anticlerical liberals and conservative catholics during the early years of the Swiss federal state. Additionally, Switzerland is recognized as having "one of the countries with the most salient rural-urban divide" (Zumbrunn 2024: 28). Despite this, the class divisions appear comparatively subdued (Lijphart 1979). In an exploration of party fragmentation within Swiss cantonal parliamentary elections spanning from 1986 to 1997, Vatter (2003: 454) discerned a pattern where fewer legislative parties tended to emerge in cantons characterized by rural-agrarian settings, lower population density, religious uniformity, and smaller district magnitudes. Conversely, urban areas, greater religious diversity, and lower electoral thresholds correlated with a higher number of parties in cantonal parliaments. Similarly, Flick-Witzig and Vatter (2017) concluded in their research that the exclusion magnitude, indicative of the permissiveness of an electoral system, emerges as the primary determinant in explaining party fragmentation in Swiss cantons, superseding socio-structural factors.

Socio-structural divisions generally hold less sway at subnational levels, with their impact dwindling from national to cantonal and district levels. This trend is attributed to reduced prevalence a/o politicization of these divisions as one moves down the administrative hierarchy (cf. Flick-Witzig & Vatter 2017: 12). Moreover, in the context of Swiss biproportional reforms, socio-structural factors are expected to have marginal to negligible impact, not only due to their reduced salience on the cantonal level and due to their relative stability over time, but also because these parameters are not targeted by reform. Consequently, the socio-structural dimensions of cantonal party systems will not be considered. Given that biproportional reforms primarily pertain to the institutional dimension of electoral systems, significant changes to electoral rules are likely to entail alterations in the structure of the cantonal party systems.

Biproportional apportionment entails various consequences, with the most significant and decisive being the simulation of a single constituency in the upper-tier allocation of seats. This is presumed to result in a significant reduction of the threshold implied by the district structure. This mechanical effect, coupled with the anticipation of a more accurate vote-seat conversion by both voters and parties, is likely to encourage party entrance and an increase in votes for small parties that were previously penalized by the psychological effect. These effects could contribute to a rise in the effective number of legislative parties. However, it is possible that this effect may not manifest immediately, as voters and political stakeholders may respond to the new institutional stimuli with a certain delay (Klöti & Widmer 1997: 207-210). This leads to the following two hypotheses:

H5a: The introduction of the biproportionality corresponds with an increase in the effective number of legislative parties in cantonal parliaments.

H5b: The introduction of the biproportionality corresponds with a delayed and gradual increase in the effective number of legislative parties in cantonal parliaments.

3.3.3 Electoral turnout

Turnout rates vary widely not only across countries but to some extent also sub-nationally as well as over time. Scholars have devoted considerable attention to understanding how features of electoral systems contribute to differences in turnout levels. Empirical studies generally suggest a positive correlation between the vote-seat proportionality and voter turnout (see, e.g., Jackman & Miller 1995, Blais & Dobrzynska 1998, Blais 2006, Geys 2006, Selb 2009). In his examination of the role of strategic mobilization in the relationship between electoral rules and turnout, Cox (2015: 50) noted that "the most abrupt, durable, and significant changes in turnout coincide with changes in electoral rules", citing the Swiss case where the adoption of PR for the election of the National Council in 1919 resulted in an average increase of 20% in voter turnout across cantons (Gosnell 1930: 126). Similarly, Selb (2009), who conducted a cross-country analysis using district-level data from 31 national elections in 20 different developed democracies, stated that "[e]vidence that turnout is higher under PR than in majoritarian elections is overwhelming." A common framework for explaining this relationship is the concept of pivotality, rooted in the economic model of voting proposed by Downs (1957) and further elaborated upon by Riker and Ordeshook (1968). It suggests that the decision to participate in elections largely hinges on the perceived ability of a voter to sway the outcome (Eggers 2013: 1, Sanz 2016: 8). The higher the expected effectiveness of one's vote, the greater the likelihood of turnout. In plurality elections, where the chance of being decisive is slim and typically only plausible in closely contested races among the top candidates, voters in multi-member districts in PR-based elections have better prospects of effectively affecting the outcome. Thus, the electoral system shapes voters' perceptions of efficacy, which in turn influences the propensity to vote. Given that the likelihood of a vote being effective is greater under PR than under MR, this positive stimulus may translate into higher turnout rates.

In another study, Gallego et al. (2012) found that systemic stimuli do not immediately alter electoral behavior but rather exhibit a certain delay. While disproportionality appears to be unrelated to voter participation in the early elections following democratization, it has a negative effect in established democracies. Their findings suggest that small parties adapt to the new structures of incentives and adjust their mobilization strategies to maximize their parliamentary representation. Citizens who feel close to small parties respond similarly to these new stimuli and are additionally influenced by the parties' new mobilization strategies. The combination of these two effects explains why disproportionality does not have an immediate effect on turnout directly after democratization but becomes apparent just after several elections. This "lagged response" also indicates that parties and voters are not able to fully anticipate the consequences of an electoral system ex ante but require some time to familiarize themselves with its new incentives and effects.

Despite the strong empirical and theoretical support suggesting that electoral systems play a significant role in stimulating electoral participation, the exact causal mechanism underlying this relationship remains largely a puzzle. This concern is echoed by Blais (2006: 116), who summarized the literature on electoral systems and turnout concluding that "[m]ost of the literature supports the view that PR fosters turnout, but there is no compelling explanation of how and why". A challenge arises from the fact that electoral systems, also tend to vary significantly along other dimensions such as district sizes, coalition types, party systems,

etc. This complicates the isolation of specific factors that explain differences in turnout (Eggers 2013: 2). In other words, higher turnout rates under PR may also, at least partially, be influenced by other variables, and the impact of these variables may change from one setting to another.[86]

In the context of biproportional reforms in Switzerland, one of the primary immediate effects is presumably a significant reduction of vote-seat disproportionality. Consequently, voters can generally be expected to have better prospects of effectively impacting the electoral result. This is particularly true for voters or sympathizers of small parties who may have previously been discouraged from casting their vote for their preferred party or from voting at all due to a high improbability of their vote being effective. The new system provides a set of incentives that enhance the effectiveness of their vote, which may fundamentally alter their cost-benefit calculus. Citizens for whom the costs of turnout previously outweighed its expected benefits may find their calculus changed due to the reform, potentially motivating them to participate in the electoral process. In contrast, voters of larger established parties, who had a higher propensity to impact the election results with their vote than their counterparts supporting smaller parties, may lose their previous bonus. However, there is no reason to believe that this loss of bonus would push such voters into abstention. Therefore, it can be assumed that the introduction of biproportional apportionment will increase turnout rates. However, again it is possible that this effect does not manifest immediately in the first post-reform election but rather occurs with a certain delay, as voter expectations are shaped by past experiences and institutional knowledge (Taagepera & Shugart 1989, Klöti & Widmer 1997, Riera 2013).

H6a: The introduction of biproportionality corresponds with an increase in voter turnout for cantonal parliamentary elections.

H6b: The introduction of biproportionality corresponds with a delayed and gradual increase in voter turnout for cantonal parliamentary elections.

3.3.4 Electoral competition

Competitive elections stand as important hallmarks of modern-day democracy (Lindberg 2004: 67). High levels of political contestation are indicative of a vital democratic culture and inclusive electoral rules, as the incentives for competition are evidently structured in a way to mobilize a high number of political contenders. The "pivotal elites model" (Selb 2009) suggests that parties mobilize their potential electorates "if the benefits gained through winning parliamentary seats, weighted by the probability that the mobilization effort will be decisive, exceed the mobilization costs" (Selb 2009: 531, see also Cox & Munger 1989, Cox 1999). Mobilization efforts are, in turn, influenced by institutional mechanisms governing the translation of votes into seats as these mechanisms "are commonly assumed to condition the chances of electoral success of competing parties" (Charvát 2023: 19). Hence, it is rational for parties to allocate their limited mobilization resources in districts where they have a higher chance of gaining seats, rather than in districts where their prospects are low or where such efforts are in vain (Selb 2009: 531, Charvát 2023: 1). Considering that district magnitude largely determines the threshold, electoral competition tends to be higher in larger electoral districts, holding other factors constant (Bochsler & Wasserfallen 2013: 252). Higher local competitiveness also correlates with increased electoral turnout. Relevant literature suggests that competitiveness is the key

[86] However, this should not bother us here, as we are not interested on the precise theoretical mechanism behind the proportionality-turnout nexus, but rather on whether this relationship also applies in the context of biproportional reform in Switzerland. With this short paragraph I simply want to highlight the fact that previous research suggests that this relationship may not be as straightforward as it might appear prima facie.

variable driving the proportionality-turnout nexus (e.g., Selb 2009). Thus, factors fostering competition likely overlap with those stimulating voter turnout, and high local competition further amplifies turnout.

Biproportional apportionment, by reducing the mechanical factors that impede the approximation of ideal vote-seat proportionality, is likely to enhance electoral competition, leading to an increase in the number of candidacies and electoral lists. With regard to the electoral lists, however, it must be noted that none of the reforming cantons provided for the possibility of apparentements, neither interparty nor intraparty apparentements. Under the d'Hondt method, apparentements were widely used as a strategic means to maximize votes and, consequently, parliamentary seats. The inflationary use of these apparentements and the resulting lack of transparency in electoral competition gave rise to voter criticism. Voters specifically criticized the challenge in determining which party ultimately benefited from votes cast for parties allied in sub-apparentements. This skepticism towards apparentements was also prevalent among parties themselves, albeit reluctant to forgo such alliances voluntarily. The redundancy of apparentements under biproportionality provided an opportunity to eliminate this widely unpopular but highly effective electoral tool. Analysis of policy processes in the second part of the thesis revealed that the abolition of apparentements was generally met with unanimous approval from political parties. The removal of this option is likely to negatively affect the total number of electoral lists.

Another characteristic of biproportionality is the mandate for list groups to have a consistent label across the entire electoral area. For instance, in the electoral law of the Canton of Aargau, the respective paragraph from the electoral law paradigmatically stipulates that "lists with the same label form a list group in the canton" (Grossratswahlgesetz §13(1)). This requirement assumingly reduces the inter-district variance of lists, particularly as list labels previously sometimes explicitly referenced the constituency. In summary, it is likely that the absolute number of electoral lists will show a significant decline immediately following the reform. The stimulating effect of biproportionality is anticipated to manifest gradually in terms of the number of electoral lists, leading to a gradual increase over time.

H7a: The introduction of biproportionality corresponds with an increase in the number of candidacies.

H7b: The introduction of biproportionality corresponds with a short-term decrease followed by a longer-term increase in the number of electoral lists.

3.4 Data, variables, and analytical strategy

This section delineates the structure of the data and the data sources. It further provides a description of the key variables, and outlines the analytical approach undertaken in this third part of the thesis addressing hypotheses 3-7.

3.4.1 Data and data sources

The units of analysis in this chapter are Swiss cantons, with each observation corresponding to a canton in a specific year between 1976 and 2023. The electoral data set comprises 1209 observations and 31 variables. Researchers working on Swiss cantons are well aware that Swiss federalism is also reflected at the data level. Typically, centralized, and systematized historical data at the cantonal level, extending beyond demographics and election results, are scarce. As a primary data reference, I rely on two publicly available data sets: the data set "Quality of Democracy in the Swiss Cantons 1979-2006" by Schaub & Dlabac (2012), and the data set "Patterns of Democracy in the Swiss Cantons 1979-2018" by Vatter et al. (2020). This data is cross-checked randomly and adjusted as necessary. For missing years, data is supplemented following the

procedures outlined in the respective codebooks. While these two data sets are referenced, most of the data in the main data set created for this study (Lüscher 2024b) is self-collected, primarily through archival research or official statistics provided by the FSO. Notably, there is no existing data on electoral competition in cantonal parliamentary elections (number of electoral lists and number of candidacies), requiring data collection through research in cantonal archives. Additionally, all district-related data is sourced from an additional data set (Lüscher 2024c) I separately created as detailed data on electoral districts for cantonal parliamentary elections does not exist – at least not systematically and for the time frame of interest. This district-related information is also gathered through a combination of archival research, data research, and historical news articles accessed through online archives, as well as outreach to libraries and cantonal chancelleries. A codebook for the main data set is provided in Table 42 the appendix.

3.4.2 Variables

As explained in the previous section, some variables are adopted from preexisting data sets, and their operationalizations are applied identically in this study for data completion. Other variables are self-collected and, in some cases, coded in a specific way to suit the analysis. This section aims to provide transparency regarding how the variables in the data set 2 (Lüscher 2024b) are operationalized and the limitations associated with these operational definitions.

Number of lists

The variable "number of lists" indicates the total number of distinct electoral lists submitted across the entire electoral area. This can be understood as the raw number of "elective parties", although not all lists are formally constituted as parties. Alternatively, one could total the submitted lists from each district. In fact, many cantonal statistical offices use this straightforward indicator in their statistical publications. For intra-cantonal comparisons, such a self-referential measure is suitable. However, for inter-cantonal, longitudinal comparisons, this approach has two shortcomings:

- The number of lists submitted in the districts contains redundant information, as most contesting parties submit identical (or nearly identical) lists in more than one district. Thus, the number of lists is highly contingent on the number of electoral districts, which varies greatly between cantons, thus distorting inter-cantonal comparisons.
- Several cantons have transitioned to biproportionality where the main tier of allocation is the canton. Consequently, the number of lists reflects the count at the cantonal level and does not represent an aggregate of district-level list numbers, as is typical in unidimensional PR elections. A straightforward before-and-after comparison without accounting for these direct reform effects would be inappropriate.

To conduct valid cross-sectional and longitudinal comparisons, a standardized measure is needed. Therefore, I use the number of electoral lists submitted across the entire electoral area. As previously mentioned, cantonal statistical offices vary in how they report electoral lists. Some sum the lists across all districts, others report lists in the manner suggested here, and some do not report them at all or do so minimally. Consequently, data is collected for the vast majority of cantons by consulting cantonal statistical offices, chancelleries, and archives to obtain official statistical results for all parliamentary elections in the period of interest. I manually counted the number of "distinct" electoral lists submitted in the electoral districts for almost all

cantons. The operationalization of "distinctiveness" of electoral lists is empirically by no means trivial. Below, I explain the criteria used to determine whether electoral lists are subsumed under a single list or treated as distinct lists.

Different lists are subsumed under a single list when:

- If list labels clearly refer to the same party, even if they carry district-specific attributes, are treated as one list. For example, if the CVP contests with a district-specific attribute in every district, e.g. "CVP Oberrheintal" and "CVP Werdenberg", etc., these lists are counted as one list. The situation differs if the CVP launches multiple nominally distinct lists within a district (as discussed below).
- Lists combining members of different parties or groups on a single list in an electoral unit are treated as single lists. This is distinct from apparentements, where parties compete with separate lists and electorally merge them. For example, if the CVP, BDP, and EVP contest elections with a common list (not apparentement) in a certain district, it is counted as one list.

Electoral lists were treated as distinct when:

- If there is no obvious nominal commonality between lists, they are counted individually.
- They are also counted separately if the same party contests with multiple distinct lists within a district. For instance, "CVP West" and "CVP Ost" within the same district are counted as two separate lists (subject to the restriction outlined in the following point).
- If a party uses different district-specific targeting strategies (e.g., geography, gender, age), these distinctions are considered only once. For example, if the CVP contests with lists labeled "CVP Men" and "CVP Women" in different districts, these distinctions are not treated as additional lists when such a differentiation is already present in another district.

This operational approach strikes a balance between completely ignoring strategic list differentiations and overvaluing them due to district-specific targeting strategies. Consequently, the raw number of electoral lists may not always match the officially reported numbers, which can be viewed as a drawback. However, by consistently applying the above criteria, longitudinal and cross-sectional comparisons of electoral competition, as measured by electoral lists, are more valid.

Number of candidacies for cantonal elections

Collecting the number of candidacies running for office for a given election required less effort than collecting data on electoral lists, as it is information of public interest and is therefore more often included in official statistical documents. However, in some cases, it was necessary to consult official statistics (official gazettes) in the cantonal archives to determine the number of candidacies. In the French-speaking part of Switzerland, relevant media reports are also taken into consideration. The number of candidacies surveyed relates exclusively to PR elections. In the case of mixed electoral systems (e.g., in the case of Appenzell Ausserrhoden), only the data for constituencies where proportional representation applies is collected and reported. In PR elections with pre-cumulated lists, cumulated candidate names are considered by subtracting them from the total number of candidate names indicated (= filled list lines).

Number of candidacies for National Council elections

This variable shows the raw number of candidacies for National Council elections. Single-seat and majoritarian cantons (UR, OW, NW, GL, AI, GR) are excluded, following the practice of the FSO.

Number of candidacies per seat for cantonal elections

This variable measures electoral competition for a political mandate. A higher number of candidacies per seat indicates greater electoral competition. The calculation involves dividing the total number of candidacies by the number of available mandates. For mixed electoral systems, the number of candidacies per seat pertains only to constituencies where PR applies. Originally, this variable was intended as the primary measure of electoral competition. However, its sensitivity to changes in parliament sizes in many cantons prompted me to exclude it from the analysis. Nevertheless, it will remain in the data set for reference. The same applies to the next variable.

Number of candidacies per seat for National Council elections

This variable represents the number of candidacies per seat and canton for National Council elections. The data is sourced from the official data set of the FSO on national elections, specifically the number of seats attributed to the cantons (FSO 2023d) and the number of candidacies per canton (FSO 2023c). The ratios are subsequently calculated. Single-seat and majoritarian cantons (UR, OW, NW, GL, AI, GR) are excluded, following the practice of the FSO. Until 2011, the data for calculating the distribution of seats among the cantons is based on the census conducted every 10 years under the old system. From 2015 onwards, the distribution of National Council seats among the cantons is based on the new annual census.

Least Squares Index

The LSq, also known as the "Gallagher index" (Gallagher 1991), is a widely used measure in electoral research. It assesses the disproportionality of an electoral system by quantifying the deviation between the vote shares v and seat shares s of contesting parties i. Specifically, it calculates the squared differences between the vote shares and seat shares for each party, sums these squared differences, and takes the square root of half of the sum. The value of 0 indicates minimal disproportionality, while 100 indicates maximal disproportionality. Mathematically, the LSq formula is expressed as:

$$\mathrm{LSq}=\sqrt{\frac{1}{2}\sum (s_i - v_i)^2}$$

or alternatively:

$$\mathrm{LSq}=\sqrt{\frac{1}{2}\sum (v_i - s_i)^2}$$

Effective number of parties

The concept of the "effective number of parties" (ENP) is a common measure used to quantify the fragmentation of a party system. Unlike a simple count of parties, the ENP accounts for the relative strength of each party based on its vote or seat share. This concept was introduced by Laakso and Taagepera in 1979 and remains widely used in political science research. The effective number of parties is calculated by squaring the seat share of each party, summing up these squared values, and then dividing 1 by that sum. The formula for calculating the ENP is:

$$\mathrm{ENP}=\frac{1}{\sum (s_i^2)}$$

Data for calculating the ENP is sourced from the "Patterns of Democracy" data set, with manual supplementation for missing years (2019-2023). Similar to the approach of previous authors, the residual group of "others" is excluded, leading to a slight overestimation of the "true" number of effective parties based on the Laakso-Taagepera Index.

Number and size of districts and other district-related data

District-related data, such as the number and size of electoral districts, are gathered from public statistics, official election results, and archival research. These data are used to determine various district statistics, including the maximum and minimum district sizes, as well as the standard deviations from the mean. These district features are calculated separately in an Excel sheet and then imported into the main database. The data set includes information on the number of districts, the mean and median district sizes, the minimum and maximum district sizes, the range between these extreme values, and the standard deviation as a measure of data dispersion. In cases where the electoral area consists of multiple tiers, the number of districts refers to the upper-tier electoral districts, consistent with the practice of the FSC, which bases its assessment of the (dis)proportionality of electoral systems on the highest allocation level within the electoral territory.

Absolute and relative number of districts with less than 9 seats (applies PR cantons/districts only)

This variable represents both the absolute and relative count of districts within a canton that have fewer than 9 parliamentary seats, relative to the total number of districts in that canton. Its aim is to indicate the number of PR districts that are deemed unconstitutional according to the current jurisdiction of the FSC. Cantons or districts applying plurality voting or biproportional apportionment are left blank. If the seat allocation follows a two-step method, where constituencies are grouped into electoral regions, the counts provided relate to the upper-tier level of these electoral regions.

Absolute and relative number of seats from districts with less than 9 seats (applies PR cantons/districts only)

This variable, similar to the previously mentioned one, measures the total number and share of seats stemming from "non-conforming" districts within a canton. Its aim is to illustrate the sum and share of seats originating from districts deemed unconstitutional according to the jurisdiction of the FSC. Cantons or districts applying plurality voting or biproportional apportionment are left blank again. If the seat allocation follows a two-step method, where constituencies are grouped into electoral regions, the counts provided relate to the upper-tier level of these electoral regions.

Turnout in cantonal elections

Voter turnout data for cantonal parliaments is extracted from the "Patterns of Democracy" data set and then manually supplemented to include the missing years (2019-2023).

Turnout in National Council elections

The turnout for elections to the National Council serves as an auxiliary variable for estimating turnout in cantonal parliamentary elections. However, a limitation arises due to the asynchrony between cantonal and national parliamentary elections. While National Council elections occur in the same year across all cantons, cantonal parliaments are elected at different times and sometimes in different rhythms. To address this issue, one could theoretically use the National Council election closest to the cantonal parliamentary election as a reference point. However, this approach would distort the date as well as the temporal reference to

the number of cantonal candidacies per National Council seat. Since the statistical models employed can accommodate complex data structures including staggered data, I decided to accept the asynchrony between elections of different tiers. For cantons that elect their National Council representatives by majority/plurality vote, electoral turnout is not considered (AR, AI, GL, NW, OW, UR).

Turnout in cantonal votes (referenda, popular initiatives, etc.)

This variable represents the annually averaged turnout in cantonal votes, distinct from the turnout for national votes held at the cantonal level. To ascertain the cantonal voter turnout, the average of all votes on cantonal matters is computed for each year. Multiple votes occurring on the same date are considered individually. Each voting issue carries equal weight in the calculation process (cf. codebook "Patterns of Democracy"). Data from 1976 to 2018 is sourced from the "Patterns of Democracy" data set, while values for the years 2019 to 2023 are supplemented using official data from cantonal authorities.

Number of seats per canton for National Council elections

This variable represents the absolute number of mandates per canton. It is important to note that starting from the 2015 federal elections, there was a change in the data basis for seat apportionment. Refer to the remarks on the variable "number of candidacies per seat for National Council elections" for further details.

Effective electoral threshold

This variable captures the effective threshold implied by the electoral rules. Given that these rules vary between cantons and sometimes within cantons over time, different formulas are applied to account for this heterogeneity. Four rules apply to differentiate whether the election occurred under biproportionality or not, whether there are several constituencies or only one constituency, and whether a statutory quorum applies or not. Rule (1) relates to Lijphart's (1994: 11) proposition of using the average district size for multimember districts to approximate the effective threshold. Hence, *M* refers to the mean district size for the for the election *i*. Rule (2) pertains to single-district elections and follows a formula suggested by relevant literature (e.g. Pukelsheim 2004: 518). Here, *M* equals the assembly size. Rule (3) applies to all biproportional elections. *M* denotes the assembly size and *L* denotes the number of elective parties, which equals the number of electoral lists. The formula also originates in Pukelsheim's work (ibid. 513-518), which references earlier work by Balinski & Young (2001: 154) and Lijphart & Gibberd (1977: 223). Rule (4) overrides all other rules, as it can be assumed that such a threshold only exists if it exceeds the effective threshold. For biproportional elections with two quorums, the global/canton-wide quorum is taken as reference.

1. $\text{eff_threshold}_i = \frac{50}{\text{M_mean}_i+1} + \frac{50}{2\ \text{M_mean}_i}$ if ref_dummy = 0 and distr_num > 1
2. $\text{eff_threshold}_i = \frac{100}{M_i + 1}$ if ref_dummy = 0 and distr_num = 1
3. $\text{eff_threshold}_i = \frac{100}{2\ M_i + 2 - L_i}$ if ref_dummy = 1
4. eff_threshold_i = statutory quorum if present

It should be noted that the effective threshold varies only marginally within cantons, as the relevant institutional parameters remain relatively constant over time. The only factors that might influence the threshold level are changes in the mean district size, adjustments in the assembly size, the adoption of biproportional apportionment, or modifications to the quorum regulations.

Figure 39 presents the density plots for the main independent and dependent variables. Apparently, not all variables follow a normal distribution. Thus, I use the option inbuilt in the used R-package to normalize all variables if normality is violated.

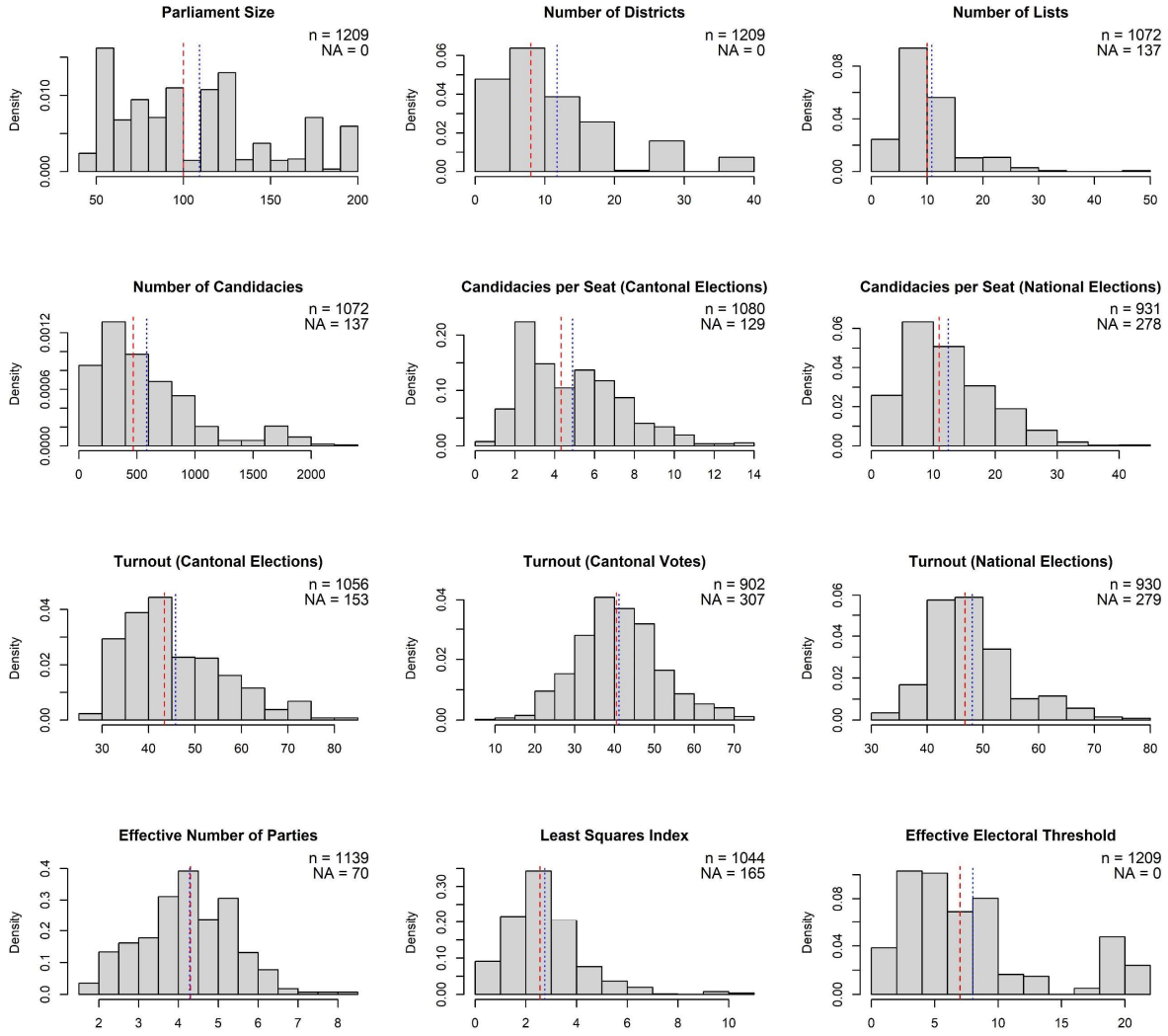


Figure 39: Density plots for the main variables. The red-dashed line shows the median, the blue-dotted line shows the mean. Unless otherwise specified, the variables pertain to the cantonal level. NA = Not Available (= missing data).

3.4.3 Analytical strategy

The analysis focuses on the twenty-six Swiss cantons, nine of which have adopted the biproportional apportionment method. The R-package used for the statistical analysis, discussed in more detail below, automatically excludes treated cases with fewer than five pre-treatment periods. Since the Canton of Graubünden lacks data for many key variables before treatment/reform, owing to its previous use of majoritarianism for parliamentary elections, this case is excluded from the analysis. The primary analytical goal of this section of the thesis is to determine the dynamic causal effect of the electoral reform on various aggregate-level variables by isolating this effect from other potential confounders, both observed and unobserved.

The method applied in this third part of the thesis is based on the framework of two-way fixed effects counterfactual estimation for causal inference (fect) with time-series cross-sectional (TSCS) data. This framework advances and outperforms traditional, more demanding models, such as those in the synthetic control method class, in several respects (Liu et al. 2024). Traditionally, the estimation of treatment effects in longitudinal studies has relied heavily on fixed effects models, which often make strong assumptions about the temporal stability of treatment effects and the absence of time-varying confounders. However, these assumptions can be unrealistic in many real-world scenarios and can lead to biased estimates and erroneous conclusions. The proposed framework addresses these limitations and offers three different (but similar) estimators:

- **FEct (fixed effects counterfactual)** : This estimator extends the traditional fixed effects model by incorporating counterfactuals to estimate treatment effects. It accounts for individual-specific effects and controls for unobserved heterogeneity, ensuring robust treatment effect estimation by leveraging within-unit variation over time.

$$Y_{it}(0) = X_{it}^{'}\beta + \alpha_i + \xi_t + \varepsilon_{it}, \text{ for all } (i,t)$$ [87]

- **IFEct (interactive fixed effects counterfactual)** : Building on the FEct model, the IFEct estimator integrates interactive fixed effects to handle unobserved time-varying confounders. This model captures complex temporal patterns by including factors that interact with individual fixed effects. It thus addresses potential biases from unobserved time-varying factors.

$$Y_{it}(0) = X_{it}^{'}\beta + \alpha_i + \xi_t + \lambda_i^{'}f_t + \varepsilon_{it}, \text{ for all } (i,t)$$ [88]

- **MC (matrix completion)** : The MC estimator employs matrix completion techniques to fill in missing data and estimate treatment effects. This method is particularly useful in contexts with substantial missing data or irregular treatment timings, as it reconstructs the missing entries by leveraging the low-rank structure of the data matrix, ensuring a robust analysis even with incomplete data sets.

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- $Y_{it}(0)$ denotes the potential outcome for unit i at time t under the control condition.
- $X_{it}^{'}$ is a vector of covariates for unit i at time t. These variables are measured characteristics that can affect the outcome $Y_{it}(0)$.
- β is the vector of coefficients associated with the covariates X_{it}. Each element of β represents the effect of one of the covariates on the outcome. Essentially, β quantifies how changes in each covariate X_{it} are expected to influence the outcome $Y_{it}(0)$.
- α_i captures the unit-specific fixed effect. It accounts for unobserved characteristics that are constant over time for each unit i but may vary across different units.
- ξ_t represents the time-specific fixed effect. It accounts for factors that are constant across all units at a given time t but may vary over time.
- ε_{it} is the error term, capturing random noise or unobserved factors that affect the outcome for unit i at time t. It represents the idiosyncratic shocks that are not explained by the covariates X_{it}, unit-specific effects α_i, or time-specific effects ξ_t.

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- $\lambda_i^{'}f_t$ is the term that introduces interactive fixed effects. $\lambda_i^{'}$ is a vector of factor loadings specific to unit i and f_t is a vector of common factors at time t.

$$Y(0) = X\beta + L + \varepsilon$$ [89]

These models offer more reliable causal estimates compared to conventional two-way fixed effects models, especially when treatment effects are heterogeneous or unobserved time-varying confounders are likely to exist.

In this framework, estimators treat the observations from the treated group as missing data and impute counterfactuals for each treated observation within a time-series cross-sectional (TSCS) data set. This imputation is achieved by fitting one of the previously discussed models using the data from untreated cases and a set of covariates associated with the outcome variable. Subsequently, the individual treatment effect for each treated observation is estimated by subtracting the predicted counterfactual outcome from its observed outcome. Finally, the average treatment effect on the treated (ATT) or, alternatively, period-specific ATTs are calculated (Liu et al. 2024).

A key innovation of this class of estimators is their independence from the observed outcomes of the treated units at the modelling stage, which allows for more flexible and robust estimates of treatment effects over time. Central to this approach is an iterative algorithm trained to identify the optimal model based on mean squared prediction errors (MSPE) via a k-fold cross-validation scheme with 2000 bootstraps. Furthermore, the estimation models based on the "fect" framework are equipped to effectively handle missing data, a common issue in panel data analysis. The framework also provides a set of inbuilt diagnostic tests to verify the validity of the identifying assumptions underpinning the estimation process. These tests include placebo tests, which check for spurious treatment effects by simulating treatment effects at placebo time points, and tests for (no) pretrend.[90] The staggered adoption of reforms renders the tests for carryover effects inapplicable. By subjecting the estimated treatment effects to these diagnostic tests, the framework helps to detect potential model misspecifications and enhances the credibility of the causal claims.

Overall, the proposed framework represents a significant advancement in causal inference methodology. It offers a powerful toolkit for analyzing complex panel data and enhances the credibility of causal claims. By combining flexible estimation techniques with rigorous diagnostic tests, the method allows for more accurate and reliable estimates of treatment effects, ultimately leading to a deeper understanding of causal relationships in panel data analysis (cf. Gobillon & Magnac 2016, Xu 2017, Athey et al. 2018, Liu et al. 2024).

Both the data analysis and the data visualization are carried out with the statistics software "R" version 4.4.0 and "R-Studio" version 2024.04.1-748. For counterfactual ATT estimation, I use the "fect" package written by Liu et al. (2022). The package effectively circumvents the problems addressed above, such as missing data, complex data structures, and staggered reform adoption. However, a drawback of the package is that it only generates data matrices for the relative ATTs and directly computes the delta between observed and estimated data. There is no matrix that stores the estimated outcome in absolute numbers. Although the ATT

[89]

- X is a vector of covariates. These variables are the measured characteristics or features that can influence the outcome $Y(0)$.
- L represents the combined effect of unobserved factors. It captures the influence of latent variables or fixed effects that are not explicitly included in the covariates X. L can encompass both unit-specific and time-specific effects that are not directly measured. For more details, see Liu et al. 2024: 163-167.

[90] For the placebo test, the three observations before the treatment are omitted (t-3, t-2, t-1). For the pretrend test, the TOST threshold is set by default to correspond to the equivalence range of 0.36 standard deviation from the mean of the outcome variable after partializing out the two-way fixed-effects.

is undoubtingly the main quantity of interest, it allows no conclusions about the substantivity of the causal effect. To circumvent this limitation, I reconstruct the estimated averages for the outcome variables of the reformed cantons over the entire available period (i.e., before and after the reform). I do so by isolating the ATTs and subtracting them from the observed averages. This subtraction provides the estimated outcomes in absolute numbers. In simpler terms, I calculate the estimated outcome by subtracting the ATTs from the observed outcome.[91] Although this process is rather inconvenient because of the complex data structures and thus requires a lot of coding work, it is the only way to quantify the substantivity of the ATT. In the next step, I plot the ATTs to better visualize the dynamic of the effect. Another impediment arises from the staggered implementation of the reforms between 2007 and 2022. The inbuilt algorithm addresses this issue by standardizing the time variable (year) relative to the time of reform ($t = 0$). However, this leads to a decreasing number of cases with increasing years into the post-reform period, implying higher statistical uncertainty.

3.5 Results

3.5.1 How biproportional reforms affect vote-seat disproportionality

One of the key goals for implementing biproportional apportionment in parliamentary elections is to minimize the discrepancy between a party's share of votes and its share of seats. These discrepancies, however, cannot be entirely explained by the mechanical features of the electoral system itself but by a combination of these features and actual voting figures (i.e., number and geographical dispersion of votes, number of competing lists, etc.). Thus, the least squares index as a measure of vote-seat disproportionality reflects not only mechanical effects but is also subject to electoral volatilities. To quantitatively assess the influence of the mechanical features of an electoral system on observed disproportionality, I conduct a regression analysis of the LSq-index against the effective electoral threshold, while controlling for unit (i.e., canton) fixed effects. The model yields an R^2 of 0.34 and a slope for the effective threshold of ≈ 0.191. An increase of five percentage points in the effective threshold thus corresponds to approximately a one percentage point increase in the LSq-index ($p < 0.001$) when controlling for canton fixed-effects.[92] This finding indicates a significant relationship between the effective threshold, used as an indirect measure of an electoral system's mechanical effect, and disproportionality, as anticipated. Moreover, the effective threshold accounts for approximately one third of the observed variance in disproportionality. However, it is worth noting that there are other factors, collectively more influential, that also impact the disproportionality level of an election.

[91] The reason for subtraction can be exemplified as follows: If the ATT is 3.5 and the mean observed outcome of the treated at time t is 10.0, the absolute estimated (counterfactual) outcome at time t would be 6.5. Thus, the difference between the observed and the estimated outcome corresponds to a positive ATT of 3.5.

[92] The following formula applies: $\text{lsq_index}_{it} = \beta_0 + \beta_1 \times \text{eff_threshold}_{it} + \sum_i \alpha_i \times canton_i + \epsilon$, where α_i are the canton fixed effects.

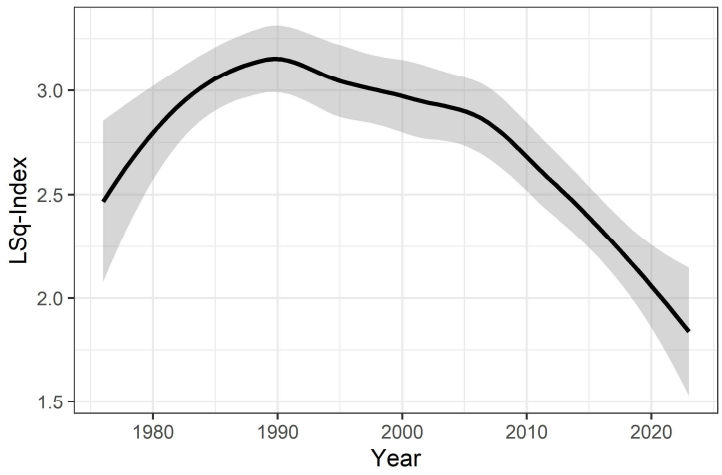


Figure 40: General time trend for vote-seat disproportionality for all PR-based cantonal parliamentary elections. The graph shows the locally weighted polynomial regression curve with 95% confidence intervals.

In the Swiss cantons, vote-seat disproportionality is generally moderate and aligns with international averages (cf. Lijphart 1999: 163, Vatter 2008: 17-18). Figure 40 illustrates an upward trend from the mid-1970s, peaking in 1990, followed by a continuous decline. This decline, both in relative and absolute terms, is notable. Particularly noteworthy is the steeper decline observed since 2005, possibly indicating the impact of biproportional reforms. During the period depicted, several electoral system reforms occurred. Notably, since 2005, eleven cantonal parliaments have undergone reductions in size, some quite substantial (e.g., SG from 180 to 120 seats per 2007). These parliamentary reductions, in turn, required adjustments at the level of electoral districts. While revisions of cantonal electoral law are not rare, the period since the turn of the millennium has seen notably high levels of cantonal reform activity, with biproportionality being one manifestation of this trend.

An examination of cantonal trends in disproportionality levels reveals significant inter- and, in some cases, intra-cantonal variations (Figure 41). This variance can mainly be attributed to differences in parliamentary sizes, constituency structures, seat allocation methods, and statutory quorums. For instance, the canton of Geneva, voting as a single electoral district, imposes a canton-wide quorum of 7%, creating a substantial electoral hurdle that impedes or even prevents the emergence of small parties and can result in high disproportionality values. It is also notable that the overall trend towards lower disproportionality levels is observed in most cantons. However, some cantons exhibit a contrary trend. Nevertheless, it is evident that all cantons adopting biproportional apportionment have witnessed a decline in disproportionality over time, with some experiencing a significant reduction.

While these descriptive statistics suggest a negative impact of biproportional reforms on vote-seat disproportionality, they do not conclusively prove an effect. If disproportionality was exclusively determined by institutional factors, it could be predicted prior to elections and would be constant if these institutional parameters remained unchanged. However, while mechanical aspects are indeed highly determinative for disproportionality and set the boundaries within which disproportionality empirically fluctuates, it is exogenous factors that ultimately determine the actual level of disproportionality of an election.

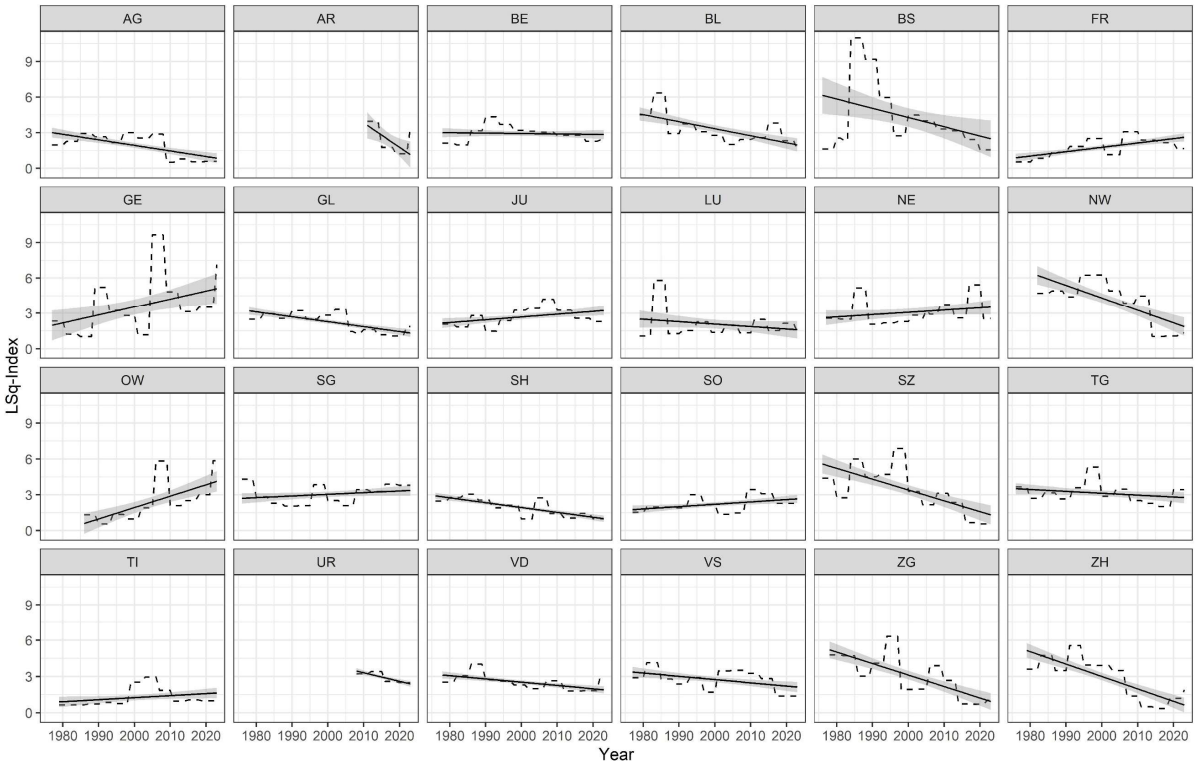


Figure 41: Trend in disproportionality levels in PR-based cantonal parliamentary elections. The dashed line shows the observed LSq values; the solid line shows the locally weighted linear regression line with 95% confidence intervals.

To estimate the causal effect of biproportional reforms on disproportionality levels, i.e. the ATT, I incorporate the following variables: the reform dummy as the binary treatment indicator; the number of effective legislative parties, the range between the largest and smallest district in terms of seats and the number of candidacies in cantonal elections as the covariates. The MSPE-based cross-validation procedure suggests using the matrix completion method (MC).[93]

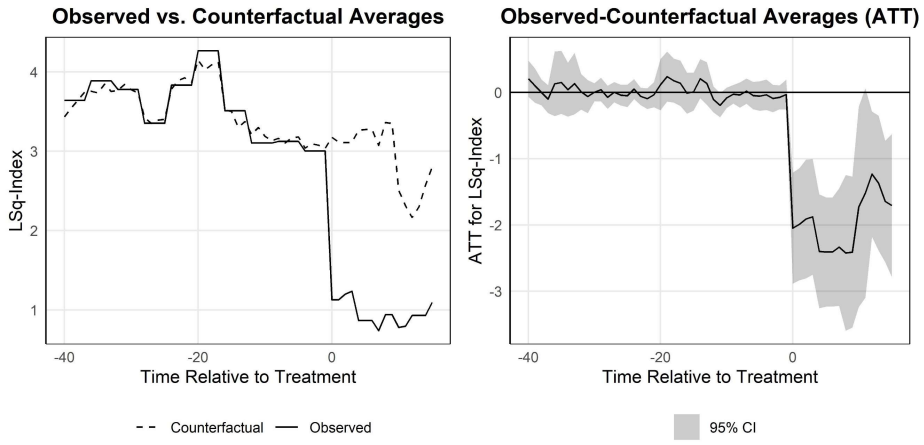


Figure 42: Observed and estimated average vote-seat disproportionality (left panel) and the corresponding ATT for treated cases (right panel) before ($t < 0$) and after ($t \geq 0$) treatment/reform. The bar plot in the figures displaying the test diagnostics shows the number of cases treated per year.

[93] Following formula: lsq_index = $\beta_0 + \beta_1 \times$ ref_dummy + $\beta_2 \times$ enp + $\beta_3 \times$ distr_range + $\beta_4 \times$ cand_num_ce + $L + \varepsilon$

Figure 42 shows the observed and counterfactual trends relative to the time of reform adoption, depicted as absolute averages (left panel) and residual averages (right panel). The left panel reveals that during the pre-treatment period, the treated cases' averages range between LSq values of 3 and 4.5, showing a strong correspondence until a sudden decline in observed averages at the time of intervention, followed by a minor increase in the final post-treatment years. It is important to note that these averages are based on only three treated cases (ZH, SH, AG). After treatment, the counterfactual values fluctuate between 2.5 and 3.5.

In the right panel we see the ATT. Residual averages in the pretreatment period remain close to zero, with a pronounced negative treatment effect on LSq values emerging immediately after treatment. The overall ATT is approximately -2.048 ($p < 0.001$) and remains negative throughout the post-treatment period.

The regression results (Table 43 in the appendix) indicate that all covariates are positively associated with the outcome variable. Noteworthily, no significant relationship appears between the effective number of parties and the LSq-index. This implies that a fragmented party system does not necessarily translate to higher disproportionality rates, nor does a lower level of fragmentation inherently correspond with lower disproportionality rates.

The placebo test (Figure 54 in the appendix) indicates that we cannot reject the null hypothesis of zero placebo effect ($p \approx 0.507$), and we cannot reject the null hypothesis of inequivalence ($p \approx 0.116$). Hence, the placebo test passed but not the test for equivalence. To test for the absence of a pretrend, the F test does not reject the null hypothesis of no pretrend ($p \approx 0.904$). Meanwhile, the TOST rejects the null hypothesis of inequivalence ($p \approx 0.015$), indicating a good pre-treatment fit.

The findings confirm the third hypothesis according to which the implementation of biproportional apportionment leads to notable reductions in disproportionality averages. Post-reform, average disproportionality levels are approximately half of the pre-reform values. Thus, despite the diverse sources contributing to vote-seat disproportionality and the variations in reform designs, the adoption of these reforms notably enhanced the electoral systems' ability to accurately reflect voter preferences.

However, does the increase in proportionality after reform show a stronger effect in cantons with lower average district sizes before the reform, given that district size is a key factor in affecting disproportionality (H4)? To address this question, the assessment includes all treated cantons except Graubünden due to missing LSq-index data prior to reform. As Uri is somewhat of an outlier, providing only one post-reform observation and due to its unique mixed electoral system with PR in larger districts (≥ 5 seats) and majority representation in smaller ones, I conduct the assessment with and without Uri.

Exploring the relationship between mean district sizes and disproportionality levels based on the treatment status, Figure 43 presents the marginal effects. While a perfect parallel trend would imply the absence of an effect, a very small effect emerges when Uri is included. However, when Uri is excluded, the effect is more substantial, albeit not overwhelming: The decline in disproportionality, indicated by a relatively large delta, is more pronounced in elections where the mean district sizes were smaller compared to elections with larger district averages. It is also noteworthy that including Uri yields a slightly negative relationship between disproportionality and mean district sizes after the reform. However, this weak relationship vanishes when Uri is excluded. The zero slope and low intercept indicate that after reform, disproportionality is low irrespective of mean district sizes. This empirical finding is in line with theoretical expectations. Biproportional apportionment mitigates the adverse impact of electoral district sizes on vote-seat proportionality by

employing an allocation mechanism that distributes seats to parties based on their overall electoral performance. Therefore, these results lend support to H4.

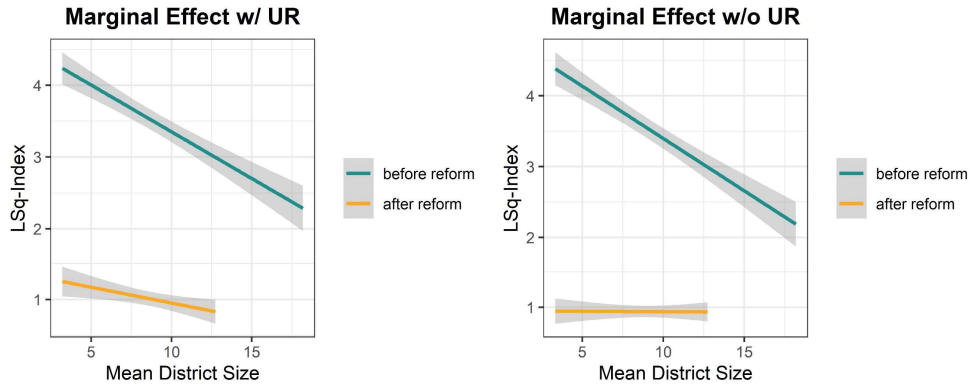


Figure 43: Marginal effects of reform on LSq-index by mean district size with (left panel) and without the Canton of Uri (right panel). The graphs show the locally weighted linear regression lines with 95% confidence intervals.

3.5.2 Proportionality as a catalyst for party system fragmentation?

The previous chapter demonstrated that, as anticipated, disproportionality decreased significantly following the implementation of the biproportional reform, albeit to varying degrees across the cantons that adopted it. This prompts an investigation into potential secondary effects, particularly the "effective number of legislative parties". The theoretical framework suggests a positive correlation between the proportionality of an electoral system and the number of effective legislative parties, leading to the hypothesis that the number of parties would increase post-reform (H5a). However, it is conceivable, and quite likely, that this effect may not manifest immediately but rather with a certain delay (H5b).

First and foremost, a gradual increase is evident in the overall trend of the effective number of parties (see Figure 44). While the effective number averaged well below four parties in the 1980s, it has since increased. Despite a brief period of stagnation just before the turn of the millennium, the positive trend has persisted. Examining individual cantonal trends (see Figure 45) confirms that this upward trend is supported by most cantons. Aargau, Schaffhausen, and Zurich show no or only a very weak increase, while Basel Stadt shows a decrease over time.

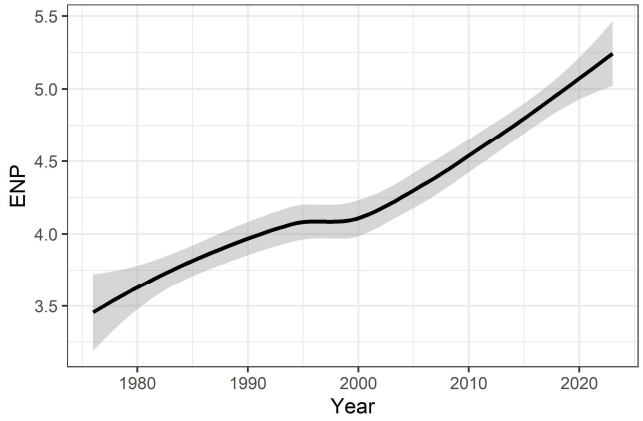


Figure 44: General time trend for the effective number of legislative parties for all cantonal parliamentary elections. The graph shows the locally weighted polynomial regression curve with 95% confidence intervals. AI is excluded due to missing data.

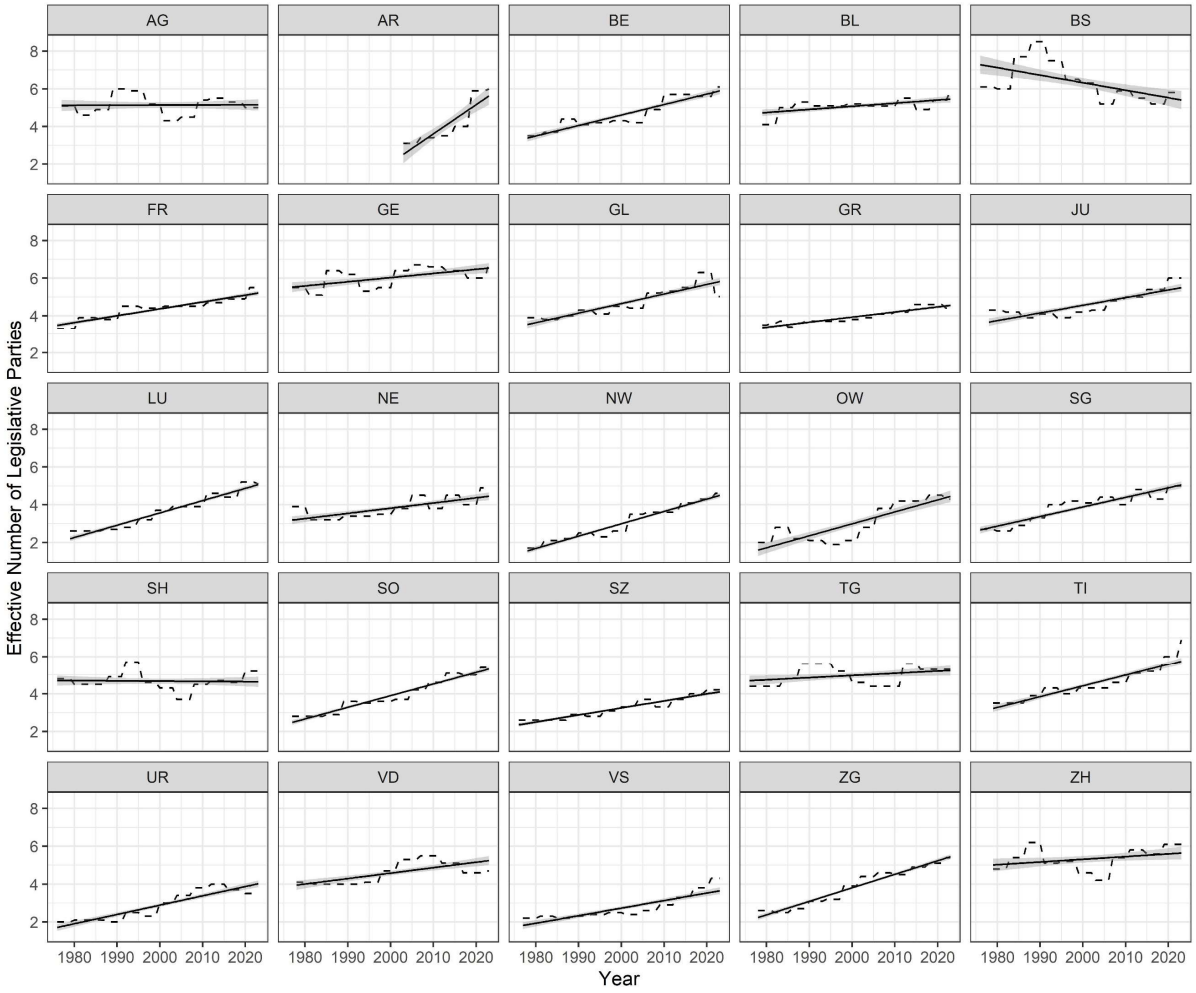


Figure 45: Trend in the effective number of legislative parties in the Swiss cantons. The dashed line shows the observed ENP values; the solid line shows the locally weighted linear regression line with 95% confidence intervals. AI is excluded due to missing data.

In his comprehensive assessment of the Swiss national and cantonal party systems, Ladner (2001: 127) concluded that "cantonal party systems become more alike, converging towards a medium effective number of parties between four and five (…)." While the descriptive graphs of cantonal trends support a gradual convergence in effective party numbers, Ladner's forecast appears somewhat conservative. The "steady increase" he had diagnosed (ibid. 2001: 125) did not plateau between an effective number of parties of four to five, as predicted, but clearly surpassed this range, and indicate a further rise towards an effective number of six parties. The evident question at this juncture is: Was this trend propelled by the cantonal biproportional reforms? Could Ladner's prediction have possibly materialized if these reforms had not been implemented in the nine cantons, which he could not possibly have foreseen?

To estimate the causal effect of biproportional reforms on the ENP, I use the following variables: the reform dummy as the binary treatment indicator, the least squares index, the range between the largest and smallest district in terms of seats, the turnout in cantonal parliamentary elections, and the number of candidacies in

cantonal elections as the covariates. The MSPE-based cross-validation procedure suggests using the interactive fixed effects counterfactual model (IFEct) and indicates three latent factors ($r = 3$).[94]

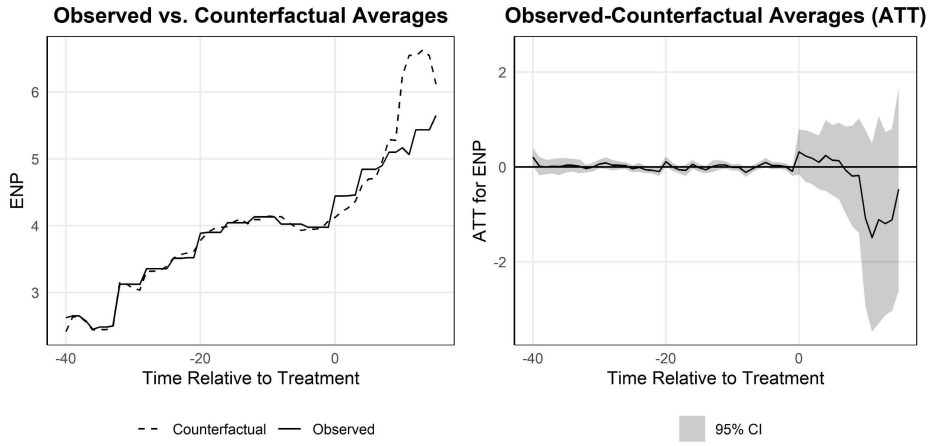


Figure 46: Observed and estimated average effective number of legislative parties (left panel) and the corresponding ATT for treated cases (right panel) before ($t < 0$) and after ($t \geq 0$) treatment/reform. The bar plot in the figures displaying the test diagnostics shows the number of cases treated per year.

Looking at Figure 46, the clear answer is "no". Although there is a slightly positive ATT immediately following the reforms, this trend is interrupted by a subsequent negative trend, which accounts for almost one effective party. The wide range of the confidence intervals, coupled with the fact that they never differ significantly from the zero ATT baseline during the entire post-treatment period, indicates that the estimated ATTs are not statistically significant (overall ATT ≈ -0.118, $p \approx 0.763$). The number of candidacies demonstrates a substantial effect on ENP, both in terms of the coefficient and statistical significance (see Table 44 in the appendix). In contrast, the other covariates do not exhibit strong effects.

Thus, the ongoing proliferation of parties in Swiss cantonal parliaments seems unrelated to the increased proportionality resulting from biproportional reforms. While this does not exclude potential reform effects within individual cantons, there is no evidence of a systematic cross-cantonal relationship. The lack of a consistent effect may be due to the quorum requirements in most cantons, which can significantly hinder party formation as well as the entry of small, non-partisan groups.

Figure 55 in the appendix shows that the null hypothesis of no differences cannot be rejected (p ≈ 0.182), indicating no significant differences between the placebo ATTs and the estimated ATTs. The placebo equivalence test is slightly above the 5% threshold ($p \approx 0.069$). Similarly, as shown in the right panel of the graph, the test for a potential pretrend yields mixed results. The high F test p-value suggests there is no statistically significant evidence of a problematic pretrend (null hypothesis = no pretrend). However, the p-value of ≈ 0.084 indicates that we cannot confidently assert that the pretrends are statistically equivalent. This ambiguity might be due to the relatively small sample size and/or the omission of other key covariates. Overall, the data indicates that there is no significant reform effect on the effective number of legislative parties. Thus, hypotheses 5a and 5b are rejected.

[94] The following formula applies: enp = $\beta_0 + \beta_1 \times$ ref_dummy + $\beta_2 \times$ lsq_index + $\beta_3 \times$ distr_range + $\beta_4 \times$ turnout_ce + $\beta_5 \times$ cand_num_ce + $L + \varepsilon$

3.5.3 Higher mapping accuracy as a stimulant for electoral turnout?

As demonstrated in the previous chapter, the significantly lower disproportionality values in cantonal biproportional elections do not have a systematic effect on the number of parties represented in parliament. How does this relate to voter turnout? Theoretically, improved prospects for an efficient allocation of votes should incentivize higher voter turnout (H6a). However, this effect is expected to manifest gradually over time (H6b), rather than immediately. This is because the mechanism is indirect, depending on voters' perceptions of and reactions to the electoral system's incentives. While the introduction of biproportional apportionment directly results in a more accurate reflection of the voters' will and enhances the chances of electoral success of smaller parties (and their voters, respectively), a broader effect on turnout is likely to emerge only over time as voters gain experience and adjust their perceptions of the electoral system and its incentives.

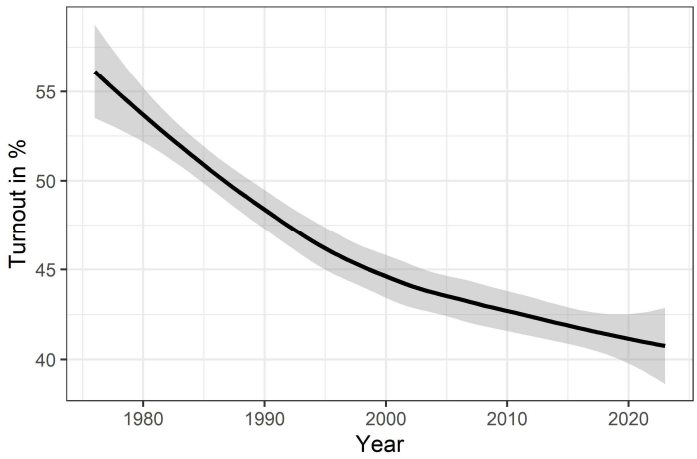


Figure 47: General time trend for the electoral turnout in cantonal parliamentary elections. The graph shows the locally weighted polynomial regression curve with 95% confidence intervals. AI and GR are excluded due to missing data.

The estimation of the causal effect of biproportional reforms on the electoral turnout includes following variables: the reform dummy as the binary treatment indicator, turnout in National Council elections, the number of candidacies in cantonal elections, the effective number of legislative parties, and the effective electoral threshold as the covariates. The MSPE-based cross-validation procedure suggests using the matrix completion method and indicates the presence of one latent factor (r = 1).[95]

As depicted in Figure 47, voter turnout has steadily declined across cantons over time. Although the rate of decline has slowed somewhat since the turn of the millennium, the trend persists. Before 1980, the average turnout was over 55%, but it has now fallen to around 40%. This widespread decline in cantonal election turnout has prompted authorities to introduce measures such as postal voting and free postage for postal votes, though these efforts have not achieved lasting success.[96]

[95] The following formula applies: $\text{turnout_ce} = \beta_0 + \beta_1 \times \text{ref_dummy} + \beta_2 \times \text{turnout_ne} + \beta_3 \times \text{cand_num_ce} + \beta_4 \times \text{enp} + \beta_5 \times \text{eff_threshold} + L + \varepsilon$. The turnout in cantonal votes (turnout_cv) was not included as an additional variable (as originally intended), as its inclusion significantly impairs the model's performance.

[96] Selb and Lachat (2004) demonstrated in their analysis of the Swiss national elections in 2003 that electoral turnout increased particularly in cantons where the option for "simplified postal voting" was provided (see also Lutz & Selb 2006: 442).

Figure 48 also highlights significant variations in voter turnout levels between cantons and a consistent decline in most. The decline is especially pronounced in cantons with traditionally high participation rates (e.g., Jura and Glarus), while cantons with generally low participation rates have experienced stagnation or only a slight decline. Schaffhausen presents a unique case, where voting is compulsory, and non-compliance incurs a symbolic fine of six Swiss francs. Despite this measure, voter turnout in Schaffhausen has also dropped by a substantial 20 percentage points over the past four decades.

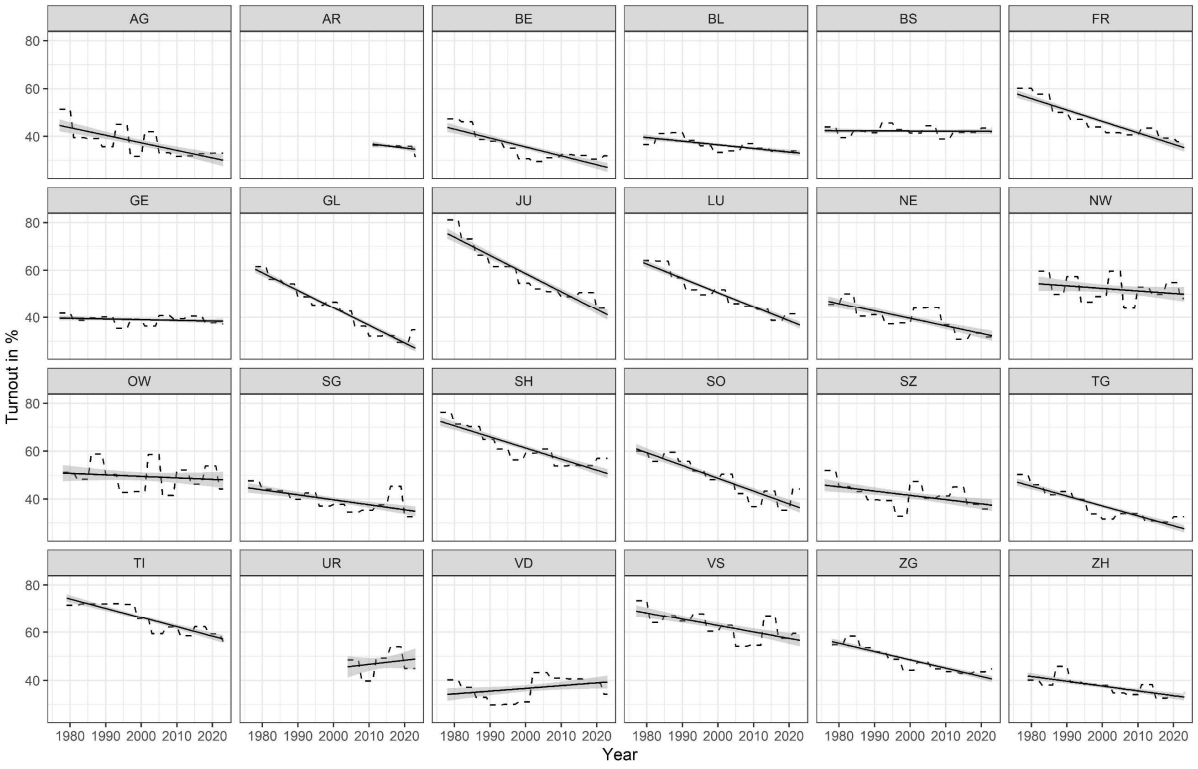


Figure 48: Time trend of turnout in cantonal parliamentary elections. The dashed line shows the observed turnout rates; the solid line shows the locally weighted linear regression line with 95% confidence intervals. AI and GR are excluded due to missing data.

The macro-level causes of this downward trend are manifold. A distinction must be made between general dynamics of alienation from institutional politics observed in many developed democracies, such as general political apathy, saturation, and overburdening effects, or socio-economic changes like increasing mobility on the one hand – and specific causes within the polity on the other hand (Freitag 2005). In the context of cantonal parliamentary elections, these include a certain relativization of cantonal identity, or the demise of local media outlets, leading to a reduced coverage of regional events and, consequently, to a depoliticization (Kübler & Goodman 2018). The question now is: would the decline in voter turnout have been even more pronounced in the reformed cantons had the biproportional reforms not taken place? Is there a positive effect on electoral turnout?

The MSPE-based cross-validation procedure suggests using the matrix completion method (MC). As Figure 49 illustrates, there is no evidence of a positive effect of biproportional reforms on voter turnout. In fact, an immediate negative effect appears in the first election following the new procedure, which stabilizes over

time and approaches the zero-effect line. The mean ATT is -4.455 ($p \approx 0.260$). However, this effect is statistically not significant at any point during the post-reform period, as the 95% confidence interval consistently overlaps the zero-effect line.

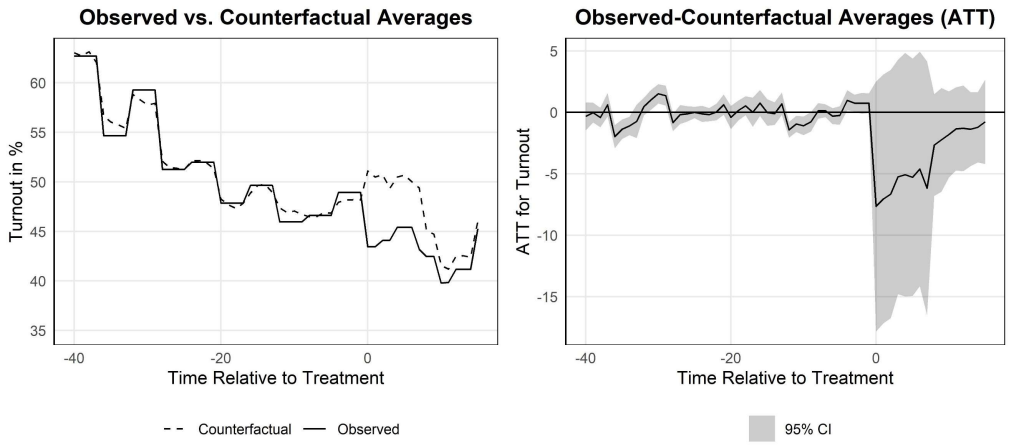


Figure 49: Observed and estimated average electoral turnout in cantonal parliamentary elections (left panel) and the corresponding ATT for treated cases (right panel) before ($t < 0$) and after ($t \geq 0$) treatment/reform. GR, NW, and UR are excluded from the counterfactual estimation due to missing data on the covariates. The bar plot in the figures displaying the test diagnostics shows the number of cases treated per year.

In terms of p-values, voter turnout in National Council elections is most strongly and positively correlated with voter turnout in cantonal parliamentary elections as shown in Table 45 in the appendix. The number of candidacies and the effective number of parties show a negative correlation with cantonal voter turnout, which is unexpected given that electoral competition and a fragmented party system typically promote political participation according to the literature. On the other hand, the negative relationship between a higher effective threshold and voter turnout aligns with theoretical expectations.

Upon examining the robustness tests (Figure 56) and comparing them with previous tests for other outcome variables, no placebo effect or problematic pretrend is observed. However, it is evident that the residual averages in the pretreatment period are not statistically equivalent to zero, as indicated by the high p-values from the TOST equivalence tests. Therefore, despite the absence of indications of problematic pretrends in the tests, the data used for estimation is not sufficient for robust effect estimation. For a more precise effect estimation, integrating additional covariates into the model is required. However, identifying supplementary variables for the entire period under study that effectively covary with electoral turnout beyond those already included poses a challenge. Overall, the model does not provide evidence of a significant positive reform effect on cantonal voter turnout, leading to the rejection of hypotheses 6a and 6b. However, the evaluation indicates a dynamic effect over time, albeit with a high degree of statistical uncertainty.

3.5.4 Biproportionality and its impact on electoral competition

Thus far, no significant positive reform effect has been found on either the effective number of parties or political participation. Initially, a positive effect on political competition was also anticipated, but these preliminary findings cast doubt on a possible effect. If biproportional reforms did not boost participation and did not substantially alter the number of parties, it would be a rather surprising finding if political competition as a variable related to the other outcome variables showed strong positive effects. Nonetheless, the

assumption is that the reforms positively influence the number of candidacies (H7a). Due to further technical adjustments in the course of reform, such as the abolition of the possibility of apparentements and the requirement for uniform list group labels, an immediate decrease in the number of lists post-reform is anticipated. A gradual positive effect on the number of lists is expected over time (H7b).

If we look at Figure 50, we can observe a continuous increase in both the number of candidacies and the number of electoral lists before the turn of the millennium. However, the increase in the number of candidacies halts before the millennium and, on average, even slightly decreases. From the turn of the millennium, the two trends decouple to some extent. While the number of candidacies shows a strong and stable increase, the number of lists stagnates.

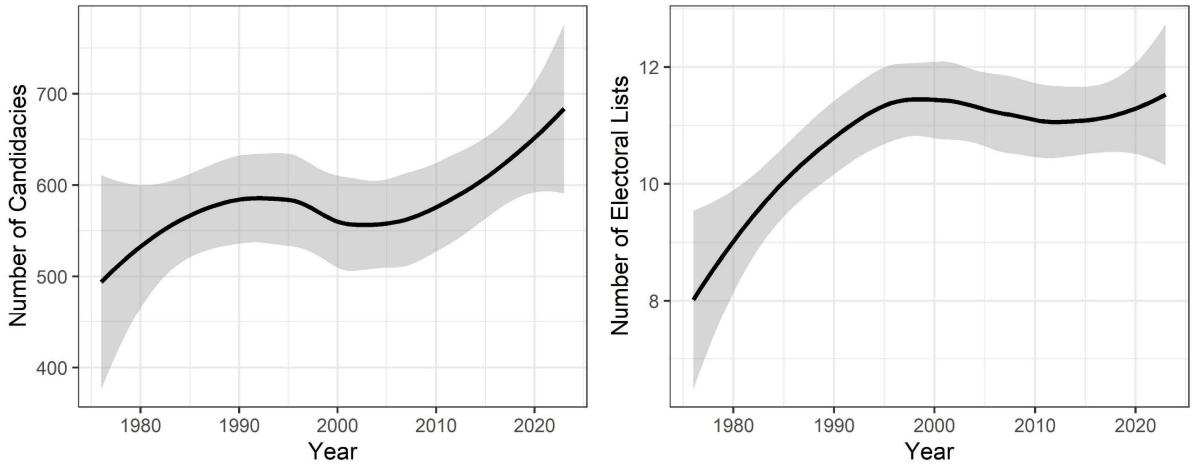


Figure 50: General time trend for the electoral competition in cantonal parliamentary elections. The graph shows the locally weighted polynomial regression curve with 95% confidence intervals for the number of candidacies (left panel) and the number of electoral lists (right panel). AI and GR are excluded due to missing data.

Why do these two figures not correlate perfectly with each other? The reason is that parties do not have to fill all list places with candidates when drawing up their party lists. For instance, in a constituency with ten seats (i.e., a maximum of ten list places), a party can field as few as one candidate. If these figures correlated perfectly, the total number of candidacies divided by the number of electoral lists should roughly equal the number of total seats, i.e., the parliament size. Thus, if we divide the ratio between the number of candidacies and electoral lists by parliament size, the resulting value should theoretically be close to one.

Figure 51 shows the actual ratio, categorized by reform status. The observed values reveal a distribution with a median below one and lower for non-reformed cases. This indicates that for a significant share of lists, not all available spots are being filled. The figure also shows a higher ratio for reformed cases, suggesting that electoral lists submitted post-reform have fewer empty spots. Further deviations can be explained by the specific operationalization of the number of lists, which adjusts the total for strategic submissions in elections with many sub-lists, providing an approximation of the "effective number" of lists. If the actual number of lists were used as a basis, the ratio would be even lower than shown.

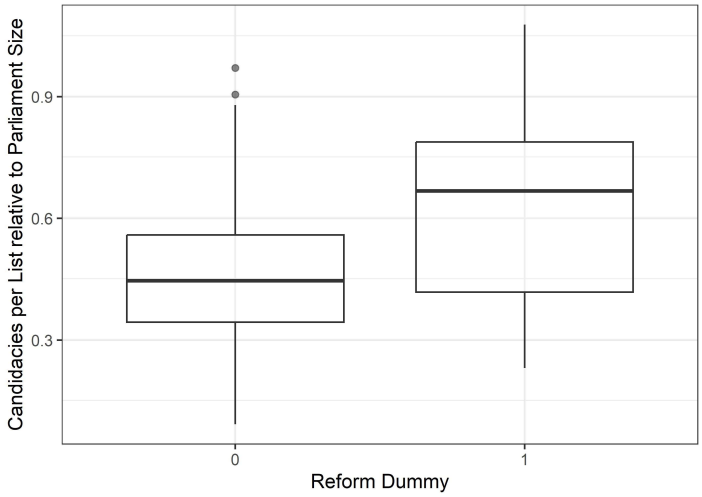


Figure 51: Total number of candidacies per electoral list, adjusted for parliament size, categorized by reform status (0 = pre-reform or no reform, 1 = post-reform). The formula used for the variable on the y-axis is (cand_num_ce / lists_num) × (1 / parl_size).

How is political competition developing in the cantons? Figure 52 illustrates trends for both the number of candidacies and the number of lists. There are significant differences across cantons. In smaller, rural cantons with traditionally small parliaments (e.g., Appenzell Ausserrhoden, Obwalden, Schwyz), political competition remains relatively stable and at a lower level over time. Conversely, in more populous cantons, there is generally higher political competition and greater dynamics over time.

The trends in the two largest cantons by population, Bern and Zurich, are particularly notable. In Bern, the number of electoral lists submitted is relatively low at around 20, yet the number of candidacies has nearly doubled over the past 40 years. This sharp increase is striking, as no other canton has experienced a similar rise within the same timeframe. Since cantonal elections have so far been largely neglected in Swiss election campaign research, the causes can only be speculated upon. It is conceivable that due to the strong mediatization and politicization of national elections, along with the significant increase in submitted lists and candidacies for National Council elections over the past five decades (Lutz & Selb 2006: 431-432) and considering that Bern is the seat of the national parliament, spillover effects may have occurred in cantonal elections in Bern and spurred cantonal competition. However, this remains just an assumption.

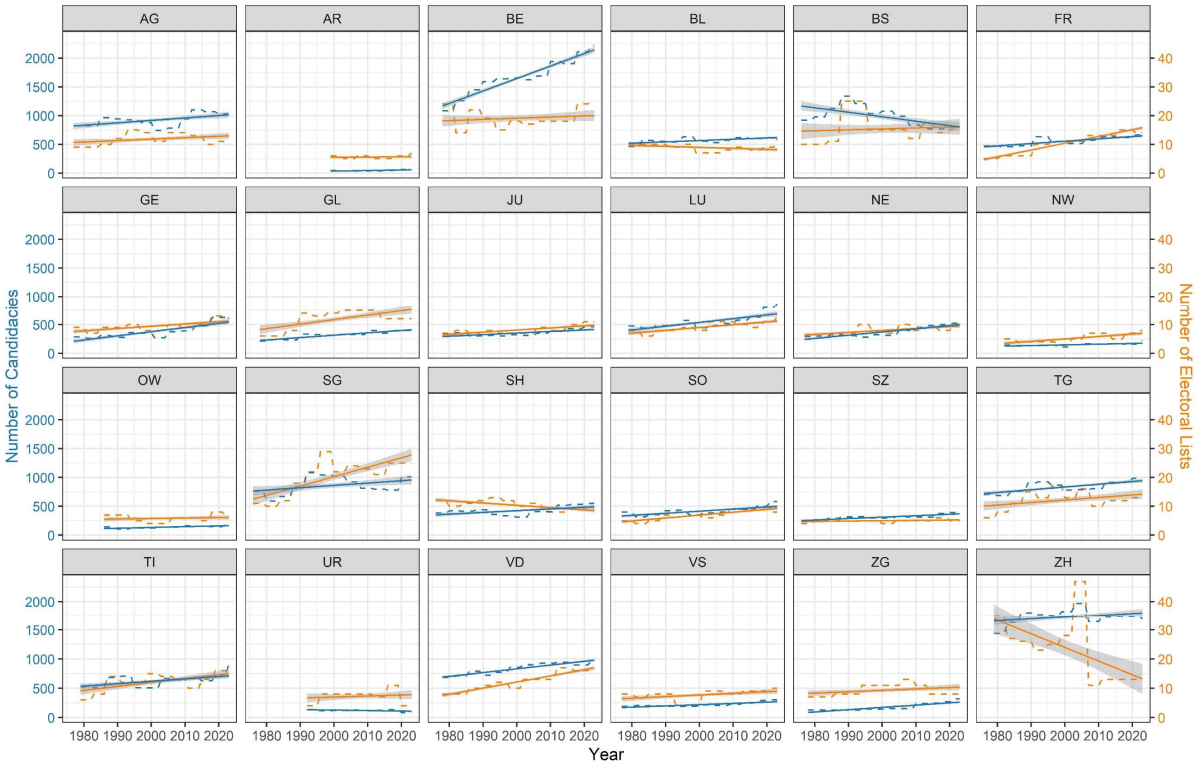


Figure 52: Time trends of number of candidacies (y-axis on the left) and electoral lists (y-axis on the right) in cantonal PR-based parliamentary elections. The dashed lines show the observed values; the solid lines show the locally weighted linear regression lines with 95% confidence intervals. AI and GR are excluded due to missing data.

In contrast, Zurich has seen a relatively stable number of candidacies, ranging between 1500 and 2000. However, the number of lists dropped sharply after 2007, coinciding with the introduction of biproportional apportionment. This decline is somewhat mitigated by the fact that the number of lists during the 2003-2007 legislative period was exceptionally high and far above the cantonal average. Nevertheless, the sharp decline in the number of lists is likely due to the reform that abolished apparentements and mandated uniform list group labels. A similar, though less pronounced, trend is observed in other reformed cantons such as Aargau, Uri, Zug, and Schaffhausen. In contrast, no effect is evident in the cases of Schwyz, Valais, and Nidwalden, possibly because the number of lists in these cantons was already relatively low before the reform, ranging between five and ten lists.

The question now arises whether the biproportional apportionment significantly increased political competition, measured by the number of candidacies and electoral lists. To estimate the causal reform effect, the following variables are included: the reform dummy as the binary treatment indicator, the effective number of legislative parties, voter turnout in cantonal elections, the range between the minimum and maximum district sizes, the least squares index, the number of candidacies for the National Council elections, and the effective electoral threshold as covariates.[97] For the first model, where the outcome variable is the number of candidacies, the MSPE-based cross-validation procedure suggests using the interactive fixed-effects

[97] The following formula applies: cand_num_ce = β_0 + β_1 × ref_dummy + β_2 × enp + β_3 × turnout_ce + β_4 × distr_range + β_5 × lsq_index + β_6 × cand_num_ce + β_7 × eff_threshold + L + ε. The second formula for the outcome variable lists_num is identical.

counterfactual model (IFEct) with two latent factors (r = 2). For the second model, with the number of electoral lists as the outcome variable, the cross-validation procedure favors the matrix completion method.

Figure 53 illustrates the reform effects on candidacies in the top two panels. In the upper left panel, the number of candidacies in reformed cantons has risen sharply since the reform – from around 500 before the reform to just over 1000 in the most recent post-reform period. However, the increase would have been even more pronounced without the reform, resulting in a negative ATT, as shown in the upper right panel. The mean ATT for the entire post-reform period is -162.6, with a p-value of 0.262. This suggests that, on average, around 163 more people would have stood for election without the reform, holding all other factors constant. However, this effect is subject to high statistical uncertainty.

The lower two panels present an even clearer picture regarding the number of lists: the number of submitted electoral lists decreased by an average of 6.71 (ATT) with a p-value of 0.063. This effect is relatively consistent over time but shows a stronger negative trend in the most recent post-reform years. The coefficients for all variables for both outcome variable are shown in Table 46 in the appendix.

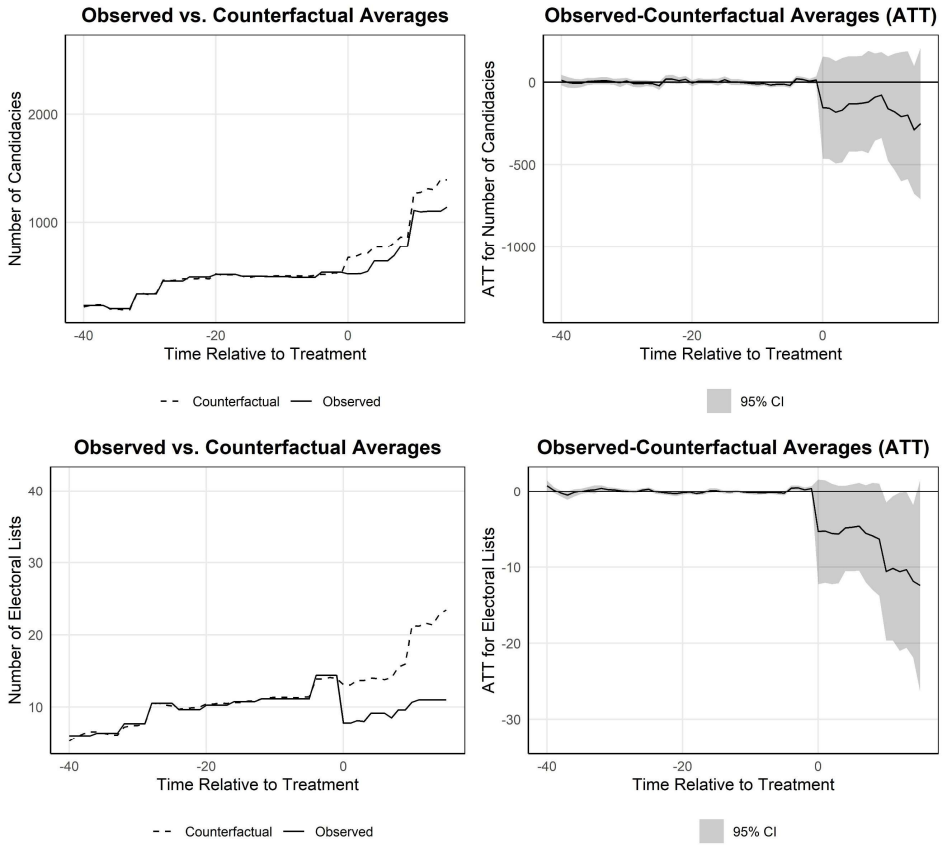


Figure 53: Observed and estimated average electoral competition in cantonal parliamentary elections (left panels) and the corresponding ATT for treated cases (right panels) before (t < 0) and after (t ≥ 0) treatment/reform. AI, GE, TI, and UR are excluded from the counterfactual estimation due to missing data on the covariates. The bar plot in the figures displaying the test diagnostics shows the number of cases treated per year.

There is a strong, positive statistical correlation between the two outcome variables and the covariates ENP and the number of candidacies in the respective canton in the National Council elections. In the first model, which considers the number of candidacies, the disproportionality index is also positively significant at the 5% level. All other covariates fall outside the range of statistical significance.

Upon examining the robustness tests (Figure 57 in the appendix) and comparing them with previous tests for other outcome variables, no placebo effect or problematic pretrend is observed. However, the residual averages in the pretreatment period deviate significantly from zero, as indicated by the high p-values from the TOST equivalence tests, especially regarding the reform effect on the number of candidacies. Although the tests do not suggest problematic pretrends, the model could benefit from incorporating more and better-correlated covariates.

Overall, the model does not indicate a positive reform effect on electoral competition, whether measured by the number of candidacies or the number of electoral lists. In fact, the results suggest that biproportional reforms might reduce electoral competition, leading to the rejection of hypothesis 7a, which assumed a positive effect. Conversely, the results support hypothesis 7b, positing a possible negative and immediate reform effect on the number of electoral lists. After the reform, the average number of lists dropped by about five, further decreasing to below ten lists in the last post-treatment period, with p-values below the 10% threshold.

The significant decrease in the number of electoral lists post-reform can likely be attributed to the reform's design, which regulates electoral competition in terms of electoral lists. Parties can no longer split their main list into multiple, differently labeled sub-lists to strategically target and mobilize voters based on specific demographics. Although parties can technically still separate a branch and run it as an independent list, empirical evidence does not support the existence of such behavior. Fractionalizing a party increases the risk of losing non-integer seat entitlements to rival parties, even with standard rounding. As a unified party, the risk of yielding a quotient of less than 0.5 occurs only once, but this risk increases with each additional split. Party strategists are likely aware of this. Given that all lists within electoral districts must conform to the same canton-wide label to be treated as a list group in the super-apportionment of seats, there is an incentive for a cross-district presence of competing parties. The potential benefit of "electoral omnipresence" outweighs the costs, even if the additional votes gained from the extra list in a district contribute only marginally to the overall result. This shift from focusing on individual districts to targeting the canton-wide electorate may, over time, lead to a "de-regionalization" or "cantonalization" of the entire party system. However, further research is needed to investigate this causal link.

3.6 Conclusion

The third part of this dissertation aimed to examine the impact of reforms on cantonal politics. The focus was on whether and how biproportional reforms influence macro-level characteristics such as voter turnout, political competition, and the effective number of political parties. The hypothesis posits that lowering the threshold implied by the constituency structure contributes to a significant increase in vote-seat proportionality. This increased mapping accuracy of the electoral system is expected to benefit marginalized voters, particularly those supporting smaller fringe parties. The assumption is that these voters are encouraged to participate more actively, either by voting when they had previously abstained or by voting more sincerely instead of settling for their second preference. Based on relevant theory, it is expected that the increased proportionality translates to a "democratic surge": higher participation rates, greater political competition, and an increase in the number of effective parties in parliaments.

The analysis shows that vote-seat proportionality significantly decreases, nearly halving on average post-reform. This effect is both immediate and consistent over time, with the proportionality increase being more

pronounced in cantons with smaller average constituency sizes. Biproportionality thus ensures a high degree of accuracy, largely independent of prior constituency structures.

Regarding secondary effects presumed to result from reduced disproportionality, the study presents mixed findings. There is no evidence that the reform significantly increases the number of effective parties. The number of parties in the cantons continues to rise, irrespective of whether a reform occurs. For political participation, a direct negative effect of the reform is observed; however, this effect is not strong and diminishes over time. A statistically insignificant negative effect on the number of candidacies is noted for political competition. The effects on the number of lists are more pronounced, with the reforms clearly leading to a decline in the number of electoral lists submitted. Thus, biproportional reforms and the improved chances of fair electoral representation do not necessarily stimulate political competition. The negative effect on the number of lists can be explained by specific reform provisions that create incentives to consolidate electoral lists at the cantonal level and encourage their presence in as many constituencies as possible. This example highlights how even seemingly minor technical changes can carry significant strategic implications for political parties, which are forced to navigate the efficient use of their limited resources.

The results indicate that political institutions, such as electoral systems, engage in a complex, reciprocal interaction with voters and party strategists. Party strategists, well-versed in the technical details of new electoral rules, anticipate political effects and react accordingly. Conversely, the average voter lacks such specialized knowledge and resources, which may limit awareness of how the reform affects their personal options and the prospects of success of these options. The absence of a democratic surge, despite improved proportionality, could be attributed to various factors, including:

- Voters are unaware of the improved proportionality.
- Voters recognize the improved proportionality but do not significantly alter their behavior.
- The reform includes elements that counteract the expected behavioral changes.
- The secondary effects may only become evident after a longer period.

At this point, I highlight some limitations of the analysis. Firstly, the number of cases is rather small for a large-N study. Although the methodology used can handle low case numbers and complex, even incomplete data structures, a higher number of cases would yield more robust results. However, increasing the number of cases is feasible only in a context beyond the Swiss case. Since the reforms in Switzerland have a very specific and unique design, extending the study to other, similar electoral reforms might face criticisms of lack of comparability. Instead of expanding the number of cases, increasing the number of observations, such as including additional election years, would also enhance robustness. Since this study has exhausted all available data, such optimizations in the longitudinal dimension will only be possible in the future with new electoral data.

Another limitation is that some non-reformed cases used for the counterfactual estimation partly respond to the FSC ruling by optimizing proportionality through alternative reforms. Additionally, other institutional reforms, such as parliamentary reductions, took place in numerous cantons. While such concurrent reforms limit the validity of the study, considering all these circumstances would significantly complicate the study design and might ultimately require the exclusion of some cases. An avenue for optimization could involve incorporating additional covariates into the analysis, particularly exogenous variables. However, the

process of data collection has revealed significant challenges in identifying suitable variables. These challenges arise from both data availability constraints and theoretical complexities, particularly in finding variables that conform to strict exogeneity.

In summary, the biproportional reforms resulted in immediate, albeit primarily mechanical, effects. However, the anticipated psychological response from voters did not materialize as expected, leading to a lack of significant political effects. This outcome is disappointing for those who anticipated that the reforms would invigorate cantonal politics. Conversely, it offers reassurance to those concerned about potential drawbacks such as an excessive fragmentation of legislative parties and the alleged inefficiencies in parliamentary work that could result from it.

4. Concluding remarks

In this final chapter, I synthesize the three parts of the dissertation to draw a cross-chapter conclusion of the thesis. This study represents the first comprehensive analysis of the biproportional apportionment method in Switzerland. I assess three central questions: How do political parties involved in the reform process affect the outcome, and what patterns emerge? How do conflict patterns during decision-making shape the final reform design? And how do the biproportional reforms impact cantonal political landscapes?

The analysis of political parties' strategic behavior highlights a critical aspect of electoral reform: the interplay of power and interest. The study confirms the hypothesis that parties benefiting from the existing electoral system tend to resist reforms, while those disadvantaged are more likely to advocate for change. This finding aligns with broader theories of institutional change, which posit that established beneficiaries often oppose alterations to maintain their advantages (Lipset & Rokkan 1968, Tsebelis 1990, Boix 1999, Rokkan 2009, Mahoney & Thelen 2010). However, the research also uncovers deviations from this pattern. For instance, certain parties, despite being advantaged by the current system, support reforms due to legal pressures, political ideals, or a commitment to democratic principles. These deviations suggest that while strategic interests play a significant role, they are not the sole determinants of party behavior during the reform process.

The study's examination of conflict patterns within the decision-making process reveals that the nature and intensity of these conflicts are pivotal in shaping the final design of the biproportional reform. Contrary to the traditional emphasis on consensus as a desirable outcome in policy-making, the research indicates that conflict can play an equally important role in driving substantive changes. The conflicts between political actors, especially between those advantaged and those disadvantaged by the existing laws, are instrumental in achieving a more equitable outcome. This illustrates how political contention, rather than consensus, can drive substantive change and lead to more equitable outcomes, ultimately aligning reforms more closely with voter preferences and democratic ideals.

Evaluating the political effects of the biproportional reforms shows that biproportionality significantly enhances vote-seat proportionality, resulting in more representative electoral outcomes – a valuable achievement in its own right. However, some anticipated secondary effects do not materialize as expected. The expected increases in voter turnout, political competition, and the number of effective political parties are limited. Instead, the reforms lead to a decrease in the number of electoral lists submitted, suggesting that political parties are fast in responding to the new guidelines and adapt their electoral strategies accordingly. This finding underscores the complex strategic implications of electoral reforms, as parties navigate these changes to enhance their positions. Additionally, the study indicates that the expected psychological impact of the reforms on voters – intended to boost engagement and trust in the electoral process – is somewhat muted. This limited effect may be attributed to factors such as insufficient public awareness or understanding of the reforms, pointing to a need for improved communication and education about electoral changes.

Several limitations of this study should be acknowledged. The focus on the biproportional method within Switzerland means that the findings may not be fully applicable to other countries or electoral systems with different political contexts and institutional structures. This concentration limits the generalizability of the results and suggests the need for comparative studies. Furthermore, the analysis relies on data from cantonal

elections where the biproportional system has been in place since 2005. This temporal limitation may constrain the study's ability to capture long-term effects and evolving trends in voter behavior and party strategies.

Future research could address these limitations by exploring the long-term political effects of biproportional apportionment. Investigating how the method influences party strategies, voter behavior, and political stability over time would provide a more in-depth understanding of its impact. Comparative analyses with other PR-based systems used globally could offer valuable insights into the relative advantages and disadvantages of the biproportional method, potentially informing the adaptation and implementation of similar systems in other electoral contexts.

Additionally, to gain more nuanced insights, it would be valuable to examine voter and party behavior at a more granular level, such as within individual constituencies. For instance, how does the voting behavior of minority party supporters in single-member districts in the canton of Graubünden change? Are fringe parties now more inclined to field candidates in such districts? Are former abstainers motivated to vote because their vote now carries more weight? Do previous strategic voters now cast their ballots according to their first preference?

Furthermore, do the procedural changes contribute to the centralization of parties' electoral efforts, as some political actors have speculated? If so, does this pooling of resources at the cantonal level undermine the practical relevance of regional party sections, possibly even leading to their disappearance as feared by some? Exploring these and other questions would offer fascinating insights into the manifold consequences of institutional reforms.

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6. Appendix

Canton[98]	Electoral System	Apportionment Method	Statutory Quorum
Appenzell Innerrhoden	Majoritarian	-	-
Appenzell Ausserrhoden	Mixed	MR / Hagenbach-Bischoff	-
Aargau	Proportional	Biproportional	5% local or 3% global
Basel Stadt	Mixed	MR / Sainte-Laguë	-
Basel Landschaft	Proportional	Hagenbach-Bischoff	-
Bern	Proportional	Hagenbach-Bischoff	-
Fribourg	Proportional	Hagenbach-Bischoff	-
Geneva	Proportional	Hagenbach-Bischoff	7% global
Glarus	Proportional	Sainte-Laguë	-
Graubünden	Proportional	Biproportional	3% global
Jura	Proportional	Hagenbach-Bischoff	-
Lucerne	Proportional	Hagenbach-Bischoff	-
Neuchâtel	Proportional	Hagenbach-Bischoff	3% global
Nidwalden	Proportional	Biproportional	-
Obwalden	Proportional	Hagenbach-Bischoff	-
Schwyz	Proportional	Biproportional	1% global
Schaffhausen	Proportional	Biproportional	-
Solothurn	Proportional	Hagenbach-Bischoff	-
St. Gallen	Proportional	Hagenbach-Bischoff	-
Ticino	Proportional	Hare-Niemeyer	1.11% (Hare quota)
Thurgau	Proportional	Hagenbach-Bischoff	-
Uri	Mixed	MR / Biproportional	-
Vaud	Proportional	Hare-Niemeyer	5% local
Valais	Proportional	Biproportional	8% local
Zug	Proportional	Biproportional	5% local or 3% global
Zurich	Proportional	Biproportional	5% local or 3% global

Table 38: Main institutional features of electoral systems for cantonal parliamentary elections.

[98] Canton names according to the English version of the Swiss Federal Constitution (1999).

Label	German Denomination	Translation	Remarks
BDP	Bürgerlich-demokratische Partei	Conservative Democratic Party of Switzerland	In 2021, CVP and BDP merged together to a new party labelled "Die Mitte". Since not all cantonal sections performed the merger (also due to the absence of the merging partner BDP in some cantons) and to guarantee data consistency, the new party label is disregarded.
CSP	Christlichsoziale Volkspartei	Christian Social People's Party	
CVP	Christlichdemokratische Volkspartei	Christian Democratic People's Party of Switzerland	See remarks above (BDP).
DN	Demokratisches Nidwalden	Democratic Nidwalden	Party confined to the Canton of Nidwalden.
DSP	Demokratisch-soziale Partei	Democratic Social Party	Party confined to the cantons of Basel Stadt, Fribourg, and Graubünden.
EDU	Eidgenössisch-demokratische Union	Federal Democratic Union of Switzerland	
EVP	Evangelische Volkspartei	Evangelical People's Party of Switzerland	
FDP	Freisinnig-demokratische Partei / Die Liberalen	Free Democratic Party of Switzerland	In 2009, FDP and FPS merged together to a new party labelled "FDP.Die Liberalen". To keep things simple, it will be labelled as FDP.
FGA	Feministische und grün-alternative Gruppierungen	Feminist-Green-Alternative Groups	
FPS	Freiheits-Partei der Schweiz	Freedom Party of Switzerland	Temporarily operated under the name «Auto-Partei» (party that advocates for motorists). Disappeared in 2009.
GLP	Grünliberale Partei	Green Liberal Party	
GPS	Grüne Partei Schweiz	Green Party of Switzerland	
LDU	Landesring der Unabhängigen	Alliance of Independents	Disappeared in 2003.
LPS	Liberale Partei der Schweiz	Liberal Party of Switzerland	
PDA	Partei der Arbeit	Swiss Party of Labour	
POCH	Progressive Organisationen der Schweiz	Progressive Organizations of Switzerland	Disappeared in 1998.
SD	Schweizer Demokraten	Swiss Democrats	
SVP	Schweizerische Volkspartei	Swiss People's Party	
Others	Andere Parteien	Other parties	Group containing all parties/MPs who do not belong to any of the above-mentioned parties (including independents).

Table 39: Overview of the relevant parties in the Swiss cantons. [99]

[99] The labels are based on the classification provided by the FSO (2023d). The "Rest" group was relabeled to "Others".

Table 40: Overview over the primary sources for the qualitative assessment of cantonal reform processes.

Canton	Institution/Instrument	Content	Document type	Date (m-d-y)	Pages	Remarks
Zurich	Government Council	Proposal for the revision of the law on political rights	Draft bill and comments	08-28-2002		
	STGK Commission	Proposal for the revision of the law on political rights	Draft bill	03-07-2003		1st part
	Cantonal Council	Revision of the law on political rights, parliamentary debate	Verbatim minute	04-07-2003	16374 – 16417	1st reading
	Drafting Committee	Proposal for the revision of the law on political rights	Draft bill	05-14-2003		
	STGK Commission	Proposal for the revision of the law on political rights	Draft bill	05-16-2003		2nd part
	Cantonal Council	Revision of the law on political rights, parliamentary debate	Verbatim minute	09-01-2003	891 – 904	2nd reading
	Cantonal Council	Revision of the apportionment method, parliamentary debate	Verbatim minute	09-01-2003	905 – 943	1st reading
	Cantonal Council	Revision of the apportionment method, parliamentary debate	Verbatim minute	17/11-2003	1761 – 1775	2nd reading
	Parliamentary Initiative	Requesting the introduction of a global 3% quorum	Written request	03-21-2016		PI 110/2016
	STGK Commission	Counterproposal for the introduction of a global 3% quorum	Draft bill	06-29-2018		KR-No. 110a/2016
	Cantonal Council	Revision of the quorum regulation, parliamentary debate	Verbatim minute	08-17-2020		1st reading
	Cantonal Council	Revision of the quorum regulation, parliamentary debate	Verbatim minute	11-30-2020		2nd reading
	Parliamentary Initiative	Requesting the correction of the apportionment formula	Written request	04-23-2018		PI 118/2018
	STGK Commission	Proposal for a modified apportionment formula	Draft bill and comments	02-07-2020		
	Drafting Committee	Proposal for a modified apportionment formula	Draft bill	03-04-2021		
	Cantonal Council	Revision of the apportionment formula, parliamentary debate	Verbatim minute	03-01-2021		1st reading
	Cantonal Council	Revision of the apportionment formula, parliamentary debate	Verbatim minute	03-22-2021		2nd reading
Aargau	Government Council	Proposal for the revision of electoral law	Commentary	09-27-2006		
	Grand Council	Revision of electoral law, parliamentary debate	Verbatim minute	03-20-2007	2092 – 2101	1st reading
	Grand Council	Revision of electoral law, parliamentary debate	Verbatim minute	09-18-2007	2581 – 2594	2nd reading
	Grand Council	Implementation of an electoral quorum, parliamentary debate	Verbatim minute	03-01-2011	2447 – 2457	1st reading
	Grand Council	Implementation of an electoral quorum, parliamentary debate	Verbatim minute	06-07-2011	2929 – 2937	2nd reading
Schaffhausen	Parliamentary Motion	Requesting a better electoral protection of small parties	Written request	08-30-2004		
	Parliamentary Motion	Requesting a general reform of electoral law	Written request	11-17-2004		
	Government Council	Proposal for the revision of the electoral law	Draft bill and comments	04-24-2007		
	Cantonal Council	Revision of electoral law, parliamentary debate	Verbatim minute	09-03-2007	658 – 666	1st reading
	Cantonal Council	Revision of electoral law, parliamentary debate	Verbatim minute	09-17-2007	684 – 701	1st reading (cont.)
	Cantonal Council	Revision of electoral law, parliamentary debate	Verbatim minute	10-29-2007	832 – 845	2nd reading
	Cantonal Council	Revision of electoral law, parliamentary debate	Verbatim minute	10-26-2015	750 – 758	
Nidwalden	Government Council	Proposal for the revision of the electoral law	Draft bill and comments	11-23-2010		1st draft
	Government Council	Proposal for the revision of the electoral law	Draft bill and comments	09-??-2011		2nd draft

	Government Council	Proposal for the revision of the electoral law	Draft bill and comments	02-07-2012		3rd draft
	Cantonal Council	Revision of electoral law, parliamentary debate	Verbatim minute	03-28-2012	713 – 731	1st reading
	Cantonal Council	Revision of electoral law, parliamentary debate	Verbatim minute	04-25-2012	745 – 746	2nd reading
Zug	Cantonal Council	Parliamentary debate on the request of the minority of the Advisory Commission to introduce biproportional apportionment	Verbatim minute	07-06-2006	2017 – 2031	
	Government Council	Proposal for the revision of the partial revision of the cantonal constitution and the revision of electoral law	Written request	07-10-2012		
	Advisory Commission	Proposal for the revision of the partial revision of the cantonal constitution and the revision of electoral law	Written request	11-08-2012		
	Cantonal Council	Partial revision of the cantonal constitution and revision of the electoral law, parliamentary debate	Verbatim minute	01-31-2013	1345 – 1363	1st reading
	Editorial Commission	Adjusted version of the draft according to the first parliamentary reading	Written request, comments, and synopses	04-09-2013		
	Cantonal Council	Partial revision of the cantonal constitution and revision of the electoral law, parliamentary debate	Verbatim minute	05-02-2013	1572 – 1577	2nd reading
	Cantonal Council	Partial revision of the cantonal constitution and revision of the electoral law, parliamentary debate	Verbatim minute	05-02-2013	1579 – 1597	2nd reading (cont.)
Schwyz	Popular Initiative	The SVP requests the introduction of plurality voting for cantonal parliamentary elections	Initiative text, draft constitution	08-30-2013		
	Popular Initiative	A multipartisan committee requests the introduction of a "fair" PR-based electoral system for cantonal parliamentary elections	Initiative text, draft constitution	10-17-2013		
	Government Council	Statement on both popular initiatives	Commentary	06-17-2014		RRB 660/2014
	Government Council	Statement on the advisory commission's counsel	Commentary	10-21-2014		RRB 1093/2014
	Cantonal Council	General debate and plenary debate on the popular initiative submitted by the SVP (plurality voting)	Verbatim minute	11-19-2014	980 – 1010	1st reading
	Cantonal Council	Plenary debate on the popular initiative submitted by the multipartisan committee (undistorted PR voting)	Verbatim minute	12-17-2014	1028 – 1038	2nd reading
	Government Council	Explanations of the Government Council on the cantonal votes of March 8, 2015	Voting brochure	03-08-2015		
Valais	Popular Initiative	A multipartisan committee successfully submits the initiative "Every Vote Counts", which calls for the introduction of biproportional apportionment	Initiative text, draft constitution	12-06-2010		
	Extra-parliamentary ad-hoc commission	Report of the commission on the reforms of the territory and institution in the Canton of Valais in the 21st century (short: R21)	Report	10-03-2012		
	Cantonal Council	Partial revision of the cantonal constitution, parliamentary debate	Verbatim minute	09-08-2014	15 – 33	1st reading
	Cantonal Council	Partial revision of the cantonal constitution, parliamentary debate	Verbatim minute	03-12-2015	21 – 40, 111 – 160	2nd reading
	Government Council	Explanations of the Government Council on the cantonal votes of June 14, 2015	Voting brochure	06-14-2015		
	Government Council	Explanations of the Government Council on the cantonal votes of November 29, 2015	Voting brochure	11-29-2015		
	Government Council	Request for the approval of a decree concerning the electoral system for cantonal parliamentary elections	Draft bill and comments	12-17-2015		
	Cantonal Council	Approval of the decree amending the Law on Political Rights	Decree	03-09-2016		
	Cantonal Council	Partial revision of the Law on Political Rights, parliamentary debate	Verbatim minute	12-14-2017		1st (and only) reading
Uri	Government Council	Report and proposal concerning the cantonal popular initiative "Election of heads instead of parties"	Report and request	04-03-2012		Nr. 2012-227 R-362-13
	Parliamentary Motion	Parliamentary motion "for a future-proof electoral system for the Cantonal Council of Uri"	Parliamentary motion	11-16-2016		LA.2016-0856
	Cantonal Council	Revision of electoral law, parliamentary debate	Minute of decisions	03-21-2018	5 – 6	
	Government Council	Report and proposal concerning the revision of the electoral law	Report and request	01-08-2019		LA.2017-0413
	Government Council	Consultation report	Report	01-08-2019		

	Judicial Commission	Statement and amendments to the government's proposal	Written request	01-28-2019		LA.2017-0413
	Cantonal Council	Revision of electoral law, parliamentary debate	Audio recording (the audio files were transcribed)	02-13-2019		1st (and only) reading
	Government Council	Explanations of the Government Council on the cantonal votes of May 19, 2019	Voting brochure	05-19-2019		
Graubünden	Government Council	Explanations of the Government Council on the cantonal votes of March 3, 2013	Voting brochure	03-03-2013		
	Government Council	Report and proposal concerning the revision of the electoral law	Report and request	08-24-2020	429 – 520	Heft 8 / 2020-2021
	Cantonal Council	Revision of electoral law, parliamentary debate	Verbatim minute	02-15-2021	829 – 855	1st reading
	Cantonal Council	Revision of electoral law, parliamentary debate	Verbatim minute	02-16-2021	856 – 882	2nd reading
Obwalden	Government Council	Government Council decision of October 23, 2012	Resolution	10-23-2012		Nr. 144
	Government Council	Government Council decision of September 22, 2014	Resolution	09-22-2014		Nr. 107
	Parliamentary Query	Query by Max Rötheli & co-signatories to the Government Council regarding the electoral system for the cantonal parliamentary elections	Written query	05-28-2015		Nr. 55.15.01
	Government Council	Response letter of the Government Council to the query of Max Rötheli and co-signers	Response letter	08-18-2015		Nr. 50
	Parliamentary Motion	Parliamentary motion by Max Rötheli & others requesting the government to provide the legal basis for the introduction biproportional apportionment	Written request	10-22-2020		52.20.04
	Government Council	Response letter of the Government Council to the motion of Max Rötheli and co-signers	Response letter	01-12-2021		Nr. 261
	Cantonal Council	Plenary debate on the parliamentary motion requesting the government to provide the legal basis for the introduction biproportional apportionment	Verbatim minute	01-28-2021	210 – 216	1st (and only) reading 52.20.04
	Parliamentary Motion	Parliamentary motion by the CVP faction requesting the government to provide the legal basis for the introduction of an electoral system mixing MR and PR elements	Written request	01-28-2021		52.21.01
	Government Council	Response letter of the Government Council to the motion submitted by the CVP faction	Response letter	04-27-2021		NR. 406
	Cantonal Council	Plenary debate on the parliamentary motion requesting the government to provide the legal basis for the introduction of a mixed electoral system	Verbatim minute		340 – 347	1st (and only) reading 52.21.01
	Parliamentary Motion	Joint parliamentary motion submitted by SP, SVP and CSP requesting the government to provide the legal basis for the introduction biproportional apportionment	Written request	05-26-2023		52.23.05
Bern	Parliamentary Postulate	Parliamentary postulate submitted by Daniel Kast, Ruedi Löffel, and 16 co-signers in which the government was instructed to examine the use of the biproportional method	Written request	01-22-2007		P 029-2007
	Parliamentary Motion	Parliamentary motion submitted by Daniel Kast, Ruedi Löffel, and 17 co-signers in which the government was instructed to implement the biproportional method	Written request	06-12-2007		M 175-2007
	Parliamentary Motion	Parliamentary motion submitted by Daniel Steiner, Ruedi Löffel, and 8 co-signers in which the government was instructed to implement the biproportional method	Written request	06-04-2014		M 127-2014
	Parliamentary Motion	Parliamentary motion submitted by Ruedi Löffel together with 6 others and 45 co-signers in which the government was instructed to implement the biproportional method	Written request	12-10-2019		M 310-2019
Thurgau	Parliamentary Motion	Parliamentary motion submitted by Anita Dähler, Matthias Müller, and 25 co-signers requesting the government to provide the legal basis for the introduction of biproportional apportionment	Written request	03-06-2008		04/MO 42/430
	Popular Initiative	A broad committee successfully submits the initiative "Fair electoral system for the elections to the Grand Council", calling for the introduction of biproportional apportionment	Initiative text	03-01-2010		08/VI 7/209
	Cantonal Council	Plenary debate on the popular initiative calling for the introduction of biproportionality	Verbatim minute	11-24-2010	23 – 36	08/VI 7/209
	Government Council	Explanations of the Government Council on the cantonal votes of May 15, 2011	Voting brochure	02-22-2011	22 – 31	

Lucerne	Government Council	Planning Report of the Government Council to the Grand Council on the division of the of the canton into electoral districts and in divisions for the decentralized performance of tasks	Report	08-22-2006		B 158
	Parliamentary Motion	Requesting the introduction of biproportional apportionment	Written request	01-21-2008		M 118
	Government Council	Response letter of the Government Council to the parliamentary motion M 118	Response letter	08-19-2008		RRB-Nr. 916
	Government Council	Explanations of the Government Council on the referendum launched by the SVP against the reorganization of the district structure for the cantonal parliamentary elections	Voting brochure	09-26-2010		
Fribourg	Government Council	Report of the State Council to the Grand Council on the amendment of the electoral law	Report and request	03-11-2014		2013-DIAF-73
	Cantonal Council	Revision of electoral law, parliamentary debate	Verbatim minute	09-14-2014		2013-DIAF-73
	Cantonal Council	Revision of electoral law, parliamentary debate	Verbatim minute	08-10-2014		2013-DIAF-73
Basel Stadt	Parliamentary Motion	Requesting the introduction of biproportional apportionment	Written request	01-13-2010		Nr. 09.5367.01
	Special Commission	Report of the Special Commission for the attention of the Grand Council	Report and request	05-11-2011		Nr. 09.1175.02
Basel Landschaft	Parliamentary Interpellation	Requesting the assessment of the biproportional method	Written request	01-15-2015		2015/023
	Government Council	Response letter to the parliamentary motion 2015-023	Response letter	04-19-2016		2015/023
	Procedural Postulate	Requesting the appointment of a special commission to review the current electoral law	Written request	03-21-2019		2019/216
	Cantonal Council	Parliamentary debate on the postulate 2019/216	Verbatim minute	10-17-2019		2019/216
	Parl. Executive Board	Report on the Cantonal Council's instruction to examine alternative electoral models	Report and request	07-01-2022		2019/733
	Cantonal Council	Parliamentary debate on the Parl. Executive Board's request to instruct the Government Council with drafting a proposal for the adoption of biproportional apportionment	Verbatim minute	09-15-2022		2019/733
Solothurn	Parliamentary Interpellation	Requesting the assessment of the biproportional method	Written request	12-14-2011		I 228/2011
	Parliamentary Motion	Requesting the implementation of biproportional apportionment	Written request	01-25-2012		A 012/2012
	Government Council	Response letter to the parliamentary interpellation I 228/2011	Response letter	02-28-2012		2012/441
	Government Council	Response letter to the parliamentary motion A 012/2012	Response letter	02-28-2012		2012/440
	Cantonal Council	Revision of electoral law, parliamentary debate	Verbatim minute	03-28-2012	263 – 277	2nd session, 5th meeting

	Zurich	Aargau	Schaffhausen	Nidwalden	Zug	Schwyz	Valais	Uri	Graubünden
Year of enactment	2005	2008	2008	2013	2013	2015	2017	2019	2021
Year of 1st application	2007	2009	2008	2014	2014	2016	2017	2020	2022
Number of elections since reform	5	4	4	3	3	2	2	1	1
Level of initiation	Executive	Legislative	Executive	Executive	Executive	Plebiscitary	Executive	Executive	Executive
Legal ground(s) of the reform	Law	Law & Constitution	Law & Constitution	Law	Law & Constitution	Law & Constitution	Decree & Law	Law & Constitution	Law & Constitution
Reform required a popular referendum?	No	Yes	Yes	No	Yes	Yes	No	Yes	Yes
Challenged by referendum?	No	No	No	Yes	No	No	No	No	No
Statutory basis of the biproportional apportionment method	Gesetz über die politischen Rechte: § 101 – 105	Gesetz über die Wahl des Grossen Rates: § 13 – 14b	Gesetz über die vom Volke vorzunehmenden Abstimmungen und Wahlen sowie über die Ausübung der Volksrechte Art. 2b – 2e Verfassung des Kantons Schaffhausen Art. 25(2)	Gesetz über die Verhältniswahl des Landrates Art. 22 – 25	Gesetz über die Wahlen und Abstimmungen § 52a – 52f Verfassung des Kantons Zug § 38(4)	Kantonsratswahlgesetz § 16 – 19	Gesetz über die politischen Rechte Art. 154 – 157	Gesetz über die Verhältniswahl des Landrats Art. 22 – 25 Verfassung des Kantons Uri Art. 88(1)	Gesetz über die Wahl des Grossen Rates Art. 25 – 29
Prior reform attempts?	No	No	No	No	Yes (2006)	No	Yes (2015)	No	No
Statutory quorum?	Yes: § 102(3)	Yes: § 13(2)	No	No	Yes: § 52c (3)	Yes: § 16(3)	Yes: Art. 154	No	Yes: Art. 26(1)
Global quorum	3%	3%	-	-	3%	1%	-	-	3%
Local quorum	5%	5%	-	-	5%	-	8%	-	-
Apparentements allowed?	No: § 93	No: § 8	No: Art. 2b (3)	No: Art. 13(2)	No: § 52c (4)	No: § 9(1)	No: Art. 149	No: Art. 12(1)	No: Art. 15(1)
Majority clause?	No	No	Yes: Art 2d (1bis)	No	Yes: § 52f (1a)	No	No	No	Yes: Art. 28(2)
Level of super-apportionment	canton	canton	canton	canton	canton	canton	6 upper-tier districts	canton (PR municipalities)	canton
Level of sub-apportionment	18 electoral districts	11 political districts	6 electoral districts	11 municipalities	11 municipalities	30 municipalities	14 lower-tier districts	4 political districts (PR municipalities)	39 electoral districts
Alliance pattern during reform process	Cross-party consensus (pro)	Center-left (pro) vs. right	SVP (contra) vs. others	SVP (contra) vs. others	Left (pro) vs. right	SVP (contra) vs. others	Center-right (pro) vs. others	Cross-party consensus (pro)	
Comments	The global quorum of 3% was added to the preexisting 5% local quorum in 2021.	Quorums were added after a referendum in 2011.			Only canton that specified the new method directly exclusively in the constitution. First canton introducing a majoritarian 'winner-take-one' clause.	Only canton in which the new method was demanded by popular initiative.	For reasons of urgency, the new method was introduced by decree. An earlier attempt to reform the electoral system in 2015 failed in a popular vote.	Mixed electoral system (MR in districts with less than 5 seats and PR in all others).	Only canton to switch directly from MR to the biproportional method.

Table 41: Synopsis of the cantonal reform processes and the reform designs (as of January 2023).

Table 42: Codebook data set 2 (Lüscher 2024b).

Variable label	Explanation	Variable type / Operationalization	Source		
			PD[100]	QD[101]	SL[102]
canton	Cantonal political system under review (26 in total)	nominal			x
year	Year under review (starting from the first cantonal election after 1976 and ending in 2023)	discrete			x
elec_dummy	Dummy variable indicating whether or whether not an election was held in the given year.	binary 0: absence election 1: presence of election	x		x
time_var	Time variable capturing the years before and after the reform (if there was one).	discrete			x
ref_dummy	Reform dummy indicating whether, for a given year, biproportional apportionment applied or not.	binary 0: absence of reform 1: presence of reform			x
parl_size	Size of the cantonal parliament in terms of seats	discrete		x	x
elec_system	Electoral system in use	categorial 1: majority/plurality 2: mixed electoral system 3: proportional system			x
app_method	Apportionment method in cantons/districts applying PR	categorial 1: Hare-Niemeyer 2: Hagenbach-Bischoff 3: Sainte-Laguë 4: Pukelsheim			x
distr_num	Number of electoral districts	discrete			x
distr_mean	Average district size (canton-wide)	continuous			x
distr_median	Median district size (canton-wide)	continuous			x
distr_min	Minimum district size	discrete			x
distr_max	Maximum district size	discrete			x
distr_range	Numeric range between the maximum and the minimum district size	discrete			x
distr_sd	Standard deviation of mean district sizes	continuous			x
distr_num_small_abs	Sum of districts with less than nine seats	discrete			x
distr_num_small_rel	Share of districts with less than nine seats	discrete			x
distr_small_abs	Sum of seats from districts with less than nine seats	discrete			x
distr_small_rel	Share of seats from districts with less than nine seats	discrete			x
lists_num	Number of distinct electoral lists submitted	discrete			x
cand_num_ce	Number of candidacies (cantonal elections)	discrete			x
cand_seat_ce	Number of candidacies per parliamentary seat (cantonal elections)	continuous			x

[100] Source: "Patterns of Democracy" data set.
[101] Source: "Quality of Democracy" data set.
[102] Source: Own data collection (either complementary or completely).

cand_num_ne	Number of candidacies (national elections)	discrete			x
cand_seat_ne	Number of candidacies per parliamentary seat (national elections)	continuous			x
lsq_index	Least-squares index of disproportionality of electoral systems	continuous	x		x
enp	Effective number of legislative parties applying the Laakso-Taagepera-Index	continuous		x	x
turnout_ce	Turnout in the cantonal election	continuous	x		x
turnout_ne	The cantonal turnout in the national elections closest to the cantonal election	continuous			x
turnout_cv	Turnout in the cantonal votes (annual averages; in case of multiple simultaneous votes, each vote was treated separately)	continuous	x		x
seats_ne	Number of National Councils eats attributed to the canton	Discrete			x
eff_threshold	Value indicating the effective electoral threshold at the cantonal level.	continuous			x

Covariate	Coefficient	Standard Error	*p*-value
ref_dummy	-2.048	0.401	3.236×10^{-7}
Enp	0.264	0.308	0.391
distr_range	0.030	0.026	0.240
cand_num_ce	2.241×10^{-3}	1.602×10^{-3}	0.162

Table 43: Regression coefficients and statistics for the outcome variable LSq-index. "ref_dummy" represents the treatment indicator.

Covariate	Coefficient	Standard Error	*p*-value
ref_dummy	-0.118	0.392	0.763
lsq_index	0.010	0.045	0.818
distr_range	-0.013	0.015	0.403
turnout_ce	7.901×10^{-5}	0.010	0.995
cand_num_ce	2.678×10^{-3}	1.071×10^{-3}	0.012

Table 44: Regression coefficients and statistics for the outcome variable ENP. "ref_dummy" represents the treatment indicator.

Covariate	Coefficient	Standard Error	*p*-value
ref_dummy	-4.455	3.951	0.260
turnout_ne	0.630	0.086	3.315×10^{-13}
cand_num_ce	-3.354×10^{-3}	4.186×10^{-3}	0.423
enp	-0.991	0.558	0.076
eff_threshold	-0.740	0.656	0.259

Table 45: Regression coefficients and statistics for the outcome variable electoral turnout. "ref_dummy" represents the treatment indicator.

Covariate	Coef.	S.E.	*p*-value
ref_dummy	-162.6	144.9	0.262
enp	63.031	14.478	1.338×10^{-5}
turnout_ce	-0.239	1.439	0.868
distr_range	1.928	2.700	0.475
lsq_index	10.114	4.416	0.022
cand_num_ne	1.195	0.225	1.109×10^{-7}
eff_threshold	-16.861	16.111	0.295

Covariate	Coef.	S.E.	*p*-value
ref_dummy	-6.71	3.614	0.063
enp	0.977	0.510	0.055
turnout_ce	-1.11×10^{-3}	0.054	0.985
distr_range	0.022	0.058	0.711
lsq_index	0.180	0.138	0.194
cand_num_ne	0.022	7.707×10^{-3}	2.342×10^{-3}
eff_threshold	-0.120	0.268	0.655

Table 46: Regression coefficients and statistics for the outcome variables number of candidacies (left-side table) and number of electoral lists (right-side table). "ref_dummy" represents the treatment indicator.

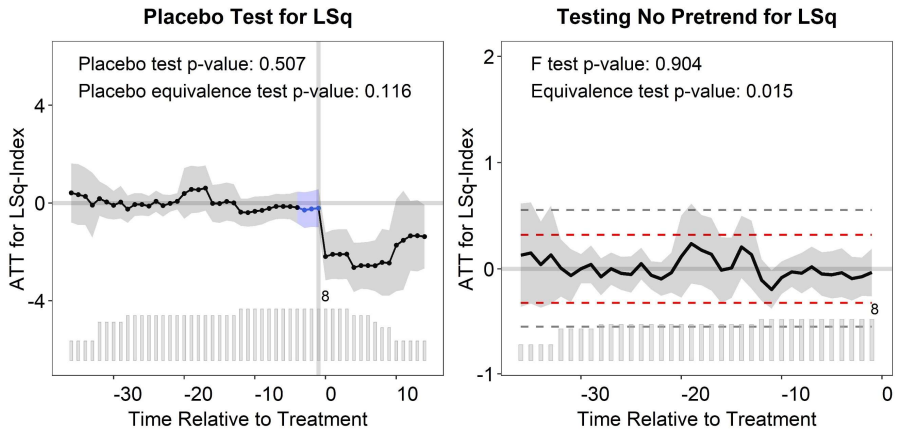


Figure 54: Placebo and no pretrend diagnostic for the LSq model. The left panel shows the results from a placebo test using the treated observations in three pretreatment periods as a placebo. The right panel shows the results of an equivalence test for no pretrend, in which the red and grey dashed lines mark the equivalence range and the minimum range, respectively.

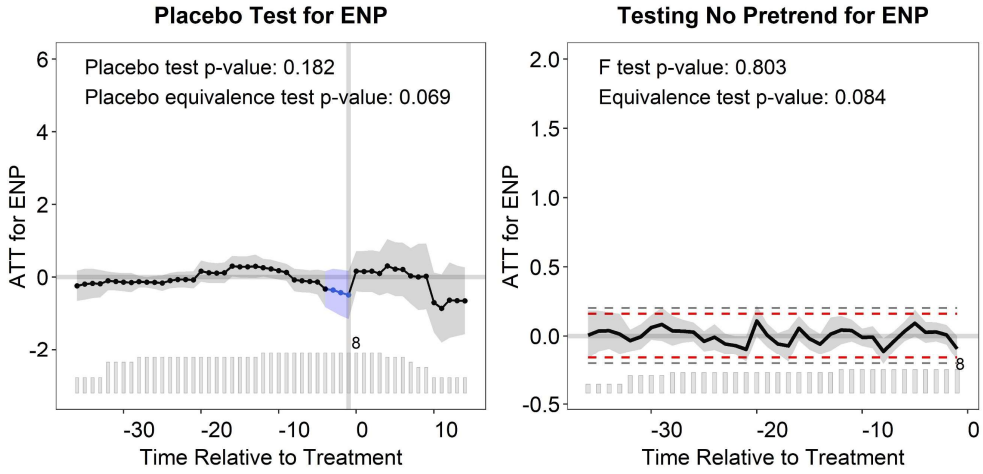


Figure 55: Placebo and no pretrend diagnostic for the ENP model.

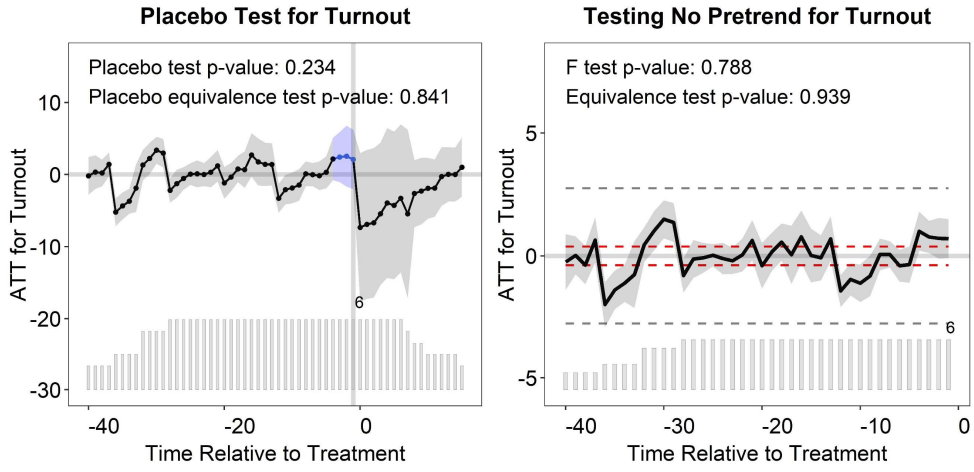


Figure 56: Placebo and no pretrend diagnostic for the turnout model.

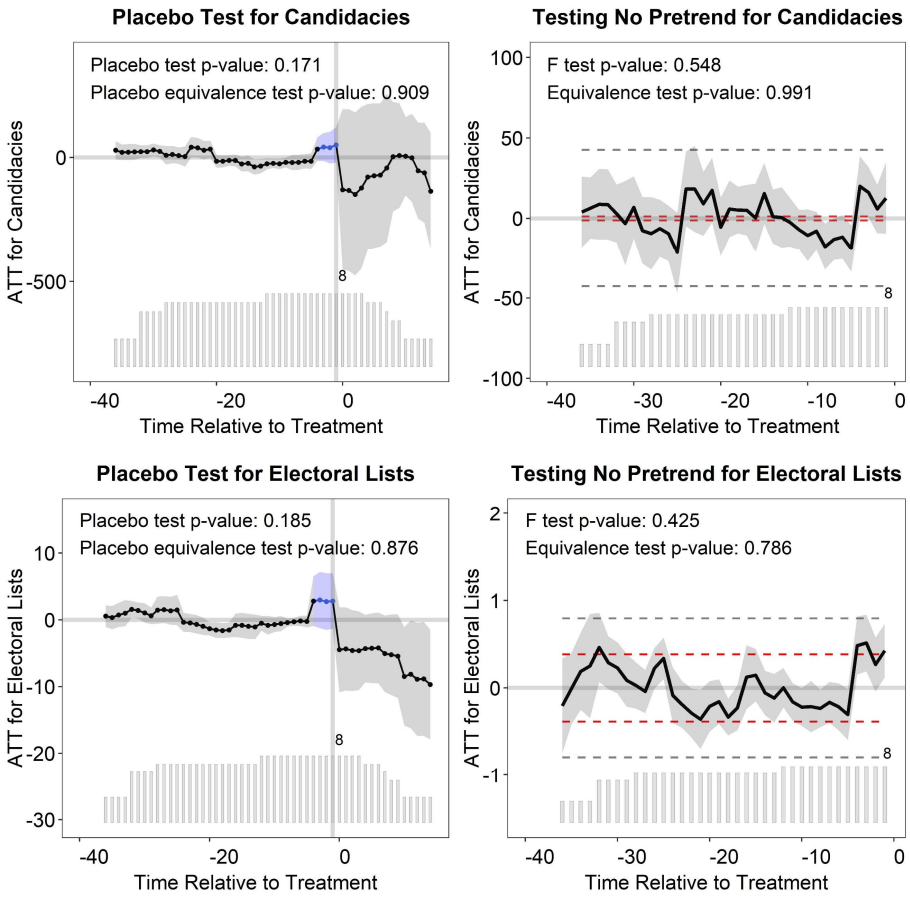


Figure 57: Placebo and no pretrend diagnostic for the models on electoral competition.